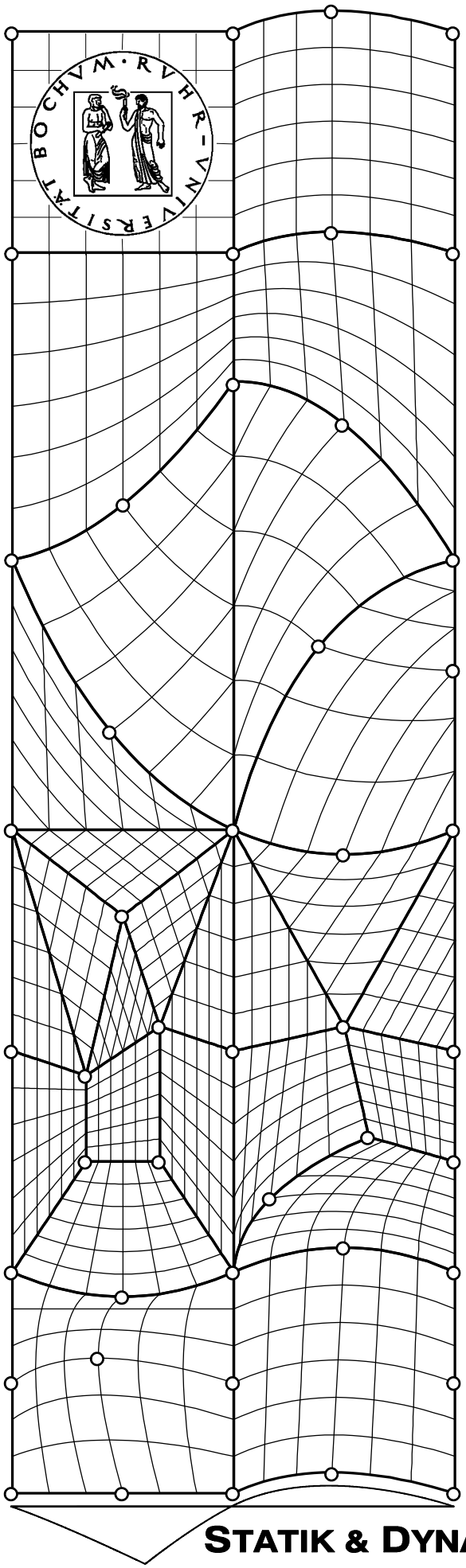

Lecture Notes

Finite Element Methods in Linear Structural Mechanics

Dr.-Ing. habil. D. Kuhl
Univ. Prof. Dr. techn. G. Meschke

May 2005

Ruhr University Bochum
Institute for Structural Mechanics

STATIK & DYNAMIK

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Preface

These lecture notes, which in actual fact are an English translation of the German lecture notes 'Finite Elemente Methoden I' of the diploma study course, were created in the context of the lecture 'Finite Element Methods I' which was first held in this form during the winter term 1998/1999. 'Finite Element Methods in Linear Structural Mechanics' thus represents the teachings of finite element methods in the area of linear structural mechanics with the focus on showing of possibilities and limits of the numerical method as well as the development of isoparametric finite elements. These notes are to support the students in following up the lecture and to prepare them for the exam. They cannot possibly substitute the lecture or the exercise entities. In addition to the lecture and the notes, mathematical programmes for deepening the lecture contents are available at the homepage of the Institute for Structural Mechanics `http://www.sd.ruhr-uni-bochum.de/`.

Here, the authors would like to thank Mr. Jörn Mosler and Mr. Stefan Jox for the excellent conduction of the theoretical and practical exercise entities accompanying the lecture 'Finite Element Methods I'. Moreover, the authors give their thanks to Ms. Barbara Kalkhoff, graphical designer, for the high quality drawings as well as to Ms. Monika Rotthaus, Ms. Wiebke Breil, Ms. Sandra Krimpmann, Ms. Julia Mergenheim, Mr. Christian Becker, Mr. Alexander Beer, Mr. Sönke Carstens and Mr. Janosch Stascheit for their indispensable efforts in creating these lecture notes.

Last but not least the authors would like to thank Mr. Ivaylo Vladimirov, Mr. Hrvoje Vucemilovic and Ms. Amelie Gray who helped to translate the notes into the English language. At the same time we would like to excuse the fact that the description of the drawings are in German. Nevertheless, we believe that the meaning becomes clear. The authors are continually working on improving the lecture notes. Therefore, please feel free to communicate your comments, ideas and corrections.

For all students who intend to continue with the lecture 'Finite Element Methods II' with the emphasis on non-linear structural mechanics, the lecture notes are complemented by the corresponding chapters 5 to 7 as well as by the indication of further literature. The chapters concerning the non-linear finite element methods are also available in the form of lecture notes ('Finite Elemente Methoden II', 3. edition, October 2002, in German language) at the Institute for Structural Mechanics, IA 6/127.

Bochum, May 2005 Günther Meschke and Detlef Kuhl

Chapter 1

Fundamentals of Linear Structural Mechanics

The purpose of this chapter is to derive the *Principle of Virtual Work* as fundamental for the formulation of the Finite Element Method. The basis of the so-called weak formulation of the Initial Boundary Value Problem of elastodynamics is characterized by the description of the deformation of a material body by means of the displacement field and the corresponding strains (Kinematics), the force equilibrium of stresses on a differential volume element (Kinetics), the formulation of geometric and static boundary conditions and the constitutive relationship between stresses and strains (Material Law).

The primary variables of elastostatics are the displacements, since the stresses can be described by means of the Constitutive Law as a function of the stresses. In case of structures in motion, the primary variables along with their second time derivatives, the accelerations, are considered. The change from the strong form of the partial differential equation and its boundary conditions to the weak form gives in the end the *Principle of Virtual Work*. In the weak form the geometric boundary conditions are strongly satisfied, whereas the balance of momentum and the static boundary conditions must only be satisfied in an integral form. This integral formulation hence allows the exact solution of the Initial Boundary Value Problem to be replaced by an approximated solution, which satisfies the integral but not the local form of the corresponding differential equation. This shows the significance of the weak formulation of the fundamental equations of structural mechanics for the design of approximation methods in general, and of the Finite Elemente Methode in particular.

The present chapter deals with the kinematic and kinetic equations of three-dimensional continua. The formulation of a linear elastic material model together with the addition of the necessary initial and boundary conditions makes possible the formulation and characterization of the Initial Boundary Value Problem of structural mechanics, which afterwards is transformed into the weak form.

Recommended additional literature: Altenbach & Altenbach [38], Başar & Weichert [40], Betten [43], de Boer [44], Bonet & Wood [46], Eriksson et al. [52], Flügge [53], Gross [54], Hjelmstad [56], Leipholz [60], Malvern [63], Marsden & Hughes [65], Smith [73], Stein & Barthold [74], Truesdell & Noll [78]

1.1 Continuum Kinematics

Continuum kinematics describes the geometry of a body, its motion in space as well as its deformation during motion. A basis for this description is the consideration of a body as an ensemble of material points as well as the characterization of their initial and current position by means of the position and displacement vectors. By considering the immediate vicinity of material points one finally gets to the concept of strains, which describe the deformation of a material body. First, the strains are described without further assumptions in a non-linear form and afterwards they are reduced to a linear description according to the deformation theory of small displacements or strains. For the description of non-linear kinematics the *material* or LAGRANGE-*ian approach* will be used, according to which the state of a point is defined as a function of its initial position and time.

1.1.1 Displacement Field

The motion of continuum in three-dimensional space is completely defined by the *position vector* of a material point $\boldsymbol{X} = [X_1\ X_2\ X_3]^T$ and its change of position at deformation under arbitrary internal or external influence. This motion of the material point from the undeformed to the deformed state is described by means of the *displacement vector* $\boldsymbol{u} = [u_1\ u_2\ u_3]^T$ as a function of the position of the material point (Fig. 1.1). The components of the position and displacement vectors are defined in the cartesian basis with the orthogonal unit vectors, *base vectors* or simply *bases* $\boldsymbol{e}_i$ for $i \in \{1, 2, 3\}$.Thus, the vectors can be described by their components and the base vectors as follows:

$$\begin{aligned} X_i &= \boldsymbol{e}_i \cdot \boldsymbol{X} & \qquad \boldsymbol{X} &= \boldsymbol{e}_i\ X_i \\ u_i(\boldsymbol{X}) &= \boldsymbol{e}_i \cdot \boldsymbol{u}(\boldsymbol{X}) & \qquad \boldsymbol{u}(\boldsymbol{X}) &= \boldsymbol{e}_i\ u_i(\boldsymbol{X}) \end{aligned} \tag{1.1}$$

where the dot $\cdot$ represents the scalar product of two vectors or tensors of the same order. Furthermore, EINSTEIN's summation convention is assumed to hold. The current position of the material point under consideration at time t is given by the position vector

$$\boldsymbol{x}(\boldsymbol{X}, t) = \boldsymbol{X} + \boldsymbol{u}(\boldsymbol{X}, t) \qquad \boldsymbol{x}(\boldsymbol{X}, 0) = \boldsymbol{X} \tag{1.2}$$

The LAGRANGE-ian approach is to be observed clearly here in the context of the dependence of the current position on the initial position and on time t. Here, time is of physical relevance only in dynamic considerations. In the static case, time is transformed into pseudo-time, which only serves to characterize the state of deformation. On the basis of this formulation the state and shape of the deformed body can be fully described, but an expression for the local *strains* or *elongations*, actually is not possible.

1.1.2 Definition of a Non-Linear Strain Measure

According to the explanations above, an expression for the local strains can be obtained by considering the immediate vicinity of a material point. Here, the motion of a body is described by its displacement field $\boldsymbol{u}(\boldsymbol{X}, t)$. Fig. 1.1 illustrates a material body in its undeformed and deformed states. These positions are designated as *reference configuration* and *current configuration*. The deformation of the body from the reference to the current configuration is described in general by means of the time-dependent mapping $\varphi(\boldsymbol{X}, t)$ of all particles of the body. The displacement vector of a point with the coordinate $\boldsymbol{X}$ is given by Eq. (1.2) as the difference

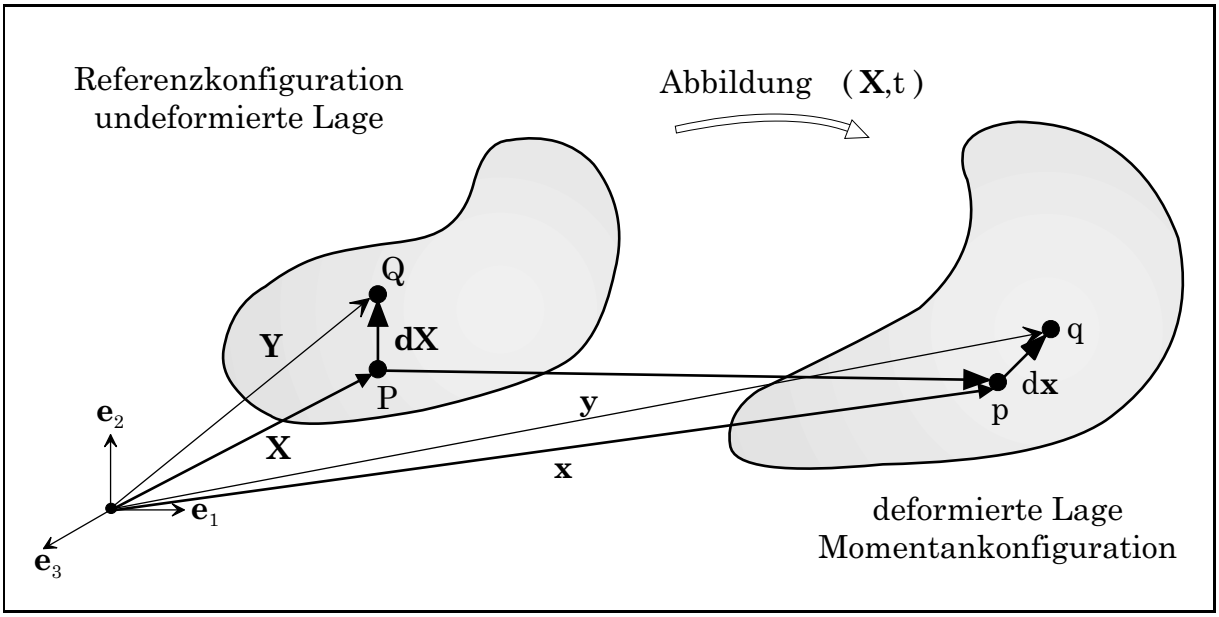


Figure 1.1: Undeformed and deformed configurations of a material body

between its deformed and undeformed positions

$$\boldsymbol{u}(\boldsymbol{X},t) = \varphi(\boldsymbol{X},t) - \varphi(\boldsymbol{X},0) = \boldsymbol{x}(\boldsymbol{X},t) - \boldsymbol{X} \tag{1.3}$$

$\boldsymbol{x}(\boldsymbol{X},t) = \varphi(\boldsymbol{X},t)$ is the current state of a particle under consideration in the deformed body, characterized by its position in the reference configuration $\boldsymbol{X}$ and by the mapping of the position in the current configuration. The behaviour of the immediate vicinity of a material point according to the mapping $\varphi(\boldsymbol{X},t)$ can be observed by means of a differential line element $d\boldsymbol{X}$. This line element is defined by the connection between two points P and Q at a differential distance from one another, expressed by the differential vector $d\boldsymbol{X} = \boldsymbol{X}_Q - \boldsymbol{X}$ in the reference configuration, and between the points p and q respectively, described by the vector $d\boldsymbol{x} = \boldsymbol{x}_q - \boldsymbol{x}$ in the current configuration of the body. By a Taylor series expansion of the current configuration $\varphi(\boldsymbol{X},t)$ with respect to the reference configuration $\boldsymbol{X}$, one obtains the differentially distant point $\boldsymbol{y} = \boldsymbol{x}(\boldsymbol{X},t) + d\boldsymbol{x}(\boldsymbol{X},t)$ on the deformed configuration.

$$\boldsymbol{x}(\boldsymbol{X},t) + d\boldsymbol{x}(\boldsymbol{X},t) = \varphi(\boldsymbol{X},t) + \frac{\partial\varphi(\boldsymbol{X},t)}{\partial\boldsymbol{X}}\,(\boldsymbol{X}_Q - \boldsymbol{X}) + \ldots = \boldsymbol{x}(\boldsymbol{X},t) + \frac{\partial\boldsymbol{x}(\boldsymbol{X},t)}{\partial\boldsymbol{X}}\,d\boldsymbol{X} + \ldots \tag{1.4}$$

By truncating the endless series after the linear term and by using Eq. (1.3) in the above equation, the mapping or transformation of the differential line element $d\boldsymbol{X}$ of the reference configuration to the current line element $d\boldsymbol{x}$ can be obtained .

$$d\boldsymbol{x} = \frac{\partial\boldsymbol{x}(\boldsymbol{X},t)}{\partial\boldsymbol{X}}\,d\boldsymbol{X} = \frac{\partial}{\partial\boldsymbol{X}}\left(\boldsymbol{u}(\boldsymbol{X},t) + \boldsymbol{X}\right)\,d\boldsymbol{X} = \left(\frac{\partial\boldsymbol{u}(\boldsymbol{X},t)}{\partial\boldsymbol{X}} + \mathbf{1}\right)\,d\boldsymbol{X} \tag{1.5}$$

Here, $\mathbf{1}$ is the second order unit tensor, the components of which represent the KRONECKER symbol δ_{ij},

$$\mathbf{1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \delta_{ij}\,\boldsymbol{e}_i \otimes \boldsymbol{e}_j \qquad\qquad \delta_{ij} = \begin{cases} = 1 & \text{für} \quad i = j \\ = 0 & \text{für} \quad i \neq j \end{cases} \tag{1.6}$$

and the first term defines the derivative of the displacement vector with respect to the position vector of the reference configuration. This term is designated as the *Displacement gradient* $\nabla \boldsymbol{u}$.

$$\frac{\partial \boldsymbol{u}(\boldsymbol{X},t)}{\partial \boldsymbol{X}} = \nabla \boldsymbol{u}(\boldsymbol{X},t) = \begin{bmatrix} \frac{\partial u_1}{\partial X_1} & \frac{\partial u_1}{\partial X_2} & \frac{\partial u_1}{\partial X_3} \\ \frac{\partial u_2}{\partial X_1} & \frac{\partial u_2}{\partial X_2} & \frac{\partial u_2}{\partial X_3} \\ \frac{\partial u_3}{\partial X_1} & \frac{\partial u_3}{\partial X_2} & \frac{\partial u_3}{\partial X_3} \end{bmatrix} = \begin{bmatrix} u_{1,1} & u_{1,2} & u_{1,3} \\ u_{2,1} & u_{2,2} & u_{2,3} \\ u_{3,1} & u_{3,2} & u_{3,3} \end{bmatrix} = u_{i,j}\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j \tag{1.7}$$

As a measure for the change in length of a line element $d\boldsymbol{X}$ during deformation, the square of the length $dS^2 = \|d\boldsymbol{X}\|^2 = d\boldsymbol{X} \cdot d\boldsymbol{X}$, or $ds^2 = \|d\boldsymbol{x}\|^2$, of the line elements is observed in the reference configuration, and in the current configuration, respectively.

$$\begin{aligned} ds^2 &= d\boldsymbol{x} \cdot d\boldsymbol{x} = (d\boldsymbol{X} + \nabla \boldsymbol{u} \cdot d\boldsymbol{X}) \cdot (d\boldsymbol{X} + \nabla \boldsymbol{u} \cdot d\boldsymbol{X}) \\ &= d\boldsymbol{X} \cdot d\boldsymbol{X} + d\boldsymbol{X} \cdot (\nabla \boldsymbol{u} \cdot d\boldsymbol{X}) + (\nabla \boldsymbol{u} \cdot d\boldsymbol{X}) \cdot d\boldsymbol{X} + (\nabla \boldsymbol{u} \cdot d\boldsymbol{X}) \cdot (\nabla \boldsymbol{u} \cdot d\boldsymbol{X}) \\ &= d\boldsymbol{X} \cdot d\boldsymbol{X} + d\boldsymbol{X} \cdot \nabla \boldsymbol{u} \cdot d\boldsymbol{X} + d\boldsymbol{X} \cdot \nabla^T \boldsymbol{u} \cdot d\boldsymbol{X} + d\boldsymbol{X} \cdot \nabla^T \boldsymbol{u} \cdot \nabla \boldsymbol{u} \cdot d\boldsymbol{X} \end{aligned} \tag{1.8}$$

For the generation of the above equation the identities $(\nabla \boldsymbol{u} \cdot d\boldsymbol{X}) \cdot d\boldsymbol{X} = d\boldsymbol{X} \cdot \nabla^T \boldsymbol{u} \cdot d\boldsymbol{X}$ and $(\nabla \boldsymbol{u} \cdot d\boldsymbol{X}) \cdot (\nabla \boldsymbol{u} \cdot d\boldsymbol{X}) = d\boldsymbol{X} \cdot \nabla^T \boldsymbol{u} \cdot \nabla \boldsymbol{u} \cdot d\boldsymbol{X}$ were used. After some additional simplifications and taking into account the definition of dS^2, half of the relative change in length can be obtained.

$$\frac{ds^2 - dS^2}{2} = d\boldsymbol{X} \cdot \frac{1}{2} \left[\nabla \boldsymbol{u} + \nabla^T \boldsymbol{u} + \nabla^T \boldsymbol{u} \cdot \nabla \boldsymbol{u} \right] \cdot d\boldsymbol{X} \tag{1.9}$$

The middle tensor in Equation (1.9) represents the strain state of continuum. It defines the *Green Lagrange Strain Tensor* $\boldsymbol{E}$.

$$\boldsymbol{E} = \frac{1}{2} \left[\nabla \boldsymbol{u} + \nabla^T \boldsymbol{u} + \nabla^T \boldsymbol{u} \cdot \nabla \boldsymbol{u} \right] \tag{1.10}$$

Through this definition of a strain measure it is guaranteed that the reference configuration ($\boldsymbol{u} = \boldsymbol{0}$), and the displacements of a rigid body ($\nabla \boldsymbol{u} = \boldsymbol{0}$) are free of strain. It should be noted, however, that this is not the only possible definition of a strain measure. Alternative strain measures can be found in literature (e.g. Altenbach & Altenbach [38], Betten [43] or Stein & Barthold [74]). Nevertheless, in this text the Green Lagrange strain tensor will be exclusively used in its original and linearized form. For simplification of Eq. (1.10), the displacement gradient $\nabla \boldsymbol{u}$ is decomposed into a symmetric and a skew-symmetric part.

$$\nabla \boldsymbol{u} = \nabla^{\text{sym}} \boldsymbol{u} + \nabla^{\text{skw}} \boldsymbol{u} = \frac{1}{2} \left[\nabla \boldsymbol{u} + \nabla^T \boldsymbol{u} \right] + \frac{1}{2} \left[\nabla \boldsymbol{u} - \nabla^T \boldsymbol{u} \right] \tag{1.11}$$

Based on this decomposition, the Green Lagrange strain tensor can be written in the following compact form:

$$\boxed{\boldsymbol{E} = \nabla^{\text{sym}} \boldsymbol{u} + \frac{1}{2} \nabla^T \boldsymbol{u} \cdot \nabla \boldsymbol{u} \qquad \nabla^{\text{sym}} \boldsymbol{u} = \frac{1}{2} \left[\nabla \boldsymbol{u} + \nabla^T \boldsymbol{u} \right] \qquad (1.12)}$$

The first term in this equation $\nabla^{\text{sym}} \boldsymbol{u}$ is a linear function of the diplacement gradient $\nabla \boldsymbol{u}$. In

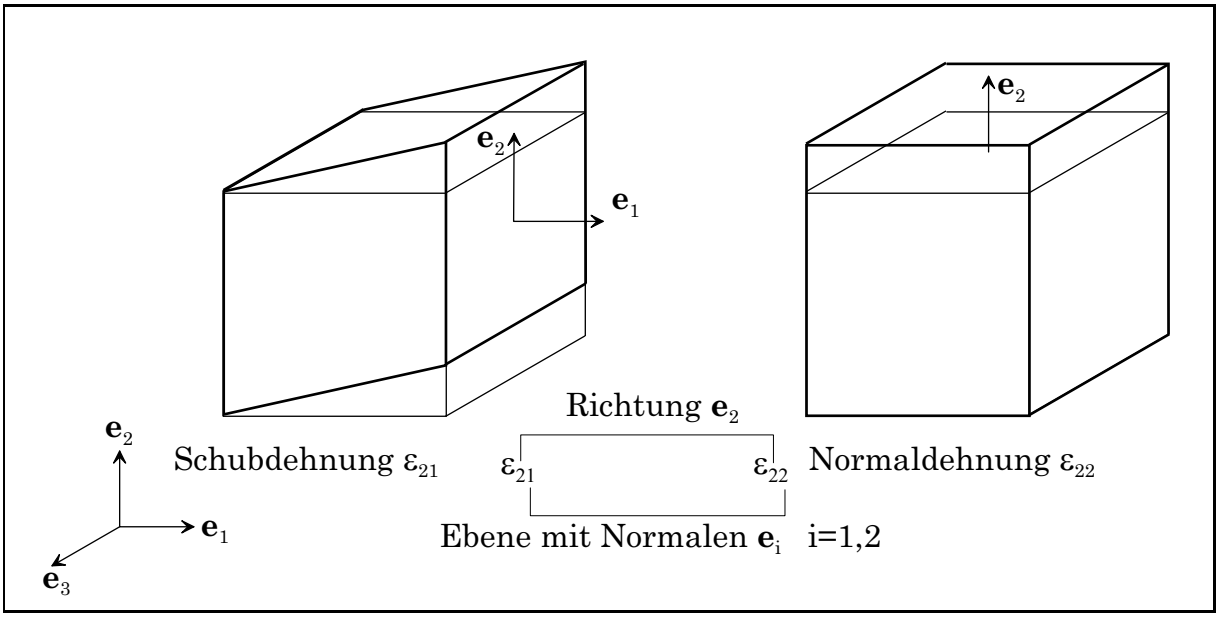


Figure 1.2: Shear and normal strains on a volume element

contrast to this, the second term $1/2\nabla^T\boldsymbol{u}\cdot\nabla\boldsymbol{u}$ is non-linear in $\nabla\boldsymbol{u}$. This non-linearity, based on the mapping of geometry from the undeformed to the deformed state, is called *geometrical non-linearity*. The non-linear term affects the strain tensor decisively only when the gradient of the displacement field is big. This can occur in slender structures like rope structures and shells or in the case of plastification or damage of materials which is of importance, for instance, in geomechanics or in the analysis of highly-loaded structural elements.

1.1.3 Definition of a Linear Strain Measure

In contrast to the previous section, the non-linear term of the strain tensor can be neglected if the deformations are very small ($1/2\ \nabla^T\boldsymbol{u}\cdot\nabla\boldsymbol{u}\approx\boldsymbol{0}$). In this case we speak of the *geometrically linear theory*, which is also known as the *theory of small strains*. The strain measure of the geometrically linear theory is thus defined by the symmetric part of the displacement gradient $\nabla\boldsymbol{u}$.

$$\boldsymbol{\varepsilon}=\nabla^{\mathrm{sym}}\boldsymbol{u} \qquad \nabla^{\mathrm{sym}}\boldsymbol{u}=\frac{1}{2}\left[\nabla\boldsymbol{u}+\nabla^T\boldsymbol{u}\right] \tag{1.13}$$

The *linear strain tensor*, which is also described as the *infinitesimal strain tensor*, is denoted with $\boldsymbol{\varepsilon}$ to represent the theory of small strains. The components of the symmetric strain tensor $\boldsymbol{\varepsilon}$ can be described by the definitions of the symmetric part of a second order tensor and the gradient.

$$\boldsymbol{\varepsilon}=\begin{bmatrix} u_{1,1} & \frac{1}{2}\left(u_{1,2}+u_{2,1}\right) & \frac{1}{2}\left(u_{1,3}+u_{3,1}\right)\\ \frac{1}{2}\left(u_{1,2}+u_{2,1}\right) & u_{2,2} & \frac{1}{2}\left(u_{2,3}+u_{3,2}\right)\\ \frac{1}{2}\left(u_{1,3}+u_{3,1}\right) & \frac{1}{2}\left(u_{2,3}+u_{3,2}\right) & u_{3,3}\end{bmatrix}=\frac{1}{2}\left(u_{i,j}+u_{j,i}\right)\ \boldsymbol{e}_i\otimes\boldsymbol{e}_j \tag{1.14}$$

The definition of the strain tensor components ε_{ij} is illustrated in Fig. 1.2, with $\varepsilon_{ij}=\varepsilon_{ji}$ being valid due to the symmetry of the strain tensor. In the chosen definition, the first index

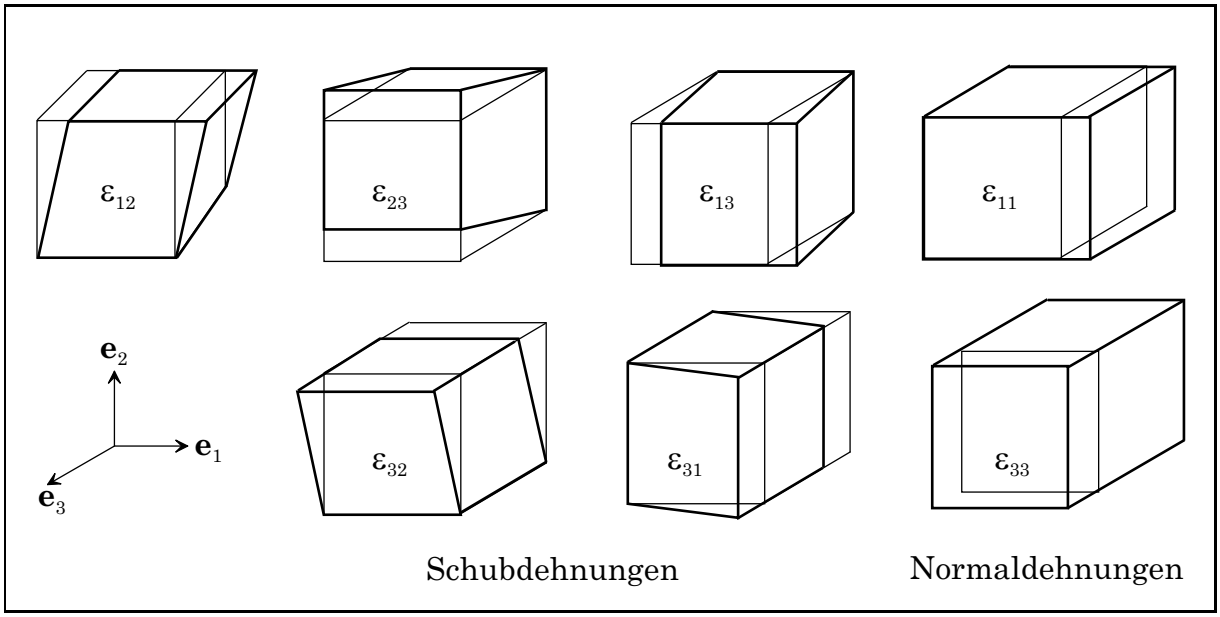


Figure 1.3: Strain components on a volume element

characterizes the strain direction. The second index characterizes the normal to the distorted surface of the representative volume element. In the context of the Finite Element Method, the strain state is characterized by means of the *strain vector* $\boldsymbol{\varepsilon}$. The strain vector defined below contains the normal strains ε_{11}, ε_{22} and ε_{33}, as well as the three differing shear strains ε_{12}, ε_{23} and ε_{13}.

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{11} & \varepsilon_{22} & \varepsilon_{33} & 2\varepsilon_{12} & 2\varepsilon_{23} & 2\varepsilon_{13} \end{bmatrix}^T \qquad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{11} & \varepsilon_{12} & \varepsilon_{13} \\ & \varepsilon_{22} & \varepsilon_{23} \\ \text{sym} & & \varepsilon_{33} \end{bmatrix} \tag{1.15}$$

The construction of the strain vector from the strain tensor is shown in the left part of Eq. (1.15). Factor two, with which the shear strain components are equipped, is of special importance. By means of this factor, the formally equivalent formulation of the specific internal energy in the tensor and vector notation ($\boldsymbol{\varepsilon} \cdot \boldsymbol{\sigma} = \boldsymbol{\varepsilon} : \boldsymbol{\sigma}$) is possible in connection with the stress tensor and vector yet to be defined. A further advantage of this definition will manifest itself in the equivalence of the differential operator and the transposed differential operator in the representation of the strains and the balance of momentum (Chapter 1.2.2) by means of differential operators. The first differential operator has to be developed as a basis for the direct calculation of the strain vector from the displacement vector. The desired kinematic relation of the strain and the displacement vectors is derived from the definition of the strain components in Eq. (1.14), whereby the components of the differential operator $\mathbf{D}_\varepsilon$ represent

rules for derivatives.

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial X_1} & 0 & 0 \\ 0 & \frac{\partial}{\partial X_2} & 0 \\ 0 & 0 & \frac{\partial}{\partial X_3} \\ \frac{\partial}{\partial X_2} & \frac{\partial}{\partial X_1} & 0 \\ 0 & \frac{\partial}{\partial X_3} & \frac{\partial}{\partial X_2} \\ \frac{\partial}{\partial X_3} & 0 & \frac{\partial}{\partial X_1} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \qquad \boldsymbol{\varepsilon} = \mathbf{D}_\varepsilon \, \boldsymbol{u} \tag{1.16}$$

The validity of the differentiation model (1.16) can be tested by the calculation of separate strain components and by comparison with their definition according to (1.14). As an example, the strain components ε_{11} and ε_{12} are computed here.

$$\varepsilon_{11} = \frac{\partial}{\partial X_1} \, u_1 = u_{1,1} \qquad\qquad 2\varepsilon_{12} = \frac{\partial}{\partial X_2} \, u_1 + \frac{\partial}{\partial X_1} \, u_2 = u_{1,2} + u_{2,1} \tag{1.17}$$

1.2 Continuum Kinetics

Kinetics describes the relation between external and internal forces acting on a material body. According to the stress principle of CAUCHY, a tensor field of stresses $\boldsymbol{\sigma}$ exists in a material body as a consequence of the external forces. Together with the static and dynamic loads acting throughout the volume, these stresses form the *local balance of momentum* or the *equilibrium of forces*. The balance of momentum must be satisfied throughout the deformed configuration. In the context of the here utilized geometrically linear theory it is admitted, however, to form the equilibrium of forces for the undeformed state.

1.2.1 Cauchy's Theorem

CAUCHY's theorem is based upon the postulate of a stress vector $\boldsymbol{t}$ on an arbitrary cross section of a material body. This stress vector is defined as the ratio of the force $\Delta \boldsymbol{f}$, acting on the section and the cross-sectional area ΔA, when the area approaches zero.

$$\boldsymbol{t} = \lim_{\Delta A \to 0} \frac{\Delta \boldsymbol{f}}{\Delta A} \tag{1.18}$$

Here, the orientation of the surface is characterized by means of its normal vector $\boldsymbol{n}$. According to the *Cauchy* Lemma, the stress vector in the interior of the body as a function of the outward directed normal is balanced with the stress vector of the inward directed normal ($\boldsymbol{t}(\boldsymbol{n}) + \boldsymbol{t}(-\boldsymbol{n}) = \mathbf{0}$). The theorem of CAUCHY now demands that a tensor field $\boldsymbol{\sigma}$ related to the vector $\boldsymbol{t}$ exists, which satisfies a linear mapping as follows:

$$\boldsymbol{t}(\boldsymbol{X}, \boldsymbol{n}) = \boldsymbol{\sigma}(\boldsymbol{X}) \cdot \boldsymbol{n} \tag{1.19}$$

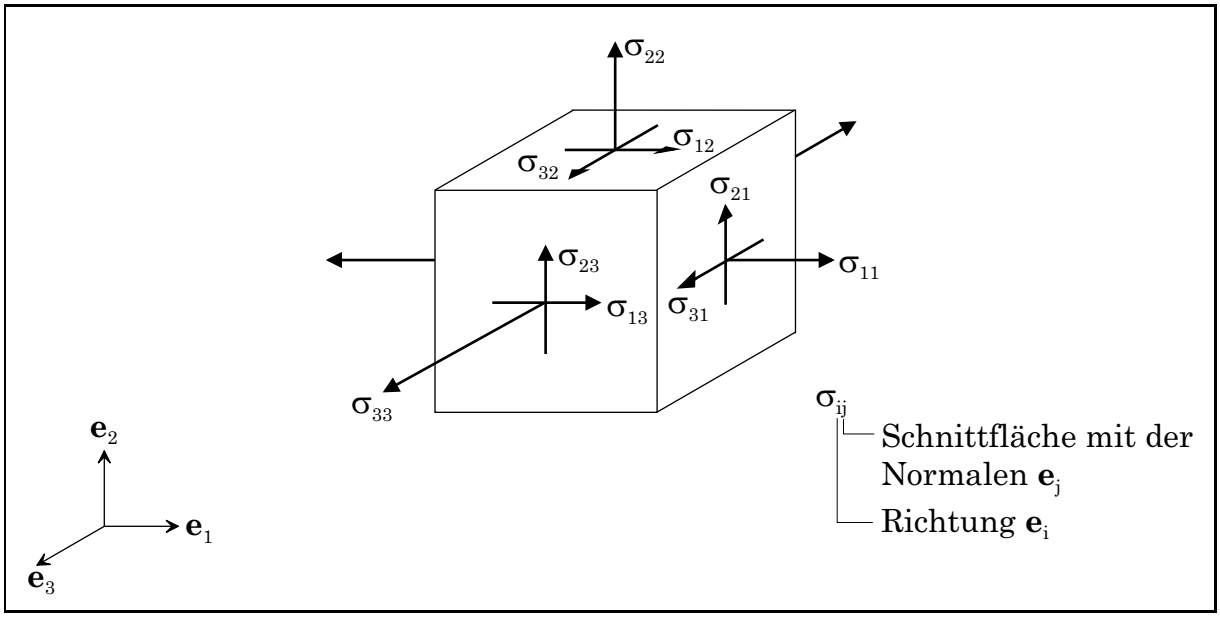


Figure 1.4: Stress components on a volume element

The so-postulated symmetric stress tensor is known as *Cauchy's stress tensor*.

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} \end{bmatrix} = \sigma_{ij}\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j \qquad \boldsymbol{\sigma} = \boldsymbol{\sigma}^T \tag{1.20}$$

The stress components σ_{ij} of the CAUCHY stress tensor are illustrated in Fig. 1.4 by means of arrows on the representative volume element. Analagously to the definition of strains, the first index indicates the stress direction and the second one the surface with the corresponding normal. (TRUESDELL & NOLL [78]). By estimating the balance of angular momentum the symmetry of the CAUCHY stress tensor $\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$ can be shown. For the continuum mechanics-based proof refer to more specific literature (z.B. ALTENBACH & ALTENBACH [38], DE BOER [44], MARSDEN & HUGHES [65]). This manuscript only intends to give an illustrative explanation by means of the sketch of the stress tensor in Fig. 1.4. If the moment equilibrium of all the stress components multiplied with the areas on which they act is formed around the middle point of the representative volume element with dimensions dX_1, dX_2 and dX_3, the symmetry of the stress tensor follows. As an example, the equilibrium of moments around the $\boldsymbol{e}_3$-coordinate axis is shown.

$$2\,\sigma_{12}\,dX_1 dX_3 \frac{dX_2}{2} - 2\,\sigma_{21}\,dX_2 dX_3 \frac{dX_1}{2} = 0 \quad \sigma_{12} - \sigma_{21} = 0 \tag{1.21}$$

1.2.2 Balance of Momentum

The balance equation of the linear momentum describes the equilibrium of the internal forces and the stresses. The forces acting on a body can be classified as:

- deformation-independent, volume-specific loads $\rho\,\boldsymbol{b} = \rho\,[b_1\; b_2\; b_3]^T$ (physical units $\frac{N}{m^3}$),
- volume-specific inertial forces, which according to the *Newton Axiom* are opposite in di-

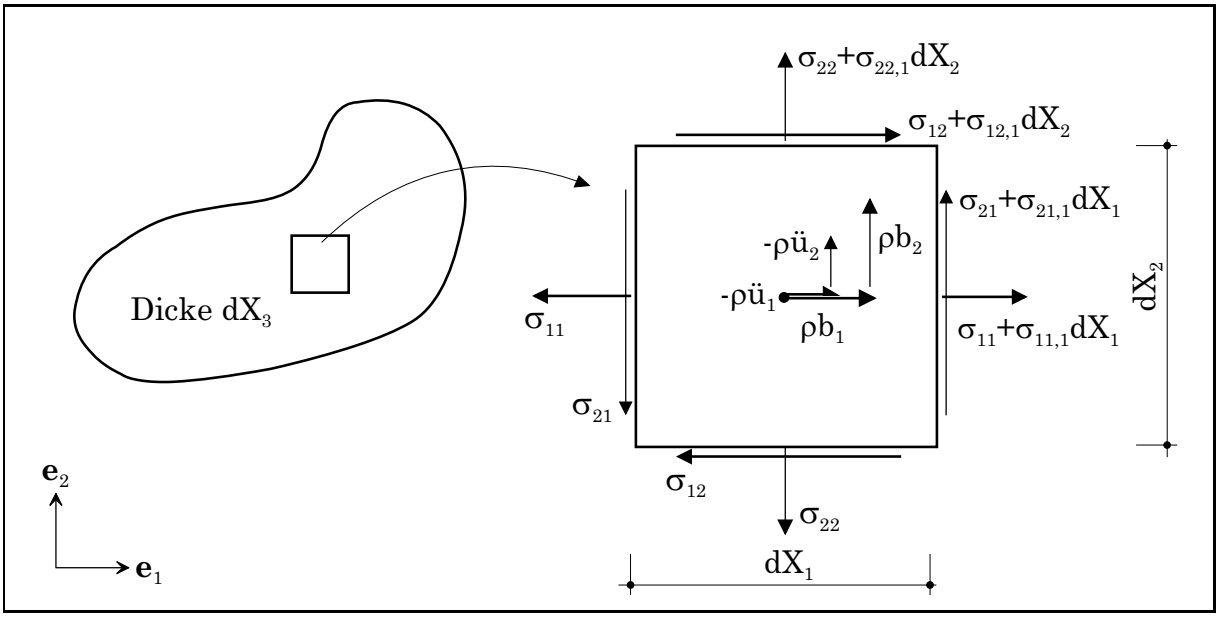


Figure 1.5: Momentum balance of a differential volume element (2D)

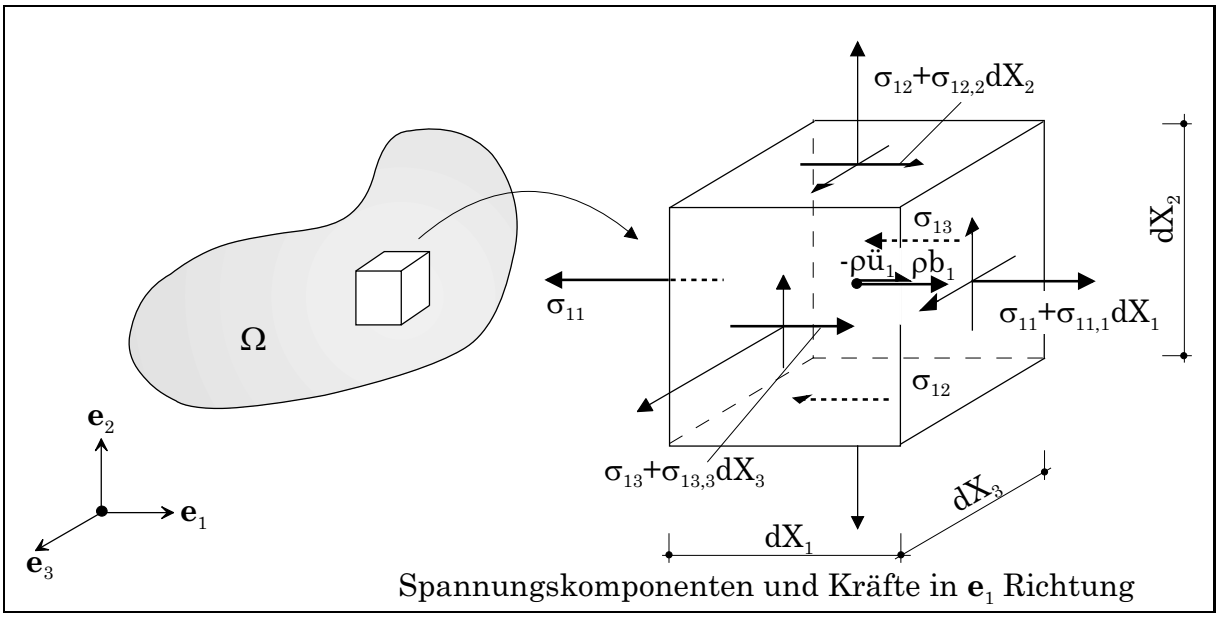


Figure 1.6: Momentum balance of a differential volume element (3D)

rection to the acceleration $-\rho\ \ddot{\boldsymbol{u}} = -\rho\ [\ddot{u}_1\ \ddot{u}_2\ \ddot{u}_3]^T$ (physical units $\frac{kg}{m^3}\frac{m}{s^2} = \frac{N}{m^3}$)

- and forces resulting from the stresses.

The local balance of momentum can be derived in accordance with continuum mechanics, based on the integral balance of momentum and under consideration of Cauchy's theorem and some mathematical simplifications, as shown for example by Altenbach & Altenbach [38], de Boer [44], or Marsden & Hughes [65]. Alternatively, a clear argumentation must lead to the equilibrium of forces.

The derivation of the internal forces equilibrium or the momentum law is limited to the two-dimensional case and afterwards is expanded for spatial considerations. Consider the differential

area element $dX_1 dX_2$ of depth dX_3, illustrated in Fig. 1.5. The volume-specific loads $\rho \boldsymbol{b}$ and $-\rho \ddot{\boldsymbol{u}}$ act in the centre. At the boundaries of the volume element, the stress components with the corresponding area elements contribute to the force equilibrium. Here, the differential changes of the stress components σ_{ij} inside the area element and the symmetry of the stress tensor ($\sigma_{ij} = \sigma_{ji}$) are taken into account. The force equilibrium in the direction of the base vector $\boldsymbol{e}_1$ contains the stress components σ_{11}, σ_{12} and the components of the volume-specific loads b_1 and $-\ddot{u}_1$.

$$\begin{aligned} 0 = & \underbrace{\left(\sigma_{11} + \frac{\partial \sigma_{11}}{\partial X_1}\, dX_1\right)}_{\text{Spannung}} \underbrace{dX_2 dX_3}_{\text{Fläche}} - \sigma_{11}\, dX_2 dX_3 \\ & + \overbrace{\left(\sigma_{12} + \frac{\partial \sigma_{12}}{\partial X_2}\, dX_2\right)} \overbrace{dX_1 dX_3} - \sigma_{12}\, dX_1 dX_3 \\ & + (\rho\, b_1 - \rho\, \ddot{u}_1)\; dX_1 dX_2 dX_3 \end{aligned} \tag{1.22}$$

The stress components σ_{11} and σ_{12} vanish, which means that only differentiated stress components take part in the equilibrium formulation. The division by the element volume $dX_1 dX_2 dX_3$ results in the local form of the momentum law in $\boldsymbol{e}_1$-direction. Analogously, the partial differential equation for the orthogonal direction $\boldsymbol{e}_2$ can be developed and expanded for three-dimensional considerations.

$$\begin{aligned} \rho\, \ddot{u}_1 &= \frac{\partial \sigma_{11}}{\partial X_1} + \frac{\partial \sigma_{12}}{\partial X_2} + \rho\, b_1 = \sigma_{11,1} + \sigma_{12,2} + \rho\, b_1 \\ & \qquad\qquad\qquad\qquad\qquad\qquad \sigma_{ij} = \sigma_{ji} \\ \rho\, \ddot{u}_2 &= \frac{\partial \sigma_{22}}{\partial X_2} + \frac{\partial \sigma_{12}}{\partial X_1} + \rho\, b_2 = \sigma_{22,2} + \sigma_{21,1} + \rho\, b_2 \end{aligned} \tag{1.23}$$

This results in the following system of partial differential equations:

$$\begin{aligned} \rho\, \ddot{u}_1 &= \sigma_{11,1} + \sigma_{12,2} + \sigma_{13,3} + \rho\, b_1 \\ \rho\, \ddot{u}_2 &= \sigma_{21,1} + \sigma_{22,2} + \sigma_{23,3} + \rho\, b_2 \qquad\qquad \sigma_{ij} = \sigma_{ji} \\ \rho\, \ddot{u}_3 &= \sigma_{31,1} + \sigma_{32,2} + \sigma_{33,3} + \rho\, b_3 \end{aligned} \tag{1.24}$$

Hence, in tensorial form, the local form of the momentum balance, the force equilibrium or the Cauchy's equation of motion is:

$$\boxed{\rho\, \ddot{\boldsymbol{u}} = \mathrm{div}\boldsymbol{\sigma} + \rho\, \boldsymbol{b} = (\sigma_{ij,j} + \rho\, b_i)\; \boldsymbol{e}_i} \tag{1.25}$$

Here div$\boldsymbol{\sigma}$ symbolizes the divergence of the *Cauchy stress tensor* $\boldsymbol{\sigma}$. The application of divergence to the second order stress tensor yields a volume-specific force vector, which according to the momentum balance (1.25) is in equilibrium with the inertial forces and the volume loads.

$$\mathrm{div}\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{11,1} + \sigma_{12,2} + \sigma_{13,3} \\ \sigma_{21,1} + \sigma_{22,2} + \sigma_{23,3} \\ \sigma_{31,1} + \sigma_{32,2} + \sigma_{33,3} \end{bmatrix} = \sigma_{ij,j}\, \boldsymbol{e}_i \qquad \sigma_{ij} = \sigma_{ji} \tag{1.26}$$

Alternatively, the momentum law can be represented in component form.

$$\rho\,\ddot{u}_i - \rho\,b_i = \frac{\partial}{\partial X_j}\,\sigma_{ij} = \sigma_{ij,j} \qquad\qquad \sigma_{ij} = \sigma_{ji} \tag{1.27}$$

The tensor $\boldsymbol{\sigma}$ is the conjugated stress magnitude to the strain tensor $\boldsymbol{\varepsilon}$ as defined in the kinematic equation (1.13). In the geometrically non-linear case, the stress tensor $\boldsymbol{\sigma}$ must be replaced by the *second Piola Kirchoff stress tensor* $\boldsymbol{S}$, conjugated to the GREEN LAGRANGE strain tensor $\boldsymbol{E}$. The latter appears in the non-linear balance of momentum in the transformed form of the material deformation gradient $\mathbf{F} = \partial\boldsymbol{x}/\partial\boldsymbol{X}$. It should be noted that in this case the density is also measured in the instantaneous configuration (see lecture notes on 'Finite Element Methodes II'). In the geometrically linear considerations of the deformations, differentiation between the stress tensors defined in the different configurations is not necessary.

Analogously to the definiton of the strain vector, the components of the stress tensor can be written in a vector. The so-defined *stress vector* contains the normal stress components σ_{11}, σ_{33} and σ_{33} as well as the shear stress components σ_{12}, σ_{23} and σ_{33}. In contrast to the strain vector, the shear components are not factorized.

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{22} & \sigma_{33} & \sigma_{12} & \sigma_{23} & \sigma_{13} \end{bmatrix}^T \qquad\qquad \boldsymbol{\sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ & \sigma_{22} & \sigma_{23} \\ \text{sym} & & \sigma_{33} \end{bmatrix} \tag{1.28}$$

By means of equations (1.24), the balance of momentum (1.25) can be formulated based on the stress vector and the definition of the differential operator $\mathbf{D}_\sigma$.

$$\boxed{\rho \begin{bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \\ \ddot{u}_3 \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial X_1} & 0 & 0 & \frac{\partial}{\partial X_2} & 0 & \frac{\partial}{\partial X_3} \\ 0 & \frac{\partial}{\partial X_2} & 0 & \frac{\partial}{\partial X_1} & \frac{\partial}{\partial X_3} & 0 \\ 0 & 0 & \frac{\partial}{\partial X_3} & 0 & \frac{\partial}{\partial X_2} & \frac{\partial}{\partial X_1} \end{bmatrix} \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{bmatrix} + \rho \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \mathbf{D}_\sigma \boldsymbol{\sigma} + \rho\,\boldsymbol{b}} \tag{1.29}$$

By comparing equations (1.16) and (1.29), the relation between the differential operators $\mathbf{D}_\varepsilon$ and $\mathbf{D}_\sigma$ is obtained.

$$\mathbf{D}_\varepsilon = \mathbf{D}_\sigma^T \tag{1.30}$$

1.2.3 Initial Stresses

Equilibrium stresses $\boldsymbol{\sigma}$, which satisfy the balance of momentum (1.25), can have different origins. The first and also the most important cause of stresses are the strains in the material body. These constitutive stresses $\boldsymbol{\sigma}^\varepsilon$ can be calculated by means of the constitutive law introduced in Chapter 1.4 on the basis of the strain state. On the other hand, *initial stresses* $\boldsymbol{\sigma}^0$ can be present in a material body in the undeformed state. These can be internal stresses, which appear, for example, in the cooling process of castings or in prestressed concrete elements or rope structures.

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^0 + \boldsymbol{\sigma}^\varepsilon \tag{1.31}$$

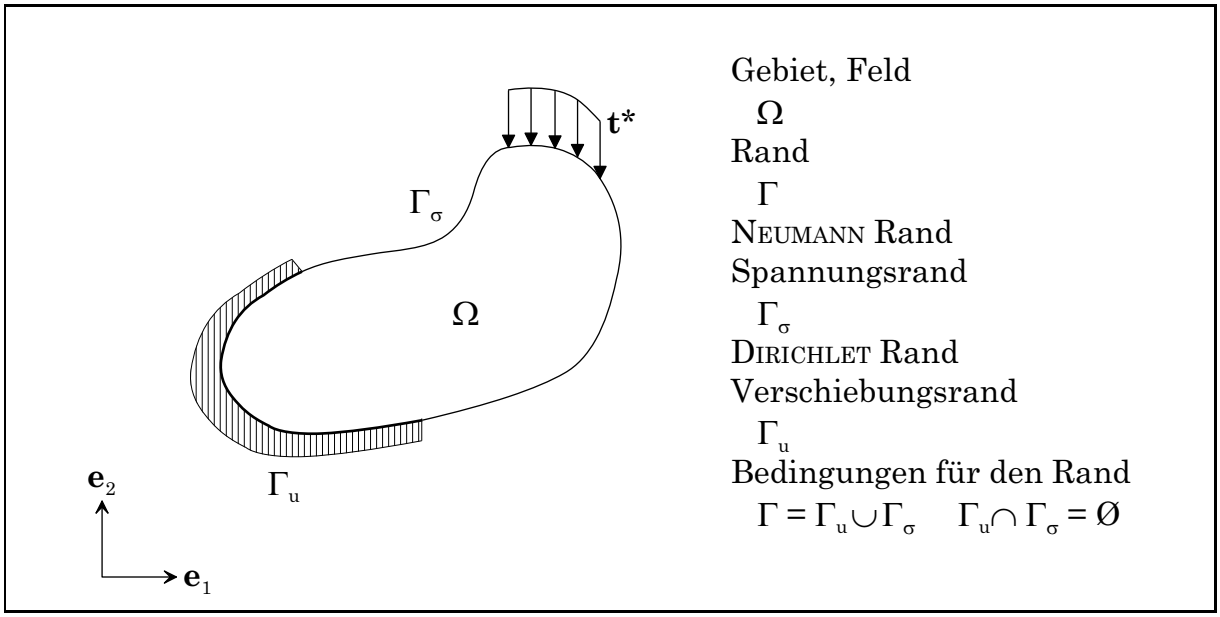


Figure 1.7: Dirichlet and Neumann boundary conditions

1.3 Initial and Boundary Conditions

The basic equations of kinematics and kinetics, derived in the previous sections are valid inside a material body or domain Ω at an arbitrary point in time. This system of equations has to be supplemented with initial conditions for the displacement or the acceleration field and with boundary conditions concerning the characteristic kinematic and kinetic size of the body's surface or the domain boundary Γ.

1.3.1 Classification of Initial and Boundary Conditions

Fig. 1.7 depicts a material body, the volume or domain Ω of which is limited by the boundary $\Gamma = \partial\Omega$. The balance of momentum (1.25), including the definition of the strain measure (1.13), holds throughout Ω. Furthermore, in the case of time-dependent problems, initial conditions in the domain Ω have to be prescribed. The domain's boundary Γ is divided into the non-overlapping Dirichlet boundary Γ_u and Neumann boundary Γ_σ.

$$\Gamma = \Gamma_u \cup \Gamma_\sigma \qquad\qquad \Gamma_u \cap \Gamma_\sigma = \emptyset \tag{1.32}$$

Here, as a rule, the primary variable is prescribed on the Dirichlet boundary, and dependent quantities are prescribed on the Neumann boundary. In the context of elastomechanics, these are the displacements $\boldsymbol{u}$ and the stress vector $\boldsymbol{t}$, respectively.

1.3.2 Dirichlet Boundary Conditions

Continuum kinematics is supplemented by the *essential*, *geometrical* or *Dirichlet boundary conditions*. Dirichlet boundary conditions are prescribed displacements (see Fig. 1.7) at a given time t for the region Γ_u of the boundary Γ.

$$\boxed{\boldsymbol{u}(\boldsymbol{X}, t) = \boldsymbol{u}^\star(\boldsymbol{X}, t) \qquad \forall \quad \boldsymbol{X} \in \Gamma_u} \tag{1.33}$$

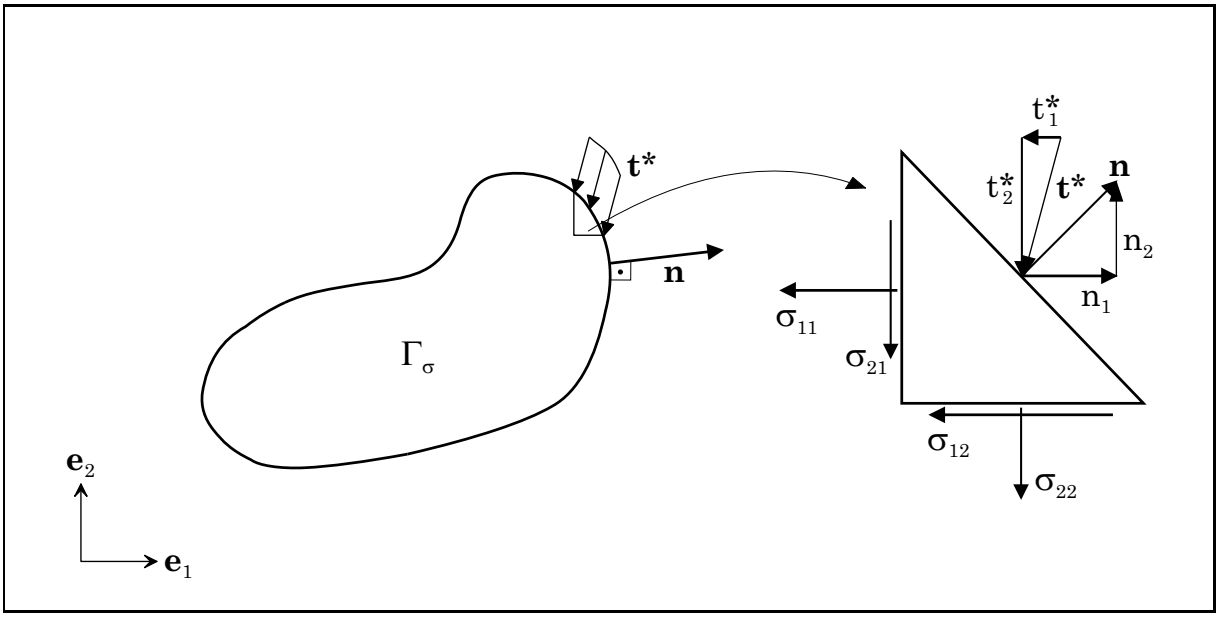


Figure 1.8: NEUMANN boundary conditions of a differential surface element (2D)

If the prescribed displacements are identical to zero, they are referred to as *homogeneous* DIRICHLET boundary conditions, which are prescribed, for instance, by supports.

$$\boldsymbol{u}(\boldsymbol{X}, t) = \mathbf{0} \qquad \forall \quad \boldsymbol{X} \in \Gamma_u \tag{1.34}$$

1.3.3 Neumann Boundary Conditions

For the derivation of the *static*, *natural* or *Neumann boundary conditions*, the two-dimensional case is considered first. Afterwards, the derived system of equations is expanded to three dimensions. Fig. 1.8 shows a surface element of a material body. The surface is characterized by the normal vector $\boldsymbol{n} = [n_1 \; n_2]^T$ with $\|\boldsymbol{n}\| = 1$. The stress vector $\boldsymbol{t}^\star = [t_1^\star \; t_2^\star]^T$ related to the line element dS is held in equilibrium by the stresses on the surface elements dX_1 und dX_2. The force equilibrium in the direction of $\boldsymbol{e}_1$

$$\sigma_{11} \; dX_2 dX_3 + \sigma_{12} \; dX_1 dX_3 = t_1^\star \; dS dX_3 \tag{1.35}$$

divided by the depth dX_3 and side length dS yields the following condition:

$$\sigma_{11} \; \frac{dX_2}{dS} + \sigma_{12} \; \frac{dX_1}{dS} = t_1^\star \tag{1.36}$$

Here, the derivatives dX_1/dS and dX_2/dS can be obtained from the similarity of the normal vector triangle with sides $n_1, \; n_2, \; \|\boldsymbol{n}\| = 1$, and the geometrical triangle with sides $dX_1, \; dX_2, \; dS$

$$\frac{dX_1}{dS} = \frac{n_2}{\|\boldsymbol{n}\|} = n_2 \qquad\qquad \frac{dX_2}{dS} = \frac{n_1}{\|\boldsymbol{n}\|} = n_1 \tag{1.37}$$

If, additionally, the force equilibrium is formed analogously in the direction of $\boldsymbol{e}_2$, one obtains

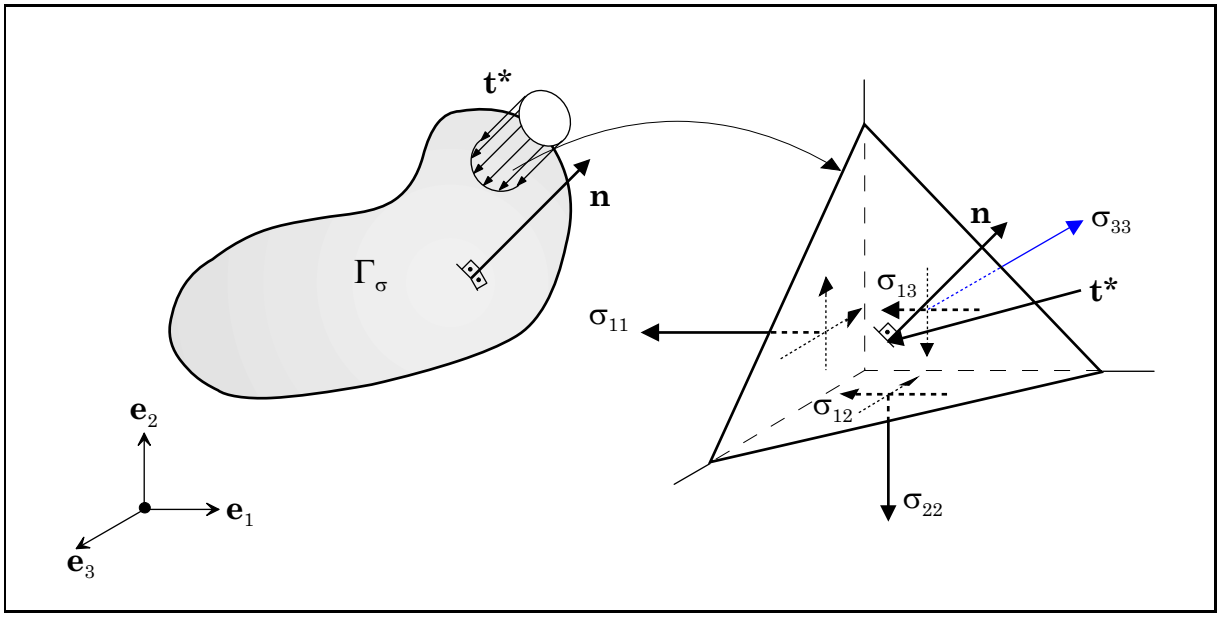


Figure 1.9: NEUMANN boundary conditions of a differential surface element (3D)

the system of equations for the two-dimensional case.

$$\begin{aligned} \sigma_{11}\, n_1 + \sigma_{12}\, n_2 &= t_1^\star \\ \sigma_{12}\, n_1 + \sigma_{22}\, n_2 &= t_2^\star \end{aligned} \tag{1.38}$$

For an expanded three-dimensional consideration (see Fig. 1.9), the force equilibrium of the surface element with normal vector $\boldsymbol{n} = [n_1 \;\; n_2 \;\; n_3]^T$ and stress vector on the surface $\boldsymbol{t}^\star = [t_1^\star \; t_2^\star \; t_3^\star]^T$ is as follows:

$$\begin{aligned} \sigma_{11}\, n_1 + \sigma_{12}\, n_2 + \sigma_{13}\, n_3 &= t_1^\star \\ \sigma_{12}\, n_1 + \sigma_{22}\, n_2 + \sigma_{23}\, n_3 &= t_2^\star \\ \sigma_{13}\, n_1 + \sigma_{23}\, n_2 + \sigma_{33}\, n_3 &= t_3^\star \end{aligned} \qquad \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} t_1^\star \\ t_2^\star \\ t_3^\star \end{bmatrix} \tag{1.39}$$

The force equilibrium at the stress or NEUMANN-boundary Γ_σ (Eq. (1.39)) can be written in a compact form in tensorial notation in the form of the *Cauchy equation*.

$$\boxed{\boldsymbol{\sigma}(\boldsymbol{X}, t) \cdot \boldsymbol{n} = \boldsymbol{t}^\star(\boldsymbol{X}, t) \qquad \forall \quad \boldsymbol{X} \in \Gamma_\sigma} \tag{1.40}$$

Here, the simple contraction of a first-order and a second-order tensor was used.

$$\sigma_{ij}\, n_j = t_i^\star \qquad\qquad \sigma_{ij} = \sigma_{ji} \tag{1.41}$$

Alternatively, Eq. (1.40) can be developed by application of CAUCHY's theorem Eq. (1.19) on a surface element of the body as a special case of an arbitrary cross section, see Section 1.2.2. Usage of the stress vector according to the definition in Eq. (1.28) results in the operator

representation of the static boundary conditions.

$$\begin{bmatrix} n_1 & 0 & 0 & n_2 & 0 & n_3 \\ 0 & n_2 & 0 & n_1 & n_3 & 0 \\ 0 & 0 & n_3 & 0 & n_2 & n_1 \end{bmatrix} \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{bmatrix} = \begin{bmatrix} t_1^\star \\ t_2^\star \\ t_3^\star \end{bmatrix} \qquad \mathbf{D}_t \boldsymbol{\sigma} = \boldsymbol{t}^\star \tag{1.42}$$

In order to find out a relationship between the differential operators $\mathbf{D}_\sigma$, Eq. (1.29), and $\mathbf{D}_t$, Eq. (1.42), the surface of the body described in the form $S(\boldsymbol{X}) = 0$ is examined. The gradient of $S(\boldsymbol{X})$ is perpendicular to the surface and consequently parallel to the normal unit vector $\boldsymbol{n}$.

$$\nabla S = \left[\frac{\partial}{\partial X_1} \; \frac{\partial}{\partial X_2} \; \frac{\partial}{\partial X_3} \right]^T S \qquad \boldsymbol{n} = \frac{\nabla S}{\|\nabla S\|}, \; n_i = \frac{1}{\|\nabla S\|} \frac{\partial}{\partial X_i} S \tag{1.43}$$

Inserting n_i in the differential operator $\mathbf{D}_t$ yields the desired relationship. $\mathbf{D}_t$ arises by application of the differential operator $\mathbf{D}_\sigma$ on the implicit representation of the surface $S(\boldsymbol{X}) = 0$ and by taking the norm $\|\nabla S\|$:

$$\mathbf{D}_t = \frac{1}{\|\nabla S\|} \mathbf{D}_\sigma \; S(\boldsymbol{X}) \tag{1.44}$$

1.3.4 Initial Conditions

Dynamic problems require in addition to the boundary conditions also knowledge of the initial state of the body at time $t = t_0$. This state is unambiguously characterized by the partial differential equations (Chapter 1.1 and 1.2), describing the deformation, and by one of the two fields of the displacements $\boldsymbol{u}(\boldsymbol{X}, t_0)$ or the accelerations $\ddot{\boldsymbol{u}}(\boldsymbol{X}, t_0)$.

$$\boldsymbol{u}(\boldsymbol{X}, t_0) = \boldsymbol{u}_0(\boldsymbol{X}) \quad \forall \quad \boldsymbol{X} \in \Omega \qquad \ddot{\boldsymbol{u}}(\boldsymbol{X}, t_0) = \ddot{\boldsymbol{u}}_0(\boldsymbol{X}) \quad \forall \quad \boldsymbol{X} \in \Omega \tag{1.45}$$

The special choice of the initial time $t_0 = 0$ results, according to Eq. (1.2), in $\boldsymbol{u}_0 = \mathbf{0}$. The types of initial conditions given in Eq. (1.45) are self-exclusive, as by prescribing the displacement field for $t = t_0$, the acceleration field follows from the evaluation of the balance of momentum (1.25) at this moment of time, and vice versa.

1.4 Hyperelastic Constitutive Laws

In the previous sections stresses and strains were defined based on the momentum balance and the displacement field, respectively. Hence, both the stress tensor and the displacement vector are variables which are needed for the unambiguous description of the continuum's state of motion. This number of variables can be reduced by the postulate of a constitutive relationship which relates the stresses on the one hand, and the strains on the other. As a consequence of this postulate, the stresses become dependent on the displacement vector. This postulate is based on the observation of material behaviour under monotonous or cyclic loading. The variety of materials and their states induces various possibilities of mathematical description or modelling of material behaviour. First, the fundamental material models can be classified as *linear* and

non-linear material models. Here, we want to restrict the variety of material models to linear models, which have proved to be representative in various engineering applications. Non-linear material models and engineering applications which require consideration of non-linear effects are discussed in the books HAUPT [55], GROSS [54] and LEMAITRE & CHABOCHE [61].

In this chapter, the fundamental assumptions for the formulation of the constitutive equations are stated first. Equally,the material models of potential character, the so-called *hyperelastic material models*, are specified. Afterwards, the generalized HOOKE's Law is formulated as a basis for the Finite Element Method in linear structural mechanics and specialized for the plane stress and strain state as well as for the classical one-dimensional *Hooke's Law*.

1.4.1 Fundamental Assumptions and Classification

Constitutive equations in the classical sense presume the existence of a relation between forces and deformation, respectively between stresses and strains, which is exclusively local, i.e., at the considered material point. In the context of this axiomatic prerequisite and assuming vanishing initial stresses ($\boldsymbol{\sigma}_0 = \mathbf{0}$), a material law sets the relation between stresses $\boldsymbol{\sigma}$, strains $\boldsymbol{\varepsilon}$, strain rates $\dot{\boldsymbol{\varepsilon}}$, which describe the velocity dependence of the stress tensor, and internal variables $\boldsymbol{\alpha}$, which represent the dependence of the stresses on the history (plastification or damage).

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}(\boldsymbol{\varepsilon}, \dot{\boldsymbol{\varepsilon}}, \boldsymbol{\alpha}) \tag{1.46}$$

This generalized material law contains a number of material models for the description of non-linear material behaviour, taking into account microstructural damage, residual plastic strains and time-dependent effects. If, however, we focus our attention on the modelling of reversible, time-independent, elastic processes, the stress state can be defined only based on the strain state, with the stress tensor turning into a null tensor in the undeformed configuration.

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}(\boldsymbol{\varepsilon}) \tag{1.47}$$

Furthermore, it is to be assumed that the material is *homogeneous* and that the material properties are not dependent on the direction. The latter restriction characterizes an *isotropic material model*. If this property is not satisfied, we speak of an *anisotropic material model*. Popular materials with distinct anisotropic characteristics are fibre-reinforced composite materials, the classical construction material timber, reinforced concrete, or rolled steel. These materials often show extreme differences when loaded parallel or transversely to the fibre direction, or to the orientation of crystals, respectively. It should be noted that the undertaken restriction to isotropic material models only has effects on the formulation of the material law in the following sections, and not on the formulation of linear finite elements.

1.4.2 Elastic Material Models

Elasticity means that the stress state only depends on the instantaneous strain state and not on the stress path. The desired path-independence is only guaranteed, if the stress tensor can be derived by differentiation of an elastic potential function $W(\boldsymbol{\varepsilon})$ with respect to the strain tensor.

$$\boxed{\boldsymbol{\sigma}(\boldsymbol{\varepsilon}) = \frac{\partial W(\boldsymbol{\varepsilon})}{\partial \boldsymbol{\varepsilon}}} \tag{1.48}$$

If one integrates from $\boldsymbol{\sigma}(\boldsymbol{\varepsilon}_1)$ to $\boldsymbol{\sigma}(\boldsymbol{\varepsilon}_2)$ along an arbitrary path in the strain space, one obtains an energy difference independent of the path.

$$\int_{\boldsymbol{\varepsilon}_1}^{\boldsymbol{\varepsilon}_2} \boldsymbol{\sigma}(\boldsymbol{\varepsilon})\, d\boldsymbol{\varepsilon} = \int_{\boldsymbol{\varepsilon}_1}^{\boldsymbol{\varepsilon}_2} \frac{\partial W}{\partial \boldsymbol{\varepsilon}}\, d\boldsymbol{\varepsilon} = W(\boldsymbol{\varepsilon}_2) - W(\boldsymbol{\varepsilon}_1) \tag{1.49}$$

If the deformation is independent of the path, the corresponding material laws are *hyperelastic*. Derivation of the stress tensor with respect to the strain tensor yields the *tangential modulus of elasticity*, *constitutive tensor* or *material tensor* $\boldsymbol{\mathcal{C}}$. On the other hand, the material tensor represents the linear mapping of the strain tensor onto the stress tensor.

$$\boxed{\boldsymbol{\mathcal{C}} = \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}} = \frac{\partial W}{\partial \boldsymbol{\varepsilon} \otimes \partial \boldsymbol{\varepsilon}} = C_{ijkl}\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_k \otimes \boldsymbol{e}_l \qquad \boldsymbol{\sigma} = \boldsymbol{\mathcal{C}} : \boldsymbol{\varepsilon} = C_{ijkl}\, \varepsilon_{kl}\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j} \tag{1.50}$$

As a consequence of the symmetry of the stress and strain tensors, the constitutive tensor satisfies the following symmetry properties:

$$C_{ijkl} = C_{jikl} = C_{jilk} = C_{ijlk} \tag{1.51}$$

If the material tensor $\boldsymbol{\mathcal{C}}$ is independent of the strains, i.e., a linear relationship exists between stresses and strains, we are talking about a *physically* or *material linear* constitutive law. All other material models are characterized correspondingly by the attributes *physically* or *material non-linear*.

1.4.3 Isotropic, Elastic Material Relation of Continuum

Based on the fundamental ideas for the formulation of material models in the previous sections, the generalized HOOKE's law is to be derived as representative of three-dimensional, linear, elastic and isotropic material models. The isotropic, elastic material law is characterized by means of two *material parameters*. The representation of the constitutive equation is realized with the so-called LAMÉ-constants μ and λ . The relation of the modulus of elasticity E, the shear modulus G and the POISSON-transverse contraction ratio ν is given by

$$\mu = \frac{E}{2\,(1+\nu)} = G \qquad\qquad \lambda = \frac{\nu\, E}{(1+\nu)(1-2\nu)} \tag{1.52}$$

Further relations between usually used elasticity constants are summarized in Table 1.1 according to the books of STEIN & BARTHOLD [74] and LEIPHOLZ [60].

The potential function $W(\boldsymbol{\varepsilon})$ of the generalized HOOKE's material law of the isotropic continuum is postulated as a quadratic function of the strain tensor and the chosen material parameters as follows:

$$W(\boldsymbol{\varepsilon}) = \mu\, \boldsymbol{\varepsilon} : \boldsymbol{\varepsilon} + \frac{1}{2}\, \lambda\, (\boldsymbol{\varepsilon} : \mathbf{1})^2 \tag{1.53}$$

By differentiation of the scalar-valued potential with respect to the strain tensor, the stress tensor is obtained according to Eq. (1.48).

$$\boldsymbol{\sigma} = 2\mu\, \boldsymbol{\varepsilon} + \lambda\, (\boldsymbol{\varepsilon} : \mathbf{1})\, \mathbf{1} = (2\mu\, \varepsilon_{ij} + \lambda\, \varepsilon_{kk}\, \delta_{ij})\ \boldsymbol{e}_i \otimes \boldsymbol{e}_j \tag{1.54}$$

	λ	$\mu = G$	E	ν	K
$\lambda \quad \mu$	λ	μ	$\dfrac{\mu(3\lambda+2\mu)}{\lambda+\mu}$	$\dfrac{\lambda}{2(\lambda+\mu)}$	$\lambda+\frac{2}{3}\mu$
$\lambda \quad E$	λ	$\dfrac{E-3\lambda+c_1}{4}$	E	$\dfrac{2\lambda}{E+\lambda+c_1}$	$\dfrac{E+3\lambda+c_1}{6}$
$\lambda \quad \nu$	λ	$\dfrac{\lambda(1-2\nu)}{2\nu}$	$\dfrac{\lambda\, c_2}{\nu}$	ν	$\dfrac{\lambda(1+\nu)}{3\nu}$
$\lambda \quad K$	λ	$\frac{3}{2}(K-\lambda)$	$\dfrac{9K(K-\lambda)}{3K-\lambda}$	$\dfrac{\lambda}{3K-\lambda}$	K
$\mu \quad E$	$\dfrac{\mu(E-2\mu)}{3\mu-E}$	μ	E	$\dfrac{E-2\mu}{2\mu}$	$\dfrac{\mu E}{3(3\mu-E)}$
$\mu \quad \nu$	$\dfrac{2\mu\nu}{1-2\nu}$	μ	$2\mu(1+\nu)$	ν	$\dfrac{2\mu(1+\nu)}{3(1-2\nu)}$
$\mu \quad K$	$K-\frac{2}{3}\mu$	μ	$\dfrac{9K\mu}{3K+\mu}$	$\dfrac{3K-2\mu}{6K+2\mu}$	K
$E \quad \nu$	$\dfrac{E\nu}{c_2}$	$\dfrac{E}{2(1+\nu)}$	E	ν	$\dfrac{E}{3(1-2\nu)}$
$E \quad K$	$\dfrac{3K(3K-E)}{9K-E}$	$\dfrac{3KE}{9K-E}$	E	$\dfrac{3K-E}{6K}$	K
$\nu \quad K$	$\dfrac{3K\nu}{1+\nu}$	$\dfrac{3K(1-2\nu)}{2(1+\nu)}$	$3K(1-2\nu)$	ν	K

λ	LAMÉ-Konstante	K	Kompressionsmodul
μ	LAMÉ-Konstante, Schubmodul, $\mu = G$	c_1	$= \sqrt{E^2+9\lambda^2+2E\lambda}$
E	Elastizitätsmodul	c_2	$= (1+\nu)(1-2\nu)$
ν	Querkontraktionszahl, POISSON-Zahl		

Table 1.1: Relationships between material parameters

In the above equation, $\mathbf{1}$ characterizes the second order identity tensor and the term $\boldsymbol{\varepsilon} : \mathbf{1}$ characterizes the trace of the tensor $\boldsymbol{\varepsilon}$. The trace of the tensor $\boldsymbol{\varepsilon}$ can be designated alternatively by $\mathrm{tr}(\boldsymbol{\varepsilon})$.

$$\mathbf{1} : \boldsymbol{\varepsilon} = \mathrm{tr}(\boldsymbol{\varepsilon}) = \delta_{ij}\, \varepsilon_{ij} = \varepsilon_{ii} = \varepsilon_{11} + \varepsilon_{22} + \varepsilon_{33} \tag{1.55}$$

The fourth order constitutive tensor $\boldsymbol{\mathcal{C}}$ is obtained by additional differentiation according to Eq. (1.50).

$$\boxed{\boldsymbol{\mathcal{C}} = 2\mu\, \boldsymbol{\mathcal{I}}^{\mathrm{sym}} + \lambda\, \mathbf{1} \otimes \mathbf{1} = [\mu\, (\delta_{il}\delta_{jk} + \delta_{ik}\delta_{jl}) + \lambda\, \delta_{ij}\delta_{kl}]\ \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_k \otimes \boldsymbol{e}_l} \tag{1.56}$$

Herein, $\boldsymbol{\mathcal{I}}^{\mathrm{sym}}$ symbolizes the symmetric fourth-order unit tensor and $\mathbf{1} \otimes \mathbf{1}$ symbolizes the dyadic product of second order unit tensors. The result of this dyadic product is a fourth order tensor.

$$\boldsymbol{\mathcal{I}}^{\mathrm{sym}} = \frac{1}{2}\, (\delta_{il}\delta_{jk} + \delta_{ik}\delta_{jl})\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_k \otimes \boldsymbol{e}_l \qquad \mathbf{1} \otimes \mathbf{1} = \delta_{ij}\, \delta_{kl}\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_k \otimes \boldsymbol{e}_l \tag{1.57}$$

The symmetrical properties $C_{ijkl} = C_{jikl} = C_{jilk} = C_{ijlk}$ of the constitutive tensor $\boldsymbol{\mathcal{C}}$ result from the symmetry of the strain and stress tensors. In order to verify Eq. (1.56), the latter can be inserted into Eq. (1.50). Using $\delta_{ik}\delta_{jl}\ \varepsilon_{kl} = \delta_{ik}\ \varepsilon_{kj} = \varepsilon_{ij}$ and the symmetry of the strain

tensor $\varepsilon_{ij} = \varepsilon_{ji}$ in the index notation yields the direct relation between strains and stresses, as demonstrated in Eq. (1.54).

$$\begin{aligned}\sigma_{ij} &= C_{ijkl}\,\varepsilon_{kl} = [\mu\,(\delta_{il}\delta_{jk} + \delta_{ik}\delta_{jl}) + \lambda\,\delta_{ij}\delta_{kl}]\;\varepsilon_{kl} \\ &= \mu\,(\delta_{il}\delta_{jk}\,\varepsilon_{kl} + \delta_{ik}\delta_{jl}\,\varepsilon_{kl}) + \lambda\,\delta_{ij}\delta_{kl}\,\varepsilon_{kl} = \mu\,(\delta_{il}\,\varepsilon_{jl} + \delta_{ik}\,\varepsilon_{kj}) + \lambda\,\delta_{ij}\,\varepsilon_{kk} \\ &= \mu\,(\varepsilon_{ji} + \varepsilon_{ij}) + \lambda\,\delta_{ij}\,\varepsilon_{kk} = 2\mu\,\varepsilon_{ij} + \lambda\,\delta_{ij}\,\varepsilon_{kk}\end{aligned} \tag{1.58}$$

Using the definition of stresses and strains in vector form in the context of the development of finite elements, one obtains the linear relation between kinematics and kinetics (Eqs. (1.50) and (1.56)), or strain and stress vectors in matrix notation, respectively,

$$\boldsymbol{\sigma} = \mathbf{C}\,\boldsymbol{\varepsilon} \tag{1.59}$$

with the components of the constitutive matrix $\mathbf{C}$ connecting the components of the strain vector $\varepsilon_{kl} = \varepsilon_{lk}$ and the stress vector $\sigma_{ij} = \sigma_{ji}$ as follows.

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{bmatrix} = \begin{bmatrix} C_{1111} & C_{1122} & C_{1133} & C_{1112} & C_{1123} & C_{1113} \\ C_{2211} & C_{2222} & C_{2233} & C_{2212} & C_{2223} & C_{2213} \\ C_{3311} & C_{3322} & C_{3333} & C_{3312} & C_{3323} & C_{3313} \\ C_{1211} & C_{1222} & C_{1233} & C_{1212} & C_{1223} & C_{1213} \\ C_{2311} & C_{2322} & C_{2333} & C_{2312} & C_{2323} & C_{2313} \\ C_{1311} & C_{1322} & C_{1333} & C_{1312} & C_{1323} & C_{1313} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{bmatrix} \tag{1.60}$$

The entries C_{ijkl} of the constitutive matrix can be developed with Eq. (1.56) using the definition of the KRONECKER symbol. As an example, the development of the components C_{1111}, C_{1122}, C_{1112} and C_{1212} is demonstrated; all other components of the material stiffness matrix $\mathbf{C}$ can be obtained accordingly.

$$\begin{aligned} C_{1111} &= \mu\,(\delta_{11}\delta_{11} + \delta_{11}\delta_{11}) + \lambda\,\delta_{11}\delta_{11} = 2\,\mu + \lambda \\ C_{1122} &= \mu\,(\delta_{12}\delta_{12} + \delta_{12}\delta_{12}) + \lambda\,\delta_{11}\delta_{22} = \lambda \\ C_{1112} &= \mu\,(\delta_{12}\delta_{11} + \delta_{11}\delta_{12}) + \lambda\,\delta_{11}\delta_{12} = 0 \\ C_{1212} &= \mu\,(\delta_{12}\delta_{21} + \delta_{11}\delta_{22}) + \lambda\,\delta_{12}\delta_{12} = \mu \end{aligned} \tag{1.61}$$

Thus, the material stiffness matrix $\mathbf{C}$ is defined with the LAMÉ-parameters μ and λ.

$$\underbrace{\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{bmatrix}}_{\boldsymbol{\sigma}} = \underbrace{\begin{bmatrix} 2\mu+\lambda & \lambda & \lambda & 0 & 0 & 0 \\ & 2\mu+\lambda & \lambda & 0 & 0 & 0 \\ & & 2\mu+\lambda & 0 & 0 & 0 \\ & & & \mu & 0 & 0 \\ & & & & \mu & 0 \\ \text{sym} & & & & & \mu \end{bmatrix}}_{\mathbf{C}} \underbrace{\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{bmatrix}}_{\boldsymbol{\varepsilon}} \tag{1.62}$$

After a transformation of the material parameters according to Eq. (1.52) or Table 1.1, the

	Annahme	Folgerung	$\boldsymbol{\sigma}$	$\mathbf{C}$	$\boldsymbol{\varepsilon}$
3d	-	-	$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{bmatrix}$	$\begin{bmatrix} 2\mu+\lambda & \lambda & \lambda & 0 & 0 & 0 \\ & 2\mu+\lambda & \lambda & 0 & 0 & 0 \\ & & 2\mu+\lambda & 0 & 0 & 0 \\ & & & \mu & 0 & 0 \\ & & & & \mu & 0 \\ \text{sym} & & & & & \mu \end{bmatrix}$	$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{bmatrix}$
es	$\sigma_{33}=0$ $\sigma_{13}=0$ $\sigma_{23}=0$	$\varepsilon_{13}=0$ $\varepsilon_{23}=0$ $\varepsilon_{33}=-\dfrac{\lambda}{2\mu+\lambda}(\varepsilon_{11}+\varepsilon_{22})$	$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix}$	$\dfrac{2\mu}{2\mu+\lambda}\begin{bmatrix} 2(\mu+\lambda) & \lambda & 0 \\ & 2(\mu+\lambda) & 0 \\ \text{sym} & & \dfrac{2\mu+\lambda}{2} \end{bmatrix}$	$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix}$
ev	$\varepsilon_{33}=0$ $\varepsilon_{13}=0$ $\varepsilon_{23}=0$	$\sigma_{13}=0$ $\sigma_{23}=0$ $\sigma_{33}=\dfrac{\lambda}{2(\mu+\lambda)}(\sigma_{11}+\sigma_{22})$	$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix}$	$\begin{bmatrix} 2\mu+\lambda & \lambda & 0 \\ & 2\mu+\lambda & 0 \\ \text{sym} & & \mu \end{bmatrix}$	$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix}$

Table 1.2: Isotropic, linear elastic, constitutive laws in matrix notation

constitutive matrix can be described by means of the modulus of elasticity E and the POISSON ratio ν.

$$\mathbf{C} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ & 1-\nu & \nu & 0 & 0 & 0 \\ & & 1-\nu & 0 & 0 & 0 \\ & & & \dfrac{1-2\nu}{2} & 0 & 0 \\ & & & & \dfrac{1-2\nu}{2} & 0 \\ \text{sym} & & & & & \dfrac{1-2\nu}{2} \end{bmatrix} \tag{1.63}$$

For the deformation analysis of two-dimensional continua, the plane stress and the plane strain states are of interest. Two application examples of these special states are depicted in Fig. 1.10. Typical applications of plane stress states are structural members of small depth, e.g. membranes, disks, plates and shells. The plane strain state is mostly used in cases where the dimension in one direction is very big with the loading in this direction remaining unchanged. The plane strain state is very common in the field of geo and soil mechanics. The constitutive equations of isotropic, linear elastic materials are summarized in Table 1.2 for the general three-dimensional stress state, plane stress and plane strain states. The derivation of these equations can be found in the following sections.

1.4.4 Plane Stress State

A representative plane element is examined, which lies in the plane spanned by the base vectors $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$. In the case of a plane stress state it is assumed that the stress components σ_{33}, σ_{13} and σ_{23} vanish

$$\sigma_{33} = \sigma_{13} = \sigma_{23} = 0 \tag{1.64}$$

with the remaining stress components being constant in the direction of the base vector $\boldsymbol{e}_3$, see Fig. 1.11 and Fig. 1.11. Eq. (1.62) can be fulfilled for the assumptions of the plane stress state

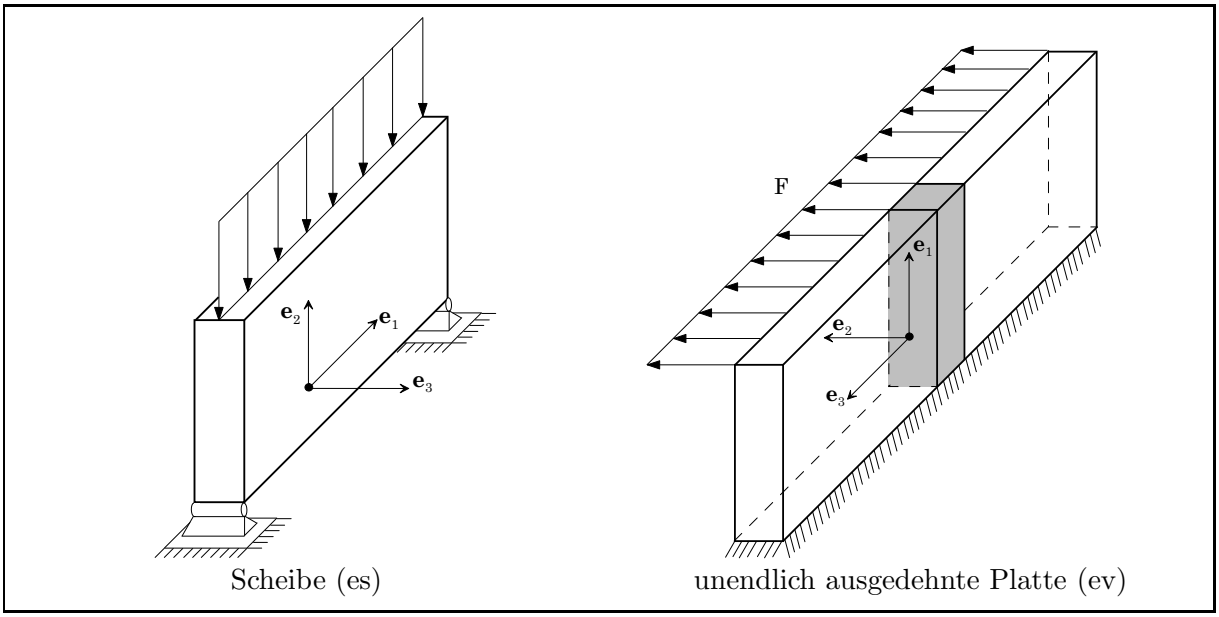


Figure 1.10: Examples for the application of plane stress and strain states

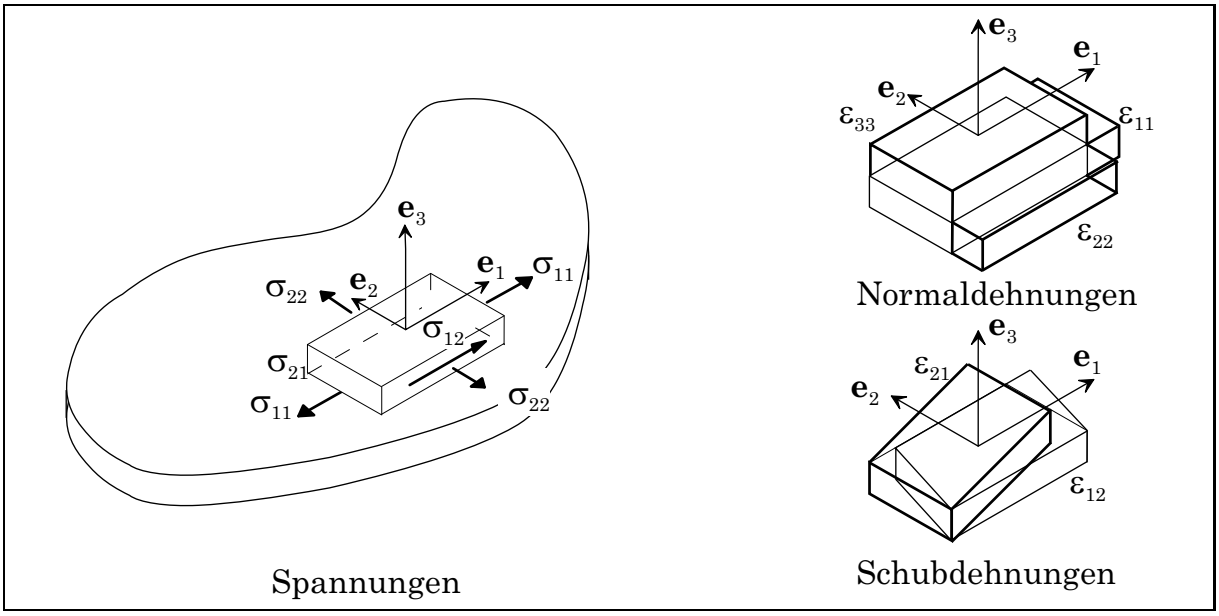


Figure 1.11: Plane stress state

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ 0 \\ \sigma_{12} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2\mu+\lambda & \lambda & \lambda & 0 & 0 & 0 \\ & 2\mu+\lambda & \lambda & 0 & 0 & 0 \\ & & 2\mu+\lambda & 0 & 0 & 0 \\ & & & \mu & 0 & 0 \\ & & & & \mu & 0 \\ \text{sym} & & & & & \mu \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{bmatrix} \tag{1.65}$$

only if the conditions

$$\varepsilon_{13} = \varepsilon_{23} = 0 \qquad\qquad \lambda\,\varepsilon_{11} + \lambda\,\varepsilon_{22} + (2\mu+\lambda)\,\varepsilon_{33} = 0 \tag{1.66}$$

hold true. This last requirement gives the normal strain ε_{33} as function of the normal strains ε_{11} and ε_{22}.

$$\varepsilon_{33} = -\frac{\lambda}{2\mu+\lambda}\left(\varepsilon_{11}+\varepsilon_{22}\right) = -\frac{\nu}{1-\nu}\left(\varepsilon_{11}+\varepsilon_{22}\right) \tag{1.67}$$

Through the conditions (1.64-1.67) the constitutive relation of the three-dimensional continuum (1.62) can be reduced as follows.

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ 0 \\ \sigma_{12} \end{bmatrix} = \begin{bmatrix} 2\mu+\lambda & \lambda & \lambda & 0 \\ & 2\mu+\lambda & \lambda & 0 \\ & & 2\mu+\lambda & 0 \\ \text{sym} & & & \mu \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ -\dfrac{\lambda}{2\mu+\lambda}\left(\varepsilon_{11}+\varepsilon_{22}\right) \\ 2\varepsilon_{12} \end{bmatrix} \tag{1.68}$$

By summarizing linearly dependent terms one obtains the linear elastic material law of the plane stress state in the form $\boldsymbol{\sigma} = \mathbf{C}^{es}\boldsymbol{\varepsilon}$.

$$\boxed{\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} = \frac{2\mu}{2\mu+\lambda} \begin{bmatrix} 2(\mu+\lambda) & \lambda & 0 \\ & 2(\mu+\lambda) & 0 \\ \text{sym} & & \dfrac{2\mu+\lambda}{2} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} \qquad (1.69)}$$

Or, alternatively, in terms of the material constants ν and E:

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ & 1 & 0 \\ \text{sym} & & \dfrac{1-\nu}{2} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} \tag{1.70}$$

1.4.5 Plane Strain State

Again, a representative plane element is examined which lies in the plane spanned by the base vectors $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$, see Fig. 1.12. For the generation of the plane strain state it is assumed that the strain components ε_{33}, ε_{13} and ε_{23} vanish.

$$\varepsilon_{33} = \varepsilon_{13} = \varepsilon_{23} = 0 \tag{1.71}$$

According to the three-dimensional constitutive relationship (1.62)

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{bmatrix} = \begin{bmatrix} 2\mu+\lambda & \lambda & \lambda & 0 & 0 & 0 \\ & 2\mu+\lambda & \lambda & 0 & 0 & 0 \\ & & 2\mu+\lambda & 0 & 0 & 0 \\ & & & \mu & 0 & 0 \\ & & & & \mu & 0 \\ \text{sym} & & & & & \mu \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 0 \\ 2\varepsilon_{12} \\ 0 \\ 0 \end{bmatrix} \tag{1.72}$$

the stress components σ_{23}, σ_{13} become zero; the stress σ_{33}, on the contrary, is different from zero.

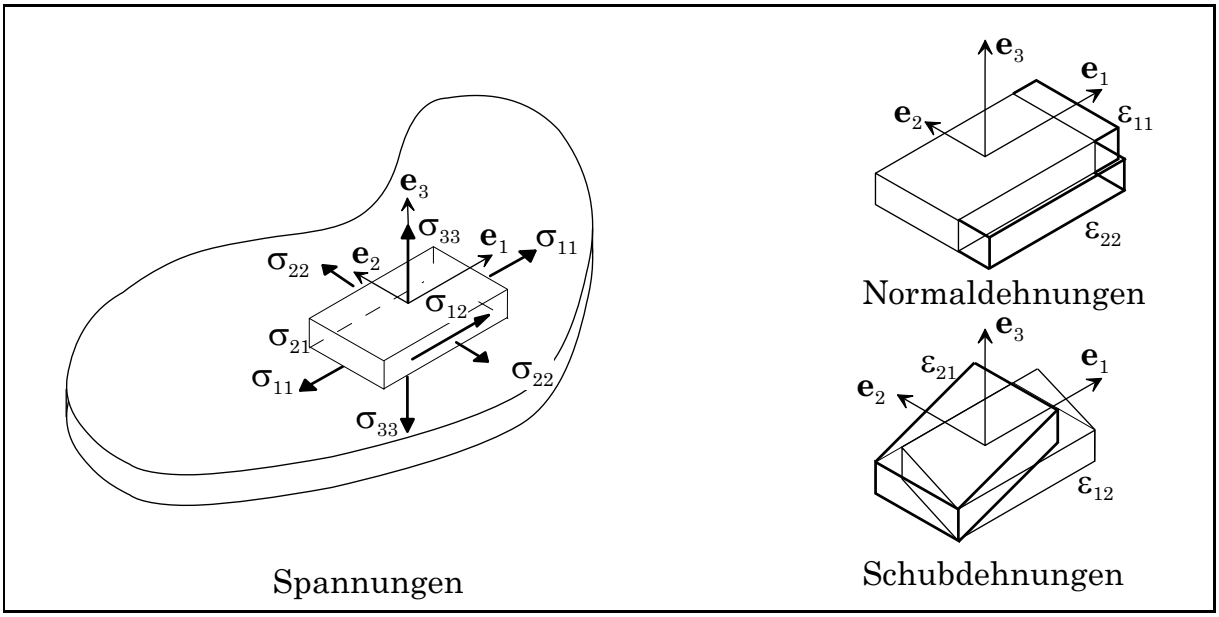


Figure 1.12: Plane strain state

$$\sigma_{23} = \sigma_{13} = 0 \qquad \sigma_{33} = \lambda\,(\varepsilon_{11}+\varepsilon_{22}) = \frac{\lambda}{2(\lambda+\mu)}\,(\sigma_{11}+\sigma_{22}) \tag{1.73}$$

Here, the transverse stress component σ_{33} could be alternatively expressed by means of the first two rows of the three-dimensional constitutive law (1.62), by the normal components σ_{11} and σ_{22}. According to Eq. (1.52), the relations (1.73) can be analogously formulated with the elasticity constants E and ν.

$$\sigma_{33} = \frac{E\,\nu}{(1+\nu)(1-2\nu)}\;(\varepsilon_{11}+\varepsilon_{22}) \qquad \sigma_{33} = \nu\;(\sigma_{11}+\sigma_{22}) \tag{1.74}$$

The stress component σ_{33} does not contribute to the internal energy or to the internal virtual work, since the conjugated strain component ε_{33} is zero according to the assumption above (1.71). Therefore, the stress-strain relationship can be represented by only three stress components in the form $\boldsymbol{\sigma} = \mathbf{C}^{ev}\boldsymbol{\varepsilon}$.

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} = \begin{bmatrix} 2\mu+\lambda & \lambda & 0 \\ & 2\mu+\lambda & 0 \\ \text{sym} & & \mu \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} \tag{1.75}$$

By alternative parametrization (see Table 1.1), the constitutive equation of plane strain turns into:

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ & 1-\nu & 0 \\ \text{sym} & & \frac{1-2\nu}{2} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} \tag{1.76}$$

1.4.6 The Classical Hooke's Law

The one-dimensional stress-strain relationship in the direction of the base vector $\boldsymbol{e}_1$, is based upon the assumptions

$$\sigma_{22} = \sigma_{33} = \sigma_{23} = 0 \tag{1.77}$$

wherefrom, by application of Eq. (1.62) and Table 1.1, the following relation between stresses and strains results.

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{12} \\ \sigma_{13} \end{bmatrix} = \begin{bmatrix} \frac{\mu(3\lambda+2\mu)}{\lambda+\mu} & 0 & 0 \\ & \mu & 0 \\ \text{sym} & & \mu \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ 2\varepsilon_{12} \\ 2\varepsilon_{13} \end{bmatrix} = \begin{bmatrix} E & 0 & 0 \\ & G & 0 \\ \text{sym} & & G \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ 2\varepsilon_{12} \\ 2\varepsilon_{13} \end{bmatrix} \tag{1.78}$$

The strain components ε_{22} and ε_{33} can be expressed as functions of the normal strain ε_{11}.

$$\varepsilon_{22} = \varepsilon_{33} = -\frac{\lambda}{2(\lambda+\mu)}\,\varepsilon_{11} = -\nu\,\varepsilon_{11} \tag{1.79}$$

The *classical Hooke's law* describes the one-dimensional stress-strain relationship of a truss element or a spring.

$$\sigma_{11} = \frac{\mu(3\lambda+2\mu)}{\lambda+\mu}\,\varepsilon_{11} = E\,\varepsilon_{11} \tag{1.80}$$

1.5 Initial Boundary Value Problem of Elastomechanics

The summary of the fundamental equations of the three-dimensional continuum, developed in the previous sections, forms the initial boundary value problem of elastomechanics. In detail, these were the description of deformation in the context of kinematics, the formulation of the force equilibrium based on kinetic considerations, the constitutive equation as well as the initial and boundary conditions.

1.5.1 Characterization

The character of the initial boundary value problem of structural mechanics depends on the type of structure and loading that have to be described, which, on the other hand, decisively affect the modelling of the load-carrying behaviour. In the previous sections, the essential modelling aspects were already discussed on a geometrical and material level. In summary, the modelling can be classified, in essence, according to the aspects of

- geometrical linearity or non-linearity,
- material linearity or non-linearity,
- and time-dependence or time-independence.

The various approximation levels differ significantly in the complexity of the numerical solution of the underlying physical problem. The correlation between the simplification of the physical problem and the complexity of the numerical solution is illustrated in Fig. 1.13. Furthermore, the dynamic or static formulation of the problem is decisive for the effort expanded on the numerical solution.

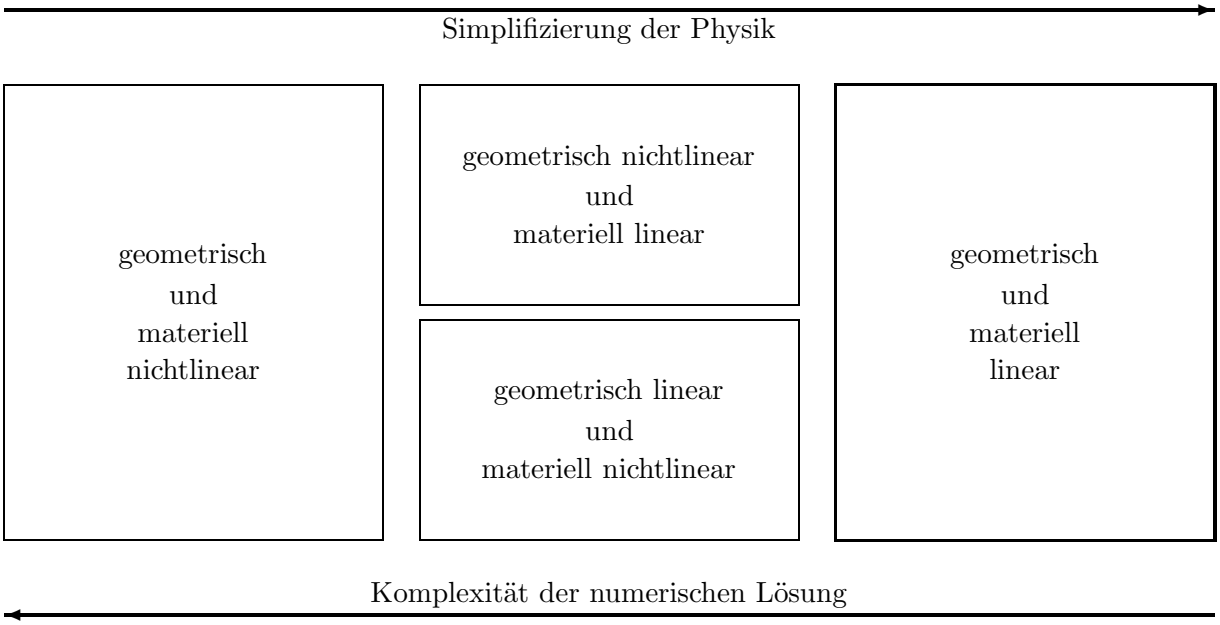


Figure 1.13: Charakterization of elastodynamics according to the type of non-linearity

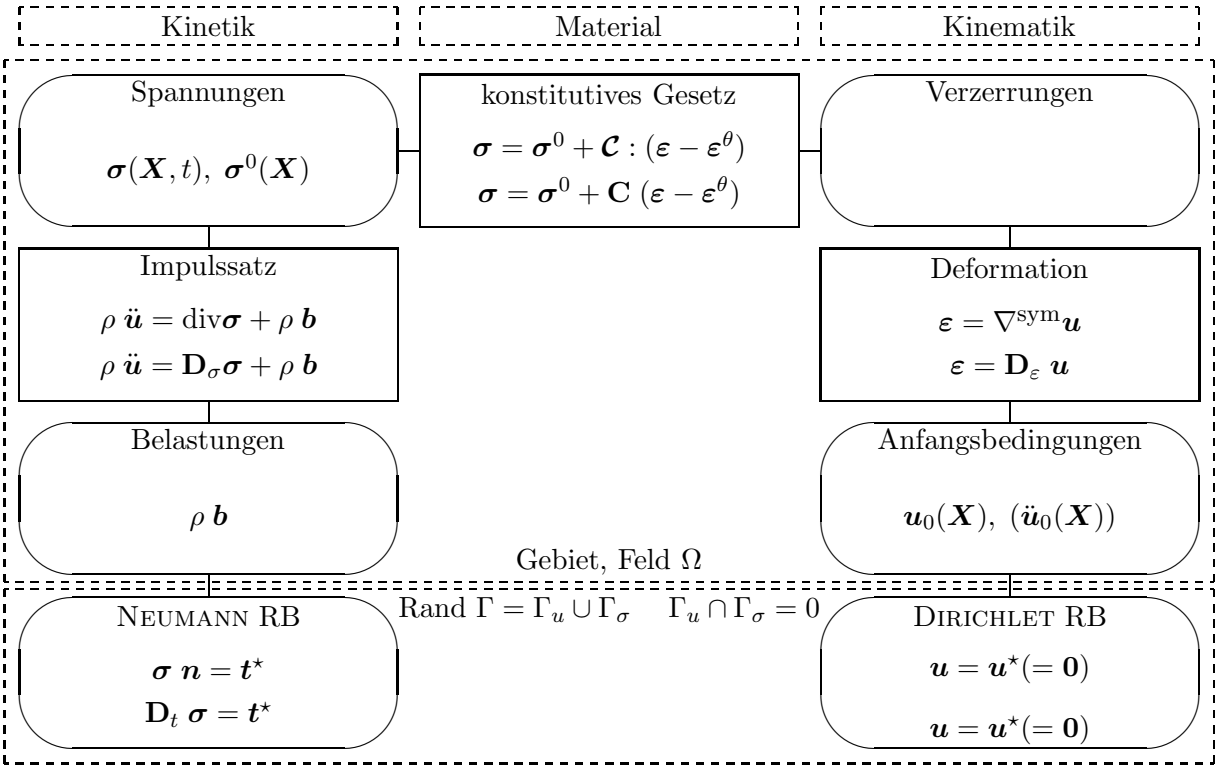


Figure 1.14: Initial boundary value problem of linear elastodynamics

1.5.2 Geometrically and Materially Linear Elastodynamics

Under corresponding prerequisites, namely small deformations and small strains, it is allowed to perform structural analyses in the context of the geometrically and materially linear theory. The field equations, initial and boundary conditions of the corresponding initial boundary value problem of linear elastodynamics are summarized in Fig. 1.14. The essential components of the description of small, linear elastic deformations make for the formulation of the relationship

between displacement and strain field, the equilibrium of forces and the constitutive equation relating the stresses and strains. All three components (in tensor notation these are Equations (1.13, 1.25, 1.56)) together form the second order partial differential equation of linear elastodynamics with the displacement field as the solution variable.

$$\begin{aligned} \rho\,\ddot{\boldsymbol{u}} &= \mathrm{div}\boldsymbol{\sigma} + \rho\,\boldsymbol{b} \\ \boldsymbol{\sigma} &= \boldsymbol{\mathcal{C}} : \boldsymbol{\varepsilon} \\ \boldsymbol{\varepsilon} &= \nabla^{\mathrm{sym}}\boldsymbol{u} \end{aligned} \qquad \rho\,\ddot{\boldsymbol{u}} - \rho\,\boldsymbol{b} = \mathrm{div}\,(\boldsymbol{\mathcal{C}} : \nabla^{\mathrm{sym}}\boldsymbol{u}) \tag{1.81}$$

Here, vanishing thermal strains $\boldsymbol{\varepsilon}^{\theta} = \mathbf{0}$ and initial stresses $\boldsymbol{\sigma}^{0} = \mathbf{0}$ were presumed. For the solution of the differential equation (1.81), the definition of the Dirichlet and Neumann boundary conditions, as well as the initial conditions of the displacement or acceleration field, have to be added (see Chapter 1.3.2, 1.3.3 und 1.3.4).

1.5.3 Geometrically and Materially Linear Elastostatics

In the case of static or quasi-static analyses of structures, the initial boundary value problem is reduced to a boundary value problem by neglecting transient effects. The resulting differential equation is given by

$$\mathbf{0} = \mathrm{div}\,(\boldsymbol{\mathcal{C}} : \nabla^{\mathrm{sym}}\boldsymbol{u}) + \rho\,\boldsymbol{b} \tag{1.82}$$

Furthermore, the Dirichlet and Neumann boundary conditions supplement the *boundary value problem* of elastostatics equivalent to the transient analysis.

1.6 Weak Form of The Initial Boundary Value Problem

The local behaviour of an elastic body was fully described in the previous sections by means of the initial boundary value problem. In general, the solution of this differential equation is not possible analytically. Therefore, approximation methods, in particular the Finite Element Method, are used in order to find an approximate solution. This method actually does not solve the so-called *strong form* of the differential equation. It merely solves its integral over the domain, the so-called *weak form* of the differential equation. This weak formulation forms the basic prerequisite for the application of approximation methods. Integral principles of mechanics are

- the *principle of virtual displacements* or *principle of virtual work*,
- the *principle of virtual forces*
- and the *principle of the minimum of total potential* or its generalization for transient considerations, the *Hamiltion's principle of continuum.*

The principle of the minimum of the total potential requires the existence of a potential, whereby its applicability remains restricted to the structural mechanics of hyperelastic materials. Applied to structural mechanics, the principle of virtual forces represents the method of force magnitudes, which turned out to be inconvenient in the computer-oriented implementation, see Argyris & Mlejnek [4]. In contrast to that, the finite element method based on the principle of virtual work is universally applicable for arbitrary materials and excellently programmable. The derivation and discussion of the principle of virtual work in linear structural mechanics is what this section focuses on.

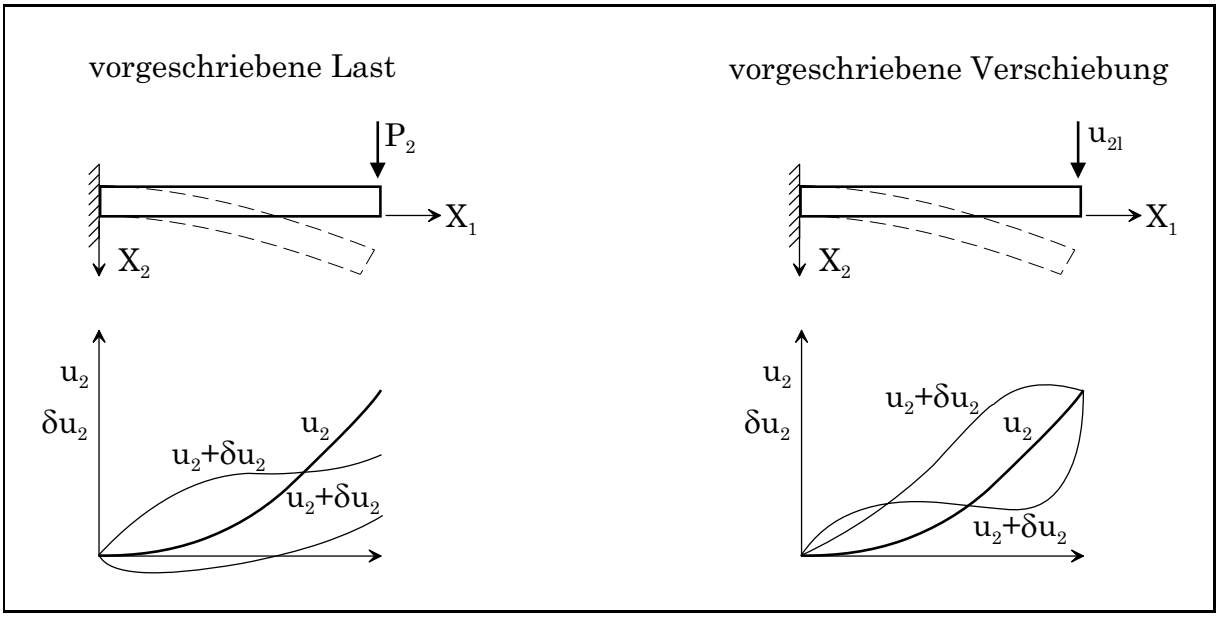


Figure 1.15: Examples for admissible test functions δu for cantilever beams

1.6.1 Principle of Virtual Work

For the generation of the principle of virtual work, the strong form of the differential equation, which corresponds with the local balance of momentum, as well as the static boundary condition are scalarly multiplied by a vector-valued test function and integrated over the volume, respectively over the NEUMANN boundary, of the body under consideration. As test function the virtual displacements $\delta\boldsymbol{u}$ are chosen. This special test function has the following properties (see Fig. 1.15):

- $\delta\boldsymbol{u}$ satisfies the geometrical boundary conditions

$$\delta\boldsymbol{u} = \boldsymbol{0} \quad \forall \quad \boldsymbol{X} \in \Gamma_u \tag{1.83}$$

- $\delta\boldsymbol{u}$ satisfies the field conditions

$$\nabla^{\text{sym}}\delta\boldsymbol{u} = \delta\boldsymbol{\varepsilon} \tag{1.84}$$

- $\delta\boldsymbol{u}$ is infinitesimal
- $\delta\boldsymbol{u}$ is arbitrary

The weak formulation of the balance of momentum (1.25) and of the static boundary condition (1.40) results from the reformulation of these fundamental equations,

$$\boldsymbol{0} = \rho\,\ddot{\boldsymbol{u}} - \text{div}\boldsymbol{\sigma} - \rho\,\boldsymbol{b} \qquad\qquad \boldsymbol{0} = \boldsymbol{\sigma}\cdot\boldsymbol{n} - \boldsymbol{t}^{\star} \tag{1.85}$$

multiplication by the test function $\delta\boldsymbol{u}$, integration over the volume, respectively over the NEUMANN boundary, and addition of the integral terms.

$$\int_{\Omega} \delta\boldsymbol{u}\cdot(\rho\,\ddot{\boldsymbol{u}} - \rho\,\boldsymbol{b})\;dV - \int_{\Omega} \delta\boldsymbol{u}\cdot\text{div}\boldsymbol{\sigma}\;dV + \int_{\Gamma_\sigma} \delta\boldsymbol{u}\cdot(\boldsymbol{\sigma}\cdot\boldsymbol{n} - \boldsymbol{t}^{\star})\;dA = 0 \tag{1.86}$$

For the further simplification of this equation, the term $\delta\boldsymbol{u}\cdot\mathrm{div}\boldsymbol{\sigma}$ is considered first. The latter can be transformed into $\mathrm{div}(\delta\boldsymbol{u}\cdot\boldsymbol{\sigma})$ by application of the product rule for divergence. Additionally, the interchangeability of the order of application of variation with the symbol δ and differentiation with the symbol ∇ is utilized for further simplification.

$$\begin{aligned}\mathrm{div}(\delta\boldsymbol{u}\cdot\boldsymbol{\sigma}) &= \delta\boldsymbol{u}\cdot\mathrm{div}\boldsymbol{\sigma} \quad + \nabla\delta\boldsymbol{u}:\boldsymbol{\sigma} = \delta\boldsymbol{u}\cdot\mathrm{div}\boldsymbol{\sigma} \quad + \delta\nabla\boldsymbol{u}:\boldsymbol{\sigma}\\ \delta\boldsymbol{u}\cdot\mathrm{div}\boldsymbol{\sigma} &= \mathrm{div}(\delta\boldsymbol{u}\cdot\boldsymbol{\sigma}) - \nabla\delta\boldsymbol{u}:\boldsymbol{\sigma} = \mathrm{div}(\delta\boldsymbol{u}\cdot\boldsymbol{\sigma}) - \delta\nabla\boldsymbol{u}:\boldsymbol{\sigma}\end{aligned} \tag{1.87}$$

This simplification can be derived or proved by means of components.

$$\underbrace{(\delta u_i\,\sigma_{ij})_{,j}}_{\mathrm{div}(\delta\boldsymbol{u}\cdot\boldsymbol{\sigma})} = \underbrace{\delta u_i\,\sigma_{ij,j}}_{\delta\boldsymbol{u}\cdot\mathrm{div}\boldsymbol{\sigma}} + \underbrace{\delta u_{i,j}\,\sigma_{ij}}_{\delta\nabla\boldsymbol{u}:\boldsymbol{\sigma}} \tag{1.88}$$

Furthermore, the Gauss theorem for the divergence of a first order tensor is applied to the volume integral of the term

$$\int_\Omega \mathrm{div}(\delta\boldsymbol{u}\cdot\boldsymbol{\sigma})\,dV = \int_\Gamma \delta\boldsymbol{u}\cdot\boldsymbol{\sigma}\cdot\boldsymbol{n}\,dA = \int_{\Gamma_\sigma} \delta\boldsymbol{u}\cdot\boldsymbol{\sigma}\cdot\boldsymbol{n}\,dA \tag{1.89}$$

It was possible to substitute the boundary Γ in the above equation by the Neumann boundary Γ_σ, since the test function $\delta\boldsymbol{u}$ is zero at the Dirichlet boundary, in accordance with (1.83). Using the equations (1.86), (1.87) and (1.89), the weak form of the momentum equation can be represented as follows:

$$\int_\Omega \delta\boldsymbol{u}\cdot(\rho\,\ddot{\boldsymbol{u}} - \rho\,\boldsymbol{b})\,dV + \int_\Omega \delta\nabla\boldsymbol{u}:\boldsymbol{\sigma}\,dV - \int_{\Gamma_\sigma} \delta\boldsymbol{u}\cdot\boldsymbol{\sigma}\cdot\boldsymbol{n}\,dA + \int_{\Gamma_\sigma} \delta\boldsymbol{u}\cdot(\boldsymbol{\sigma}\cdot\boldsymbol{n} - \boldsymbol{t}^\star)\,dA = 0 \tag{1.90}$$

Finally, the term $\delta\nabla\boldsymbol{u}:\boldsymbol{\sigma}$ is examined and rewritten in an alternative form. This term represents the double contraction of the symmetrical stress tensor $\boldsymbol{\sigma}$ (see Eq. (1.20)) and the non-symmetrical tensor $\delta\nabla\boldsymbol{u}$. $\delta\nabla\boldsymbol{u}$ can be substituted by the symmetrical part of this tensor. The latter again can be substituted by using the definition of the strain tensor (1.13).

$$\delta\nabla\boldsymbol{u}:\boldsymbol{\sigma} = (\delta\nabla\boldsymbol{u})^{\mathrm{sym}}:\boldsymbol{\sigma} = \delta\nabla^{\mathrm{sym}}\boldsymbol{u}:\boldsymbol{\sigma} = \delta\boldsymbol{\varepsilon}:\boldsymbol{\sigma} \tag{1.91}$$

This reformulation can be proved by component representation.

$$\underbrace{\frac{1}{2}\,(\delta u_{i,j} + \delta u_{j,i})\;\sigma_{ij}}_{\delta\boldsymbol{\varepsilon}:\boldsymbol{\sigma}} = \frac{1}{2}\,(\delta u_{i,j}\,\sigma_{ij} + \delta u_{j,i}\,\sigma_{ji}) = \frac{1}{2}\,(\delta u_{i,j}\,\sigma_{ij} + \delta u_{i,j}\,\sigma_{ij}) = \underbrace{\delta u_{i,j}\,\sigma_{ij}}_{\delta\nabla\boldsymbol{u}:\boldsymbol{\sigma}} \tag{1.92}$$

After the first equality sign, the symmetry of the stress tensor ($\sigma_{ij} = \sigma_{ji}$) was introduced and after the second one - the interchangeability of dumb indices.

Thus, the principle of virtual work is derived in its usual form, with the scalar product of the variation of the strain tensor and the stress tensor.

$$\int_\Omega \delta\boldsymbol{u}\cdot\ddot{\boldsymbol{u}}\,\rho\,dV + \int_\Omega \delta\boldsymbol{\varepsilon}:\boldsymbol{\sigma}\,dV = \int_\Omega \delta\boldsymbol{u}\cdot\boldsymbol{b}\,\rho\,dV + \int_{\Gamma_\sigma} \delta\boldsymbol{u}\cdot\boldsymbol{t}^\star\,dA \tag{1.93}$$

The separate summands in Eq. (1.94) are described as *virtual work of the inertial forces* δW_{dyn}, *internal virtual work* δW_{int} and *virtual work of the external forces* or *external virtual work* δW_{ext}.

$$\delta W_{\text{dyn}} + \delta W_{\text{int}} = \delta W_{\text{ext}} \qquad \begin{aligned} \delta W_{\text{dyn}} &= \int_{\Omega} \delta \boldsymbol{u} \cdot \ddot{\boldsymbol{u}}\, \rho\, dV \\ \delta W_{\text{int}} &= \int_{\Omega} \delta \boldsymbol{\varepsilon} : \boldsymbol{\sigma}\; dV \\ \delta W_{\text{ext}} &= \int_{\Omega} \delta \boldsymbol{u} \cdot \boldsymbol{b}\, \rho\, dV + \int_{\Gamma_\sigma} \delta \boldsymbol{u} \cdot \boldsymbol{t}^{\star}\, dA \end{aligned} \tag{1.94}$$

Componentwise, the principle of virtual work reads as follows:

$$\int_{\Omega} \delta u_i\, \ddot{u}_i\, \rho\, dV + \int_{\Omega} \delta \varepsilon_{ij} \sigma_{ij}\, dV = \int_{\Omega} \delta u_i \cdot b_i\, \rho\, dV + \int_{\Gamma_\sigma} \delta u_i\, t_i^{\star}\, dA \tag{1.95}$$

If, alternatively, the definition of stresses and strains as vectors is used for the generation of finite elements, and additionally the kinematic equation (1.16) and the constitutive law (1.62) are taken into account, one obtains the internal virtual work as function of the displacement vector $\boldsymbol{u}$, the constitutive matrix $\mathbf{C}$ and the differential operator $\mathbf{D}_{\varepsilon}$.

$$\delta W_{\text{int}} = \int_{\Omega} \delta \boldsymbol{\varepsilon} \cdot \boldsymbol{\sigma}\, dV = \int_{\Omega} \delta \boldsymbol{u} \cdot \mathbf{D}_{\varepsilon}^{T}\, \mathbf{C}\, \mathbf{D}_{\varepsilon}\, \boldsymbol{u}\, dV \tag{1.96}$$

1.6.2 Properties of The Principle of Virtual Work

Since the stresses $\boldsymbol{\sigma}(\boldsymbol{\varepsilon})$ are functions of the strains and since these, in turn, depend on the displacements through the geometrical relation $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}(\boldsymbol{u})$, the equations (1.94)-(1.96) represent a conditional equation for the unknown displacements $\boldsymbol{u}$. If one knows the solution of these equations, then this is also the solution of the corresponding strong form, namely the equilibrium condition (1.25). Since the equations (1.94)-(1.96) must hold for arbitrary test functions $\delta \boldsymbol{u}$, they contain the differential equations of the momentum law and the static boundary conditions.

If, however, the principle of virtual work is not solved exactly but by means of approximation functions (as is the case with the Finite Element Method), the solutions of the weak and the strong form are not identical. The approximate solution for the displacements, when introduced into the strong form of the momentum law, results in an error, the so-called residuum. This means that the strong and the weak forms are identical in the continuous case but not in the discrete one. Since the integral form of the equilibrium equation and the NEUMANN boundary condition allows local errors, it forms the basis for the development of the Finite Element Method.

As a consequence of the choice of the virtual displacement $\delta \boldsymbol{u}$ as a special type of a test function which satisfies the geometrical boundary conditions (1.83), the geometrical boundary conditions are strongly fulfilled in the principle of virtual work. On the other hand, the equilibrium and

the static boundary conditions are only weakly fulfilled, i.e. in an integral sense, because of their multiplication by the test function and integration over the volulme.

Due to the significance of the principle of virtual work for the development of the Finite Element Method, the obtained notions should be summarized in the end:

- The DIRICHLET boundary conditions are strongly fulfilled in the principle of virtual work.
- The NEUMANN boundary conditions and the equilibrium equation must be fulfilled only weakly in the principle of virtual work.
- The advantage of the integral form over the differential form lies in the fact that the weak form 'forgives' local errors which can arise during approximation, as long as the differential equation is fulfilled in an integral sense.
- Because of this reason, the weak formulation forms the basis for the Finite Element Method.

Chapter 2

Spatial Isoparametric Truss Elements

The practical application of finite truss elements is of secondary importance. Pure truss structures can also be computed in a simple and straightforward manner by the classical methods of statics. Only the combination with other finite elements, such as volume or plate elements, for the modelling of complex structures, distinguishes the finite truss element from the classical methods of static analysis of truss structures. As the example of the 'tetraeder' shows, in the computation of spatial truss structures, it is dealt with the modelling of real structures as an ensemble of straight, massless and buckling-free connected truss elements.

The development of isoparametric finite truss elements presented here is methodically oriented towards the generation of multidimensional isoparametric finite elements, and of structural elements, such as beam and plate elements. Since the truss element is considered as the simplest element with respect to the element development and the mechanical and mathematical understanding related to it, it is superbly suited to represent the fundamental methodics and relationships of the Finite Element Method. Therefore, the didactic significance of the truss element is to be evaluated quite differently from its practical relevance. It is excellently suited to 'enlighten' the students in the secrets of modelling, approximation and finite element generation.

Based on the motion of the elastic three-dimensional continuum formulated in the previous chapter, the fundamental equations of one-dimensional continua are derived first. In analogy to essentially more complicated structural models, such as disks, plates, beams or shells, this derivation is based, beside the definiton of geometry, only on two additional assumptions related to kinetics and kinematics. These are the assumptions of the emerging one-dimensional stress state and the constant displacement field over the cross section of the truss. The necessary steps for the finite element analysis of a structure will be explained by means of spatial truss structures and the resulting differential equation of separate truss elements. In particular, this can be decomposed into the three main processes of domain partitioning or localization, finite element discretization, assembly and solution of the resulting system equation. The main emphasis of the finite element development is placed here on the discretization, which in turn is decomposed into the approximation of the primary variables, the dependent variables and, in the end, the virtual work. Through the last step, the internal virtual work, the external virtual work and the virtual work of the inertial forces can be calculated in an approximative way by means of the element stiffness matrix, the consistent load vector, the element mass matrix and a finite number of element parameters, the element displacement vector and the element acceleration vector.

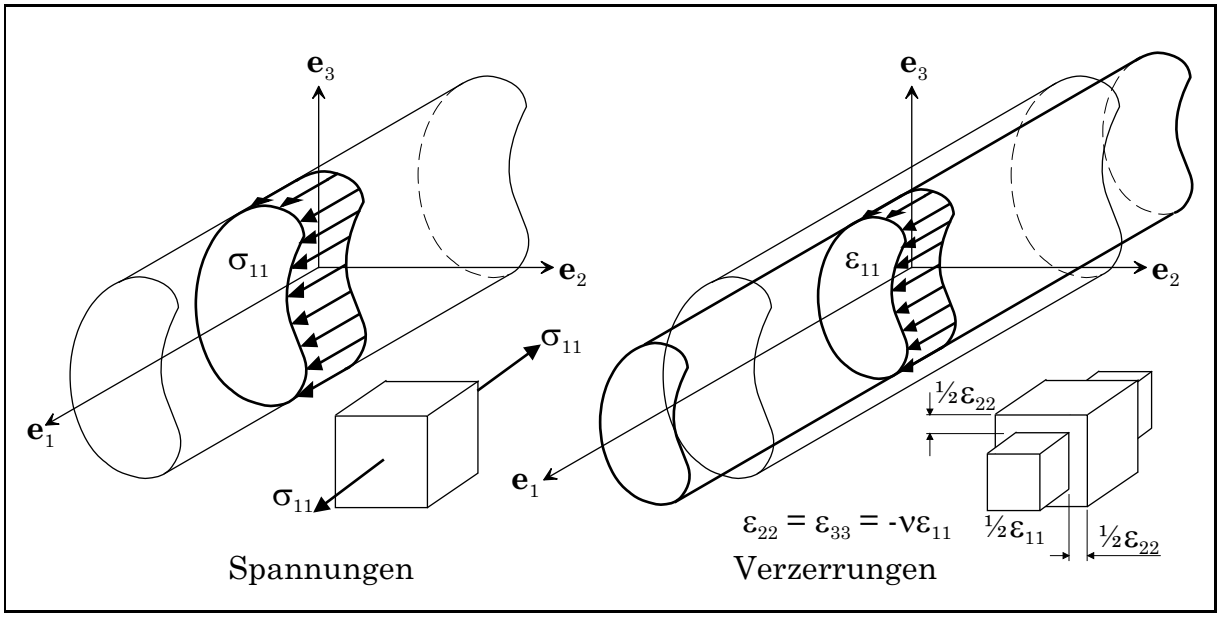


Figure 2.1: Kinetics and Kinematics of the spatial truss element

Recommended literature: Quoted finite element books (especially Bathe [7, 8], Cook & Malkus [17], Knothe & Wessels [24], Krätzig & Başar [25], Ottosen & Petersson [28], Szabó & Babuška [32], Zienkiewicz & Taylor [36]).

2.1 Fundamental Equations of One-dimensional Continua

2.1.1 Geometry

A truss element is characterized geometrically by the aspect of slenderness. This means that the length L of the truss in one direction is essentially greater than the dimensions in orthogonal directions. Furthermore, for the simplification of the following derivations, a truss element with a constant cross-sectional area A is assumed. Without loss of generality, the coordinate system is chosen in such a way, so that the base vector $\boldsymbol{e}_1$ points in the direction of the longitudinal axis of the truss.

2.1.2 Kinetics

The first *fundamental assumption* for the modelling of the truss is that five of the six stress components of the stress tensor are zero.

$$\sigma_{22} = \sigma_{33} = \sigma_{12} = \sigma_{23} = \sigma_{13} = 0 \tag{2.1}$$

The remaining stress state $\sigma_{11} \neq 0$ is sketched in Fig.2.1, where the constant normal stress distribution follows by means of the material law from the kinematic assumption in Chapter 2.1.3. From the requirement postulated in Eq. (2.1), and taking into consideration the Cauchy Eq. (1.40)

$$\begin{bmatrix} \sigma_{11} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} t_1^\star \\ t_2^\star \\ t_3^\star \end{bmatrix} \qquad \boldsymbol{\sigma}\, \boldsymbol{n} = \boldsymbol{t}^\star \tag{2.2}$$

it follows directly that only the component $t_1^\star$ of the surface stress vector $\boldsymbol{t}^\star$ could be different from zero.

$$t_1^\star = \sigma_{11}\, n_1 \qquad t_2^\star = t_3^\star = 0 \tag{2.3}$$

For a truss of a constant cross-sectional area A the normal vector $\boldsymbol{n}$ of the outer surface of the truss is parallel to the plane, spanned by the base vectors $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$, i.e., the component n_1 of the normal vector being identically zero on the outer surface. Hence, also the stress component $t_1^\star$ vanishes on the outer surface of the truss. However, on the faces this stress component is present.

$$t_1^\star = 0 \qquad \forall \quad \boldsymbol{X} \in \Gamma_\sigma \setminus X_1 = \pm\frac{L}{2} \tag{2.4}$$

2.1.3 Kinematics

The second *fundamental assumption* for the truss concerns the displacement field: all material points of a truss cross section experience the same displacement u_1 in the longitudinal direction.

$$\boxed{u_1 = u_1(X_1) \qquad (2.5)}$$

Consequently, the strain component ε_{11} is also constant across the cross section, according to the defining equation of the strain tensor (1.13).

$$\varepsilon_{11} = \varepsilon_{11}(X_1) = u_{1,1}(X_1) \tag{2.6}$$

The other components of the strain vector of a truss can be obtained using the assumption postulated in Eq. (2.1) and by application of the constitutive law of the three-dimensional continuum (1.62)

$$\begin{bmatrix} \sigma_{11} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2\mu+\lambda & \lambda & \lambda & 0 & 0 & 0 \\ & 2\mu+\lambda & \lambda & 0 & 0 & 0 \\ & & 2\mu+\lambda & 0 & 0 & 0 \\ & & & \mu & 0 & 0 \\ & & & & \mu & 0 \\ \text{sym} & & & & & \mu \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{bmatrix} \tag{2.7}$$

This equation is fulfilled for the prescribed stress state when the shear terms of the strain vector vanish $\varepsilon_{12} = \varepsilon_{23} = \varepsilon_{13} = 0$. For the longitudinal normal strain ε_{11} and the transverse normal strains ε_{22} and ε_{33} the following relationship results.

$$\varepsilon_{22} = \varepsilon_{33} = -\frac{\lambda}{2(\lambda+\mu)}\, \varepsilon_{11} = -\nu\, \varepsilon_{11} \tag{2.8}$$

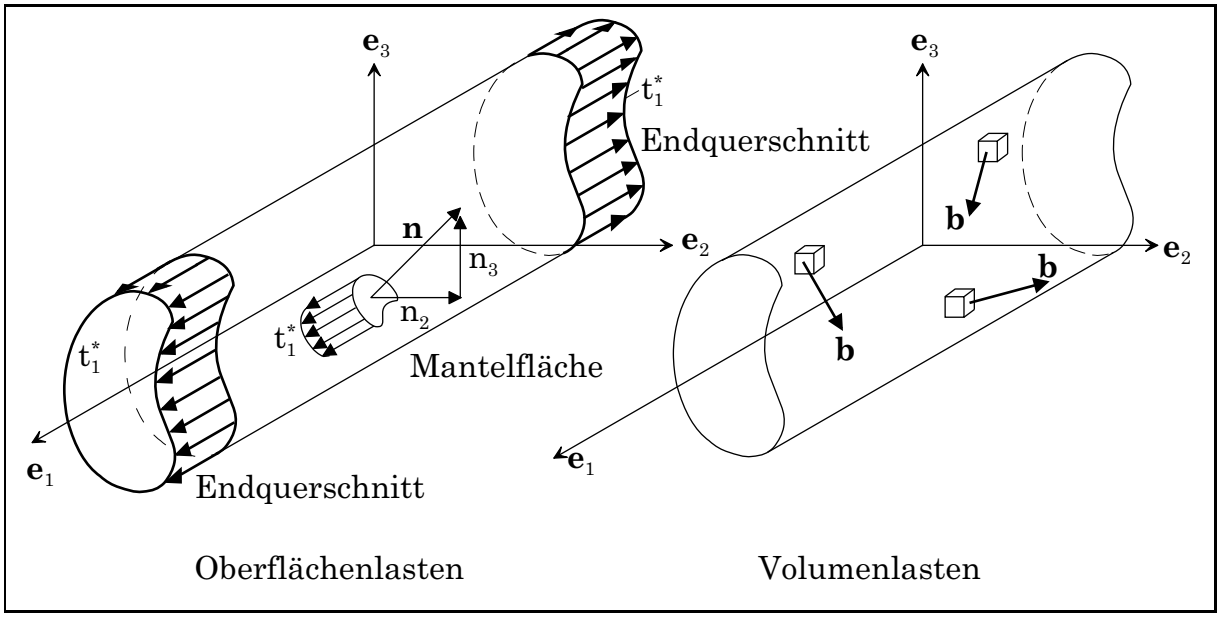


Figure 2.2: Surface and volume loads of the spatial truss element

2.1.4 Constitutive Equation

The constitutive equation of the truss relates the longitudinal normal stresses and strains. Based on the three-dimensional linear elastic constitutive law (1.50, 1.56), the classical HOOKE's law can be developed by application of Eq. (2.8) and the transition to the modulus of elasticity as the characteristic material parameter (Table 1.1)

$$\sigma_{11} = (2\mu + \lambda)\, \varepsilon_{11} + \lambda\, (\varepsilon_{22} + \varepsilon_{33}) = \frac{\mu(3\lambda + 2\mu)}{\lambda + \mu}\, \varepsilon_{11} = E\, \varepsilon_{11} \tag{2.9}$$

As already mentioned, it follows from here that the normal stresses σ_{11} are constant across the truss cross section (see Fig 2.1). From the static boundary condition in Eq. (2.2) also the stress vector $t_1^\star$ is constant at the end cross sections (for all other regions of the outer surface $t_1^\star = 0$, see Eq. (2.4))

$$\sigma_{11} = \sigma_{11}(X_1) \qquad\qquad t_1^\star = t_1^\star(X_1) \quad \forall \quad X_1 = \pm\frac{L}{2} \tag{2.10}$$

2.1.5 Principle of Virtual Work

With kinematics, kinetics and the material law at hand, the essential relations for the formulation of the Principle of Virtual Work of a truss are now available. The fundamental assumptions concerning the stress state and the displacement field, formulated in Eqs. (2.1) and (2.5),together with the introduction of the resulting consequences for the strain state into the principle of virtual work of the three-dimensional continuum (Eq. (1.96)) allow the shear-free stress and

strain representation of the internal virtual work.

$$
\begin{aligned}
&\delta W_{\text{dyn}} + \delta W_{\text{int}} = \delta W_{\text{ext}} = \delta W_{\text{ext}}^{\Gamma} + \delta W_{\text{ext}}^{\Omega} \\
&\delta W_{\text{dyn}} + \delta W_{\text{int}} = \int_{\Omega} \begin{bmatrix} \delta u_1 \\ \delta u_2 \\ \delta u_3 \end{bmatrix} \cdot \begin{bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \\ \ddot{u}_3 \end{bmatrix} \rho \, dV + \int_{\Omega} \begin{bmatrix} \delta\varepsilon_{11} \\ \delta\varepsilon_{22} \\ \delta\varepsilon_{33} \end{bmatrix} \cdot \begin{bmatrix} \sigma_{11} \\ 0 \\ 0 \end{bmatrix} dV \\
&\delta W_{\text{ext}}^{\Gamma} + \delta W_{\text{ext}}^{\Omega} = \int_{\Gamma_\sigma} \begin{bmatrix} \delta u_1 \\ \delta u_2 \\ \delta u_3 \end{bmatrix} \cdot \begin{bmatrix} t_1^{\star} \\ 0 \\ 0 \end{bmatrix} dA \quad + \int_{\Omega} \begin{bmatrix} \delta u_1 \\ \delta u_2 \\ \delta u_3 \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \rho \, dV
\end{aligned}
\tag{2.11}
$$

The internal virtual work δW_{int} and the virtual work of the surface loads $\delta W_{\text{ext}}^{\Gamma}$ contain only terms in the longitudinal direction. Since the accelerations and the volume-specific loads exhibit transverse components different from zero, further detailed studies should provide information for their physical meaning and relevance.

2.1.5.1 Virtual Work of Inertial Forces

For a detailed analysis of the virtual work of inertial forces δW_{dyn}, the integration over the volume element dV is split into the integration over the area element of the cross section dA and the line element of the length of the truss dX_1. Before integration of the virtual work of inertial forces, a functional dependence of the transverse accelerations $\ddot{u}_2$ and $\ddot{u}_3$ and the distribution of the transverse components of the test function $\delta\boldsymbol{u}$ from the coordinates X_2 and X_3 have to be explored. From the definition of the strain components ε_{22} and ε_{33} in Eq. (2.6), the fundamental assumption Eq. (2.5) and Eq. (2.8), and by integrating, one obtains a linear distribution of the displacement and acceleration components across the thickness or width of the truss.

$$
\begin{aligned}
\varepsilon_{22} = u_{2,2} = \frac{\partial u_2}{\partial X_2} = -\nu\, \varepsilon_{11} = -\nu\, u_{1,1} \qquad & u_2 = -\nu\, u_{1,1}\, X_2, \quad \ddot{u}_2 = -\nu\, \ddot{u}_{1,1}\, X_2 \\
\varepsilon_{33} = u_{3,3} = \frac{\partial u_3}{\partial X_3} = -\nu\, \varepsilon_{11} = -\nu\, u_{1,1} \qquad & u_3 = -\nu\, u_{1,1}\, X_3, \quad \ddot{u}_3 = -\nu\, \ddot{u}_{1,1}\, X_3
\end{aligned}
\tag{2.12}
$$

Since the displacements u_2 and u_3 depend linearly on the coordinates X_2 und X_3, the variation of these displacement components is independent of the cross-sectional coordinates X_2 and X_3.

$$
\delta u_2 = \delta u_2(X_1) \qquad\qquad \delta u_3 = \delta u_3(X_1) \tag{2.13}
$$

Further, the acceleration component $\ddot{u}_1(X_1)$ and the variation $\delta u_1(X_1)$ are constant across the cross section A, according to the kinematic assumption (2.5),

$$
\delta u_1 = \delta u_1(X_1) \qquad\qquad \ddot{u}_1 = \ddot{u}_1(X_1) \tag{2.14}
$$

wherefrom the virtual work of the inertial forces can be split into a longitudinal and a transverse part, as follows:

$$
\delta W_{\text{dyn}} = \int_{-\frac{L}{2}}^{\frac{L}{2}} \delta u_1\, \ddot{u}_1\, \rho \underbrace{\int_A dA}_{A}\, dX_1 - \int_{-\frac{L}{2}}^{\frac{L}{2}} \nu\, \rho\, \ddot{u}_{1,1} \begin{bmatrix} \delta u_2 \\ \delta u_3 \end{bmatrix} \cdot \underbrace{\int_A \begin{bmatrix} X_2 \\ X_3 \end{bmatrix} dA}_{\mathbf{0}}\, dX_1 \tag{2.15}
$$

The integral $\int dA$ gives the cross-sectional area A of the truss and the integration of X_2 and X_3 across the cross section is zero when the coordinate origin is in the centre of gravity of the cross section. Under these prerequisites, only the acceleration in the longitudinal direction is contained in the virtual work of the inertial forces.

$$\delta W_{\text{dyn}} = \int_{-\frac{L}{2}}^{\frac{L}{2}} \delta u_1 \; \ddot{u}_1 \; \rho \; A \; dX_1 \tag{2.16}$$

2.1.5.2 Virtual Work of The Surface Loads

The examination of the virtual work of the surface loads $t_1^\star$ is based upon the utilization of Eq. (2.11). The integral over the surface of the truss is split into integrals over the outer surface and the faces of the truss, where the first integral results in zero, since on the outer surface the component $t_1^\star$ of the stress vector $\boldsymbol{t}^\star$ is identical to zero, according to Eqs. (2.3) and (2.4). The virtual work of the stress component $t_1^\star$, which is constant across the cross section, remains to be determined at the end sections of the truss $X_1 = -L/2$ and $X_1 = L/2$.

$$\delta W_{\text{ext}}^{\Gamma} = \left[\delta u_1(\frac{L}{2}) t_1^\star(\frac{L}{2}) + \delta u_1(-\frac{L}{2}) t_1^\star(-\frac{L}{2})\right] \underbrace{\int_A dA}_{A} = \delta u_1(\frac{L}{2}) N_1^\star(\frac{L}{2}) + \delta u_1(-\frac{L}{2}) N_1^\star(-\frac{L}{2}) \tag{2.17}$$

In the last simplification of this equation, the integral $\int dA = A$ was introduced as well as the definition of the prescribed normal forces at the boundary, resulting from the integration of $t_1^\star$ across the cross section.

$$N_1^\star(-\frac{L}{2}) = t_1^\star(-\frac{L}{2}) \; A \qquad\qquad N_1^\star(\frac{L}{2}) = t_1^\star(\frac{L}{2}) \; A \tag{2.18}$$

2.1.5.3 Virtual Work of The Volume Loads

In comparison to Eq. (2.11), the virtual work of the volume loads $\delta W_{\text{ext}}^{\Omega}$ cannot be simplified directly. If one actually applies the Fundamental Lemma of Variational Calculus to Eq. (2.11), taking into account Eq. (2.16), then b_2 and b_3 have to become zero for arbitrary variations δu_2 and δu_3, as the virtual work of the inertial forces (Eq. (2.16)) and the internal virtual work(Eq. (2.11)) as well as the virtual work of the surface loads (Eq. (2.17)) do not contain any terms in δu_2 and δu_3. This means that the assumptions for the truss allow no volume loads normal to the longitudinal axis.

$$b_2 = b_3 = 0 \tag{2.19}$$

The simplest case of a volume load, the self-weight, can be therefore realized only for the special case of identical longitudinal and gravitation vectors consistent with the truss theory. Therefore, the component b_1 gives the only contribution to the virtual work of the volume loads.

$$\delta W_{\text{ext}}^{\Omega} = \int_{-\frac{L}{2}}^{\frac{L}{2}} \delta u_i \; p_1(X_1) \; dX_1 \qquad\qquad p_1(X_1) = \int_A \rho \; b_1(X_1, X_2, X_3) \; dA \tag{2.20}$$

Here, the local balance of momentum $\rho\ \ddot{u}_1 = \sigma_{11,1} + \rho b_1$, according to Eq. (1.24), was used for the definition of the *distributed load* $p_1(X_1)$ as an integral of the volume load ρb_1 over the cross section of the truss. According to it, b_1 must also be independent of the coordinates X_2 and X_3. In order to realize the influence of the volume loads ρb_2 and ρb_3 on a multi-truss structure within the framework of structural analysis, these quantities are introduced as kinematically equivalent node loads r_{b2} and r_{b3}. The kinematically equivalent node loads must satisfy the condition

$$\delta u_i(-\frac{L}{2}) \cdot r_{bi}(-\frac{L}{2}) + \delta u_i(\frac{L}{2}) \cdot r_{bi}(\frac{L}{2}) = \int_{-\frac{L}{2}}^{\frac{L}{2}} \int_A \delta u_i\ b_i\ \rho\ dA\ dX_1 \quad i = 2,3 \tag{2.21}$$

If, in addition, constant specific loads are assumed across the cross section, which is the case, for instance in the formulation of the self-weight, the equivalent node loads result in:

$$\delta u_i(-\frac{L}{2}) \cdot r_{bi}(-\frac{L}{2}) + \delta u_i(\frac{L}{2}) \cdot r_{bi}(\frac{L}{2}) = \int_{-\frac{L}{2}}^{\frac{L}{2}} \delta u_i\ b_i\ \rho\ A\ dX_1 \quad i = 2,3 \tag{2.22}$$

2.1.5.4 Internal Virtual Work

The internal virtual work is formed according to Eq. (2.11) only with the normal components of the strains and stresses in $\boldsymbol{e}_1$ direction. With the splitting of the volume integral, analogously to the virtual work of the inertial forces, the internal virtual work follows as

$$\delta W_{\text{int}} = \int_{-\frac{L}{2}}^{\frac{L}{2}} \delta\varepsilon_{11}\ \sigma_{11} \int_A dA\ dX_1 = \int_{-\frac{L}{2}}^{\frac{L}{2}} \delta\varepsilon_{11}\ \sigma_{11}\ A\ dX_1 \tag{2.23}$$

where care was taken, so that the normal strains and stresses would be independent of the coordinates X_2 and X_3 (Eqs. (2.6) and (2.10)).

2.1.5.5 Principle of Virtual Work of The Truss

The summary of Eqs. (2.11), (2.16), (2.17), (2.20) and (2.23) results in the principle of virtual work for the variation of the displacement δu_1, for the development of a finite truss element.

$$\boxed{\int_{-\frac{L}{2}}^{\frac{L}{2}} \delta u_1\ \ddot{u}_1\ \rho\ A\ dX_1 + \int_{-\frac{L}{2}}^{\frac{L}{2}} A\ \delta\varepsilon_{11}\ \sigma_{11}\ dX_1 = \delta u_1^{e1}\ N_1^{e1} + \delta u_1^{e2}\ N_1^{e2} + \int_{-\frac{L}{2}}^{\frac{L}{2}} \delta u_1\ p_1\ dX_1} \tag{2.24}$$

In Eq. (2.24), the loads and the virtual displacements at the truss ends were defined in the following manner:

$$N_1^{e1} = N_1^{\star}(-\frac{L}{2}), \quad N_1^{e2} = N_1^{\star}(\frac{L}{2}) \qquad\qquad \delta u_1^{e1} = \delta u_1(-\frac{L}{2}), \quad \delta u_1^{e2} = \delta u_1(\frac{L}{2}) \tag{2.25}$$

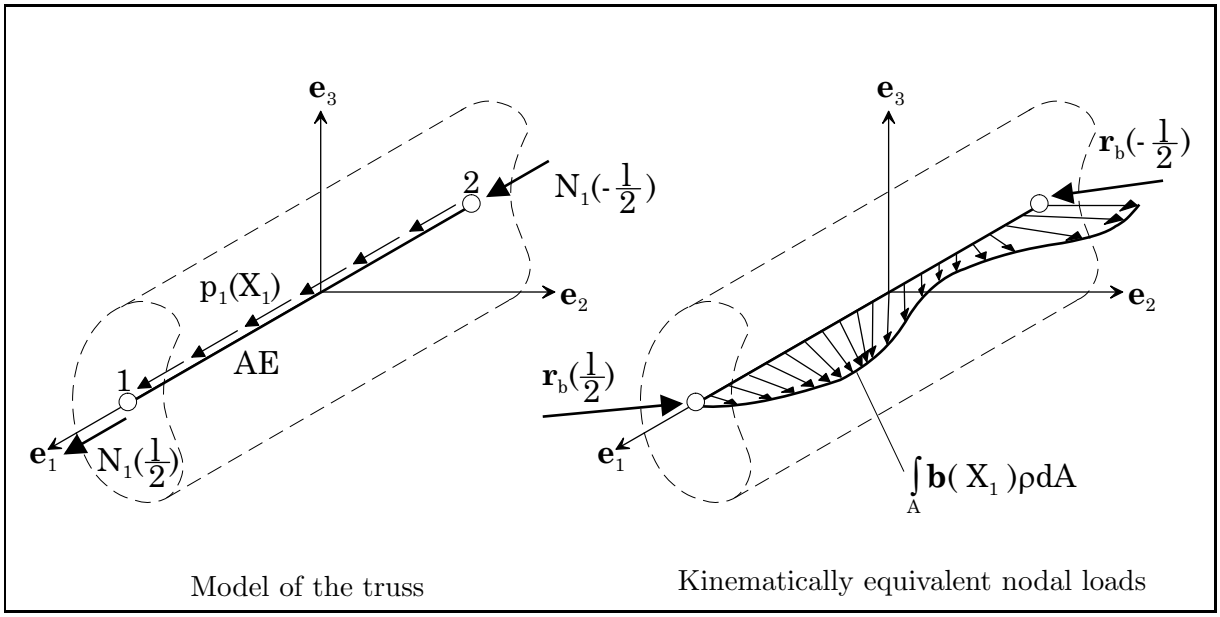


Figure 2.3: Model of the truss and volume loads

2.1.6 Euler Differential Equation and Neumann Boundary Conditions

In order to derive the EULER differential equation and the NEUMANN boundary condition of the truss element by means of the fundamental lemma of variational calculus, the variation of the strain in the term of the internal virtual work, according to Eq. (2.23), must be replaced by the variation of the displacement component. Therefore, the derivative

$$(\delta u_1\ \sigma_{11})_{,1} = \delta u_{1,1}\ \sigma_{11} + \delta u_1\ \sigma_{11,1} = \delta\varepsilon_{11}\ \sigma_{11} + \delta u_1\ \sigma_{11,1} \tag{2.26}$$

was obtained with the help of the product rule and was introduced in Eq. (2.23)[1].

$$\delta W_{\text{int}} = \int_{-\frac{L}{2}}^{\frac{L}{2}} (\delta u_1\ \underbrace{\sigma_{11}\ A}_{N_1})_{,1}\ dX_1 - \int_{-\frac{L}{2}}^{\frac{L}{2}} \delta u_1\ \underbrace{\sigma_{11,1}\ A}_{N_{1,1}}\ dX_1 \tag{2.27}$$

Here, the definition of the normal force of the truss N_1 was introduced as a stress resultant.

$$N_1 = \sigma_{11}\ A \qquad\qquad N_{1,1} = \sigma_{11,1}\ A \tag{2.28}$$

The first integrant in Eq. (2.27) is now transformed by application of the basic law of integral calculus (see e.g. BRONSTEIN & SEMENDJAJEW [48]).[2].

$$\delta W_{\text{int}} = \delta u_1(\frac{L}{2})N_1(\frac{L}{2}) - \delta u_1(-\frac{L}{2})N_1(-\frac{L}{2}) - \int_{-\frac{L}{2}}^{\frac{L}{2}} \delta u_1\ N_{1,1}\ dX_1 \tag{2.29}$$

[1]This corresponds to the one-dimensional representation of Eq. (2.10)
[2]Application of GAUSS' integral theorem in one-dimensional space (2.10)

Together with Eq. (2.23), the weak form of the differential equation and of the NEUMANN boundary condition of the truss is produced.

$$\begin{aligned}\int_{-\frac{L}{2}}^{\frac{L}{2}} \delta u_1 \; \ddot{u}_1 \; \rho \; A \; dX_1 \quad - \quad \int_{-\frac{L}{2}}^{\frac{L}{2}} \delta u_1 \; N_{1,1} \; dX_1 \quad + \quad \delta u_1(\frac{L}{2}) N_1(\frac{L}{2}) \quad - \quad \delta u_1(-\frac{L}{2}) N_1(-\frac{L}{2}) \\ = \quad \int_{-\frac{L}{2}}^{\frac{L}{2}} \delta u_1 \; p_1 \; dX_1 \quad + \quad \delta u_1(\frac{L}{2}) N_1^\star(\frac{L}{2}) \quad + \quad \delta u_1(-\frac{L}{2}) N_1^\star(-\frac{L}{2})\end{aligned} \tag{2.30}$$

The EULER differential equation or the kinetics of the truss follows from the consideration of Eq. (2.30) for arbitrary test functions δu_1. The differential equation is supplemented by the constitutive law according to Eq. (2.9), the kinematics are supplemented according to Eq. (2.6).

$$\begin{aligned} N_{1,1}(X_1) &= \ddot{u}_1(X_1) \; \rho \; A - p_1(X_1) && \text{Kinetik (EULER Differentialgleichung)} \\ N_{1,1}(X_1) &= E \; A \quad \varepsilon_{11,1}(X_1) && \text{konstitutive Gleichung} \\ \varepsilon_{11,1}(X_1) &= u_{1,11}(X_1) && \text{Kinematik} \end{aligned} \tag{2.31}$$

The NEUMANN boundary conditions result from Eq. (2.30) for arbitrary variations at the truss ends $\delta u_1(\frac{L}{2})$ and $\delta u_1(-\frac{L}{2})$.

$$N_1(\frac{L}{2}) = N_1^\star(\frac{L}{2}) \qquad\qquad N_1(-\frac{L}{2}) = -N_1^\star(-\frac{L}{2}) \tag{2.32}$$

2.2 Finite Element Discretization

2.2.1 Partitioning of The Structure into Elements

From the remarks to the principle of virtual work in Chapter 1.6.1 it follows that during the transition from the infinite element to the total body, or from the differential to the integral from, respectively, local errors in the solution are tolerated. The integral balance of the virtual work, however, is fulfilled. This fact motivates the subdivision of a structure into *finite elements*, in which the principle of virtual work is always fulfilled in an integral sense. However, local errors in the fundamental balance equation and NEUMANN boundary conditions are accepted. It is obvious that with diminishing finite elements the exact solution is approximated more accurately. In the limiting case of an infinitely great number of finite elements, the finite element turns into the infinite or differential element and the approximate solution becomes the exact one.

Mathematically, the structure or the domain Ω is formed by the union of domains of finite dimensions Ω^e. Furthermore, the partial domains may not overlap.

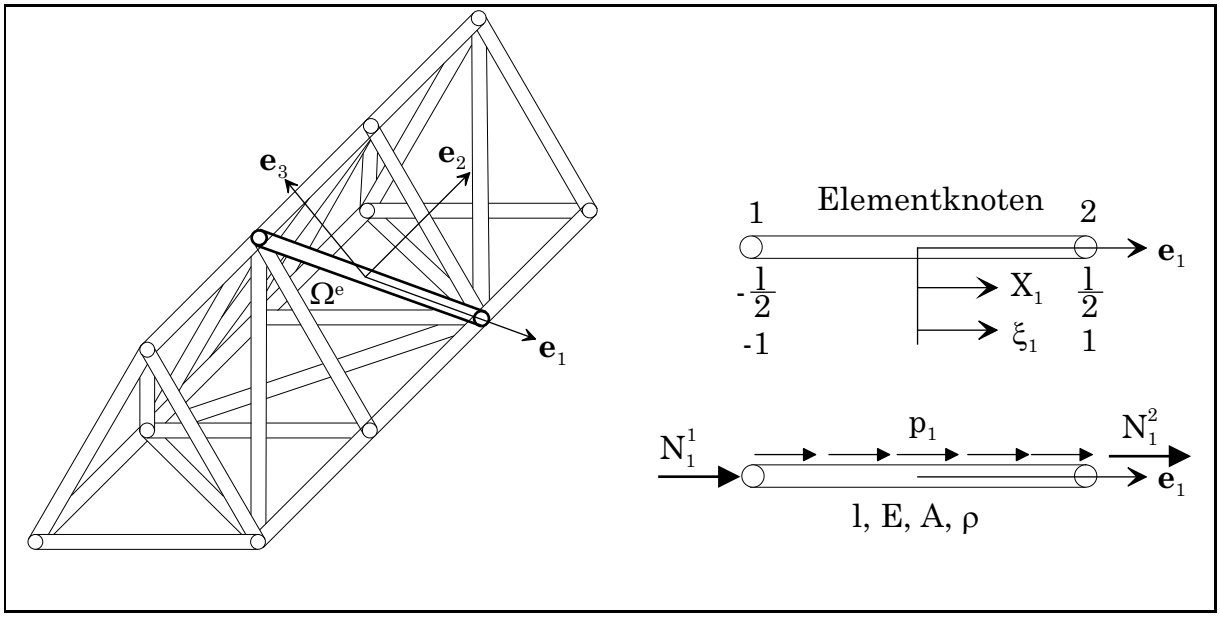


Figure 2.4: Truss frame and finite truss element e

$$\Omega = \bigcup_{e=1}^{NE} \Omega^e \qquad \Omega^i \cap \Omega^j = \emptyset \quad \text{für } i \neq j \tag{2.33}$$

As shown in Fig. 2.4 for the example of a spatial truss structure, the partial domain Ω^e corresponds to a single truss. The principle of virtual work must be fulfilled for the domain Ω and for each domain Ω^e,

$$\delta W_{\text{dyn}} + \delta W_{\text{int}} = \delta W_{\text{ext}} \qquad \delta W^e_{\text{dyn}} + \delta W^e_{\text{int}} = \delta W^e_{\text{ext}} \tag{2.34}$$

where the sum of the virtual work terms of all finite elements must result in the corresponding virtual work terms of the structure.

$$\delta W_{\text{dyn}} = \sum_{e=1}^{NE} \delta W^e_{\text{dyn}}, \quad \delta W_{\text{int}} = \sum_{e=1}^{NE} \delta W^e_{\text{int}}, \quad \delta W_{\text{ext}} = \sum_{e=1}^{NE} \delta W^e_{\text{ext}} \tag{2.35}$$

2.2.2 Approximation of Variables of One-dimensional Continua

In the finite element discretization of a truss element, the (unknown) exact displacement field u_1 is replaced by an approximation $\tilde{u}_1$ (the displacement distribution u_1 depending on the physical coordinate X_1, or on the natural coordinate ξ_1, respectively). The corresponding approximation (shape) function is defined by few parameters and the assumption for its qualititive form. Because of their favourable properties, polynomial functions have established themselves to be used as shape functions. For the systematic representation of such approximation functions of finite elements of similar geometry, shape functions are defined in the natural parameter space and afterwards the developed finite element is transformed to the particular element geometry. Within the framework of the isoparametric finite element concept, the discretization of position $X_1(\xi_1)$, of variation of displacement $\delta u_1(\xi_1)$ and of acceleration $\ddot{u}_1(\xi_1)$ follow directly from the dicretization of displacement. Here, it is sufficient to discuss at first the approximation of displacement field and afterwards to apply it analogously to the other quantities that are to be

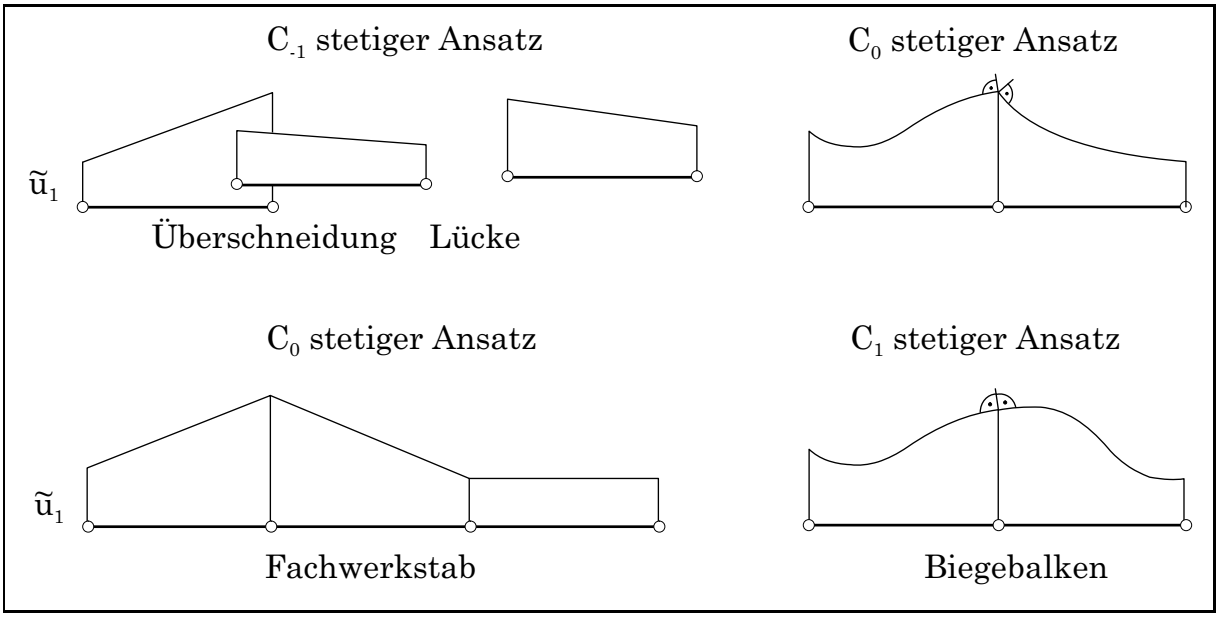


Figure 2.5: Compatibility condition of one-dimensional finite elements

approximated.

2.2.2.1 Choice of A Shape Function

For the formulation of finite elements the shape functions must

- be conformable,
- supply at least constant strains
- and may not supply any strains during rigid body displacements.

The first viewpoint refers to the interaction and compatibility of neighbouring finite elements, whereas the other conditions are restricted locally to each finite element of a structure.

In order for no gaps or overlappings to occur at the element ends during deformation of a structure, the displacements must be conformable. This is guaranteed for a finite element with displacement approximations which satisfy the Dirichlet boundary conditions of the element. In the example of the truss, this requires that the displacements of neighbouring elements have the same value at the common node. Since the transition condition must be fulfilled only for the solution variable u_1 itself, i.e. only for the zero-th derivative of the solution variable, the shape functions are characterized as C_0 conformable (see Fig. 2.5). In contrast to this, elements, which along with the displacement field also use the derivative of the displacement field $u_{1,1}$ as a solution variable (e.g. beam, plate or shell elements), are equipped with C_1 conformable or C_1 continuous shape functions in order to also satisfy the derivatives of the transition condition. If in these cases non-conformable shape functions are used, the external virtual work terms of the nodal loads do not vanish during the assembly of the structure (building-up of the structure from the finite elements, see Chapter 2.3).

Constant strains are required, so that the variation of the strains or the strains themselves, and hence also the internal virtual work, do not become zero. This statement can be verified in the general three-dimensional case by taking into account Eq. (2.6) and the principle of virtual work of a truss (Eq. (2.24)).

The third criterium, which the shape functions must satisfy, is the requirement to be able to describe strain-free rigid body motion. This means that during a deformation-free motion of an element no internal virtual work may arise due to the shape functions.

2.2.2.2 Physical and Natural Coordinates

In the case of a truss element, the position of a point relative to the longitudinal axis is measured in terms of the physical coordinate X_1 or the natural coordinate ξ_1 (see Fig. 2.4).

$$X_1 \in \left[-\frac{L}{2}, \frac{L}{2} \right] \qquad \xi_1 \in \left[-1, 1 \right] \tag{2.36}$$

The relationship between the coordinates X_1 and $\xi_1$is described by the equation

$$X_1 = \frac{L}{2}\, \xi_1 \tag{2.37}$$

wherefrom also the transformation of the corresponding differential line elements dX_1 and $d\xi_1$ is determined by the derivative of the physical coordinate with respect to the natural one.

$$\mathbf{J} = \frac{\partial X_1}{\partial \xi_1} = X_{1;1} = \frac{L}{2} \qquad dX_1 = \frac{L}{2}\, d\xi_1 = |\mathbf{J}|\ d\xi_1 \tag{2.38}$$

Here, the index ; 1 was defined to symbolize the derivative with respect to the natural coordinate ξ_1. Furthermore, in Eq. (2.38) the JACOBI or *functional* matrix $\mathbf{J}$ and its determinant, the JACOBI or *functional determinant* $|\mathbf{J}|$ were defined. In the one-dimensional special case of a truss, the JACOBI matrix is a (1×1) matrix or a scalar. Thus, in the one-dimensional case the JACOBI determinant is identical to the JACOBI matrix.

2.2.2.3 Local Polynomial Approximation of One-dimensional Functions

In general, every arbitrary function $u_1(\xi_1)$ can be *approximated*, or *interpolated*, by means of a LAGRANGE polynomial $\tilde{u}_1(\xi_1)$ of degree p.

$$u_1(\xi_1) \approx \tilde{u}_1(\xi_1) = \sum_{j=0}^{p} \alpha^j\ (\xi_1)^j = \alpha^0 + \alpha^1\ (\xi_1) + \alpha^2\ (\xi_1)^2 + \alpha^3\ (\xi_1)^3 + \cdots + \alpha^p\ (\xi_1)^p \tag{2.39}$$

Here, the coefficients α^j are to be chosen in such a way that $\tilde{u}_1$, at the element node i described by ξ_1^i, be identical to the displacements $\tilde{u}_1(\xi_1^i) = u_1^{ei}$. As an example the condition correlated to the element node 1 is given here.

$$u_1(\xi_1^1) = u_1^{e1} = \tilde{u}_1(\xi_1^1) = \sum_{j=0}^{p} \alpha^j\ \left(\xi_1^1\right)^j = \alpha^0 + \alpha^1 \left(\xi_1^1\right) + \alpha^2 \left(\xi_1^1\right)^2 + \alpha^3 \left(\xi_1^1\right)^3 + \cdots + \alpha^p \left(\xi_1^1\right)^p \tag{2.40}$$

The corresponding $p+1$ conditions can be summarized in matrix form as follows:

$$\begin{bmatrix} u_1^{e1} \\ u_1^{e2} \\ \vdots \\ u_1^{ep+1} \end{bmatrix} = \begin{bmatrix} 1 & \left(\xi_1^1\right) & \cdots & \left(\xi_1^1\right)^p \\ 1 & \left(\xi_1^2\right) & \cdots & \left(\xi_1^2\right)^p \\ \vdots & \vdots & & \vdots \\ 1 & \left(\xi_1^{p+1}\right) & \cdots & \left(\xi_1^{p+1}\right)^p \end{bmatrix} \begin{bmatrix} \alpha^0 \\ \alpha^1 \\ \vdots \\ \alpha^p \end{bmatrix} \tag{2.41}$$

The matrix used in this equation is refered to as VANDERMONDE matrix in the respective literature. By inverting Eq. (2.41) one obtains the parameters α^j, wherefrom the LAGRANGE polynomial (2.39), defined by discrete pairs of values $(\xi_1^i, u_1(\xi_1^i))$, can be adapted to arbitrary functions $u_1(\xi_1)$.

With the described approach any function can be interpolated by means of a LAGRANGE polynomial. However, the determination of the abstract parameters α^j seems a little cumbersome. If one considers the generalized form of the VANDERMONDE matrix in Eq. (2.41), it is straightforward to carry out also its inversion in general form and to describe Eq. (2.39) directly with the nodal displacements u_1^{ei} as parameters. This results in the approximation of an arbitrary displacement distribution $u_1(\xi_1)$ by means of the nodal displacements u_1^{ei} and the *Lagrangian interpolation polynomial* $N^i(\xi_1)$.

$$u_1(\xi_1) \approx \tilde{u}_1(\xi_1) = \sum_{i=1}^{p+1} u_1^{ei}\, N^i(\xi_1) = \mathbf{N}(\xi_1)\, \boldsymbol{u}^e \tag{2.42}$$

In the last reformulation in Eq. (2.42), the summation was replaced by the matrix product of the *matrix of shape functions* $\mathbf{N}(\xi_1)$ and the *element displacement vector* $\boldsymbol{u}^e$, where $\mathbf{N}(\xi_1)$ and $\boldsymbol{u}^e$ are defined in the following way:

$$\mathbf{N}(\xi_1) = \Big[\, N^1(\xi_1)\; N^2(\xi_1)\; \cdots\; N^{NN}(\xi_1) \,\Big] \qquad \boldsymbol{u}^e = \Big[\, u_1^{e1}\; u_1^{e2}\; \cdots\; u_1^{eNN} \,\Big]^T \tag{2.43}$$

$NN = p + 1$ symbolizes the number of element nodes or points of support. For the complete description of Eq. (2.42) the computation of the LAGRANGE interpolation polynomials $N^i(\xi_1)$ is needed. Generally, a LAGRANGE'ian interpolation polynomial of degree p can be formed by the product

$$N^i(\xi_1) = \prod_{\substack{k=1 \\ k \neq i}}^{p+1} \frac{\xi_1^k - \xi_1}{\xi_1^k - \xi_1^i} \tag{2.44}$$

with ξ_1^k characterizing the position of the node k and N^i - the shape function associated to the node i. It should be noted that the LAGRANGE interpolation polynomials exhibit the *interpolation property*. This means that the shape functions $N^i(\xi_1)$ take the value one at the node i and are zero at the remaining nodes $k \neq i$.

$$N^i(\xi_1^k) = \begin{cases} 1 & \text{für} \quad i = k \\ 0 & \text{für} \quad i \neq k \end{cases} \tag{2.45}$$

This statement can be verified by introducing $\xi_1 = \xi_1^k$ and $\xi_1 = \xi_1^i$ into Eq. (2.44).

2.2.2.4 Approximation of The Derivative of The Primary Variables

Together with the approximation of the primary variables, the approximation of the derivative of the variables with respect to the natural coordinate ξ_1 is also of significance within the framework of the finite element formulation of undeformable truss elements. This derivative forms the basis for the approximation of the strain $\tilde{\varepsilon}_{11}$ in the finite truss element. In Eq. (2.42), only the shape functions N^i are functions of the natural coordinate ξ_1, the displacements u_1^{ei}, on the contrary, are discrete nodal displacements and are hence independent of ξ_1.

$$\frac{\partial u_1(\xi_1)}{\partial \xi_1} = u_{1;1}(\xi_1) \approx \tilde{u}_{1;1}(\xi_1) = \sum_{i=1}^{p+1} u_1^{ei}\, N^i_{;1}(\xi_1) = \mathbf{N}_{;1}(\xi_1)\, \boldsymbol{u}^e \tag{2.46}$$

2.2.2.5 Linear, Quadratic and Cubic Interpolation Polynomials

Here, the LAGRANGE interpolation polynomials used for one-dimensional finite elements must be defined for $p = 1$, $p = 2$ and $p = 3$. For the illustration of the method of generation of shape functions, the linear shape functions are discussed in detail, whereas for the quadratic and cubic approximation only the result is given. The LAGRANGE polynomial, according to Eq. (2.39), results for $p = 1$ in the polynomial approximation of the linear truss element.

$$u_1(\xi_1) \approx \tilde{u}_1(\xi_1) = \sum_{j=0}^{1} \alpha^j\ (\xi_1)^j = \alpha^0 + \alpha^1\, \xi_1 \tag{2.47}$$

By substituting the support points $\xi_1 = -1$ and $\xi_1 = 1$, one obtains the VANDERMONDE matrix through Eq. (2.41), and by its inversion - the coefficients α^0 and α^1.

$$\begin{bmatrix} u_1^{e1} \\ u_1^{e2} \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha^0 \\ \alpha^1 \end{bmatrix} \qquad \begin{bmatrix} \alpha^0 \\ \alpha^1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} u_1^{e1} \\ u_1^{e2} \end{bmatrix} \tag{2.48}$$

Inserting the coefficients α^0 and α^1 into Eq. (2.47) yields, by comparison with Eq. (2.42), the shape functions $N^1(\xi_1)$ und $N^2(\xi_2)$ of the linear truss element.

$$\begin{aligned} u_1(\xi_1) \approx \tilde{u}_1(\xi_1) = \frac{1}{2}(u_1^{e1} + u_1^{e2}) - \frac{1}{2}(u_1^{e1} - u_1^{e2})\, \xi_1 &= \frac{1}{2}(1 - \xi_1)\, u_1^{e1} + \frac{1}{2}(1 + \xi_1)\, u_1^{e2} \\ &= \quad N^1(\xi_1)\, u_1^{e1} + \quad N^2(\xi_1)\, u_1^{e2} \end{aligned} \tag{2.49}$$

Alternatively, the already developed shape functions $N^1(\xi_1)$ and $N^2(\xi_2)$ can be computed directly according to Eq. (2.44). With the nodal coordinates $\xi_1 = -1$ and $\xi_1 = 1$ the following expressions result:

$$N^1(\xi_1) = \frac{1 - \xi_1}{1 - (-1)} = \frac{1}{2}(1 - \xi_1) \qquad N^2(\xi_1) = \frac{(-1) - \xi_1}{(-1) - 1} = \frac{1}{2}(1 + \xi_1) \tag{2.50}$$

Analogously,the quadratic and cubic LAGRANGE interpolation polynomials can be generated. A summary of the shape functions is to be found in Table 2.1, their visualization as well as the illustration of the displacement approximation are shown in Fig. 2.6. The derivative of the shape functions can follow based on Eqs. (2.50)

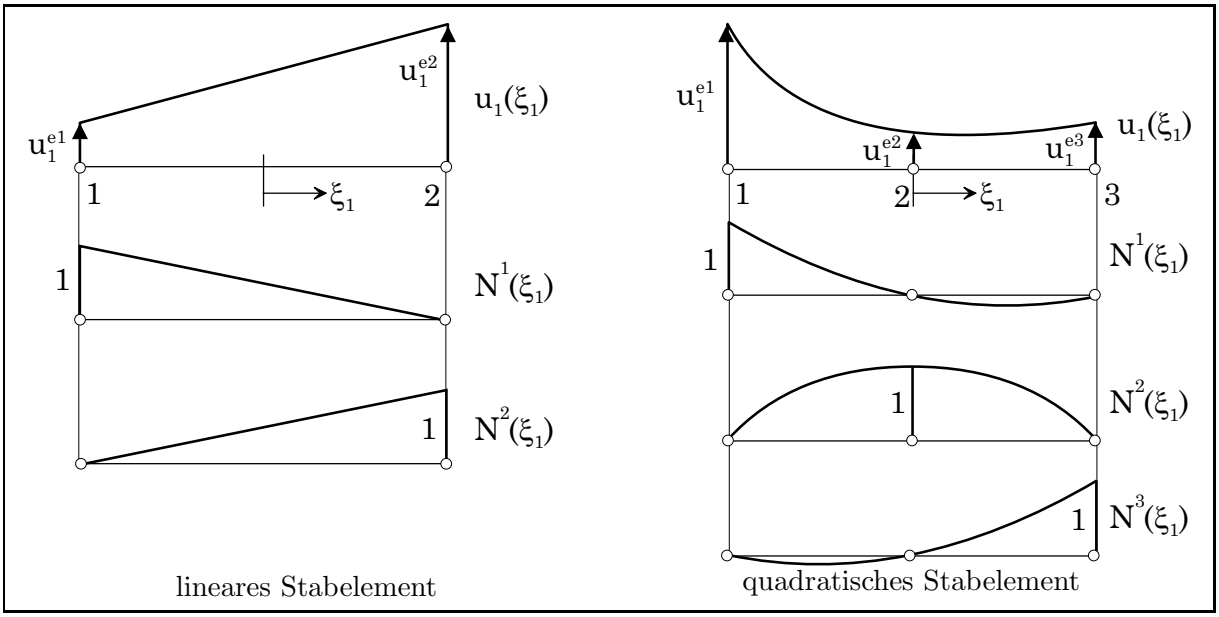


Figure 2.6: Linear, quadratic and cubic shape functions of the truss element

$$N^1_{;1}(\xi_1) = -\frac{1}{2} \qquad\qquad N^2_{;1}(\xi_1) = \frac{1}{2} \tag{2.51}$$

The derived linear, quadratic and cubic shape functions are summarized in Table 2.1.

2.2.3 Truss Element with Linear Shape Functions

The generation of the linear truss element is based on the principle of virtual work on an element level (2.34), where the virtual work terms are defined in Eq. (2.24), as well as the approximation of the displacement field and its derivative in terms of the natural coordinates ξ_1. Thus, the

$N^i(\xi_1)$	linear $p = 1$ 1 — 2	quadratisch $p = 2$ 1 — 2 — 3	kubisch $p = 3$ 1 — 2 — 3 — 4
$N^1(\xi_1)$	$\frac{1}{2}(1-\xi_1)$	$\frac{1}{2}(\xi_1 - 1)\,\xi_1$	$\frac{9}{16}(1-\xi_1)(\xi_1^2 - \frac{1}{9})$
$N^2(\xi_1)$	$\frac{1}{2}(1+\xi_1)$	$1-\xi_1^2$	$\frac{27}{16}(\xi_1^2 - 1)(\xi_1 - \frac{1}{3})$
$N^3(\xi_1)$		$\frac{1}{2}(1+\xi_1)\,\xi_1$	$\frac{27}{16}(1-\xi_1^2)(\xi_1 + \frac{1}{3})$
$N^4(\xi_1)$			$\frac{9}{16}(\xi_1 + 1)(\xi_1^2 - \frac{1}{9})$
$N^1_{;1}(\xi_1)$	$-\frac{1}{2}$	$\xi_1 - \frac{1}{2}$	$\frac{1}{16}(-27\,\xi_1^2 + 18\,\xi_1 + 1)$
$N^2_{;1}(\xi_1)$	$\frac{1}{2}$	$-2\,\xi_1$	$\frac{9}{16}(9\,\xi_1^2 - 2\,\xi_1 - 3)$
$N^3_{;1}(\xi_1)$		$\xi_1 + \frac{1}{2}$	$\frac{9}{16}(-9\,\xi_1^2 - 2\,\xi_1 + 3)$
$N^4_{;1}(\xi_1)$			$\frac{1}{16}(-27\,\xi_1^2 + 18\,\xi_1 - 1)$

Table 2.1: Shape functions of one-dimensional finite elements and their derivatives

finite element is defined by the choice of nodal degrees of freedom u_1^{e1} and u_1^{e2} and the shape functions N^1 and N^2.

2.2.3.1 Linear Shape Functions

The linear shape functions of the truss element were made available already in Section 2.2.2.5. They were introduced for the approximation of the physical coordinate $X_1(\xi_1)$, the displacement $u_1(\xi_1)$, the variation of the displacement $\delta u_1(\xi_1)$ and the acceleration $\ddot{u}_1(\xi_1)$. For the matrix notation of the approximation relations, the LAGRANGE interpolation polynomials were put together in the matrix of shape functions.

$$N^1(\xi_1) = \frac{1}{2}(1-\xi_1) \qquad N^2(\xi_1) = \frac{1}{2}(1+\xi_1) \qquad \mathbf{N}(\xi_1) = \begin{bmatrix} N^1(\xi_1) & N^2(\xi_1) \end{bmatrix} \tag{2.52}$$

2.2.3.2 Approximation of Variables

The approximation of the continuous fields $u_1(\xi_1)$, $\delta u_1(\xi_1)$ and $\ddot{u}_1(\xi_1)$ is realized by means of the matrix of shape functions $\mathbf{N}(\xi_1)$, the *element displacement vector* $\boldsymbol{u}^e$, the *variation of the element displacement vector* $\delta\boldsymbol{u}^e$ and the *element acceleration vector* $\ddot{\boldsymbol{u}}^e$.

$$\begin{aligned} u_1(\xi_1) &\approx \tilde{u}_1(\xi_1) &&= \mathbf{N}(\xi_1)\ \boldsymbol{u}^e & \boldsymbol{u}^e &= \begin{bmatrix} u_1^{e1} & u_1^{e2} \end{bmatrix}^T \\ \delta u_1(\xi_1) &\approx \delta\tilde{u}_1(\xi_1) &&= \mathbf{N}(\xi_1)\ \delta\boldsymbol{u}^e & \delta\boldsymbol{u}^e &= \begin{bmatrix} \delta u_1^{e1} & \delta u_1^{e2} \end{bmatrix}^T \\ \ddot{u}_1(\xi_1) &\approx \tilde{\ddot{u}}_1(\xi_1) &&= \mathbf{N}(\xi_1)\ \ddot{\boldsymbol{u}}^e & \ddot{\boldsymbol{u}}^e &= \begin{bmatrix} \ddot{u}_1^{e1} & \ddot{u}_1^{e2} \end{bmatrix}^T \end{aligned} \tag{2.53}$$

2.2.3.3 Isoparametric Approximation of Coordinates

In preparing the *isoparametric* finite element concept, the typical approach of this concept in the example of the linear truss element is to be demonstrated here. By the isoparametric concept the physical coordinate X_1 is described, in analogy to the displacement $u_1(\xi_1)$, by means of the shape functions $N^i(\xi_1)$ in the parameter space and the physical coordinates of the element nodes X_1^{e1} and X_1^{e2}.

$$X_1(\xi_1) \approx \tilde{X}_1(\xi_1) = \mathbf{N}(\xi_1)\ \boldsymbol{X}^e \qquad \boldsymbol{X}^e = \begin{bmatrix} X_1^{e1} & X_1^{e2} \end{bmatrix}^T \tag{2.54}$$

In the case of a linear truss element, this results in the mapping of coordinates from the parameter space ξ_1, identical to Eq. (2.37), into the physical space X_1.

$$\tilde{X}_1(\xi_1) = X_1^{e1}\ \frac{1-\xi_1}{2} + X_1^{e2}\ \frac{1+\xi_1}{2} = -\frac{L}{2}\ \frac{1-\xi_1}{2} + \frac{L}{2}\ \frac{1+\xi_1}{2} = \frac{L}{2}\ \xi_1 \tag{2.55}$$

Consequently, also the approximated JACOBI matrix

$$\mathbf{J} = \frac{\partial \tilde{X}_1}{\partial \xi_1} = \frac{L}{2} \tag{2.56}$$

corresponds to the exact JACOBI matrix, defined in Eq. (2.38).

2.2.3.4 Approximation of Strains

For the approximation of the strain field $\varepsilon_{11}(\xi_1)$, the coordinate transformation between the physical and natural coordinates (X_1 and ξ_1) must be taken into account in the derivation of the displacement field $u_{1,1} = \partial u_1/\partial X_1$. This means that the derivative of the approximated displacement field $u_1(\xi_1)$ with respect to the coordinate X_1 in Eq. (2.6) must be computed by the application of the chain rule.

$$\varepsilon_{11}(\xi_1) = u_{1,1}(\xi_1) = \frac{\partial u_1(\xi_1(X_1))}{\partial X_1} = \underbrace{\frac{\partial u_1(\xi_1)}{\partial \xi_1}}_{u_{1;1}(\xi_1)} \underbrace{\frac{\partial \xi_1}{\partial X_1}}_{\mathbf{J}^{-1}} = u_{1;1}(\xi_1)\ \frac{2}{L} \tag{2.57}$$

In the last reformulation, the inverse of the JACOBI matrix $\mathbf{J}^{-1} = 2/L$ according to Eq. (2.38) was introduced. Thus, the approximation of the strain field is given by means of the approximation of the displacement field (2.42).

$$\varepsilon_{11}(\xi_1) \approx \tilde{\varepsilon}_{11}(\xi_1) = \frac{2}{L}\ \frac{\partial}{\partial \xi_1}\left[\sum_{i=1}^{2} u_1^{ei}\ N^i(\xi_1)\right] = \frac{2}{L}\ \sum_{i=1}^{2} u_1^{ei}\ \frac{\partial N^i(\xi_1)}{\partial \xi_1} = \frac{2}{L}\ \sum_{i=1}^{2} u_1^{ei}\ N^i_{;1}(\xi_1) \tag{2.58}$$

In forming the derivative with respect to ξ_1 it was born in mind that the nodal degrees of freedom u_1^{ei} do not depend on the variables ξ_1, they are discrete and moreover are defined only at the nodes k. Equally, for the approximation of the strain field, the vector notation can be used instead of summation

$$\varepsilon_{11}(\xi_1) \approx \tilde{\varepsilon}_{11}(\xi_1) = \frac{2}{L}\ \begin{bmatrix} N^1_{;1}(\xi_1) & N^2_{;1}(\xi_1) \end{bmatrix} \begin{bmatrix} u_1^{e1} \\ u_1^{e2} \end{bmatrix} = \mathbf{B}(\xi_1)\ \boldsymbol{u}^e \tag{2.59}$$

with the *B-operator* or *differential operator* $\mathbf{B}$

$$\mathbf{B}(\xi_1) = \frac{2}{L}\ \begin{bmatrix} N^1_{;1}(\xi_1) & N^2_{;1}(\xi_1) \end{bmatrix} = \begin{bmatrix} -\frac{1}{L} & \frac{1}{L} \end{bmatrix} \tag{2.60}$$

being defined. In general, this differential operator is a matrix and in the special case of one-dimensional linear elements, it is a transposed vector or a (2×1) matrix. It relates the approximation of local continuous strains ε_{11} with the discrete element displacement vector $\boldsymbol{u}^e$. Analogously to Eqs. (2.58) and (2.59), the variation of the normal strain $\delta\varepsilon_{11}$ can also be approximated.

$$\delta\varepsilon_{11}(\xi_1) \approx \delta\tilde{\varepsilon}_{11}(\xi_1) = \frac{2}{L}\ \sum_{i=1}^{2} \delta u_1^{ei}\ N^i_{;1}(\xi_1) = \frac{2}{L}\ \begin{bmatrix} N^1_{;1}(\xi_1) & N^2_{;1}(\xi_1) \end{bmatrix} \begin{bmatrix} \delta u_1^{e1} \\ \delta u_1^{e2} \end{bmatrix} = \mathbf{B}(\xi_1)\ \delta\boldsymbol{u}^e \tag{2.61}$$

For the generation of the differential operator (2.60), the shape functions must be differentiated with respect to the coordinate ξ_1. The derivative of the shape functions with respect to the natural coordinate ξ_1 can be obtained from Eq. (2.51)

$$N^1_{;1}(\xi_1) = -\frac{1}{2} \qquad\qquad N^2_{;1}(\xi_1) = \frac{1}{2} \tag{2.62}$$

of the linear truss element. Thus, in the special case of a linear truss element, the differential operator $\mathbf{B}$ is independent of the position ξ_1.

2.2.3.5 Approximated Internal Virtual Work

The approximated internal virtual work is obtained from Eq. (2.23) by plugging-in the approximation of the variation of the strain $\delta\tilde{\varepsilon}_{11}$ and the material law $\sigma_{11} = E\ \varepsilon_{11}$, in which, on the other hand, the approximation of the strain $\tilde{\varepsilon}_{11}$ is inserted.

$$\delta\tilde{W}^e_{\text{int}} = \int_{-\frac{L}{2}}^{\frac{L}{2}} \delta\tilde{\varepsilon}_{11}\ E\ A\ \tilde{\varepsilon}_{11}\ dX_1 = \int_{-1}^{1} \delta\tilde{\varepsilon}_{11}\ E\ A\ \tilde{\varepsilon}_{11}\ |\mathbf{J}|\ d\xi_1 = \int_{-1}^{1} \delta\tilde{\varepsilon}_{11}\ E\ A\ \tilde{\varepsilon}_{11}\ \frac{L}{2}\ d\xi_1 \tag{2.63}$$

In Eq. (2.63),the determinant of the Jacobi matrix, the so-called *Jacobi determinant* or *functional determinant* $|\mathbf{J}|$ according to Eq. (2.38), was inserted for the transformation of the line element dX_1 to $d\xi_1$. For one-dimensional finite elements, the Jacobi matrix and Jacobi determinant are certainly identical. The distinction is made here only in order to clarify the similarities to multi-dimensional elements, discussed in the following chapters. Introducing the approximations of the strain and the variation of strain according to Eqs. (2.58) and (2.61) results in sum notation in:

$$\begin{aligned}
\delta\tilde{W}^e_{\text{int}} &= \int_{-1}^{1} \left(\frac{2}{L} \sum_{i=1}^{2} \delta u_1^{ei}\ N^i_{;1}(\xi_1) \right)\ E\ A\ \left(\frac{2}{L} \sum_{j=1}^{2} u_1^{ej}\ N^j_{;1}(\xi_1) \right)\ \frac{L}{2}\ d\xi_1 \\
&= \frac{2\ E\ A}{L} \sum_{i=1}^{2} \sum_{j=1}^{2} \int_{-1}^{1} \delta u_1^{ei}\ N^i_{;1}(\xi_1)\ N^j_{;1}(\xi_1)\ u_1^{ej}\ d\xi_1 \\
&= \frac{2\ E\ A}{L} \sum_{i=1}^{2} \sum_{j=1}^{2} \delta u_1^{ei} \int_{-1}^{1} N^i_{;1}(\xi_1)\ N^j_{;1}(\xi_1)\ d\xi_1\ u_1^{ej}
\end{aligned} \tag{2.64}$$

In the reformulations, it was taken into account that the displacement components u_1^{ei} as well as the variations δu_1^{ej} are discrete nodal quantities and hence are independent of the position ξ_1. Due to this reason, it is possible to take the displacement degrees of freedom out of the integral. Alternatively, Eq. (2.64) can be formulated in matrix notation with the matrix of shape functions $\mathbf{N}(\xi_1)$ (Eqs. (2.59) and (2.61)).

$$\begin{aligned}
\delta\tilde{W}^e_{\text{int}} &= \begin{bmatrix} \delta u_1^{e1} \\ \delta u_1^{e2} \end{bmatrix} \cdot \frac{2\ E\ A}{L} \int_{-1}^{1} \begin{bmatrix} N^1_{;1}(\xi_1) \\ N^2_{;1}(\xi_1) \end{bmatrix} \begin{bmatrix} N^1_{;1}(\xi_1) & N^2_{;1}(\xi_1) \end{bmatrix}\ d\xi_1 \begin{bmatrix} u_1^{e1} \\ u_1^{e2} \end{bmatrix} \\
&= \begin{bmatrix} \delta u_1^{e1} \\ \delta u_1^{e2} \end{bmatrix} \cdot \frac{2\ E\ A}{L} \int_{-1}^{1} \begin{bmatrix} N^1_{;1}(\xi_1)\ N^1_{;1}(\xi_1) & N^1_{;1}(\xi_1)\ N^2_{;1}(\xi_1) \\ N^2_{;1}(\xi_1)\ N^1_{;1}(\xi_1) & N^2_{;1}(\xi_1)\ N^2_{;1}(\xi_1) \end{bmatrix}\ d\xi_1 \begin{bmatrix} u_1^{e1} \\ u_1^{e2} \end{bmatrix}
\end{aligned} \tag{2.65}$$

Since the shape functions $N^i(\xi)$ are linear it follows that their derivatives $N^i_{;1} = \pm 1/2$ are constant (see Eqs. (2.51)). Because of this, the integral in Eq. (2.65) must be formed only over

the line element $d\xi_1$ with $\int_{-1}^{1} d\xi_1 = 2$.

$$\begin{aligned}
\delta\tilde{W}^e_{\text{int}} &= \begin{bmatrix} \delta u_1^{e1} \\ \delta u_1^{e2} \end{bmatrix} \cdot \frac{4\,E\,A}{L} \begin{bmatrix} N^1_{;1}\,N^1_{;1} & N^1_{;1}\,N^2_{;1} \\ N^2_{;1}\,N^1_{;1} & N^2_{;1}\,N^2_{;1} \end{bmatrix} \begin{bmatrix} u_1^{e1} \\ u_1^{e2} \end{bmatrix} \\
&= \underbrace{\begin{bmatrix} \delta u_1^{e1} \\ \delta u_1^{e2} \end{bmatrix}}_{\delta \boldsymbol{u}^e} \cdot \underbrace{\frac{E\,A}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}}_{\mathbf{k}^e} \underbrace{\begin{bmatrix} u_1^{e1} \\ u_1^{e2} \end{bmatrix}}_{\boldsymbol{u}^e}
\end{aligned} \tag{2.66}$$

The matrix $\mathbf{k}^e$ is the *element stiffness matrix* with respect to the coordinate system $\boldsymbol{e}_i$ and the element displacement vector $\boldsymbol{u}^e$. Thus, the approximated internal virtual work can be computed with the element displacement vector, the variation of the element displacement vector and the element stiffness matrix.

$$\boxed{\delta\tilde{W}^e_{\text{int}} = \delta\boldsymbol{u}^e \cdot \mathbf{k}^e\,\boldsymbol{u}^e \qquad \mathbf{k}^e = \frac{E\,A}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix},\ \boldsymbol{u}^e = \begin{bmatrix} u_1^{e1} \\ u_1^{e2} \end{bmatrix}} \tag{2.67}$$

It should be noted at this place once more that the integration for the generation of the element stiffness matrix $\mathbf{k}^e$ had to be performed only over constant coefficients in ξ_1.

Alternatively to the above method for development of the discrete virtual work, now the B-operator (Eq. (2.60)) must be used for the computation of the approximated virtual work. When introduced in Eq. (2.63), for the substitution of the approximated strains and their variation (Eqs. (2.59) and (2.61)) with the element displacement vector $\boldsymbol{u}^e$ and its variation $\delta\boldsymbol{u}^e$, respectively, the differential operator $\mathbf{B}$ results in:

$$\delta\tilde{W}^e_{\text{int}} = \int_{-1}^{1} (\mathbf{B}\,\delta\boldsymbol{u}^e) \cdot\ E\,A\ (\mathbf{B}\,\boldsymbol{u}^e)\ \frac{L}{2}\,d\xi_1 = \int_{-1}^{1} \delta\boldsymbol{u}^e \cdot \mathbf{B}^T\,E\,A\,\mathbf{B}\,\boldsymbol{u}^e\ |\mathbf{J}|\ d\xi_1 \tag{2.68}$$

As before, the element displacement vector and its variation can also be taken out of the integral here. If now the term EA is interpreted as generalized 'material stiffness matrix', one obtains from Eq. (2.68) a formulation for the Finite Element Method of the internal virtual work.

$$\boxed{\delta\tilde{W}^e_{\text{int}} = \delta\boldsymbol{u}^e \cdot \int_{-1}^{1} \mathbf{B}^T\,E\,A\,\mathbf{B}\ |\mathbf{J}|\ d\xi_1\ \boldsymbol{u}^e = \delta\boldsymbol{u}^e \cdot \mathbf{k}^e\,\boldsymbol{u}^e} \tag{2.69}$$

Of course, the integration of Eq. (2.69) likewise results in the approximate internal virtual work from Eq. (2.67). The advantage of Eq. (2.69) lies in its standardized computational pattern of the element stiffness matrix $\mathbf{k}^e$, which is further found in generalized form in the formulation of two- and three-dimensional finite elements.

2.2.3.6 Approximated External Virtual Work

The virtual work of the external loads can be described, according to Eq. (2.24), as function of the prescribed nodal loads N_1^i and the prescribed distributed load $p_1(\xi_1)$

$$\delta W_{\text{ext}}^e = \begin{bmatrix} \delta u_1^{e1} \\ \delta u_1^{e2} \end{bmatrix} \cdot \begin{bmatrix} N_1^{e1} \\ N_1^{e2} \end{bmatrix} + \int_{-1}^{1} \delta u_1(\xi_1)\, p_1(\xi_1)\, \frac{L}{2}\, d\xi_1 \tag{2.70}$$

If the distributed load p_1 is given as function of the physical coordinates, it must be transformed to the natural coordinates. Therefore, from Eq. (2.70) only the variation of displacements has to be approximated according to Eq. (2.53).

$$\begin{aligned} \delta \tilde{W}_{\text{ext}}^e &= \begin{bmatrix} \delta u_1^{e1} \\ \delta u_1^{e2} \end{bmatrix} \cdot \begin{bmatrix} N_1^{e1} \\ N_1^{e2} \end{bmatrix} + \int_{-1}^{1} \left(\sum_{i=1}^{2} \delta u_1^{ei}\, N^i(\xi_1) \right) p_1(\xi_1)\, \frac{L}{2}\, d\xi_1 \\ &= \begin{bmatrix} \delta u_1^{e1} \\ \delta u_1^{e2} \end{bmatrix} \cdot \underbrace{\begin{bmatrix} N_1^{e1} \\ N_1^{e2} \end{bmatrix}}_{\boldsymbol{r}_n^e} + \begin{bmatrix} \delta u_1^{e1} \\ \delta u_1^{e2} \end{bmatrix} \cdot \underbrace{\int_{-1}^{1} \begin{bmatrix} N^1(\xi_1) \\ N^2(\xi_1) \end{bmatrix} p_1(\xi_1)\, \frac{L}{2}\, d\xi_1}_{\boldsymbol{r}_p^e} \end{aligned} \tag{2.71}$$

In Eq. (2.71), the load vector of the nodal loads $\boldsymbol{r}_n^e$ and the *consistent element load vector* $\boldsymbol{r}_p^e$ were defined.

$$\boxed{\boldsymbol{r}_p^e = \begin{bmatrix} r_p^{e1} \\ r_p^{e2} \end{bmatrix} = \int_{-1}^{1} \begin{bmatrix} N^1(\xi_1) \\ N^2(\xi_1) \end{bmatrix} p_1(\xi_1)\, \frac{L}{2}\, d\xi_1 = \int_{-1}^{1} \mathbf{N}^T(\xi_1)\, p_1(\xi_1)\, |\mathbf{J}|\, d\xi_1 \qquad (2.72)}$$

The element nodal loads $\boldsymbol{r}_n^e$ and the consistent element loads $\boldsymbol{r}_p^e$ in the element load vector $\boldsymbol{r}^e$ finally yield the virtual work of the external loads.

$$\delta \tilde{W}_{\text{ext}}^e = \delta \boldsymbol{u}^e \cdot \boldsymbol{r}_n^e + \delta \boldsymbol{u}^e \cdot \boldsymbol{r}_p^e = \delta \boldsymbol{u}^e \cdot \boldsymbol{r}^e \tag{2.73}$$

For the special case of a constant uniform load, the consistent element load vector

$$\boldsymbol{r}_p^e = \begin{bmatrix} r_p^{e1} \\ r_p^{e2} \end{bmatrix} = \frac{p_1\, L}{4} \int_{-1}^{1} \begin{bmatrix} 1 - \xi_1 \\ 1 + \xi_1 \end{bmatrix} d\xi_1 = \frac{p_1\, L}{4} \begin{bmatrix} \xi_1 - \frac{1}{2}\xi_1^2 \\ \xi_1 + \frac{1}{2}\xi_1^2 \end{bmatrix}_{-1}^{1} = \frac{p_1\, L}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \tag{2.74}$$

represents the integral load $L\, p_1$, distributed equally onto the element nodes.

2.2.3.7 Approximated Virtual Work of The Inertial Forces

The virtual work of the inertial forces in Eq. (2.24) can be approximated by substitution of $\delta u_1(\xi_1)$ and $\ddot{u}_1(\xi_1)$ with the Eqs. (2.53) and afterwards can be given in vector form.

$$
\begin{aligned}
\delta\tilde{W}^e_{\text{dyn}} &= \int_{-1}^{1} \delta\tilde{u}_1 \; \tilde{\ddot{u}}_1 \; \rho A \, \frac{L}{2} \, d\xi_1 = \int_{-1}^{1} \left(\sum_{i=1}^{2} \delta u_1^{ei} \; N^i(\xi_1) \right) \left(\sum_{j=1}^{2} \ddot{u}_1^{ej} \; N^j(\xi_1) \right) \rho A \, \frac{L}{2} \, d\xi_1 \\
&= \sum_{i=1}^{2} \sum_{j=1}^{2} \delta u_1^{ei} \int_{-1}^{1} N^i(\xi_1) \; N^j(\xi_1) \; \rho \; A \, \frac{L}{2} \, d\xi_1 \; \ddot{u}_1^{ej} \\
&= \begin{bmatrix} \delta u_1^{e1} \\ \delta u_1^{e2} \end{bmatrix} \cdot \frac{\rho \; A \; L}{2} \int_{-1}^{1} \begin{bmatrix} N^1(\xi_1) \\ N^2(\xi_1) \end{bmatrix} \begin{bmatrix} N^1(\xi_1) & N^2(\xi_1) \end{bmatrix} d\xi_1 \begin{bmatrix} \ddot{u}_1^{e1} \\ \ddot{u}_1^{e2} \end{bmatrix} \\
&= \begin{bmatrix} \delta u_1^{e1} \\ \delta u_1^{e2} \end{bmatrix} \cdot \frac{\rho \; A \; L}{2} \int_{-1}^{1} \begin{bmatrix} N^1(\xi_1) \; N^1(\xi_1) & N^1(\xi_1) \; N^2(\xi_1) \\ N^1(\xi_1) \; N^2(\xi_1) & N^2(\xi_1) \; N^2(\xi_1) \end{bmatrix} d\xi_1 \begin{bmatrix} \ddot{u}_1^{e1} \\ \ddot{u}_1^{e2} \end{bmatrix}
\end{aligned}
\tag{2.75}
$$

For the derivation of the above equation, constant cross section area A and density ρ were assumed. With the definition of the shape functions $N^i(\xi_1)$ according to Eq. (2.52), $\delta\tilde{W}^e_{\text{dyn}}$ can be further developed.

$$
\begin{aligned}
\delta\tilde{W}^e_{\text{dyn}} &= \begin{bmatrix} \delta u_1^{e1} \\ \delta u_1^{e2} \end{bmatrix} \cdot \frac{\rho \; A \; L}{8} \int_{-1}^{1} \begin{bmatrix} (1-\xi_1)^2 & 1-\xi_1^2 \\ 1-\xi_1^2 & (1+\xi_1)^2 \end{bmatrix} d\xi_1 \begin{bmatrix} \ddot{u}_1^{e1} \\ \ddot{u}_1^{e2} \end{bmatrix} \\
&= \underbrace{\begin{bmatrix} \delta u_1^{e1} \\ \delta u_1^{e2} \end{bmatrix}}_{\delta \boldsymbol{u}^e} \cdot \underbrace{\frac{\rho \; A \; L}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}}_{\mathbf{m}^e} \underbrace{\begin{bmatrix} \ddot{u}_1^{e1} \\ \ddot{u}_1^{e2} \end{bmatrix}}_{\ddot{\boldsymbol{u}}^e}
\end{aligned}
\tag{2.76}
$$

In comparison to the generation of the element stiffness matrix in Eq. (2.66), quadratic terms in ξ_1 had to be integrated for the integration of the *element mass matrix* $\mathbf{m}^e$ in Eq. (2.76), whereas the integrants of the element stiffness matrix were constant.

If the element acceleration vector $\ddot{\boldsymbol{u}}^e$ and the element mass matrix $\mathbf{m}^e$ are defined, then one obtains from Eq. (2.76) the approximated virtual work of the inertial forces.

$$
\boxed{\delta\tilde{W}^e_{\text{dyn}} = \delta\boldsymbol{u}^e \cdot \mathbf{m}^e \; \ddot{\boldsymbol{u}}^e \qquad \mathbf{m}^e = \frac{\rho \; A \; L}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \; \ddot{\boldsymbol{u}}^e = \begin{bmatrix} \ddot{u}_1^{e1} \\ \ddot{u}_1^{e2} \end{bmatrix}}
\tag{2.77}
$$

The mass of the truss element can be computed as the product of the density, the cross- section area and the length. The sum of the components of the mass matrix results in the mass of the truss element.

$$m = \rho\, A\, L \qquad\qquad \sum_{i=1}^{2}\sum_{j=1}^{2} m^{eij} = \rho\, A\, L = m \tag{2.78}$$

Alternatively to the derivation of Eq. (2.77) the virtual work of the inertial forces can be given in a generalized form by means of the matrix of shape functions $\mathbf{N}(\xi)$

$$\boxed{\delta\tilde{W}^e_{\text{dyn}} = \delta\boldsymbol{u}^e \cdot \int_{-1}^{1} \mathbf{N}^T\, \mathbf{N}\, \rho\, A\, |\mathbf{J}|\, d\xi_1\, \ddot{\boldsymbol{u}}^e = \delta\boldsymbol{u}^e \cdot \mathbf{m}^e\, \ddot{\boldsymbol{u}}^e} \tag{2.79}$$

The equivalence between Eqs. (2.77) and (2.79) can be shown by introduction of the matrix of shape functions, the JACOBI determinant and integration over ξ_1.

2.2.4 Truss Element with Quadratic Shape Functions

The element vectors and matrices of the three-node truss element with quadratic displacement approximations can be derived in accordance with the previous sections. The truss element and the quadratic LAGRANGE approximation polynomials are visualized in Fig. 2.6.

2.2.4.1 Quadratic Shape Functions

The shape functions of the quadratic truss element are derived from the general LAGRANGE'an interpolation polynomial in Eq.(2.44) for the polynomial degree $p = 2$ (see Table 2.1)and are assembled in the (1×3) matrix of shape functions

$$\begin{aligned} N^1(\xi_1) &= \frac{1}{2}\,(\xi_1 - 1)\;\xi_1 \\ N^2(\xi_1) &= (1-\xi_1)\;(1+\xi_1) = 1 - \xi_1^2 \\ N^3(\xi_1) &= \frac{1}{2}\,(\xi_1 + 1)\;\xi_1 \end{aligned} \qquad \mathbf{N}(\xi_1) = \begin{bmatrix} N^1(\xi_1) & N^2(\xi_1) & N^3(\xi_1) \end{bmatrix} \tag{2.80}$$

2.2.4.2 Approximation of Coordinates

According to the isoparametric element concept the physical coordinate X_1 is computed with the shape functions $N^i(\xi_1)$ and the positions of the element nodes in the physical space X_1^i (see Eq. (2.54))

$$\begin{aligned} X_1(\xi_1) \approx \tilde{X}_1(\xi_1) &= \mathbf{N}(\xi_1)\; \boldsymbol{X}^e = \sum_{i=1}^{3} X_1^{ei}\; N^i(\xi_1) \\ &= -\frac{L}{2}\frac{\xi_1 - 1}{2}\;\xi_1 + 0(1-\xi_1^2) + \frac{L}{2}\frac{\xi_1+1}{2}\;\xi_1 \\ &= \frac{L}{4}\;\left(-\xi_1^2 + \xi_1 + \xi_1^2 + \xi_1\right) = \frac{L}{2}\;\xi_1 \end{aligned} \tag{2.81}$$

Thus, the JACOBI matrix is also determined. The JACOBI matrix/determinant is identical to the JACOBI matrix/determinant of the linear truss element in Eq. (2.37), or (2.55) respectively.

$$\mathbf{J} = \frac{\partial \tilde{X}_1}{\partial \xi_1} = \frac{L}{2} \tag{2.82}$$

2.2.4.3 Approximation of Variables

The displacement, the variation of the displacement and the acceleration can be determined by means of the matrix of shape functions $\mathbf{N}(\xi_1)$ and the corresponding element vectors $\boldsymbol{u}^e$, $\delta\boldsymbol{u}^e$ and $\ddot{\boldsymbol{u}}^e$.

$$\begin{aligned}
u_1(\xi_1) &\approx \tilde{u}_1(\xi_1) &&= \mathbf{N}(\xi_1)\ \boldsymbol{u}^e & \boldsymbol{u}^e &= \begin{bmatrix} u_1^{e1} & u_1^{e2} & u_1^{e3} \end{bmatrix}^T \\
\delta u_1(\xi_1) &\approx \delta\tilde{u}_1(\xi_1) &&= \mathbf{N}(\xi_1)\ \delta\boldsymbol{u}^e & \delta\boldsymbol{u}^e &= \begin{bmatrix} \delta u_1^{e1} & \delta u_1^{e2} & \delta u_1^{e3} \end{bmatrix}^T \\
\ddot{u}_1(\xi_1) &\approx \tilde{\ddot{u}}_1(\xi_1) &&= \mathbf{N}(\xi_1)\ \ddot{\boldsymbol{u}}^e & \ddot{\boldsymbol{u}}^e &= \begin{bmatrix} \ddot{u}_1^{e1} & \ddot{u}_1^{e2} & \ddot{u}_1^{e3} \end{bmatrix}^T
\end{aligned} \tag{2.83}$$

2.2.4.4 Element Stiffness Matrix

For the generation of the element stiffness matrix according to Eq. (2.69), it is necessary to formulate the differential operator $\mathbf{B}(\xi_1)$ by extending Eq. (2.60) with the degree of freedom of the additional element node.

$$\mathbf{B}(\xi_1) = \frac{2}{L} \begin{bmatrix} N^1_{;1}(\xi_1) & N^2_{;1}(\xi_1) & N^3_{;1}(\xi_1) \end{bmatrix} \tag{2.84}$$

The derivatives of the shape functions $N^i_{;1}(\xi_1)$ are linear in ξ_1.

$$\begin{aligned}
N^1_{;1}(\xi_1) &= \xi_1 - \frac{1}{2} \\
N^2_{;1}(\xi_1) &= -2\,\xi_1 \\
N^3_{;1}(\xi_1) &= \xi_1 + \frac{1}{2}
\end{aligned} \tag{2.85}$$

The element stiffness matrix is obtained by inserting Eqs. (2.84) and (2.85) into Eq. (2.69). In contrast to the integration of the linear truss element, here the B-operator is a function of ξ_1 and therefore cannot be taken out of the integral. Integrations of polynomials of second degree

are needed.

$$
\begin{aligned}
\mathbf{k}^e &= \int_{-1}^{1} \mathbf{B}^T\, E\, A\, \mathbf{B}\, |\mathbf{J}|\, d\xi_1 = \frac{E\, A\, L}{2} \int_{-1}^{1} \mathbf{B}^T \mathbf{B}\, d\xi_1 = \frac{2\, E\, A}{L} \int_{-1}^{1} \mathbf{N}_{;1}^T \mathbf{N}_{;1}\, d\xi_1 \\
&= \frac{2\, E\, A}{L} \int_{-1}^{1} \begin{bmatrix} N_{;1}^1\, N_{;1}^1 & N_{;1}^1\, N_{;1}^2 & N_{;1}^1\, N_{;1}^3 \\ N_{;1}^1\, N_{;1}^2 & N_{;1}^2\, N_{;1}^2 & N_{;1}^2\, N_{;1}^3 \\ N_{;1}^1\, N_{;1}^3 & N_{;1}^2\, N_{;1}^3 & N_{;1}^3\, N_{;1}^3 \end{bmatrix} d\xi_1 \\
&= \frac{2\, E\, A}{L} \int_{-1}^{1} \begin{bmatrix} \frac{1}{4} - \xi_1 + \xi_1^2 & \xi_1 - 2\xi_1^2 & -\frac{1}{4} + \xi_1^2 \\ & 4\xi_1^2 & -\xi_1 - 2\xi_1^2 \\ \text{sym} & & \frac{1}{4} + \xi_1 + \xi_1^2 \end{bmatrix} d\xi_1
\end{aligned}
\tag{2.86}
$$

The integration over the natural coordinate $\xi_1 \in [-1, 1]$ yields the element stiffness matrix $\mathbf{k}^e$ of the quadratic three-node element.

$$
\mathbf{k}^e = \frac{E\, A}{3\, L} \begin{bmatrix} 7 & -8 & 1 \\ & 16 & -8 \\ \text{sym} & & 7 \end{bmatrix}
\tag{2.87}
$$

2.2.4.5 Consistent Element Load Vector

The consistent element load vector $\boldsymbol{r}_p^e$ is generated by inserting the matrix of shape functions $\mathbf{N}(\xi_1)$, according to Eq. (2.80), into Eq. (2.72).

$$
\boldsymbol{r}_p^e = \int_{-1}^{1} \mathbf{N}^T(\xi_1)\, p_1(\xi_1)\, \frac{L}{2}\, d\xi_1 = \frac{1}{2} \int_{-1}^{1} \begin{bmatrix} (\xi_1 - 1)\ \xi_1 \\ 2\ \left(1 - \xi_1^2\right) \\ (\xi_1 + 1)\ \xi_1 \end{bmatrix} p_1(\xi_1)\, \frac{L}{2}\, d\xi_1
\tag{2.88}
$$

In the special case of a constant uniform load p_1, one obtains the consistent element load vector

$$
\boldsymbol{r}_p^e = \frac{p_1\, L}{4} \int_{-1}^{1} \begin{bmatrix} (\xi_1 - 1)\ \xi_1 \\ 2\ \left(1 - \xi_1^2\right) \\ (\xi_1 + 1)\ \xi_1 \end{bmatrix} d\xi_1 = \frac{p_1\, L}{6} \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix}
\tag{2.89}
$$

in which the essential part of 2/3 of the integrated distributed load $L\, p_1$ is applied at the middle node 2 of the truss element.

2.2.4.6 Element Mass Matrix

The element mass matrix $\mathbf{m}^e$ of the quadratic truss element results from the substitution of the matrix of shape functions $\mathbf{N}(\xi_1)$, according to Eq. (2.80), into Eq. (2.79).

$$
\begin{aligned}
\mathbf{m}^e &= \int_{-1}^{1} \mathbf{N}^T\, \mathbf{N}\, \rho\, A\, |\mathbf{J}|\, d\xi_1 \\
&= \frac{\rho\, A\, L}{2} \int_{-1}^{1} \begin{bmatrix} N^1(\xi_1) \\ N^2(\xi_1) \\ N^3(\xi_1) \end{bmatrix} \begin{bmatrix} N^1(\xi_1) & N^2(\xi_1) & N^3(\xi_1) \end{bmatrix} d\xi_1 \\
&= \frac{\rho\, A\, L}{8} \int_{-1}^{1} \begin{bmatrix} (\xi_1 - 1)\ \xi_1 \\ 2\left(1 - \xi_1^2\right) \\ (\xi_1 + 1)\ \xi_1 \end{bmatrix} \begin{bmatrix} (\xi_1 - 1)\ \xi_1 & 2\left(1 - \xi_1^2\right) & (\xi_1 + 1)\ \xi_1 \end{bmatrix} d\xi_1 \\
&= \frac{\rho\, A\, L}{8} \int_{-1}^{1} \begin{bmatrix} \xi_1^2 - 2\,\xi_1^3 + \xi^4 & 2\left(-\xi_1 + \xi_1^2 + \xi_1^3 - \xi_1^4\right) & -\xi_1^2 + \xi_1^4 \\ & 4\left(1 - 2\,\xi_1^2 + \xi_1^4\right) & 2\left(\xi_1 + \xi_1^2 - \xi_1^3 - \xi_1^4\right) \\ \text{sym} & & \xi_1^2 + 2\,\xi_1^3 + \xi_1^4 \end{bmatrix} d\xi_1
\end{aligned}
\tag{2.90}
$$

The integration of polynomials of fourth order in Eq. (2.90) yields the element mass matrix of the quadratic three-node truss element.

$$
\mathbf{m}^e = \frac{\rho\, A\, L}{30} \begin{bmatrix} 4 & 2 & -1 \\ & 16 & 2 \\ \text{sym} & & 4 \end{bmatrix}
\tag{2.91}
$$

Analogously to Eq. (2.78), the summation of the components of the mass matrix yields the mass of the truss.

$$
\sum_{i=1}^{3} \sum_{i=1}^{3} m^{eij} = \rho\, A\, l = m
\tag{2.92}
$$

2.2.5 Truss Element with Cubic Shape Functions

The observed cubic truss element must have four nodes and be of length L. The nodes are chosen in such a way that three regions of length $L/3$ are formed. The coordinates X_1 and ξ_1 have their origin in the middle of the truss. As already demonstrated by the quadratic truss element in Chapter 2.2.4, the element vectors and matrices of the four-node truss element can be analogously derived with cubic displacement approximations.

2.2.5.1 Cubic Shape Functions

The shape functions of the cubic truss element are derived from the general LAGRANGE'an approximation polynomial in Eq. (2.44) for the polynomial degree $p = 3$ (see Table 2.1).

$$
\begin{aligned}
N^1(\xi_1) &= -\frac{9}{16}\,(\xi_1 - 1)\,(\xi_1 - \frac{1}{3})\,(\xi_1 + \frac{1}{3}) &&= \frac{1}{16}\,(-9\,\xi_1^3 + 9\,\xi_1^2 + \xi_1 - 1) \\
N^2(\xi_1) &= \frac{27}{16}\,(\xi_1 + 1)\,(\xi_1 - 1)\,(\xi_1 - \frac{1}{3}) &&= \frac{9}{16}\,(3\,\xi_1^3 - \xi_1^2 - 3\,\xi_1 + 1) \\
N^3(\xi_1) &= -\frac{27}{16}\,(\xi_1 + 1)\,(\xi_1 - 1)\,(\xi_1 + \frac{1}{3}) &&= \frac{9}{16}\,(-3\,\xi_1^3 - \xi_1^2 + 3\,\xi_1 + 1) \\
N^4(\xi_1) &= \frac{9}{16}\,(\xi_1 + 1)\,(\xi_1 - \frac{1}{3})\,(\xi_1 + \frac{1}{3}) &&= \frac{1}{16}\,(9\,\xi_1^3 + 9\,\xi_1^2 - \xi_1 - 1)
\end{aligned}
\tag{2.93}
$$

The shape functions $N^i(\xi_1)$ can be assembled as usual in the (1×4) matrix of shape functions $\mathbf{N}(\xi_1)$.

$$
\mathbf{N}(\xi_1) = \begin{bmatrix} N^1(\xi_1) & N^2(\xi_1) & N^3(\xi_1) & N^4(\xi_1) \end{bmatrix}
\tag{2.94}
$$

2.2.5.2 Approximation of Coordinates

According to the isoparametric concept, the physical coordinate X_1 is computed with the shape functions $N^i(\xi_1)$ and the positions of the element nodes X_1^i (see Eq. (2.54)).

$$
\begin{aligned}
X_1(\xi_1) &\approx \tilde{X}_1(\xi_1) = \mathbf{N}(\xi_1)\,\boldsymbol{X}^e &&= \sum_{i=1}^{4} X_1^{ei}\,N^i(\xi_1) \\
&= -\frac{L}{2}\,\frac{1}{16}\left(-9\,\xi_1^3 + 9\,\xi_1^2 + \xi_1 - 1\right) &&- \frac{L}{6}\,\frac{9}{16}\left(3\,\xi_1^3 - \xi_1^2 - 3\,\xi_1 + 1\right) \\
&+ \frac{L}{6}\,\frac{9}{16}\left(-3\,\xi_1^3 - \xi_1^2 + 3\,\xi_1 + 1\right) &&+ \frac{L}{2}\,\frac{1}{16}\left(9\,\xi_1^3 + 9\,\xi_1^2 - \xi_1 - 1\right) \\
&= \frac{L}{16}\,\left(9\,\xi_1^3 - \xi_1 - 9\,\xi_1^3 + 9\,\xi_1\right) &&= \frac{L}{2}\,\xi_1
\end{aligned}
\tag{2.95}
$$

Thus, the JACOBI matrix is also determined. The JACOBI matrix/determinant is identical to the JACOBI matrix/determinant of the linear truss element in Eq. (2.37), or Eq. (2.55) respectively, if the element nodes are positioned, as prescribed, at the points $X_1^{e1} = -\frac{L}{2}$, $X_1^{e2} = -\frac{L}{6}$, $X_1^{e3} = \frac{L}{6}$ and $X_1^{e4} = \frac{L}{2}$.

$$
\mathbf{J} = \frac{\partial \tilde{X}_1}{\partial \xi_1} = \frac{L}{2}
\tag{2.96}
$$

2.2.5.3 Approximation of Variables

The displacement, the variation of displacement and the acceleration can be determined by means of the matrix of shape functions $\mathbf{N}(\xi_1)$ and the corresponding element vectors $\boldsymbol{u}^e$, $\delta\boldsymbol{u}^e$ and $\ddot{\boldsymbol{u}}^e$.

$$
\begin{array}{rcccl}
u_1(\xi_1) & \approx & \tilde{u}_1(\xi_1) & = & \mathbf{N}(\xi_1)\ \boldsymbol{u}^e \\
\delta u_1(\xi_1) & \approx & \delta\tilde{u}_1(\xi_1) & = & \mathbf{N}(\xi_1)\ \delta\boldsymbol{u}^e \\
\ddot{u}_1(\xi_1) & \approx & \ddot{\tilde{u}}_1(\xi_1) & = & \mathbf{N}(\xi_1)\ \ddot{\boldsymbol{u}}^e
\end{array}
\qquad
\begin{array}{rcl}
\boldsymbol{u}^e & = & \left[\ u_1^{e1}\ \ u_1^{e2}\ \ u_1^{e3}\ \ u_1^{e4}\ \right]^T \\
\delta\boldsymbol{u}^e & = & \left[\ \delta u_1^{e1}\ \delta u_1^{e2}\ \delta u_1^{e3}\ \delta u_1^{e4}\ \right]^T \\
\ddot{\boldsymbol{u}}^e & = & \left[\ \ddot{u}_1^{e1}\ \ddot{u}_1^{e2}\ \ddot{u}_1^{e3}\ \ddot{u}_1^{e4}\ \right]^T
\end{array}
\tag{2.97}
$$

2.2.5.4 Element Stiffness Matrix

For the generation of the element stiffness matrix according to Eq. (2.69), it is necessary to transform the differential operator $\mathbf{B}(\xi_1)$ by extending (2.60) to include the two additional degrees of freedom.

$$
\mathbf{B}(\xi_1) = \frac{2}{L}\left[\ N_{;1}^1(\xi_1)\quad N_{;1}^2(\xi_1)\quad N_{;1}^3(\xi_1)\quad N_{;1}^4(\xi_1)\ \right] \tag{2.98}
$$

The derivatives of the shape functions $N_{;1}^i(\xi_1)$ are quadratic in ξ_1.

$$
\begin{array}{ll}
N_{;1}^1(\xi_1) = \dfrac{1}{16}\left(-27\,\xi_1^2 + 18\,\xi_1 + 1\right) & N_{;1}^3(\xi_1) = \dfrac{9}{16}\left(-9\,\xi_1^2 - 2\,\xi_1 + 3\right) \\
N_{;1}^2(\xi_1) = \dfrac{9}{16}\left(9\,\xi_1^2 - 2\,\xi_1 - 3\right) & N_{;1}^4(\xi_1) = \dfrac{1}{16}\left(27\,\xi_1^2 + 18\,\xi_1 - 1\right)
\end{array}
\tag{2.99}
$$

The element stiffness matrix is obtained by inserting Eqs. (2.98) and (2.99) into Eq. (2.69). Like the integration of the quadratic truss element, the B-operator here is also a function of ξ_1. Integrations of polynomials of fourth order are necessary.

$$
\begin{aligned}
\mathbf{k}^e &= \int_{-1}^{1} \mathbf{B}^T\ E\ A\ \mathbf{B}\ |\mathbf{J}|\ d\xi_1 = \frac{E\ A\ L}{2}\int_{-1}^{1}\mathbf{B}^T\mathbf{B}\ d\xi_1 = \frac{2\ E\ A}{L}\int_{-1}^{1}\mathbf{N}_{;1}^T\mathbf{N}_{;1}\ d\xi_1 \\
&= \frac{2\ E\ A}{L}\int_{-1}^{1}\begin{bmatrix}
N_{;1}^1\ N_{;1}^1 & N_{;1}^1\ N_{;1}^2 & N_{;1}^1\ N_{;1}^3 & N_{;1}^1\ N_{;1}^4 \\
N_{;1}^2\ N_{;1}^1 & N_{;1}^2\ N_{;1}^2 & N_{;1}^2\ N_{;1}^3 & N_{;1}^2\ N_{;1}^4 \\
N_{;1}^3\ N_{;1}^1 & N_{;1}^3\ N_{;1}^2 & N_{;1}^3\ N_{;1}^3 & N_{;1}^3\ N_{;1}^4 \\
N_{;1}^4\ N_{;1}^1 & N_{;1}^4\ N_{;1}^2 & N_{;1}^4\ N_{;1}^3 & N_{;1}^4\ N_{;1}^4
\end{bmatrix} d\xi_1
\end{aligned}
\tag{2.100}
$$

The components of the above equation are given as follows:

$$
\begin{aligned}
N^1_{;1}(\xi_1)\; N^1_{;1}(\xi_1) &= \frac{1}{256}\left(729\;\xi_1^4 - 972\;\xi_1^3 + 270\;\xi_1^2 + 36\;\xi_1 + 1\right)\\
N^1_{;1}(\xi_1)\; N^2_{;1}(\xi_1) &= \frac{1}{256}\left(-2187\;\xi_1^4 + 1944\;\xi_1^3 + 486\;\xi_1^2 - 504\;\xi_1 - 27\right)\\
N^1_{;1}(\xi_1)\; N^3_{;1}(\xi_1) &= \frac{1}{256}\left(2187\;\xi_1^4 - 972\;\xi_1^3 - 1134\;\xi_1^2 + 468\;\xi_1 + 27\right)\\
N^1_{;1}(\xi_1)\; N^4_{;1}(\xi_1) &= \frac{1}{256}\left(-729\;\xi_1^4 + 378\;\xi_1^2 - \xi_1\right)\\
N^2_{;1}(\xi_1)\; N^2_{;1}(\xi_1) &= \frac{1}{256}\left(6561\;\xi_1^4 - 2916\;\xi_1^3 - 4050\;\xi_1^2 + 972\;\xi_1 + 729\right)
\end{aligned}
\tag{2.101}
$$

$$
\begin{aligned}
N^2_{;1}(\xi_1)\; N^3_{;1}(\xi_1) &= \frac{1}{256}\left(-6561\;\xi_1^4 + 4698\;\xi_1^2 - 729\right)\\
N^2_{;1}(\xi_1)\; N^4_{;1}(\xi_1) &= \frac{1}{256}\left(2187\;\xi_1^4 + 972\;\xi_1^3 - 1134\;\xi_1^2 - 468\;\xi_1 + 27\right)\\
N^3_{;1}(\xi_1)\; N^3_{;1}(\xi_1) &= \frac{1}{256}\left(6561\;\xi_1^4 + 2916\;\xi_1^3 - 4050\;\xi_1^2 - 972\;\xi_1 + 729\right)\\
N^3_{;1}(\xi_1)\; N^4_{;1}(\xi_1) &= \frac{1}{256}\left(-2187\;\xi_1^4 - 1944\;\xi_1^3 + 486\;\xi_1^2 + 504\;\xi_1 - 27\right)\\
N^4_{;1}(\xi_1)\; N^4_{;1}(\xi_1) &= \frac{1}{256}\left(729\;\xi_1^4 - 972\;\xi_1^3 + 270\;\xi_1^2 + 36\;\xi_1 + 1\right)
\end{aligned}
\tag{2.101}
$$

Integration of the polynomials given in Eq. (2.101) along the length of the truss element yields the element stiffness matrix of the cubic four-node truss element.

$$
\mathbf{k}^e = \frac{E\;A}{40\;L}\begin{bmatrix} 148 & -189 & 54 & -13 \\ & 432 & -297 & 54 \\ & & 432 & -189 \\ \text{sym} & & & 148 \end{bmatrix}
\tag{2.102}
$$

2.2.5.5 Consistent Element Load Vector

The consistent element load vector $\boldsymbol{r}^e_p$ is generated by inserting the matrix of shape functions $\mathbf{N}(\xi_1)$, according to Eq. (2.93), into Eq. (2.72).

$$
\boldsymbol{r}^e_p = \int_{-1}^{1} \mathbf{N}^T(\xi_1)\; p_1(\xi_1)\; \frac{L}{2}\; d\xi_1 = \frac{1}{16}\int_{-1}^{1} \begin{bmatrix} -9\;\xi_1^3 + 9\;\xi_1^2 + \xi_1 - 1 \\ 27\;\xi_1^3 - 9\;\xi_1^2 - 27\;\xi_1 + 9 \\ -27\;\xi_1^3 - 9\;\xi_1^2 + 27\;\xi_1 + 9 \\ 9\;\xi_1^3 + 9\;\xi_1^2 - \xi_1 - 1 \end{bmatrix} p_1(\xi_1)\; \frac{L}{2}\; d\xi_1
\tag{2.103}
$$

In the special case of a constant uniform load p_1 one obtains the consistent element load vector

$$\boldsymbol{r}_p^e = \frac{p_1\ L}{32} \int_{-1}^{1} \begin{bmatrix} -9\ \xi_1^3 + 9\ \xi_1^2 + \xi_1 - 1 \\ 27\ \xi_1^3 - 9\ \xi_1^2 - 27\ \xi_1 + 9 \\ -27\ \xi_1^3 - 9\ \xi_1^2 + 27\ \xi_1 + 9 \\ 9\ \xi_1^3 + 9\ \xi_1^2 - \xi_1 - 1 \end{bmatrix} d\xi_1 = \frac{p_1\ L}{8} \begin{bmatrix} 1 \\ 3 \\ 3 \\ 1 \end{bmatrix} \tag{2.104}$$

in which the essential parts of 3/8 of the integrated distributed load $L\ p_1$ are applied at the middle nodes 2 and 3 of the truss element.

2.2.5.6 Element Mass Matrix

The element mass matrix $\mathbf{m}^e$ of the cubic truss element results from the substitution of the matrix of shape functions $\mathbf{N}(\xi_1)$, according to Eq. (2.93), into Eq. (2.79).

$$\begin{aligned} \mathbf{m}^e &= \int_{-1}^{1} \mathbf{N}^T\ \mathbf{N}\ \rho\ A\ \ |\mathbf{J}|\ d\xi_1 \\ &= \frac{\rho\ A\ l}{2} \int_{-1}^{1} \begin{bmatrix} N^1(\xi_1) \\ N^2(\xi_1) \\ N^3(\xi_1) \\ N^4(\xi_1) \end{bmatrix} \begin{bmatrix} N^1(\xi_1) & N^2(\xi_1) & N^3(\xi_1) & N^4(\xi_1) \end{bmatrix} d\xi_1 \\ &= \frac{\rho\ A\ L}{2} \int_{-1}^{1} \begin{bmatrix} N^1\ N^1 & N^1\ N^2 & N^1\ N^3 & N^1\ N^4 \\ N^2\ N^1 & N^2\ N^2 & N^2\ N^3 & N^2\ N^4 \\ N^3\ N^1 & N^3\ N^2 & N^3\ N^3 & N^3\ N^4 \\ N^4\ N^1 & N^4\ N^2 & N^4\ N^3 & N^4\ N^4 \end{bmatrix} d\xi_1 \end{aligned} \tag{2.105}$$

As an example, here, the calculation of the components

$$N^1(\xi_1)\ N^1(\xi_1) = 81\ \xi_1^6 - 162\ \xi_1^5 + 63\ \xi_1^4 + 36\ \xi_1^3 - 17\ \xi_1^2 - 2\ \xi_1 + 1 \tag{2.106}$$

of the matrix that is to be integrated is demonstrated. The terms that are to be integrated along the longitudinal axis of the truss are polynomials of the sixth degree. Carrying out the integration in Eq. (2.105) yields the element matrix of the cubic four-node truss element.

$$\mathbf{m}^e = \frac{\rho\ A\ L}{2000} \begin{bmatrix} 152 & 118 & -43 & 23 \\ & 771 & -96 & -43 \\ & & 771 & 118 \\ \text{sym} & & & 152 \end{bmatrix} \tag{2.107}$$

2.2.6 Numerical Integration

The integration of the components of the element stiffness matrix, the element mass matrix and the consistent element load vector was analytically carried out in the previous chapters. In order to have in the development of complex finite elements an adequate tool for the integration of element quantities that are hardly integrable analytically, numerical integration is introduced and examined for the truss example. By means of numerical integration, it is possible to integrate arbitrary functions in an approximate way. The essential advantages of numerical integration are summarized as follows:

- Simplification of the integration
- Integration of analytically non-integrable functions
- Selective subintegration for elimination of defects from the element formulation

In opposition to these advantages there are actually limitations, too:

- The generation of element matrices and vectors is numerically costly
- The element matrices and vectors are integrated inexactly[3]

Within the framework of finite element methods, the so-called *Gauß-Legendre Quadrature* has established itself. The Gauss-Legendre Quadrature of a function $f(\xi_1)$ over the parameter space $\xi_1 \in [-1, 1]$ is given by the sum

$$\int_{-1}^{1} f(\xi_1)\, d\xi_1 = \sum_{i=1}^{n} \alpha^i \; f(\xi_1^i) \tag{2.108}$$

In Eq. (2.108), α^i are the weight coefficients to the function values f at the supports ξ_1^i and n is the number of integration points, or so-called *Gauß points*. Polynomials of polynomial degree

$$p \leq 2n - 1 \tag{2.109}$$

can be integrated exactly by Gauss-Legendre Quadrature, higher-order polynomials and other functions can be integrated approximately. The linear, quadratic and cubic truss elements are developed in this section by Gauss-Legendre integration with one and two supports. A summary of ξ_1^i and α^i for $n = 1, 2, 3$ is found in Table 2.2. For a representation of the fundamentals of numerical integration, the way of obtaining of the weight factors α^i and the Gauss points ξ_1^i, refer to the literature of *numerical mathematics*, e.g. Deuflhard & Hohmann [51].

2.2.6.1 Numerical Integration of the Linear Truss Element

If all element quantities ($\mathbf{k}^e$, $\boldsymbol{r}_p^e$, $\mathbf{m}^e$) of the truss element have to be integrated, it must be differentiated between the requirements of the integrands. If the exact numerical solution of all element quantities is required, one must at least use the Gauss-Legendre quadratures summarized in Table 2.3. Often, the integration order is adapted to the needs of integration of the element stiffness matrix, hence the mass matrix is integrated inexactly and the consistent load vector is integrated exactly only in special cases.

[3] This can be also an advantage for some finite elements, see *selective subintegration*

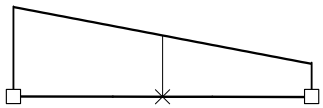
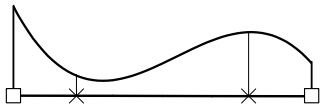
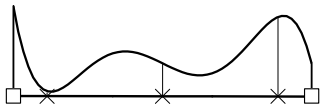
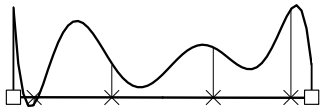
n	$2n-1$	$f(\xi_1)$	ξ_1^i	α^i
1	1		$\xi_1^1 = 0$	$\alpha^1 = 2$
2	3		$\xi_1^1 = -1/\sqrt{3}$ $\xi_1^2 = 1/\sqrt{3}$	$\alpha^1 = 1$ $\alpha^2 = 1$
3	5		$\xi_1^1 = -\sqrt{3/5}$ $\xi_1^2 = 0$ $\xi_1^3 = \sqrt{3/5}$	$\alpha^1 = 5/9$ $\alpha^2 = 8/9$ $\alpha^3 = 5/9$
4	7		$\xi_1^{1,4} = \mp 0.86114$ $\xi_1^{2,3} = \mp 0.33998$	$\alpha^{1,4} = 0.34785$ $\alpha^{2,3} = 0.65241$

Table 2.2: Gauss points ξ_1^i and weight factors α^i of the Gauss-Legendre quadrature

Element Quantity			$p=1$	$p=2$	$p=3$
Stiffness matrix	$\mathbf{k}^e$		1	2	3
Load vector	$\boldsymbol{r}_p^e$	$p_1(\xi_1)$: constant	1	2	2
	$\boldsymbol{r}_p^e$	$p_1(\xi_1)$: linear	2	2	3
	$\boldsymbol{r}_p^e$	$p_1(\xi_1)$: quadratic	2	3	3
	$\boldsymbol{r}_p^e$	$p_1(\xi_1)$: cubic	3	3	4
Mass matrix	$\mathbf{m}^e$		2	3	4

Table 2.3: Necessary number of Gauss points n for the exact Gauss-Legendre quadrature of element quantities of linear, quadratic and cubic truss elements

For the exact integration of the stiffness matrix of the linear truss element with Gauss-Legendre quadrature, it is enough, according to Eqs. (2.51) and (2.65), to carry out a single-point integration, since according to Eq. (2.109) with $n = 1$ a linear function can be exactly integrated.

In the generation of the vector of consistent nodal, loads the exact integration with one-point-Gauss-Legendre-integration is possible according to Eqs. (2.52) and (2.72) only for a constant uniform load p_1. By application of higher-order functions $p_1(\xi_1)$, the consistent load vector is inexactly integrated. For exact integration, the number of Gauss points must be adapted to the polynomial degree of the load function (Eq. (2.109)). Load functions, which cannot be represented by means of a polynomial, are in any case integrated only approximately.

According to Eq. (2.76) for the exact integration of the element mass matrix, quadratic functions have to be integrated, which requires a two-point-Gauss-Legendre-integration, according to Eq. (2.109). A single-point-Gauss-Legendre-integration, carried out in equivalence with the stiffness matrix, results in an inexact integration of the mass matrix.

2.2.6.2 Numerical Integration of Higher-Order Truss Elements

In order to define the necessary integration order for the exact integration of the element quantities ($\mathbf{k}^e$, $\boldsymbol{r}_p^e$, $\mathbf{m}^e$) of the quadratic truss element, the generation of the stiffness matrix, the

mass matrix and the vector of element loads is considered (see KUHL & MESCHKE [26]) and the fulfillment of Eq. (2.109) is verified. Herefrom, the minimal requirements, summarized in Table 2.3 for an exact GAUSS-LEGENDRE integration of the quadratic truss element, result. The corresponding support points of the function evaluation and the weighting factors of the GAUSS-LEGENDRE quadrature are found in Table 2.2.

The minimal requirements for the exact numerical integration of the element stiffness matrix, the element mass matrix and the consistent element load vector of the cubic truss element are summarized in Table 2.3

2.3 Assembly of the Structure

After the generation of the element quantities of all finite elements found in a system, they must be grouped into an ensemble. Basis for this assembly of elements is the domain partitioning, carried out in Eq. (2.33), and the requirements towards the virtual work terms, which were formulated in Eqs. (2.35), in connection with the principle of virtual work (2.24).

The generation of the system or structure of finite elements representing the building blocks of the structure is essentially divided into two steps:

- At the connections of element degrees of freedom of two or more elements at a common node of the structure, the degrees of freedom must be compatible. This means that each element degree of freedom needs during the assembly an adequate partner of the neighbour element, which, on the other hand, requires physical equivalence of the degrees of freedom and the representation in the same coordinate system.
  - The first requirement is automatically fulfilled for an ensemble of truss elements, since for these elements only translational displacements are present as degrees of freedom.
  - The second requirement is fulfilled by the transformation of the degrees of freedom of neighbour elements at a common system node to the same coordinate system.
- Neighbouring finite elements are connected to the system or structure through their compatible degrees of freedom at a common structure node; this is effected by assembling the element degrees of freedom into system degrees of freedom.

2.3.1 Transformation of the Element Matrices and Vectors

The transformation of the element quantities $\mathbf{k}^e$, $\boldsymbol{r}_p^e$ and $\mathbf{m}^e$ from the element-specific coordinate system, spanned by the basis vectors $(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3)$, to an arbitrarily-oriented cartesian coordinate system, characterized by the bases $(\boldsymbol{e}_1', \boldsymbol{e}_2', \boldsymbol{e}_3')$, is based on the transformation of the element displacement vector $\boldsymbol{u}^e$. Since the coordinate system $(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3)$ was chosen so that only the displacement in the direction of the basis vector $\boldsymbol{e}_1$ is different from zero, the element displacement vector $\boldsymbol{u}^e$ consists only of the components u_1^{ei} for $i = 1, \ldots, p+1$. Furthermore, the displacement vector contains only one component u_1^{ei} at the element nodes i. The element displacement vector in an arbitrarily-chosen coordinate system $(\boldsymbol{e}_1', \boldsymbol{e}_2', \boldsymbol{e}_3')$ consists, on the contrary, of three components per element node i.

$$\boldsymbol{u}^{ei'} = \begin{bmatrix} u_1^{ei'} & u_2^{ei'} & u_3^{ei'} \end{bmatrix}^T \qquad \boldsymbol{u}^{e'} = \begin{bmatrix} \boldsymbol{u}^{e1'\,T} & \ldots & \boldsymbol{u}^{e(p+1)'\,T} \end{bmatrix}^T \tag{2.110}$$

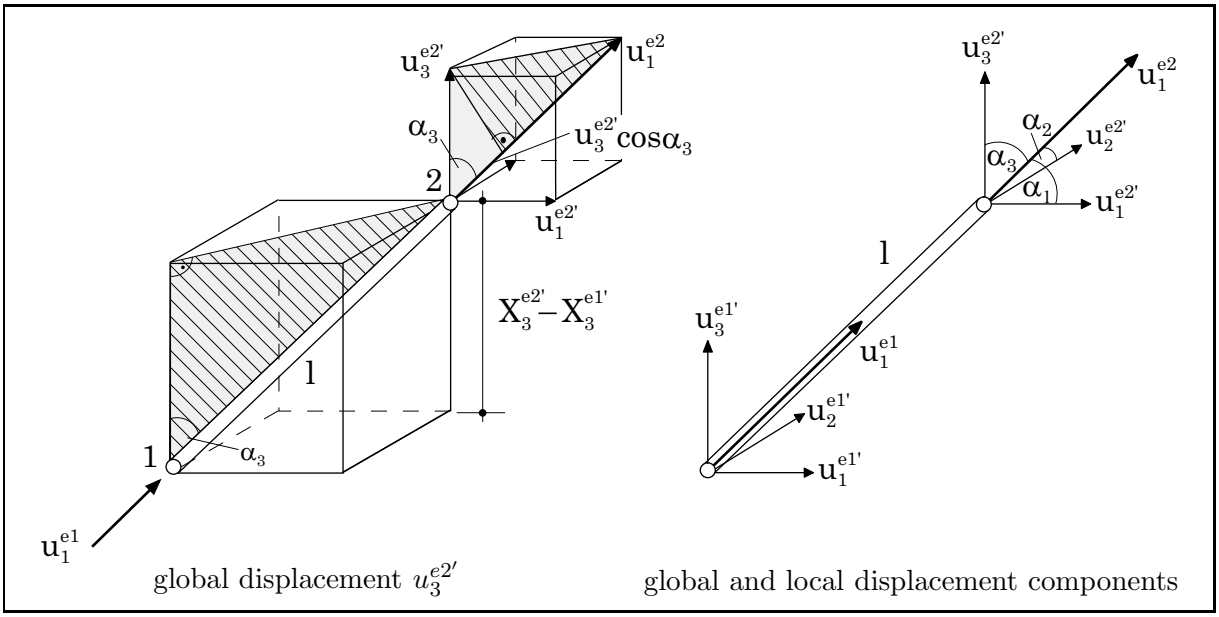


Figure 2.7: Transformation of the element displacements II

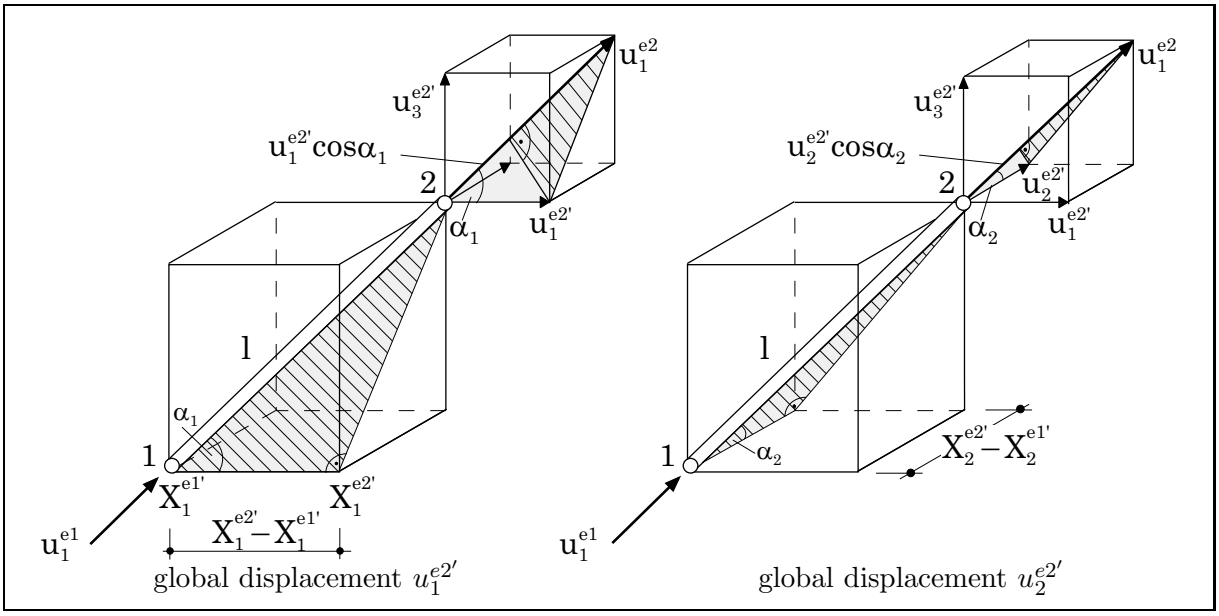


Figure 2.8: Transformation of the element displacements I

The transformation relation has to be applied nodewise and can be derived hence for the polynomial degree $p = 1$ and generalized for arbitrary polynomial degrees p. Figures 2.7 and 2.8 illustrate the element degrees of freedom in both coordinate systems $(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3)$, $(\boldsymbol{e}_1', \boldsymbol{e}_2', \boldsymbol{e}_3')$, and the generation of the element displacement u_1^{e2} by the sum of the displacement components of the vector $\boldsymbol{u}^{e2'}$, projected upon the basis $\boldsymbol{e}_1$. The projection of $u_1^{e2'}$ is given by the directional cosine $\cos\alpha_1$, which can in turn be obtained from the orientation of the truss element in space. If the other projections are analogously carried out, one obtains the desired displacement component

$$u_1^{e2} = u_1^{e2'} \cos\alpha_1 + u_2^{e2'} \cos\alpha_2 + u_3^{e2'} \cos\alpha_3 \tag{2.111}$$

with the directional cosines $\cos\alpha_j$ for $j = 1, 2, 3$, determined from the similarity of the geometry of displacement and truss orientation in space.

$$\cos\alpha_j = \frac{X_j^{e2'} - X_j^{e1'}}{L} \qquad\qquad L = \left\| \boldsymbol{X}^{e2'} - \boldsymbol{X}^{e1'} \right\| \tag{2.112}$$

Here, the element position vector was defined in analogy to the element displacement vector in Eq. (2.110).

$$\boldsymbol{X}^{ei'} = \begin{bmatrix} X_1^{ei'} & X_2^{ei'} & X_3^{ei'} \end{bmatrix}^T \qquad\qquad \boldsymbol{X}^{e'} = \begin{bmatrix} \boldsymbol{X}^{e1'\,T} & \dots & \boldsymbol{X}^{e(p+1)'\,T} \end{bmatrix}^T \tag{2.113}$$

It should be noted here that the directional cosines should not necessarily be defined relative to the global cartesian basis, spanned by the basis vectors $\boldsymbol{e}_1$, $\boldsymbol{e}_2$ and $\boldsymbol{e}_3$. It is enough to choose an arbitrary coordinate system which is identical for all element nodes that have to be connected at the observed system node. In this case, the directional cosines cannot be computed according to Eq. (2.112) but have to be given explicitly by means of geometrical considerations. This formal generalization is especially important when oblique support conditions or oblique planes of symmetry have to be taken into account. Then, the degrees of freedom of the finite element are transformed to local coordinates (a basis vector is perpendicular to the plane of motion of a gliding support) or to a coordinate system with one basis vector normal to the plane of symmetry.

The component u_1^{e1} of the element displacement vector $\boldsymbol{u}^e$ can be determined in analogy to the component u_1^{e2} by means of the directional cosines $\cos\alpha_j$ and the components $u_j^{e1'}$ of the element displacement vector $\boldsymbol{u}^{e'}$.

$$u_1^{e1} = u_1^{e1'}\ \cos\alpha_1 + u_2^{e1'}\ \cos\alpha_2 + u_3^{e1'}\ \cos\alpha_3 \tag{2.114}$$

Eqs. (2.111) and (2.114) are summarized now in matrix form.

$$\underbrace{\begin{bmatrix} u_1^{e1} \\ u_1^{e2} \end{bmatrix}}_{\boldsymbol{u}^e} = \underbrace{\begin{bmatrix} \cos\alpha_1 & \cos\alpha_2 & \cos\alpha_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos\alpha_1 & \cos\alpha_2 & \cos\alpha_3 \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} u_1^{e1'} \\ u_2^{e1'} \\ u_3^{e1'} \\ u_1^{e2'} \\ u_2^{e2'} \\ u_3^{e2'} \end{bmatrix}}_{\boldsymbol{u}^{e'}} \tag{2.115}$$

$\mathbf{T}$ represents the 2×6 transformation matrix. For truss elements of the polynomial degree p, the transformation matrix takes the dimension $(p+1) \times 3(p+1)$. Since the transformations of separate element nodes are decoupled, the transformation matrix expanded for the polynomial degree p can be directly given,

$$\boxed{\mathbf{T} = \begin{bmatrix} \mathbf{T}_1 & & & \\ & \mathbf{T}_2 & & \\ & & \ddots & \\ & & & \mathbf{T}_{p+1} \end{bmatrix} \qquad \begin{aligned} \mathbf{T}_i &= \begin{bmatrix} \cos\alpha_1 & \cos\alpha_2 & \cos\alpha_3 \end{bmatrix} \\ \cos\alpha_j &= \frac{X_j^{e2'} - X_j^{e1'}}{\left\| \boldsymbol{X}^{e2'} - \boldsymbol{X}^{e1'} \right\|} \end{aligned}} \tag{2.116}$$

whereas for the already discussed general transformation, the transformation matrices $\mathbf{T}_i$ do not need to be identical. The element displacement vector, its variation and the element acceleration vector can be mapped from the coordinate system $(\boldsymbol{e}'_1, \boldsymbol{e}'_2, \boldsymbol{e}'_3)$ into the coordinate system $(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3)$ by the transformation matrix $\mathbf{T}$.

$$\begin{aligned} \boldsymbol{u}^e &= \mathbf{T} \;\; \boldsymbol{u}^{e'} \\ \delta\boldsymbol{u}^e &= \mathbf{T}\, \delta\boldsymbol{u}^{e'} \\ \ddot{\boldsymbol{u}}^e &= \mathbf{T} \;\; \ddot{\boldsymbol{u}}^{e'} \end{aligned} \tag{2.117}$$

The inverse transformation can be realized directly from Fig. 2.53. The components $u_j^{ei'}$ result from the projection of the component u_1^{ei} (or the 1×1 displacement vector $\boldsymbol{u}^{ei}$) of the element node i onto the basis vector $\boldsymbol{e}_j$.

$$u_j^{ei'} = u_1^{ei} \cos\alpha_j \qquad\qquad \boldsymbol{u}^{ei'} = \mathbf{T}_i^T \; \boldsymbol{u}^{ei} \tag{2.118}$$

Thus, the transformations of the element vectors described in Eq. (2.117) are also given by local element coordinates to global system coordinates.

$$\begin{aligned} \boldsymbol{u}^{e'} &= \mathbf{T}^T \;\; \boldsymbol{u}^e \\ \delta\boldsymbol{u}^{e'} &= \mathbf{T}^T\, \delta\boldsymbol{u}^e \\ \ddot{\boldsymbol{u}}^{e'} &= \mathbf{T}^T \;\; \ddot{\boldsymbol{u}}^e \end{aligned} \tag{2.119}$$

The corresponding transformations of the element stiffness matrix $\mathbf{k}^e$ and the element load vector $\boldsymbol{r}^e$ can be obtained by observing the internal virtual work and the virtual work of the external loads, respectively. It is noted that the virtual work is a scalar and is thus invariant of the coordinates. Inserting the transformations of the element displacement vector and its variation according to Eqs. (2.117), into the internal virtual work of the truss element according to (2.67),

$$\delta\tilde{W}^e_{\text{int}} = \delta\boldsymbol{u}^e \cdot \mathbf{k}^e\, \boldsymbol{u}^e = \delta\boldsymbol{u}^{e\,T}\mathbf{k}^e\, \boldsymbol{u}^e = \left(\mathbf{T}\, \delta\boldsymbol{u}^{e'}\right)^T \mathbf{k}^e \left(\mathbf{T}\, \boldsymbol{u}^{e'}\right) = \delta\boldsymbol{u}^{e'} \cdot \underbrace{\mathbf{T}^T\mathbf{k}^e\mathbf{T}}_{\mathbf{k}^{e'}}\, \boldsymbol{u}^{e'} = \delta\tilde{W}^{e'}_{\text{int}} \tag{2.120}$$

yields the transformation relation for the element stiffness matrix.

$$\mathbf{k}^{e'} = \mathbf{T}^T\, \mathbf{k}^e\, \mathbf{T} \tag{2.121}$$

The transformation of the element loads results from the consideration of the virtual work of the external loads (2.73) and the transformation of the variation of the element displacement vector, according to Eq. (2.117)

$$\delta\tilde{W}^e_{\text{ext}} = \delta\boldsymbol{u}^e \cdot \boldsymbol{r}^e = \delta\boldsymbol{u}^{e'} \cdot \underbrace{\mathbf{T}^T\, \boldsymbol{r}^e}_{\boldsymbol{r}^{e'}} = \delta\boldsymbol{u}^{e'} \cdot \boldsymbol{r}^{e'} = \delta\tilde{W}^{e'}_{\text{ext}} \tag{2.122}$$

in:

$$\boldsymbol{r}^{e'} = \mathbf{T}^T\, \boldsymbol{r}^e \tag{2.123}$$

With a procedure analogous to the development of Eq. (2.120), the transformation relation of the mass matrix can be derived by means of the virtual work of the inertial forces (2.79) and the transformation of the element acceleration vector (2.117).

$$\mathbf{m}^{e'} = \mathbf{T}^T \, \mathbf{m}^e \, \mathbf{T} \tag{2.124}$$

2.3.2 Assembly of the Elements to the System

The assembly is realized according to four strategies building upon one another. They differ essentially in the mathematical formulation which can be characterized as follows:

- Direct Addition of Components
- Transformation with an element-specific compatibility matrix $\mathbf{a}^e$
- Transformation with a system-specific compatibility matrix $\mathbf{a}$
- Symbolic assembly with the *union operator* $\bigcup$

As a result of the assembly, one obtains the principle of virtual work, formulated in system vectors and matrices. In the transient case, it is transformed into the *semi-discrete differential equation of motion* with initial conditions, by application of the fundamental lemma of variational calculus. In the static case, the application of the fundamental lemma of variational calculus leads to a linear system of equations with the system degrees of freedom (system displacement vector) as solution vector.

The formulation of the assembly procedure is demonstrated with the example of a general spatial truss frame, generated from the two-node truss elements shown in Fig. 2.9. As sketched in the graph, for the symbolic representation of the procedure only the connection of the element degrees of freedom of nodes two of the elements d, e, f and g at the common system node k is considered. All other element connections are not taken into account in the following representation. It is furthermore assumed that the cartesian basis, formed by the vectors $\boldsymbol{e}_1$, $\boldsymbol{e}_2$ and $\boldsymbol{e}_3$, represents the global basis (the notation $'$ is omitted).

2.3.2.1 Direct Addition of Components

The direct addition of the corresponding element quantities to system quantities is based upon the virtual works of the system, which are composed additively from the virtual works of all the elements (Eq. (2.35)). Thus, the assembly of the stiffness matrix results from the observation of the summation of the internal virtual work of the elements over the number of finite truss elements NE present in the system, according to Eq. (2.67),

$$\delta\tilde{W}_{\text{int}} = \sum_{e=1}^{NE} \delta\tilde{W}_{\text{int}}^{e} = \sum_{e=1}^{NE} \delta\boldsymbol{u}^e \cdot \mathbf{k}^e \, \boldsymbol{u}^e = \sum_{e=1}^{NE} \begin{bmatrix} \delta\boldsymbol{u}^{e1} \\ \delta\boldsymbol{u}^{e2} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{k}^{e11} & \mathbf{k}^{e12} \\ \mathbf{k}^{e12} & \mathbf{k}^{e22} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}^{e1} \\ \boldsymbol{u}^{e2} \end{bmatrix} \tag{2.125}$$

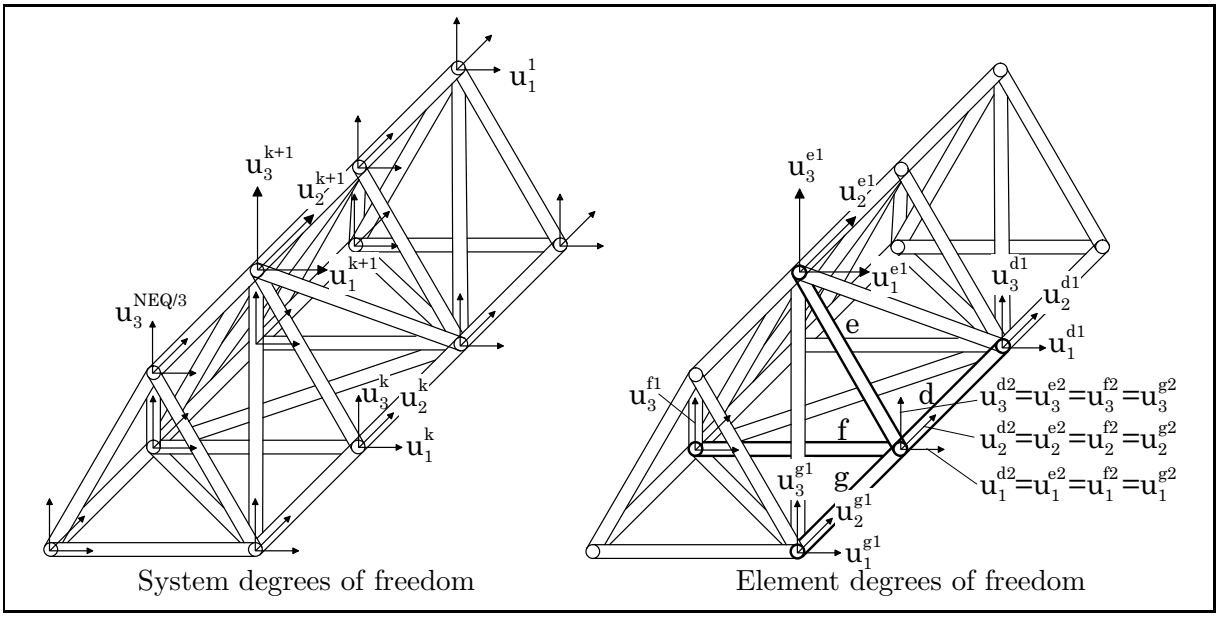


Figure 2.9: Element and system degrees of freedom in global coordinates

Here, the summation can be replaced by the arrangement of the element vectors and the element stiffness matrix into hypervectors and a hypermatrix, respectively.

$$\delta\tilde{W}_{\text{int}} = \begin{bmatrix} \delta\boldsymbol{u}^1 \\ \vdots \\ \delta\boldsymbol{u}^d \\ \delta\boldsymbol{u}^e \\ \delta\boldsymbol{u}^f \\ \delta\boldsymbol{u}^g \\ \vdots \\ \delta\boldsymbol{u}^{NE} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{k}^1 & & & & & & & \\ & \ddots & & & & & & \\ & & \mathbf{k}^d & & & & & \\ & & & \mathbf{k}^e & & & & \\ & & & & \mathbf{k}^f & & & \\ & & & & & \mathbf{k}^g & & \\ & & & & & & \ddots & \\ & & & & & & & \mathbf{k}^{NE} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}^1 \\ \vdots \\ \boldsymbol{u}^d \\ \boldsymbol{u}^e \\ \boldsymbol{u}^f \\ \boldsymbol{u}^g \\ \vdots \\ \boldsymbol{u}^{NE} \end{bmatrix} \tag{2.126}$$

If, additionally, the element vectors and the element stiffness matrix are partitioned with respect to the element nodes, as implied in Eq. (2.125), the internal virtual work results as a function of the element nodal displacements and the hypermatrix of the element stiffness matrices.

$$\delta\tilde{W}_{\text{int}} = \begin{bmatrix} \vdots \\ \delta\boldsymbol{u}^{d1} \\ \boxed{\delta\boldsymbol{u}^{d2}} \\ \delta\boldsymbol{u}^{e1} \\ \boxed{\delta\boldsymbol{u}^{e2}} \\ \delta\boldsymbol{u}^{f1} \\ \boxed{\delta\boldsymbol{u}^{f2}} \\ \delta\boldsymbol{u}^{g1} \\ \boxed{\delta\boldsymbol{u}^{g2}} \\ \vdots \end{bmatrix} \cdot \begin{bmatrix} \ddots & & & & & & & \\ & \mathbf{k}^{d11} & \boxed{\mathbf{k}^{d12}} & & & & & \\ & \boxed{\mathbf{k}^{d12}} & \boxed{\mathbf{k}^{d22}} & & & & & \\ & & & \mathbf{k}^{e11} & \boxed{\mathbf{k}^{e12}} & & & \\ & & & \boxed{\mathbf{k}^{e12}} & \boxed{\mathbf{k}^{e22}} & & & \\ & & & & & \mathbf{k}^{f11} & \boxed{\mathbf{k}^{f12}} & \\ & & & & & \boxed{\mathbf{k}^{f12}} & \boxed{\mathbf{k}^{f22}} & \\ & & & & & & \mathbf{k}^{g11} & \boxed{\mathbf{k}^{g12}} \\ & & & & & & \boxed{\mathbf{k}^{g12}} & \boxed{\mathbf{k}^{g22}} \\ & & & & & & & \ddots \end{bmatrix} \begin{bmatrix} \vdots \\ \boldsymbol{u}^{d1} \\ \boxed{\boldsymbol{u}^{d2}} \\ \boldsymbol{u}^{e1} \\ \boxed{\boldsymbol{u}^{e2}} \\ \boldsymbol{u}^{f1} \\ \boxed{\boldsymbol{u}^{f2}} \\ \boldsymbol{u}^{g1} \\ \boxed{\boldsymbol{u}^{g2}} \\ \vdots \end{bmatrix} \tag{2.127}$$

At the element node 2, each of the elements d, e, f and g is coupled to the system node k, which means that the corresponding displacements have to be identical. The element degrees of freedom of the studied elements are assigned to the system degrees of freedom $k-1$, $k+1$, $k+2$ and $k+3$. The identities $\boldsymbol{u}^{e1} = \boldsymbol{u}^{k+1}$ and $\boldsymbol{u}^{e2} = \boldsymbol{u}^{k}$ for the element e follow from the system and element degrees of freedom, sketched in Fig. 2.9. After supplementing the respective relationships for the other elements, the assignment of the element and system degrees of freedom and the corresponding variations is given as follows:

$$
\begin{aligned}
\boldsymbol{u}^{k-1} &= \boldsymbol{u}^{d1} = \cdots & \delta\boldsymbol{u}^{k-1} &= \delta\boldsymbol{u}^{d1} = \cdots \\
\boldsymbol{u}^{k} &= \boldsymbol{u}^{d2} = \boldsymbol{u}^{e2} = \boldsymbol{u}^{f2} = \boldsymbol{u}^{g2} & \delta\boldsymbol{u}^{k} &= \delta\boldsymbol{u}^{d2} = \delta\boldsymbol{u}^{e2} = \delta\boldsymbol{u}^{f2} = \delta\boldsymbol{u}^{g2} \\
\boldsymbol{u}^{k+1} &= \boldsymbol{u}^{e1} = \cdots & \delta\boldsymbol{u}^{k+1} &= \delta\boldsymbol{u}^{e1} = \cdots \\
\boldsymbol{u}^{k+2} &= \boldsymbol{u}^{f1} = \cdots & \delta\boldsymbol{u}^{k+2} &= \delta\boldsymbol{u}^{f1} = \cdots \\
\boldsymbol{u}^{k+3} &= \boldsymbol{u}^{g1} = \cdots & \delta\boldsymbol{u}^{k+3} &= \delta\boldsymbol{u}^{g1} = \cdots
\end{aligned}
\tag{2.128}
$$

The three dots $(\cdots)$ in the Eqs. (2.128) represent the contributions from other truss elements, not considered in the explanation of the assembly process. Based on the identity of the displacement degrees of freedom of the four neighbour elements at element node 2, the corresponding four rows and columns of the matrix equation (2.127) can be put into one row or column, respectively. That means that the stiffness terms that correspond to the element node 2 (e.g. $\mathbf{k}^{e22}$) have to be added to the position k of the system matrix. Stiffness terms that connect the element nodes 1 and 2 (e.g. $\mathbf{k}^{e12}$) have to be entered relative to the assignment of the element and system degrees of freedom (see Eq. (2.128)) into the matrix that has to be generated. Denoting

$$
\mathbf{K}^{kk} = \mathbf{k}^{d22} + \mathbf{k}^{e22} + \mathbf{k}^{f22} + \mathbf{k}^{g22} \tag{2.129}
$$

yields the resulting internal virtual work of the system.[4]

$$
\delta\tilde{W}_{\text{int}} = \underbrace{\begin{bmatrix} \vdots \\ \delta\boldsymbol{u}^{k-1} \\ \boxed{\delta\boldsymbol{u}^{k}} \\ \delta\boldsymbol{u}^{k+1} \\ \delta\boldsymbol{u}^{k+2} \\ \delta\boldsymbol{u}^{k+3} \\ \vdots \end{bmatrix}}_{\delta\boldsymbol{u}} \cdot \underbrace{\begin{bmatrix} \ddots & & & & & & \\ & \mathbf{k}^{d11} & \mathbf{k}^{d12} & & & & \\ & \mathbf{k}^{d12} & \boxed{\mathbf{K}^{kk}} & \mathbf{k}^{e12} & \mathbf{k}^{f12} & \mathbf{k}^{g12} & \\ & & \mathbf{k}^{e12} & \mathbf{k}^{e11} & & & \\ & & \mathbf{k}^{f12} & & \mathbf{k}^{f11} & & \\ & & \mathbf{k}^{g12} & & & \mathbf{k}^{g11} & \\ & & & & & & \ddots \end{bmatrix}}_{\mathbf{K}} \underbrace{\begin{bmatrix} \vdots \\ \boldsymbol{u}^{k-1} \\ \boxed{\boldsymbol{u}^{k}} \\ \boldsymbol{u}^{k+1} \\ \boldsymbol{u}^{k+2} \\ \boldsymbol{u}^{k+3} \\ \vdots \end{bmatrix}}_{\boldsymbol{u}} \tag{2.130}
$$

With the definition of the *system* or *structure stiffness matrix* $\mathbf{K}$ and the *system* or *structure displacement vector* $\boldsymbol{u}$, the internal virtual work at the system level can be compactly given.

$$
\delta\tilde{W}_{\text{int}} = \delta\boldsymbol{u} \cdot \mathbf{K}\,\boldsymbol{u} \tag{2.131}
$$

If analogous arguments and equivalent strategies are applied to the sum of the virtual work of the inertial forces in the above description of the assembly of the stiffness matrix, they yield the

[4] When considering the influence of other elements the additional terms $\mathbf{k}^{e12} + \cdots$ and $\mathbf{k}^{e11} + \cdots$ have to be taken into account

dynamic part of the virtual work, formulated in terms of the *system* or *structure mass matrix* $\mathbf{M}$ and the *system* or *structure acceleration vector* $\ddot{\boldsymbol{u}}$.

$$\delta\tilde{W}_{\text{dyn}} = \delta\boldsymbol{u} \cdot \mathbf{M}\,\ddot{\boldsymbol{u}} \tag{2.132}$$

Since in the assembly of the element loads $\boldsymbol{r}^e$ the element nodal loads $\boldsymbol{r}_n^e$ from neighbour elements vanish, as is still to be shown, this aspect must be clarified more precisely in what follows. The sum of the virtual works of the element loads $\boldsymbol{r}^e$ can be written in hypervector form, in analogy to Eq. (2.127).

$$\begin{aligned}
\delta\tilde{W}_{\text{ext}} &= \sum_{e=1}^{NE} \delta\tilde{W}_{\text{ext}}^e = \sum_{e=1}^{NE} \delta\boldsymbol{u}^e \cdot \boldsymbol{r}^e = \sum_{e=1}^{NE} \begin{bmatrix} \delta\boldsymbol{u}^{e1} \\ \delta\boldsymbol{u}^{e2} \end{bmatrix} \cdot \begin{bmatrix} \boldsymbol{r}^{e1} \\ \boldsymbol{r}^{e2} \end{bmatrix} \\
&= \begin{bmatrix} \delta\boldsymbol{u}^1 \\ \vdots \\ \delta\boldsymbol{u}^d \\ \delta\boldsymbol{u}^e \\ \delta\boldsymbol{u}^f \\ \delta\boldsymbol{u}^g \\ \vdots \\ \delta\boldsymbol{u}^{NE} \end{bmatrix} \cdot \begin{bmatrix} \boldsymbol{r}^1 \\ \vdots \\ \boldsymbol{r}^d \\ \boldsymbol{r}^e \\ \boldsymbol{r}^f \\ \boldsymbol{r}^g \\ \vdots \\ \boldsymbol{r}^{NE} \end{bmatrix} = \begin{bmatrix} \vdots \\ \delta\boldsymbol{u}^{d1} \\ \boxed{\delta\boldsymbol{u}^{d2}} \\ \delta\boldsymbol{u}^{e1} \\ \boxed{\delta\boldsymbol{u}^{e2}} \\ \delta\boldsymbol{u}^{f1} \\ \boxed{\delta\boldsymbol{u}^{f2}} \\ \delta\boldsymbol{u}^{g1} \\ \boxed{\delta\boldsymbol{u}^{g2}} \\ \vdots \end{bmatrix} \cdot \begin{bmatrix} \vdots \\ \boldsymbol{r}^{d1} \\ \boxed{\boldsymbol{r}^{d2}} \\ \boldsymbol{r}^{e1} \\ \boxed{\boldsymbol{r}^{e2}} \\ \boldsymbol{r}^{f1} \\ \boxed{\boldsymbol{r}^{f2}} \\ \boldsymbol{r}^{g1} \\ \boxed{\boldsymbol{r}^{g2}} \\ \vdots \end{bmatrix}
\end{aligned} \tag{2.133}$$

If now the relationships at the element and system level are inserted for the variation of the element displacement vectors $\delta\boldsymbol{u}^{e1}$ and $\delta\boldsymbol{u}^{e2}$ according to Eqs. (2.128), the dimension of the hypervectors in (2.133) can be reduced. For this, the parts of the load corresponding to the displacements are added.

$$\begin{aligned}
\boldsymbol{r}^{k-1} &= \boldsymbol{r}^{d1} + \cdots \\
\boldsymbol{r}^{k} &= \boldsymbol{r}^{d2} + \boldsymbol{r}^{e2} + \boldsymbol{r}^{f2} + \boldsymbol{r}^{g2} \\
\boldsymbol{r}^{k+1} &= \boldsymbol{r}^{e1} + \cdots
\end{aligned} \qquad
\begin{aligned}
\boldsymbol{r}^{k+2} &= \boldsymbol{r}^{f1} + \cdots \\
\boldsymbol{r}^{k+3} &= \boldsymbol{r}^{g1} + \cdots
\end{aligned} \tag{2.134}$$

The three dots $(\cdots)$ symbolize the element loads not considered in the description of the assembly. Thus, the external virtual work is defined with the *system* or *structure load vector* $\boldsymbol{r}$.

$$\delta\tilde{W}_{\text{ext}} = \underbrace{\begin{bmatrix} \vdots \\ \delta\boldsymbol{u}^{k-1} \\ \boxed{\delta\boldsymbol{u}^{k}} \\ \delta\boldsymbol{u}^{k+1} \\ \delta\boldsymbol{u}^{k+2} \\ \delta\boldsymbol{u}^{k+3} \\ \vdots \end{bmatrix}}_{\delta\boldsymbol{u}} \cdot \underbrace{\begin{bmatrix} \vdots \\ \boldsymbol{r}^{k-1} \\ \boxed{\boldsymbol{r}^{k}} \\ \boldsymbol{r}^{k+1} \\ \boldsymbol{r}^{k+2} \\ \boldsymbol{r}^{k+3} \\ \vdots \end{bmatrix}}_{\boldsymbol{r}} = \delta\boldsymbol{u} \cdot \boldsymbol{r} \tag{2.135}$$

The property of the element nodal loads $\boldsymbol{r}_n^e$ mentioned already during the assembly can be analyzed now by means of the system load vector $\boldsymbol{r}^k$, according to Eq. (2.134). The element loads can be split according to Eq. (2.73) into the consistent element loads $\boldsymbol{r}_p^e$ and the vector of the element nodal loads $\boldsymbol{r}_n^e$.

$$\boldsymbol{r}^k = \boldsymbol{r}^{d2} + \boldsymbol{r}^{e2} + \boldsymbol{r}^{f2} + \boldsymbol{r}^{g2} = \boldsymbol{r}_p^{d2} + \boldsymbol{r}_p^{e2} + \boldsymbol{r}_p^{f2} + \boldsymbol{r}_p^{g2} + \boldsymbol{r}_n^{d2} + \boldsymbol{r}_n^{e2} + \boldsymbol{r}_n^{f2} + \boldsymbol{r}_n^{g2} \tag{2.136}$$

The section loads of the truss ends $e1$ and $e2$ were defined in Eq. (2.71) as the load column $\boldsymbol{r}_n^e$ of the finite element. Therefore, it is about loads that act on the end cross sections of the truss from outside. These loads must now fulfill the equilibrium condition at the system node, i.e., in the assembly of all neighbour elements to a system node. Furthermore, the equation

$$\mathbf{0} = \boldsymbol{r}_n^{d2} + \boldsymbol{r}_n^{e2} + \boldsymbol{r}_n^{f2} + \boldsymbol{r}_n^{g2} \tag{2.137}$$

must be satisfied. Therefore, the element nodal loads $\boldsymbol{r}_n^e$ have a contribution to the virtual work at the element level but not at the structure level. Due to this reason, these loads should not be computed in the generation of the element load vector at the element level for the purpose of the subsequent assembly of the system load vector $\boldsymbol{r}$.

$$\boldsymbol{r}^k = \boldsymbol{r}_p^{d2} + \boldsymbol{r}_p^{e2} + \boldsymbol{r}_p^{f2} + \boldsymbol{r}_p^{g2} \tag{2.138}$$

Actually, if the principle of virtual work at the element level is of interest, the contribution of the element nodal loads $\boldsymbol{r}_n^e$ has to be taken into account.

2.3.2.2 Element-specific Compatibility Matrix

In the previous section, the principle procedure for the assembly of the system out of *NE* finite elements was established on the basis of the virtual works of the internal, dynamic and external loads and demonstrated with the help of the truss frame shown in Fig. 2.9. In order to mathematically realize this procedure and mainly to systematize the presentation of the Eqs. (2.128), the compatibility matrix (2.128) of a finite element is introduced. The compatibility matrix extracts the element degrees of freedom of the element e from the vector of system degrees of freedom. In order to realize this, it is enough to use a BOOL'ean matrix which contains a one at the positions where a system degree of freedom corresponds to an element degree of freedom, and which otherwise contains zeros.

The generation of such an element-specific BOOL'ean matrix must be made clear for the example of the elements e and g, already considered in the previous section. $\mathbf{a}^e$ relates the structure degrees of freedom of the node k: u_1^k, u_2^k and u_3^k and the system node $k+1$: u_1^{k+1}, u_2^{k+1} and u_3^{k+1} to the element degrees of freedom of the element nodes $e2$: u_1^{e2}, u_2^{e2} and u_3^{e2} and $e1$: u_1^{e1}, u_2^{e1} and u_3^{e1}, respectively. This means that the corresponding vectors are connected by the 3×3 unity

matrix $\mathbf{I}$. Analogously, the construction of the compatibility matrix $\mathbf{a}^g$ can be described.

$$\underbrace{\begin{bmatrix} \boldsymbol{u}^{e1} \\ \boldsymbol{u}^{e2} \end{bmatrix}}_{\boldsymbol{u}^e} = \underbrace{\begin{bmatrix} \cdots & \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \cdots \\ \cdots & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots \end{bmatrix}}_{\mathbf{a}^e} \underbrace{\begin{bmatrix} \vdots \\ \boldsymbol{u}^{k-1} \\ \boldsymbol{u}^{k} \\ \boldsymbol{u}^{k+1} \\ \boldsymbol{u}^{k+2} \\ \boldsymbol{u}^{k+3} \\ \vdots \end{bmatrix}}_{\boldsymbol{u}} = \mathbf{a}^e\, \boldsymbol{u}$$

$$\underbrace{\begin{bmatrix} \boldsymbol{u}^{g1} \\ \boldsymbol{u}^{g2} \end{bmatrix}}_{\boldsymbol{u}^g} = \underbrace{\begin{bmatrix} \cdots & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} & \cdots \\ \cdots & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots \end{bmatrix}}_{\mathbf{a}^g} \underbrace{\begin{bmatrix} \vdots \\ \boldsymbol{u}^{k-1} \\ \boldsymbol{u}^{k} \\ \boldsymbol{u}^{k+1} \\ \boldsymbol{u}^{k+2} \\ \boldsymbol{u}^{k+3} \\ \vdots \end{bmatrix}}_{\boldsymbol{u}} = \mathbf{a}^g\, \boldsymbol{u} \tag{2.139}$$

Corresponding relations are obtained for the extraction of the variation of the element displacements $\delta\boldsymbol{u}^e$ and the element accelerations $\ddot{\boldsymbol{u}}^e$ from the respective system vectors $\delta\boldsymbol{u}$ and $\delta\ddot{\boldsymbol{u}}$.

$$\begin{aligned} \delta\boldsymbol{u}^e &= \mathbf{a}^e\, \delta\boldsymbol{u} & \qquad \delta\boldsymbol{u}^g &= \mathbf{a}^g\, \delta\boldsymbol{u} \\ \ddot{\boldsymbol{u}}^e &= \mathbf{a}^e\ \ddot{\boldsymbol{u}} & \qquad \ddot{\boldsymbol{u}}^g &= \mathbf{a}^g\ \ddot{\boldsymbol{u}} \end{aligned} \tag{2.140}$$

The algorithm for the assembly of the system stiffness matrix can be obtained by means of the internal virtual work on the basis of the compatibility matrices of all finite elements.

$$\delta\tilde{W}_{\text{int}} = \sum_{e=1}^{NE} \delta\tilde{W}^e_{\text{int}} = \sum_{e=1}^{NE} \delta\boldsymbol{u}^e \cdot \mathbf{k}^e\, \boldsymbol{u}^e = \delta\boldsymbol{u} \cdot \underbrace{\left(\sum_{e=1}^{NE} \mathbf{a}^{e\,T}\, \mathbf{k}^e\, \mathbf{a}^e \right)}_{\mathbf{K}} \boldsymbol{u} = \delta\boldsymbol{u} \cdot \mathbf{K}\, \boldsymbol{u} \tag{2.141}$$

Analogously, the system mass matrix and the system load vector can be constructed.

$$\begin{aligned} \mathbf{K} &= \sum_{e=1}^{NE} \mathbf{a}^{e\,T}\, \mathbf{k}^e\, \mathbf{a}^e \\ \mathbf{M} &= \sum_{e=1}^{NE} \mathbf{a}^{e\,T}\, \mathbf{m}^e\, \mathbf{a}^e \end{aligned} \qquad \boldsymbol{r} = \sum_{e=1}^{NE} \mathbf{a}^{e\,T}\, \boldsymbol{r}^e = \sum_{e=1}^{NE} \mathbf{a}^{e\,T}\, \boldsymbol{r}^e_p \tag{2.142}$$

2.3.2.3 Compatibility Matrix of the Total System

Alternatively to the elementwise generation of the compatibiity matrix, the latter can be formulated specifically to the system. In this case, the compatibility matrix $\mathbf{a}$ relates the system

degrees of freedom to the degrees of freedom of the hypervector of element displacements, as described in Eq. (2.127).

$$\begin{bmatrix} \boldsymbol{u}^{11} \\ \boldsymbol{u}^{12} \\ \vdots \\ \boldsymbol{u}^{d1} \\ \boldsymbol{u}^{d2} \\ \boldsymbol{u}^{e1} \\ \boldsymbol{u}^{e2} \\ \boldsymbol{u}^{f1} \\ \boldsymbol{u}^{f2} \\ \boldsymbol{u}^{g1} \\ \boldsymbol{u}^{g2} \\ \vdots \\ \boldsymbol{u}^{NE1} \\ \boldsymbol{u}^{NE2} \end{bmatrix} = \underbrace{\begin{bmatrix} \ddots & & & & & & \\ & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \\ & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \\ & \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \\ & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \\ & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} & \\ & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \\ & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} & \\ & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \\ & & & & & & \ddots \end{bmatrix}}_{\mathbf{a}} \underbrace{\begin{bmatrix} \boldsymbol{u}^{1} \\ \vdots \\ \boldsymbol{u}^{k-1} \\ \boldsymbol{u}^{k} \\ \boldsymbol{u}^{k+1} \\ \boldsymbol{u}^{k+2} \\ \boldsymbol{u}^{k+3} \\ \vdots \\ \boldsymbol{u}^{NEQ/3} \end{bmatrix}}_{\boldsymbol{u}} = \begin{bmatrix} \mathbf{a}^{1} \\ \vdots \\ \mathbf{a}^{d} \\ \mathbf{a}^{e} \\ \mathbf{a}^{f} \\ \mathbf{a}^{g} \\ \vdots \\ \mathbf{a}^{NE} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}^{1} \\ \vdots \\ \boldsymbol{u}^{k-1} \\ \boldsymbol{u}^{k} \\ \boldsymbol{u}^{k+1} \\ \boldsymbol{u}^{k+2} \\ \boldsymbol{u}^{k+3} \\ \vdots \\ \boldsymbol{u}^{NEQ/3} \end{bmatrix} \tag{2.143}$$

Comparing Eqs. (2.139) and (2.143), one obtains the relationship between the element-specific and the system-specific compatibility matrices $\mathbf{a}^e$ and $\mathbf{a}$. The matrix $\mathbf{a}$ is the hypervector of matrices $\mathbf{a}^e$ for $e = 1, NE$.

$$\begin{bmatrix} \mathbf{u}^{1} \\ \vdots \\ \mathbf{u}^{NE} \end{bmatrix} = \mathbf{a}\,\boldsymbol{u} \qquad\qquad \mathbf{a} = \begin{bmatrix} \mathbf{a}^{1\,T} & \cdots & \mathbf{a}^{e\,T} & \cdots & \mathbf{a}^{NE\,T} \end{bmatrix}^T \tag{2.144}$$

By means of the virtual works, the transformation relations of the hypermatrices of the element stiffness and mass matrices, as well as the hypervector of the element load vectors, can be derived to the corresponding system quantities. Alternatively, the given relations can be also directly obtained with the help of Eq. (2.144) by transforming the Eqs. (2.142)

$$\begin{aligned} \mathbf{K} &= \mathbf{a}^T \begin{bmatrix} \mathbf{k}^{1} & & \\ & \ddots & \\ & & \mathbf{k}^{NE} \end{bmatrix} \mathbf{a} \\ \mathbf{M} &= \mathbf{a}^T \begin{bmatrix} \mathbf{m}^{1} & & \\ & \ddots & \\ & & \mathbf{m}^{NE} \end{bmatrix} \mathbf{a} \end{aligned} \qquad\qquad \boldsymbol{r} = \mathbf{a}^T \begin{bmatrix} \mathbf{r}^{1} \\ \vdots \\ \mathbf{r}^{NE} \end{bmatrix} = \mathbf{a}^T \begin{bmatrix} \mathbf{r}_p^{1} \\ \vdots \\ \mathbf{r}_p^{NE} \end{bmatrix} \tag{2.145}$$

2.3.2.4 Symbolic Assembly

Finally, the assembly of the element matrices can be formulated symbolically. To do this, the union operator $\bigcup$ is introduced, behind which the assembly process, explained in the previous sections, is hidden.

$$\mathbf{K} = \bigcup_{e=1}^{NE} \mathbf{k}^e \qquad \mathbf{M} = \bigcup_{e=1}^{NE} \mathbf{m}^e \qquad \boldsymbol{r} = \bigcup_{e=1}^{NE} \boldsymbol{r}^e = \bigcup_{e=1}^{NE} \boldsymbol{r}_p^e \tag{2.146}$$

2.3.2.5 Dynamic and Static System Equation

The discretized principle of virtual work can be formulated after the assembly by means of the system displacement vector $\boldsymbol{u}$, its variation $\delta\boldsymbol{u}$, the system acceleration vector $\ddot{\boldsymbol{u}}$, the system stiffness matrix $\mathbf{K}$, the system mass matrix $\mathbf{M}$ and the system load vector $\boldsymbol{r}$ (see Eqs. (1.94), (2.131), (2.132) and (2.135)).

$$\begin{aligned} \delta\tilde{W}_{\text{dyn}} \quad &+ \quad \delta\tilde{W}_{\text{int}} \quad = \quad \delta\tilde{W}_{\text{ext}} \\ \delta\boldsymbol{u}\cdot\mathbf{M}\,\ddot{\boldsymbol{u}} \quad &+ \quad \delta\boldsymbol{u}\cdot\mathbf{K}\,\boldsymbol{u} \quad = \quad \delta\boldsymbol{u}\cdot\boldsymbol{r} \end{aligned} \tag{2.147}$$

If one requires Eq. (2.147) to be fulfilled for arbitrary variations of the system displacement vector $\delta\boldsymbol{u}$ (fundamental lemma of variational calculus), one obtains the *semi-discrete equation of motion.*

$$\boxed{\mathbf{M}\,\ddot{\boldsymbol{u}} + \mathbf{K}\,\boldsymbol{u} = \boldsymbol{r}} \tag{2.148}$$

The equation of motion (2.148) is called *semi-discrete* because a discretization of the partial differential equation of elastodynamics is carried out only in space. The result is a space-discretized, second-order differential equation in time. For the solution of this differential equation, the discrete initial conditions of the displacements and the accelerations at time $t = 0$ are necessary. Here, only one of the two system quantities $\ddot{\boldsymbol{u}}(t=0)$ and $\boldsymbol{u}(t=0)$ can be prescribed, the other one results from the solution of the equation of motion (2.148) at time $t = 0$.

$$\begin{aligned} &\ddot{\boldsymbol{u}}(0) = \ddot{\boldsymbol{u}}^\star && \boldsymbol{u}(0) = \boldsymbol{u}^\star \\ &\boldsymbol{u}(0) = \mathbf{K}^{-1}\,\left[\boldsymbol{r}(0) - \mathbf{M}\,\ddot{\boldsymbol{u}}(0)\right] && \ddot{\boldsymbol{u}}(0) = \mathbf{M}^{-1}\,\left[\boldsymbol{r}(0) - \mathbf{K}\,\boldsymbol{u}(0)\right] \end{aligned} \tag{2.149}$$

The equation of motion (2.148) and the initial conditions (2.149) form the *semi-discrete initial-value problem* of structural dynamics. This initial-value problem can be directly integrated by numerical methods (or in special cases also analytically).

In the static case characterized by ($\ddot{\boldsymbol{u}} = \mathbf{0}$), one obtains from Eq. (2.148) the linear equation system of the solution vector $\boldsymbol{u}$, the *static system equation* or the *static structure equation.*

$$\boxed{\mathbf{K}\,\boldsymbol{u} = \boldsymbol{r}} \tag{2.150}$$

In contrast to structural dynamics, in static structural analyses as many DIRICHLET boundary conditions have to be inserted into the system equation, until no rigid-body motions are possible. Permitted rigid-body motions would lead to the system stiffness matrix being singular, i.e., it cannot be inverted, wherefrom Eq. (2.148) is not solvable.

2.3.2.6 Dirichlet Boundary Conditions at the System Level

In general, both in the static and the dynamic case, DIRICHLET boundary conditions have to be applied to the system equation. They are usually given in the form of support conditions for the problem that has to be modelled and must be rearranged now in discrete form. In the numerical rearrangement it must be differentiated between

- homogeneous DIRICHLET boundary conditions
- and inhomogeneous DIRICHLET boundary conditions.

In the first case, the prescribed displacements are zero and in the second one they are different from zero. In transient problem statements, the displacements at the DIRICHLET boundary are prescribed in the observed time intervals, whereas, in principle, both the DIRICHLET boundary and the NEUMANN boundary can be variable in time.

In order to show the realization of homogeneous DIRICHLET boundary conditions, the example of the truss frame, represented in Fig. 2.10, is taken into consideration. Any displacement of the system node $k+3$ must be hindered in all three coordinate directions.[5]. For the simplification of the discussion, all other support conditions of the truss frame are not taken into account in the following considerations. As a consequence of the homogeneous DIRICHLET boundary conditions, the variation of the displacements and the accelerations of the corresponding degrees of freedom are also zero.

$$\boldsymbol{u}^{k+3} = \delta\boldsymbol{u}^{k+3} = \ddot{\boldsymbol{u}}^{k+3} = \mathbf{0} \tag{2.151}$$

If the DIRICHLET boundary conditions (2.151) are inserted in the principle of virtual work at the system level (2.147), it becomes clear that the corresponding equations have no contribution to the virtual work ($\delta\tilde{W}_{\text{int}} + \delta\tilde{W}_{\text{dyn}} = \delta\tilde{W}_{\text{ext}}$).

$$\delta\tilde{W}_{\text{int}} = \begin{bmatrix} \delta\boldsymbol{u}^{1} \\ \vdots \\ \delta\boldsymbol{u}^{k+1} \\ \delta\boldsymbol{u}^{k+2} \\ \boxed{\mathbf{0}} \\ \delta\boldsymbol{u}^{k+4} \\ \vdots \\ \delta\boldsymbol{u}^{NEQ/3} \end{bmatrix} \cdot \begin{bmatrix} \ddots & & & & & \\ & \mathbf{K}^{k+1k+1} & \mathbf{K}^{k+1k+2} & \boxed{\mathbf{K}^{k+1k+3}} & \mathbf{K}^{k+1k+4} & \\ & \mathbf{K}^{k+2k+1} & \mathbf{K}^{k+2k+2} & \boxed{\mathbf{K}^{k+2k+3}} & \mathbf{K}^{k+2k+4} & \\ & \boxed{\mathbf{K}^{k+3k+1}} & \boxed{\mathbf{K}^{k+3k+2}} & \boxed{\mathbf{K}^{k+3k+3}} & \boxed{\mathbf{K}^{k+3k+4}} & \\ & \mathbf{K}^{k+4k+1} & \mathbf{K}^{k+4k+2} & \boxed{\mathbf{K}^{k+4k+3}} & \mathbf{K}^{k+4k+4} & \\ & & & & & \ddots \end{bmatrix} \begin{bmatrix} \boldsymbol{u}^{1} \\ \vdots \\ \boldsymbol{u}^{k+1} \\ \boldsymbol{u}^{k+2} \\ \boxed{\mathbf{0}} \\ \boldsymbol{u}^{k+4} \\ \vdots \\ \boldsymbol{u}^{NEQ/3} \end{bmatrix} \tag{2.152}$$

[5]It is not imperatively necessary for all displacements of a node to be hindered. Usually, displacements in single coordinate directions are obstructred

$$\delta\tilde{W}_{\text{dyn}} = \begin{bmatrix} \delta\boldsymbol{u}^1 \\ \vdots \\ \delta\boldsymbol{u}^{k+1} \\ \delta\boldsymbol{u}^{k+2} \\ \boxed{\mathbf{0}} \\ \delta\boldsymbol{u}^{k+4} \\ \vdots \\ \delta\boldsymbol{u}^{NEQ/3} \end{bmatrix} \cdot \begin{bmatrix} \ddots & & & & \\ & \mathbf{M}^{k+1k+1} & \mathbf{M}^{k+1k+2} & \boxed{\mathbf{M}^{k+1k+3}} & \mathbf{M}^{k+1k+4} \\ & \mathbf{M}^{k+2k+1} & \mathbf{M}^{k+2k+2} & \boxed{\mathbf{M}^{k+2k+3}} & \mathbf{M}^{k+2k+4} \\ & \boxed{\mathbf{M}^{k+3k+1}} & \boxed{\mathbf{M}^{k+3k+2}} & \boxed{\mathbf{M}^{k+3k+3}} & \boxed{\mathbf{M}^{k+3k+4}} \\ & \mathbf{M}^{k+4k+1} & \mathbf{M}^{k+4k+2} & \boxed{\mathbf{M}^{k+4k+3}} & \mathbf{M}^{k+4k+4} \\ & & & & \ddots \end{bmatrix} \begin{bmatrix} \ddot{\boldsymbol{u}}^1 \\ \vdots \\ \ddot{\boldsymbol{u}}^{k+1} \\ \ddot{\boldsymbol{u}}^{k+2} \\ \boxed{\mathbf{0}} \\ \ddot{\boldsymbol{u}}^{k+4} \\ \vdots \\ \ddot{\boldsymbol{u}}^{NEQ/3} \end{bmatrix} \tag{2.152}$$

$$\delta\tilde{W}_{\text{ext}} = \begin{bmatrix} \delta\boldsymbol{u}^1 \\ \vdots \\ \delta\boldsymbol{u}^{k+1} \\ \delta\boldsymbol{u}^{k+2} \\ \boxed{\mathbf{0}} \\ \delta\boldsymbol{u}^{k+4} \\ \vdots \\ \delta\boldsymbol{u}^{NEQ/3} \end{bmatrix} \cdot \begin{bmatrix} \boldsymbol{r}^1 \\ \vdots \\ \boldsymbol{r}^{k+1} \\ \boldsymbol{r}^{k+2} \\ \boxed{\boldsymbol{r}^{k+3}} \\ \boldsymbol{r}^{k+4} \\ \vdots \\ \boldsymbol{r}^{NEQ/3} \end{bmatrix} \tag{2.152}$$

Due to this reason, it is possible to ignore the terms corresponding to the node $k+3$ in the computation of the virtual work.

$$\delta\tilde{W}_{\text{int}} = \begin{bmatrix} \delta\boldsymbol{u}^1 \\ \vdots \\ \delta\boldsymbol{u}^{k+1} \\ \delta\boldsymbol{u}^{k+2} \\ \hline \delta\boldsymbol{u}^{k+4} \\ \vdots \\ \delta\boldsymbol{u}^{NEQ/3} \end{bmatrix} \cdot \left[\begin{array}{ccc|c} \ddots & & & \\ & \mathbf{K}^{k+1k+1} & \mathbf{K}^{k+1k+2} & \mathbf{K}^{k+1k+4} \\ & \mathbf{K}^{k+2k+1} & \mathbf{K}^{k+2k+2} & \mathbf{K}^{k+2k+4} \\ \hline & \mathbf{K}^{k+4k+1} & \mathbf{K}^{k+4k+2} & \mathbf{K}^{k+4k+4} \\ & & & \ddots \end{array}\right] \left[\begin{array}{c} \boldsymbol{u}^1 \\ \vdots \\ \boldsymbol{u}^{k+1} \\ \boldsymbol{u}^{k+2} \\ \hline \boldsymbol{u}^{k+4} \\ \vdots \\ \boldsymbol{u}^{NEQ/3} \end{array}\right] \tag{2.153}$$

$$\delta\tilde{W}_{\text{dyn}} = \begin{bmatrix} \delta\boldsymbol{u}^1 \\ \vdots \\ \delta\boldsymbol{u}^{k+1} \\ \delta\boldsymbol{u}^{k+2} \\ \hline \delta\boldsymbol{u}^{k+4} \\ \vdots \\ \delta\boldsymbol{u}^{NEQ/3} \end{bmatrix} \cdot \left[\begin{array}{ccc|c} \ddots & & & \\ & \mathbf{M}^{k+1k+1} & \mathbf{M}^{k+1k+2} & \mathbf{M}^{k+1k+4} \\ & \mathbf{M}^{k+2k+1} & \mathbf{M}^{k+2k+2} & \mathbf{M}^{k+2k+4} \\ \hline & \mathbf{M}^{k+4k+1} & \mathbf{M}^{k+4k+2} & \mathbf{M}^{k+4k+4} \\ & & & \ddots \end{array}\right] \left[\begin{array}{c} \ddot{\boldsymbol{u}}^1 \\ \vdots \\ \ddot{\boldsymbol{u}}^{k+1} \\ \ddot{\boldsymbol{u}}^{k+2} \\ \hline \ddot{\boldsymbol{u}}^{k+4} \\ \vdots \\ \ddot{\boldsymbol{u}}^{NEQ/3} \end{array}\right] \tag{2.153}$$

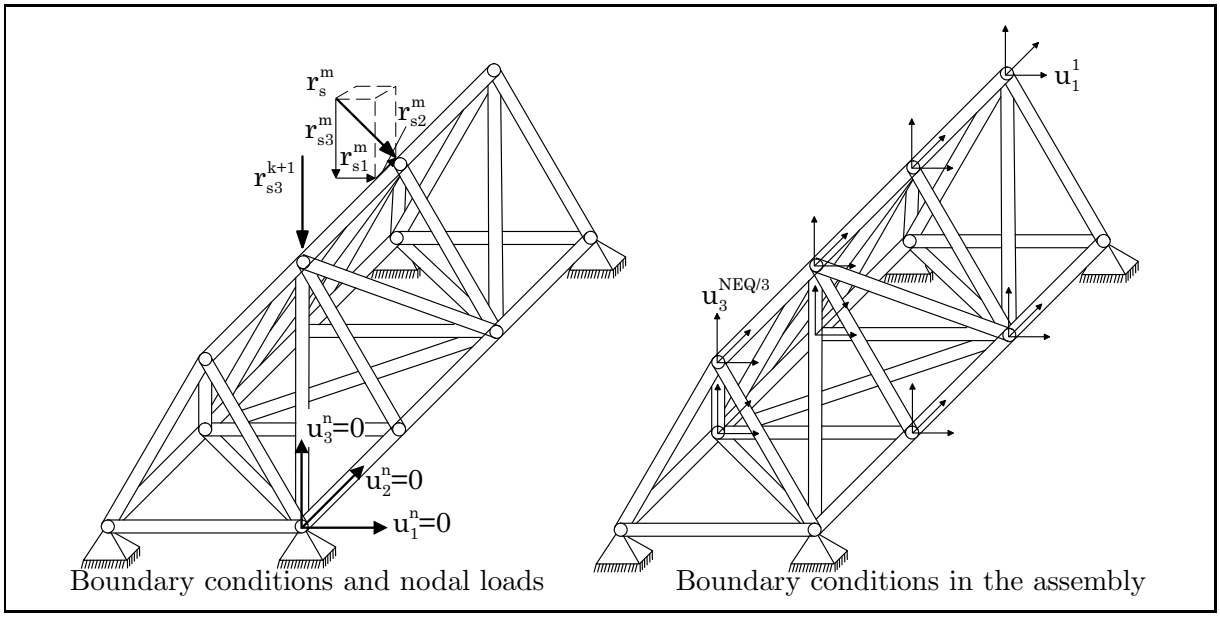


Figure 2.10: Application of DIRICHLET boundary conditions and point loads

$$\delta\tilde{W}_{\text{ext}} = \begin{bmatrix} \delta\boldsymbol{u}^1 \\ \vdots \\ \delta\boldsymbol{u}^{k+1} \\ \delta\boldsymbol{u}^{k+2} \\ \hline \delta\boldsymbol{u}^{k+4} \\ \vdots \\ \delta\boldsymbol{u}^{NEQ/3} \end{bmatrix} \cdot \begin{bmatrix} \boldsymbol{r}^1 \\ \vdots \\ \boldsymbol{r}^{k+1} \\ \boldsymbol{r}^{k+2} \\ \hline \boldsymbol{r}^{k+4} \\ \vdots \\ \boldsymbol{r}^{NEQ/3} \end{bmatrix} \tag{2.153}$$

As a consequence of this, it is possible to take out the corresponding equations (rows and columns) of the linear semi-discrete differential equation system (2.148), evolving from the fundamental lemma of variational calculus, or the linear equation system of static system analyses (2.150). By this measure, the dimension of the Eqs. (2.148) and (2.150) is reduced. Nevertheless, for the number of equations the symbol *NEQ* must still be used unchanged. This is based on the fact that the DIRICHLET boundary conditions practically are taken into account already in the definiton of the degrees of freedom of the structure (at these positions of hindered element displacements, no system degrees of freedom are defined, as shown in Fig. 2.10) and are, therefore, applied before the assembly. This means that the element quantities corresponding to the supports are not added during the assembly process, like it is done at free nodes, but are ignored. For the example of the element g, shown in Fig. 2.9, this means that the components of the system displacement vector $\boldsymbol{u}$ and the compatibility matrix $\mathbf{a}^g$, corresponding to the support nodes, do not exist (see Eq. (2.139) or Eq. (2.143)).

$$\boldsymbol{u}^g = \begin{bmatrix} \boldsymbol{u}^{g1} \\ \boldsymbol{u}^{g2} \end{bmatrix} = \left[\begin{array}{ccccc|c} \cdots & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots \\ \cdots & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \cdots \end{array}\right] \begin{bmatrix} \vdots \\ \boldsymbol{u}^{k-1} \\ \boldsymbol{u}^{k} \\ \boldsymbol{u}^{k+1} \\ \boldsymbol{u}^{k+2} \\ \hline \vdots \end{bmatrix} = \mathbf{a}^g\,\boldsymbol{u} \tag{2.154}$$

Inhomogeneous DIRICHLET boundary conditions are usually inserted into the already reduced equation system (2.148) after the homogeneous DIRICHLET boundary conditions have been inserted. For this purpose, the system equation is sorted in such a way that the system displacement vector could be partitioned into the displacement vector of the prescribed displacements $\boldsymbol{u}_u$ and the displacement vector of the unkown displacements $\boldsymbol{u}_r$.

$$\boldsymbol{u} = \begin{bmatrix} \boldsymbol{u}_u^T & \boldsymbol{u}_r^T \end{bmatrix}^T \tag{2.155}$$

The linear equation system of the static problem or the effective system equation of the dynamic problem (see Chapter 2.4.2) is partitioned respectively.

$$\begin{bmatrix} \mathbf{K}_{uu} & \mathbf{K}_{ur} \\ \mathbf{K}_{ru} & \mathbf{K}_{rr} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}_u \\ \boldsymbol{u}_r \end{bmatrix} = \begin{bmatrix} \boldsymbol{r}_u \\ \boldsymbol{r}_r \end{bmatrix} \qquad \mathbf{K}_{ur} = \mathbf{K}_{ru}^T \tag{2.156}$$

Solution variables of this partitioned equation are the displacements $\boldsymbol{u}_r$ and the loads $\boldsymbol{r}_u$, with the displacements being determined by the second equation and the loads are determined with the already solved displacements $\boldsymbol{u}_r$ from the first equation.

$$\begin{aligned} \mathbf{K}_{rr}\, \boldsymbol{u}_r &= \boldsymbol{r}_r - \mathbf{K}_{ru}\, \boldsymbol{u}_u \\ \boldsymbol{r}_u &= \mathbf{K}_{uu}\, \boldsymbol{u}_u + \mathbf{K}_{ur}\, \boldsymbol{u}_r \end{aligned} \tag{2.157}$$

Eq. $(2.157)_2$ yields the loads at the degrees of freedom of prescribed displacements. If alternatively to the presentation of the homogeneous DIRICHLET boundary conditions in the beginning of this chapter, homogeneous DIRICHLET boundary conditions are also included in the vector of the prescribed displacements $\boldsymbol{u}_u$ as a special case of the inhomogeneous DIRICHLET boundary conditions, the solution of Eq. $(2.157)_2$ contains the support reactions of the system.[6].

2.3.2.7 Consideration of Point Loads at the System Level

Beside consistent element loads, it is especially important in truss or also beam structures to be able to record external point loads at system nodes.[7]. This occurs by addition of the corresponding point load components r_{sj}^i to the system load vector $\boldsymbol{r}$. The system load vector is composed in this case of the sum of the vector $\boldsymbol{r}$ and the system vector of point loads $\boldsymbol{r}_s$. For the example of the point loads sketched in Fig. 2.10, the system vector of point loads takes the following form:

$$\boldsymbol{r}_s = \Big[0 \;\cdots\; 0 \;\; \underbrace{0 \;\; 0 \;\; r_{s3}^{k+1}}_{\boldsymbol{r}_s^{k+1\,T}} \;\; 0 \;\cdots\; 0 \;\; \underbrace{r_{s1}^m \;\; r_{s2}^m \;\; r_{s3}^m}_{\boldsymbol{r}_s^{m\,T}} \;\; 0 \;\cdots\; 0 \Big]^T \tag{2.158}$$

Correspondingly, an additional term of the virtual work of the point loads results, which has to be added to $\delta\tilde{W}_{\text{ext}}$.

$$\delta\tilde{W}_{\text{ext}}^s = \delta\boldsymbol{u} \cdot \boldsymbol{r}_s \tag{2.159}$$

[6]Usually, the support reactions are computed within the context of postprocessing, treated in Chapter 2.5.2

[7]The nonconsistent element volume loads, discussed in Chapter 2.1.5, but also life-loads or similar can be brought in here

2.4 Solution of the System Equation

The static or dynamic system equations generated in the previous chapter must be solved now for the unknowns, or in other words, their properties must be characterized. As shown in what follows, for this purpose the solution of a linear equation system or of a standard eigenvalue problem is essentially required.

2.4.1 Linear Statics

In the static case of structural mechanics, the linear equation system (2.150) has to be solved to determine the system displacements. This has to be realized symbolically by the left multiplication of this equation with the inverse stiffness matrix $\mathbf{K}^{-1}$.

$$\mathbf{K}^{-1}\,\mathbf{K}\,\boldsymbol{u} = \mathbf{I}\,\boldsymbol{u} = \boldsymbol{u} = \mathbf{K}^{-1}\,\boldsymbol{r} \tag{2.160}$$

The practical solution is carried out with methods of numerical mathematics, as discussed in Section 2.4.3

2.4.2 Linear Dynamics

For the semi-discrete initial-value problem, formed by the Eqs. (2.148) and (2.149),

- the system solution is directly integrated by means of numerical methods
- or the characteristic system behaviour is analyzed by computation of the eigenvalues and eigenvectors

For a detailed study of the thematics, refer to more specific literature (Newmark [68], Hughes [22], Argyris & Mlejnek [5], Zienkiewicz & Taylor [37], Bathe [7], Wood [81] and Bathe [8]) or the lecture *Non-linear structural dynamics.*

2.4.3 Solution of the Linear System of Equations

The solution methods of problems in structural mechanics require the solution of the linear system of equations.

$$\mathbf{A}\,\boldsymbol{x} = \boldsymbol{b} \tag{2.161}$$

For the solution of this equation system

- direct
- and iterative

solvers can be used. The classical direct solution by Gauss elimination is applicable only for a small number of degrees of freedom, whereas the iterative Gauss-Seidel method can be applied also for larger equation systems. For a study of the equation solvers refer to more specific literature (e.g. Bathe [8] and Zurmühl & Falk [82]). Within the framework of the lecture, the solution of the equation must be accepted as 'Black-Box'.

2.5 Postprocessing

With the methods presented in Chapter 2.4, the system displacement vector $\boldsymbol{u}$ can be computed, whereby the solution of the static or dynamic (time-discrete) problem, and therefore the deformation pattern, is fully described. For the application of the method of finite elements in the engineering practice (design, measurement and testing), the derived quantities (strains and stresses) are usually of greater significance than the deformation pattern. Due to this reason

- the local approximated displacements,
- the local approximated strains
- and the local approximated stresses

are computed and visualized within the context of *postprocessing*.

2.5.1 Separation and Transformation of the Element Degrees of Freedom

The first step in postprocesssing is the separation of the element displacement vector $\boldsymbol{u}^{e'}$ in global coordinates ($\boldsymbol{e}'_1$,$\boldsymbol{e}'_2$,$\boldsymbol{e}'_3$) by means of the element-specific compatibility matrix $\mathbf{a}^e$ from Eq. (2.139).

$$\boldsymbol{u}^{e'} = \mathbf{a}^e \, \boldsymbol{u} \tag{2.162}$$

Afterwards, the element displacement vector is transformed to the local element coordinates with the help of Eq. (2.117)

$$\boldsymbol{u}^e = \mathbf{T} \, \boldsymbol{u}^{e'} \tag{2.163}$$

and the continuous displacement $u_1(\xi_1)$ is computed with the matrix of shape functions, according to Eq. (2.42).

$$\tilde{u}_1(\xi_1) = \sum_{i=1}^{2} u_1^{ei} \, N^i(\xi_1) = \mathbf{N}(\xi_1) \, \boldsymbol{u}^e \tag{2.164}$$

2.5.2 Computation of Strains, Stresses and Section Loads

With the approximation of the continuous displacement, the approximated continuous strains and stresses are also given over the kinematics and the material law. The approximated strains can be directly determined with the B-operator, according to Eq. (2.59).

$$\tilde{\varepsilon}_{11}(\xi_1) = \mathbf{B}(\xi_1) \, \boldsymbol{u}^e \tag{2.165}$$

The application of the constitutive equation (2.9) finally yields the approximated stress distribution and hence, the distribution of the section loads.

$$\tilde{\sigma}_{11}(\xi_1) = E \, \tilde{\varepsilon}_{11}(\xi_1) \qquad \tilde{N}_1(\xi_1) = A \, \tilde{\sigma}_{11}(\xi_1) \tag{2.166}$$

All approximated quantities, represented in the previous Eqs. (2.164) to (2.166) in dependence of the natural coordinate ξ_1, can be mapped to the physical coordinate X_1 within the context of the isoparametric element concept by means of the transformation relation (2.54).

With Eq. $(2.166)_2$ it is further possible to compute the element load vector $\boldsymbol{r}_n^e$ with the components $N_1^{e1} = \tilde{N}_1(-1)$ and $N_1^{e2} = \tilde{N}_1(1)$.

$$\boldsymbol{r}_n^e = \begin{bmatrix} N_1^{e1} \\ N_1^{e2} \end{bmatrix} = \begin{bmatrix} \tilde{N}_1(-1) \\ \tilde{N}_1(1) \end{bmatrix} \tag{2.167}$$

The transformation of the element load vectors according to Eq. (2.123) and their assembly according to Eq. (2.142), respectively, (2.142),

$$\boldsymbol{r}_n = \sum_{e=1}^{NE} \mathbf{a}^{e\,T}\, \boldsymbol{r}_n^{e'} = \sum_{e=1}^{NE} \mathbf{a}^{e\,T}\, \mathbf{T}^T\, \boldsymbol{r}_n^e \tag{2.168}$$

finally yields an equation for the computation of the system support reactions.

$$\boldsymbol{r}_l = \boldsymbol{r} + \boldsymbol{r}_n \tag{2.169}$$

2.5.3 Aspects of Visualization

In visualization, care should be taken to the form, in which the approximated quantities are repesented. Common is the representation of

- element quantities,
- smoothed system quantities
- and element quantities, extrapolated from GAUSS points (smoothed/unsmoothed).

In particular, smoothed stress distributions trickily provide an accuracy, not present in the FEM computation. C_0-continuous displacement approximations are discontinuous in the first derivative. This means that also the strains or stresses, evolving from the first derivatives, are discontinuous beyond the element boundaries.

Chapter 3

Plane Finite Elements

Plane finite elements can be classified according to the type of mechanical problem to be solved, the element form, the polynomial degree of the shape functions and the type of the shape functions:

- mechanical problem
  - plane stress state
  - plane strain state
  - axisymmetric stress-strain state
- element form
  - triangular elements
  - rectangular elements
- polynomial degree of the shape functions
  - linear
  - quadratic
  - cubic
  - ...
- type of the shape functions
  - LAGRANGE shape functions
  - serendipity shape functions
  - hierarchical shape functions

In the following chapter first the basic equations of plane finite elements are derived from the consideration of the three-dimensional continuum with the insertion of the fundamental assumptions of planar continua. Afterwards, plane elements are classified according to the element form and the order of the shape functions and the discretization of planar continua is explained in details with the help of the four-node, linear, isoparametric LAGRANGE element. Furthermore, retangular elements of higher approximation order and triangular elements in natural triangle coordinates are discussed.

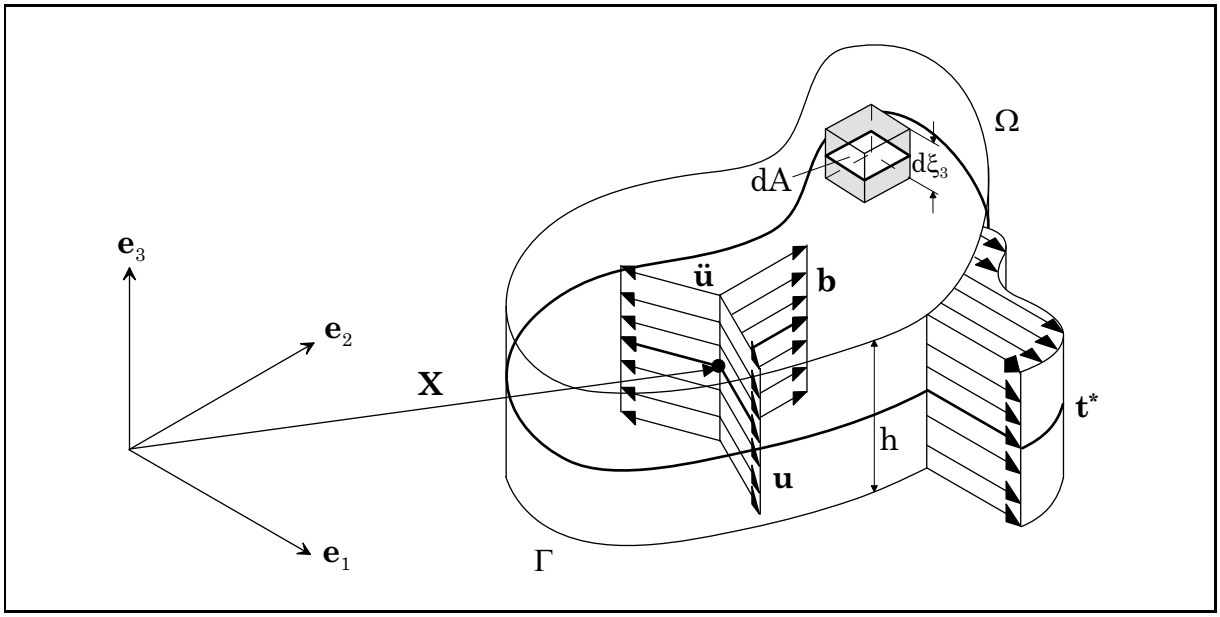


Figure 3.1: Representation of the three-dimensional continuum in plane elements

3.1 Basic Equations of Planar Continua

3.1.1 Geometry

A planar structure, or a two-dimensional continuum, is generated by the degeneration of the three-dimensional continuum onto a middle surface A and the thickness h. Due to this reason, the properties of the three-dimensional continuum are represented by the properties of the middle surface within the context of the modelling of a plane finite element (see Fig. 3.1). Two-dimensional elements of structural mechanics are

- plane elements which are characterized by a small thickness h relative to the dimensions of the middle surface and are characterized by the plane stress state (Chapter 1.4.4),
- plane elements which describe an infinite dimension of the mechanical problem in $\boldsymbol{e}_3$-direction and are, thus, described by the plane strain state (Chapter 1.4.5).

The thickness h represents in the case of plane strain state the infinite thickness or the unit thickness; therefore the thickness of finite elements of the plane strain state is often given the value one. Within the context of this lecture the element thickness must be considered a parameter also for the plane strain state in order to guarantee a generalized description of elements of the plane stress and strain state.

The geometrical description of such a two-dimensional element follows from the position vector of a material point on the middle surface

$$\boldsymbol{X} = \begin{bmatrix} X_1 & X_2 \end{bmatrix}^T \tag{3.1}$$

and its motion when a load is applied.

3.1.2 Kinetics

The first fundamental assumption for the realization of the degeneration of the three- dimensional continuum to the two-dimensional continuum is the assumption of the stress and strain state, where it is distinguished between the plane stress state (ES) and the plane strain state (EV).

$$\text{ES: } \sigma_{33} = \sigma_{23} = \sigma_{13} = 0 \qquad \text{EV: } \sigma_{33} = \nu\,(\sigma_{11} + \sigma_{22}),\ \sigma_{23} = \sigma_{13} = 0 \tag{3.2}$$

When the plane stress state is assumed to hold the internal kinetics is uniquely described by the components σ_{11}, σ_{22} and σ_{12} (see Fig.1.11). In the case of plane strain state additionally the normal stress component σ_{33} is present (see Fig. 1.12). For a complete and unique characterization of the stress state this normal component of the stress tensor is actually not necessary, as it can be computed directly from the stress components σ_{11} and σ_{22} according to Eq. (1.74). For the formulation of the internal virtual work the component σ_{33} is also of no significance, for the conjugated strain component ε_{33} is zero according to the assumptions of the plane strain state. Therefore, it is enough, both for the plane stress state and for the plane strain state, to define the stress vector

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{22} & \sigma_{12} \end{bmatrix}^T \tag{3.3}$$

with three components for the formulation of the internal virtual work and, thus, also for the generation of the discretized stiffness relation of plane finite elements. In accordance woth the above-mentioned kinetic assumptions the volulme loads b_3 and the NEUMANN boundary conditions $t_3^\star$ vanish in the $\boldsymbol{e}_3$-direction, wherefrom also the corresponding vectors are described by two components each (see Fig. 3.1).

$$\boldsymbol{b} = \begin{bmatrix} b_1 & b_2 \end{bmatrix}^T \qquad \boldsymbol{t}^\star = \begin{bmatrix} t_1^\star & t_2^\star \end{bmatrix}^T \tag{3.4}$$

3.1.3 Kinematics

The second fundamental assumption of plane finite elements concerns the displacement field: all material points at the normal of the middle surface exhibit under deformation the same displacements in the directions $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$.

$$u_1 = u_1(X_1, X_2) \qquad u_2 = u_2(X_1, X_2) \tag{3.5}$$

The displacement component u_3 is zero in the case of plane strain state. For the plane stress state u_3 is different from zero. However, it follows from the assumptions for the volume and surface loads in Eq. (3.4) that this displacement component has no contribution to the virtual work of the external loads. Here, it is assumed without a proof[1] that also the portion of this

[1]For a proof of this circumstance refer to the analogy of degeneration of the three-dimensional continuum to the one-dimensional one in the corresponding derivation of the EULER equations of spatial truss elements in Chapter 2.1.5 and to Chapter 3.1.5

component in the virtual wotk of the inertial forces vanishes during deformation. Thus, the deformation of the degenerated two-dimensional continuum can be described uniquely by the displacement vector

$$\boldsymbol{u}(X_1, X_2) = \boldsymbol{u}(\boldsymbol{X}) = \begin{bmatrix} u_1(X_1, X_2) & u_2(X_1, X_2) \end{bmatrix}^T \tag{3.6}$$

expressed, according to Eq. (3.5), only in terms of the coordinates of the plane spanned by the basis vectors $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$. From this follow directly the variation of the displacement vector and, by twice differentiating Eq. (3.6) with respect to time, the acceleration vector.

$$\delta\boldsymbol{u}(\boldsymbol{X}) = \begin{bmatrix} \delta u_1(X_1, X_2) & \delta u_2(X_1, X_2) \end{bmatrix}^T \qquad \ddot{\boldsymbol{u}}(\boldsymbol{X}) = \begin{bmatrix} \ddot{u}_1(X_1, X_2) & \ddot{u}_2(X_1, X_2) \end{bmatrix}^T \tag{3.7}$$

The complete description of the degenerated two-dimensional continuum requires furthermore the formulation of the Dirichlet boundary conditions

$$\boldsymbol{u}(\boldsymbol{X}, t) = \boldsymbol{u}^\star(\boldsymbol{X}, t) \qquad \forall \quad \boldsymbol{X} \in \Gamma_u \tag{3.8}$$

and the initial conditions of the displacement vector and the acceleration vector, where it should be noted that only one of the two initial conditions has to be prescribed, as the second one results from the evaluation of the equation of motion of the degenerated continuum at time $t = 0$ (see Eqs. (2.149)).

$$\begin{aligned} \boldsymbol{u}(\boldsymbol{X}, t = 0) &= \boldsymbol{u}^\star(\boldsymbol{X}) \\ \ddot{\boldsymbol{u}}(\boldsymbol{X}, t = 0) &= \ddot{\boldsymbol{u}}^\star(\boldsymbol{X}) \end{aligned} \qquad \forall \quad \boldsymbol{X} \in \Omega \tag{3.9}$$

It remains to describe the strain state of the degenerated continuum by the strain vector $\boldsymbol{\varepsilon}$. In accordance with the already discussed assumptions of plane stress or plane strain states this vector can be obtained by the assembly of the strain components ε_{11}, ε_{22} and $2\varepsilon_{12}$ alone.

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{11} & \varepsilon_{22} & 2\varepsilon_{12} \end{bmatrix}^T \tag{3.10}$$

The component ε_{33} that is different from zero in the plane stress state is not needed for the unique characterization of the strain state, for this strain component can be combined linearly with the components ε_{11} and ε_{22}, according to Eq. (1.67). In addition, the same argumentation for the vanishing virtual work of the strain component ε_{33} as in the definition of the stress vector $\boldsymbol{\sigma}$, applies here. The strain tensor can be computed in tensor notation from Eq. (1.13)

$$\boldsymbol{\varepsilon} = \nabla^{\text{sym}}\boldsymbol{u} \qquad \varepsilon_{\alpha\beta} = \frac{1}{2}\left(u_{\alpha,\beta} + u_{\beta,\alpha}\right) \tag{3.11}$$

where it should be considered that the indices α and β can take only the values one or two due to the degeneration ($\alpha, \beta = 1, 2$). Besides, the strain vector can be obtained by application of the differential operator $\mathbf{D}_\varepsilon$ to the displacement vector $\boldsymbol{u}$.

$$\boldsymbol{\varepsilon} = \mathbf{D}_\varepsilon \, \boldsymbol{u} \qquad \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} = \begin{bmatrix} \dfrac{\partial}{\partial X_1} & 0 \\ 0 & \dfrac{\partial}{\partial X_2} \\ \dfrac{\partial}{\partial X_2} & \dfrac{\partial}{\partial X_1} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \tag{3.12}$$

Thus, it follows directly from the Eqs. (3.5) that the components of the strain vector $\varepsilon_{\alpha\beta}$ are constant along the thickness of the degenerated continuum.

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}(X_1, X_2) \tag{3.13}$$

3.1.4 Constitutive Equation

The constitutive equation of the plane stress state is formulated by means of the constitutive matrix $\mathbf{C}^{es}$ expressed in terms of the material constants E and ν (see Eq. (1.69) and Table 1.1).

$$\boldsymbol{\sigma} = \mathbf{C}^{es} \, \boldsymbol{\varepsilon} \, , \quad \mathbf{C}^{es} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ & 1 & 0 \\ \text{sym} & & \dfrac{1-\nu}{2} \end{bmatrix} \tag{3.14}$$

Analogously, the constitutive equation of the plane strain state is described with the constitutive matrix $\mathbf{C}^{ev}$ (see Eq. (1.75) and Table 1.1).

$$\boldsymbol{\sigma} = \mathbf{C}^{ev} \, \boldsymbol{\varepsilon} \, , \quad \mathbf{C}^{ev} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ & 1-\nu & 0 \\ \text{sym} & & \dfrac{1-2\nu}{2} \end{bmatrix} \tag{3.15}$$

Whenever the symbol $\mathbf{C}$ is used for the constitutive matrix in the following explanations, $\mathbf{C}^{es}$ or $\mathbf{C}^{ev}$ can be used instead according to the regarded problem statement.

$$\boldsymbol{\sigma} = \mathbf{C} \, \boldsymbol{\varepsilon} \qquad \mathbf{C} = \mathbf{C}^{es} \quad \wedge \quad \mathbf{C} = \mathbf{C}^{ev} \tag{3.16}$$

Concerning the distribution of the stress components $\sigma_{\alpha\beta}$ along the thickness of the degenerated continuum, the statement that the stresses are also constant along the thickness can be concluded from the constant strain distribution along the thickness, according to Eq. (3.13), and from the constitutive equations (3.16).

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}(X_1, X_2) \tag{3.17}$$

3.1.5 Principle of Virtual Work

With the two-dimensional quantities of kinematics, kinetics and the material law, represented in the previous sections, the principle of virtual work of the degenerated two-dimensional continuum can be derived directly from its three-dimesional form, shown in Eq. (1.96). For this, the volume element is replaced by the area element of the middle surface dA and the line element dX_3 in the direction of the basis vector $\boldsymbol{e}_3$ (see Fig. 3.1). Likewise, the boundary of the three-dimensional continuum is split up into the boundary Γ_σ and the thickness coordinate dX_3,

where the boundary of the faces, described by $X_3 = \pm h/2$, should not be taken into account, for at this boundary the stress vector $\boldsymbol{t}^\star$ is identical to the zero vector.

$$\int_A \int_{-\frac{h}{2}}^{\frac{h}{2}} \delta\boldsymbol{u} \cdot \ddot{\boldsymbol{u}}\, \rho\, dX_3 dA + \int_A \int_{-\frac{h}{2}}^{\frac{h}{2}} \delta\boldsymbol{\varepsilon} \cdot \boldsymbol{\sigma}\, dX_3 dA = \int_A \int_{-\frac{h}{2}}^{\frac{h}{2}} \delta\boldsymbol{u} \cdot \boldsymbol{b}\, \rho\, dX_3 dA + \int_{\Gamma_\sigma} \int_{-\frac{h}{2}}^{\frac{h}{2}} \delta\boldsymbol{u} \cdot \boldsymbol{t}^\star\, dX_3 d\Gamma_\sigma \quad (3.18)$$

For the generation of the above equation it was assumed that the specific virtual work of the inertial forces in $\boldsymbol{e}_3$-direction ($\delta u_3\ \ddot{u}_3$) vanishes (a proof follows). Since all integrants in Eq. (3.18) are functions only of the coordinates X_1 and X_2, the integration over the thickness can be performed first.

$$\boxed{\int_A \delta\boldsymbol{u} \cdot \ddot{\boldsymbol{u}}\, \rho\, h\, dA + \int_A \delta\boldsymbol{\varepsilon} \cdot \boldsymbol{\sigma}\, h\, dA = \int_A \delta\boldsymbol{u} \cdot \boldsymbol{b}\, \rho\, h\, dA + \int_{\Gamma_\sigma} \delta\boldsymbol{u} \cdot \boldsymbol{t}^\star h\, d\Gamma_\sigma \qquad (3.19)}$$

The model of the degenerated two-dimensional continuum, found to be as such with the kinematic and kinetic assumptions and the pre-integration over the thickness coordinate X_3, is depicted in Fig. 3.2.

It should be noted here once again that in the derivation of the virtual work of the inertial forces of the plane stress state (first term in Eq. (3.19)) it was assumed that the integral of the term $\delta u_3\ \ddot{u}_3$ over the thickness X_3 vanishes. This assumption must be proved now. By Eq. (1.67) the strain component ε_{33} is given as a linear combination of the components ε_{11} and ε_{22}, which are constant along the thickness coordinate, multiplied with the factor $-\nu/(1-\nu)$. Furthermore, the normal strains ε_{ii} are formed according to Eq. (1.14) by differentiating the displacement u_i with respect to X_i, where a linear distribution of the displacement $u_3(X_3)$ for a fixed position (X_1, X_2) of the middle surface can be determined by separation of the variables.

$$\varepsilon_{33} = u_{3,3} = -\frac{\nu}{1-\nu}\,(u_{1,1} + u_{2,2}) \qquad\qquad u_3(X_3) = -\frac{\nu}{1-\nu}\,(u_{1,1} + u_{2,2})\, X_3 \qquad (3.20)$$

By variation, or double differentiation of Eq. $(3.20)_2$ with respect to time, respectively, one gets the variation δu_3 that is constant along the thickness and the linearly-varying acceleration component $\ddot{u}_3(X_3)$. The substitution of these quantities in the term of the dynamic virtual work yields the relationship that was to be proved.

$$\begin{aligned} \int_{-\frac{h}{2}}^{\frac{h}{2}} \delta u_3\, \ddot{u}_3\, dX_3 &= -\int_{-\frac{h}{2}}^{\frac{h}{2}} \delta u_3\, \frac{\nu}{1-\nu}\,(\ddot{u}_{1,1} + \ddot{u}_{2,2})\, X_3\, dX_3 \\ &= -\frac{\nu}{1-\nu}\,(\ddot{u}_{1,1} + \ddot{u}_{2,2})\, \delta u_3 \int_{-\frac{h}{2}}^{\frac{h}{2}} X_3\, dX_3 = 0 \end{aligned} \qquad (3.21)$$

3.1.6 Euler Differential Equation and Neumann Boundary Conditions

In order to determine the Euler differential equation and the Neumann boundary condition of the two-dimensional continuum, the variation of the strain vector $\delta\boldsymbol{\varepsilon}$ has to be replaced by the variation of the displacement vector $\delta\boldsymbol{u}$ during the formulation of the internal virtual work,

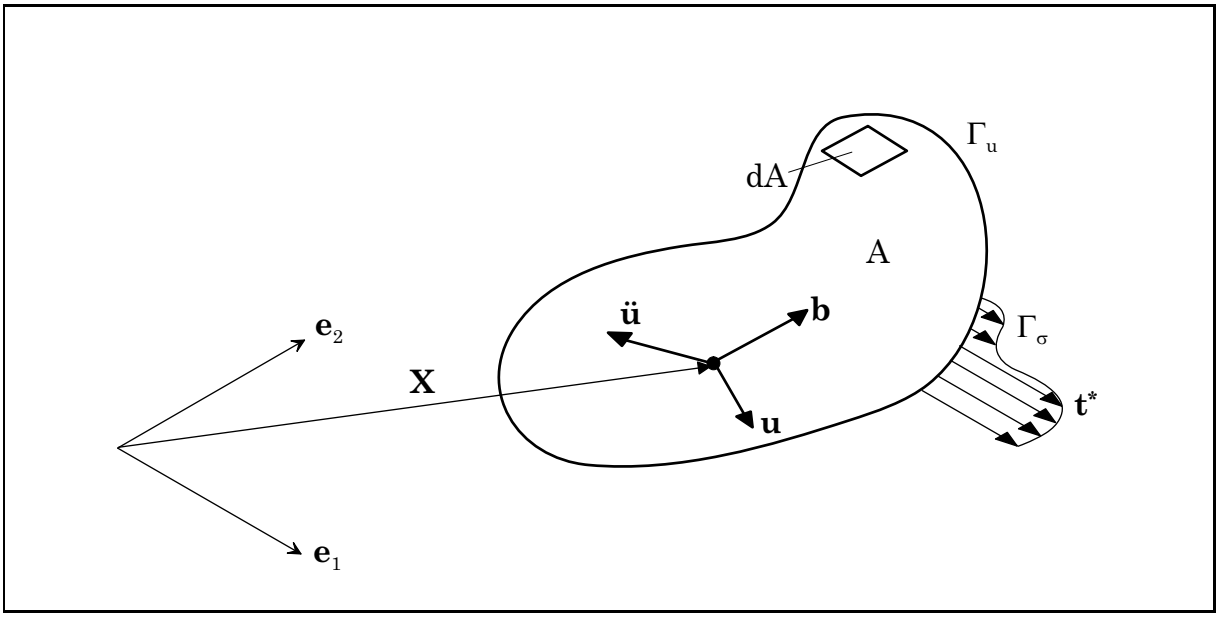


Figure 3.2: Model of a degenerated two-dimensional continuum

according to Eq. (3.19). This happens with the help of Eqs. (1.86) to (1.91), which were used for the transformation of the internal virtual work of the three-dimensional continuum. Transfering the methodology of application of the chain rule to the divergence of the stress tensor and the GAUSS integral theorem to the degenerated continuum, yields the expression for the internal virtual work of the plane mechanical problem.

$$\begin{aligned} \delta W_{\text{int}} &= \int\limits_A \delta\boldsymbol{\varepsilon}\cdot\boldsymbol{\sigma}\; h\; dA = \int\limits_{\Gamma_\sigma} \delta\boldsymbol{u}\cdot\boldsymbol{\sigma}\;\boldsymbol{n}\; h\; d\Gamma_\sigma - \int\limits_A \delta\boldsymbol{u}\cdot\text{div}\boldsymbol{\sigma}\; h\; dA \\ &= \int\limits_{\Gamma_\sigma} \delta\boldsymbol{u}\cdot\boldsymbol{t}\; h\; d\Gamma_\sigma - \int\limits_A \delta\boldsymbol{u}\cdot\mathbf{D}_\varepsilon^T\;\boldsymbol{\sigma}\; h\; dA \end{aligned} \tag{3.22}$$

For the transformaiton of the divergence operator into the differential operator $\mathbf{D}_\sigma = \mathbf{D}_\varepsilon^T$, the equations (1.29) and (1.30) were used into a reduced for the two-dimensional case form. The summary of the Eqs. (3.19) and (3.22) yields the principle of virtual work of the plane continuum, formulated completely in terms of the variation of the displacement vector.

$$\int\limits_A \delta\boldsymbol{u}\cdot\ddot{\boldsymbol{u}}\;\rho\; h\; dA + \int\limits_{\Gamma_\sigma} \delta\boldsymbol{u}\cdot\boldsymbol{t}\; h\; d\Gamma_\sigma = \int\limits_A \delta\boldsymbol{u}\cdot\mathbf{D}_\varepsilon^T\;\boldsymbol{\sigma}\; h\; dA + \int\limits_A \delta\boldsymbol{u}\cdot\boldsymbol{b}\;\rho\; h\; dA + \int\limits_{\Gamma_\sigma} \delta\boldsymbol{u}\cdot\boldsymbol{t}^\star h\; d\Gamma_\sigma \tag{3.23}$$

Should the above equation be fulfilled inside the domain A for arbitrary variations $\delta\boldsymbol{u}$, the EULER equations of the plane mechanical problem result (see Fig.3.2).

$$\ddot{\boldsymbol{u}}\;\rho\; h = \mathbf{D}_\varepsilon^T\;\underbrace{\boldsymbol{\sigma}\; h}_{\boldsymbol{N}} + \boldsymbol{b}\;\rho\; h = \mathbf{D}_\varepsilon^T\;\boldsymbol{N} + \boldsymbol{b}\;\rho\; h \tag{3.24}$$

Here, the stress resultants $\boldsymbol{N}$ of the physical units N/m were defined. In component notation one obtains the EULER equations of the classical plate theory.

$$\ddot{u}_\alpha\;\rho\; h = N_{\alpha\beta,\beta} + b_\alpha\;\rho\; h \qquad \alpha,\beta = 1,2 \tag{3.25}$$

The NEUMANN boundary conditions result from Eq. (3.23) for arbitrary variations $\delta \boldsymbol{u}$ at the boundary of the domain.

$$\boldsymbol{t}\, h = \boldsymbol{t}^{\star}\, h \qquad\qquad t_\alpha\, h = t_\alpha^{\star}\, h \tag{3.26}$$

In summary, the deformation field of plane continua can be described by the differential equation system, defined with the Eqs. (3.24), (3.16) and (3.12).

$$\boxed{\begin{array}{rclcll} \mathbf{D}_\varepsilon^T\, \boldsymbol{N} & = & \ddot{\boldsymbol{u}}\, \rho\, h & - & \boldsymbol{b}\, \rho\, h & & \text{Kinetik (EULER Differentialgleichung)} \\ \boldsymbol{N} & = & h\, \boldsymbol{\sigma} & = & h\, \mathbf{C} & \boldsymbol{\varepsilon} & \text{konstitutive Gleichung} \\ & & & & & \boldsymbol{\varepsilon} = \mathbf{D}_\varepsilon\, \boldsymbol{u} & \text{Kinematik} \end{array}} \tag{3.27}$$

3.2 Finite Elemente Discretization

The finite element discretization and analysis of plane continua consists of the partitioning of the structure, or the domain under consideration, into finite elements and the approximation of continuously distributed physical quantities (e.g. displacements) by discrete nodal degrees of freedom and the assumption of their distribution over the element area. This assumption is associated with the choice of shape functions, which depend on the variables ξ_1 and ξ_2 for the case of plane elements.

3.2.1 Partitioning into Elements and Discretization

In contrast to the spatial truss frame, for which a constructively discrete structure was available already before the mathematical discretization, now a two-dimensional continuum Ω must be subdivided into finite subdomains Ω^e.

$$\Omega = \bigcup_{e=1}^{NE} \Omega^e \qquad \text{with} \quad \Omega^i \cap \Omega^j = \emptyset \quad \text{for } i \neq j \tag{3.28}$$

Inside these finite subdomains Ω^e, or finite elements e, the continuous field variables are approximated by means of shape functions and discrete nodal degrees of freedom. A basis for the development of plane finite elements is the requirement that the principle of virtual work should be fulfilled for every finite element e.

$$\delta W^e_{\text{dyn}} + \delta W^e_{\text{int}} = \delta W^e_{\text{ext}} \tag{3.29}$$

Fig. 3.3 exemplifies the partitioning into elements of the two-dimensional continuum with three- and four-node finite elements. The topological element structure, formed by the subdivision into subdomains Ω^e, is called *finite element mesh* and the process of its generation is *meshing* or *mesh generation*.

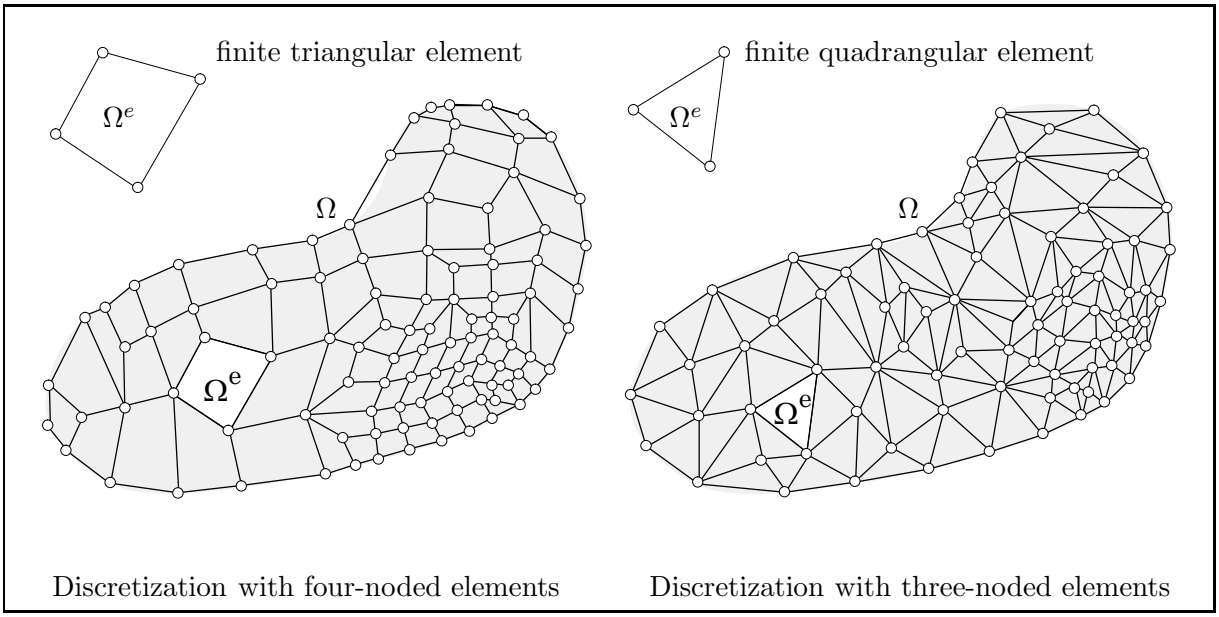


Figure 3.3: Discretization of a two-dimensional continuum

3.2.2 Classification of Plane Elements

Plane finite elements can be classified according to their form into

- triangular elements
- and quadrilateral elements,

and according to the polynomial degree of the shape functions and the existence of internal nodes into

- LAGRANGE elements
- and serendipity elements,

Despite some disadvantages when considering the aspect of accuracy, triangular elements are more popular, for arbitrary planar geometries can be easily discretized by triangles. Quadrilateral meshes of arbitrary geometry are based on the meshing with triangles and the assembly of every two neighbouring triangular elements to a quadrilateral one. It is possible after these meshing strategies that several singular elements remain present, which are very difficult to eliminate by element subdivision strategies. Nonetheless, the effort of optimizing the element subdivisions, which is inherent in this elimination process, is worth it because discretizations with quadrilateral meshes prove to be the more exact finite element analysis in comparison to triangular meshes with the same number of nodes. In application fields, such as mechanical or civil engineering, structured quadrilateral meshes can be used due to the relatively simple geometries that have to be computed, whereas the disadvantage of complex mesh generation has rarely effect for quadrilateral meshes.

As already demonstrated in the development of the finite truss element (Chapter 2), the accuracy of the finite element analysis increases with increasing the polynomial degree p, for which reason the basic elements (3- or 4-node elements) are often substituted by elements of higher polynomial degree. Furthermore, higher-order elements are essentially more suitable than the

basic elements for discretizing domains Ω with curved boundaries, due to the possibility to describe curved element boundaries. As is to be expected from the previous discussions, LAGRANGE elements with an internal node provide more accurate results than serendipity elements without an internal node. The difference in the accuracy is, however, negligible and cannot level out the shortcoming of the additionally needed internal degrees of freedom of the LAGRANGE element (which are of importance particularly by the assembled system). As a disadvantage of serendipity elements, it should be added that they are restricted to a polynomial degree $p < 4$ (see KRÄTZIG & BAŞAR [25]) and are, thus, suitable for adaptive structural analyses according to the p-method (adapting the polynomial degree to the required quality of the finite element analysis) only for several adaptive steps.

3.2.3 Shape Functions of Plane Elements

Within the framework of development of plane finite elements, shape functions of the varying natural coordinates ξ_1 and ξ_2 are used. These shape functions represent polynomials in the two-dimensional space of the form

$$\boldsymbol{u}(\xi_1, \xi_2) \approx \tilde{\boldsymbol{u}}(\xi_1, \xi_2) = \sum_{i=0}^{p} \sum_{j=0}^{p} \alpha^{ij} \ (\xi_1)^i \, (\xi_2)^j \tag{3.30}$$

where the terms of complete two-dimensional polynomials, which correspond to one polynomial degree and are exempt from the coefficients α^{ij}, can be obtained from the PASCALE's triangle in Fig. 3.4. If the terms of the PASCALE's triangle, which are to be used for the generation of a shape function, are known, the coefficients α^{ij} can be determined by generation and inversion of the VANDERMONDE matrix, as shown in Chapter 2.2. Within the context of development of plane finite elements, the shape functions $\boldsymbol{N}^i(\xi_1, \xi_2)$, however, must be generated by multiplication of one-dimensional shape functions or on the basis of interpolation properties. In order to attain already an idea of the qualitative differences of the finite elements developed in the following chapters, these elements are characterized by means of the PASCALE's triangle, despite the planned use of alternative methods for element development.

3.2.3.1 Complete Polynomials

For the development of triangular finite elements, complete LAGRANGE polynomials, according to Eq. (3.30), are used. The components $(\xi_1)^i(\xi_2)^j$ of these approximations are depicted in Fig. 3.4. The triangular elements corresponding to the polynomial degree p are also sketched in this figure, together with the respective number and position of the element nodes.

3.2.3.2 Complete Bipolynomials

Complete polynomials of the degree p are derived by multiplication of two one-dimensional polynomials, also of the degree p (see equation (2.39)).

$$\boldsymbol{u}(\xi_1, \xi_2) \approx \tilde{\boldsymbol{u}}(\xi_1, \xi_2) = \left(\sum_{i=0}^{p} \alpha^i \ (\xi_1)^i \right) \left(\sum_{j=0}^{p} \alpha^j \ (\xi_2)^j \right) \tag{3.31}$$

Accordingly, the terms $(\xi_1)^i(\xi_2)^j$ contained in the PASCALE triangle are characterised by the polynomial degree p of one of the two components paired with all potencies $\leq p$ of the other

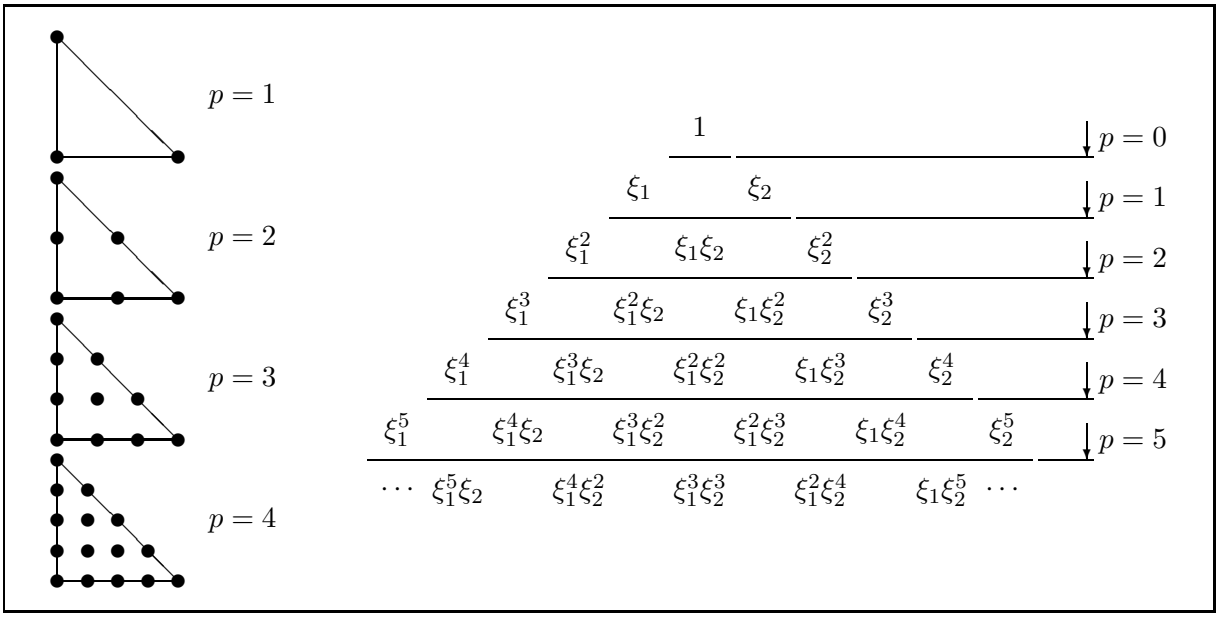


Figure 3.4: Complete two-dimensional polynomials and LAGRANGE ansatz polynomials of triangular elements in the PASCAL triangle

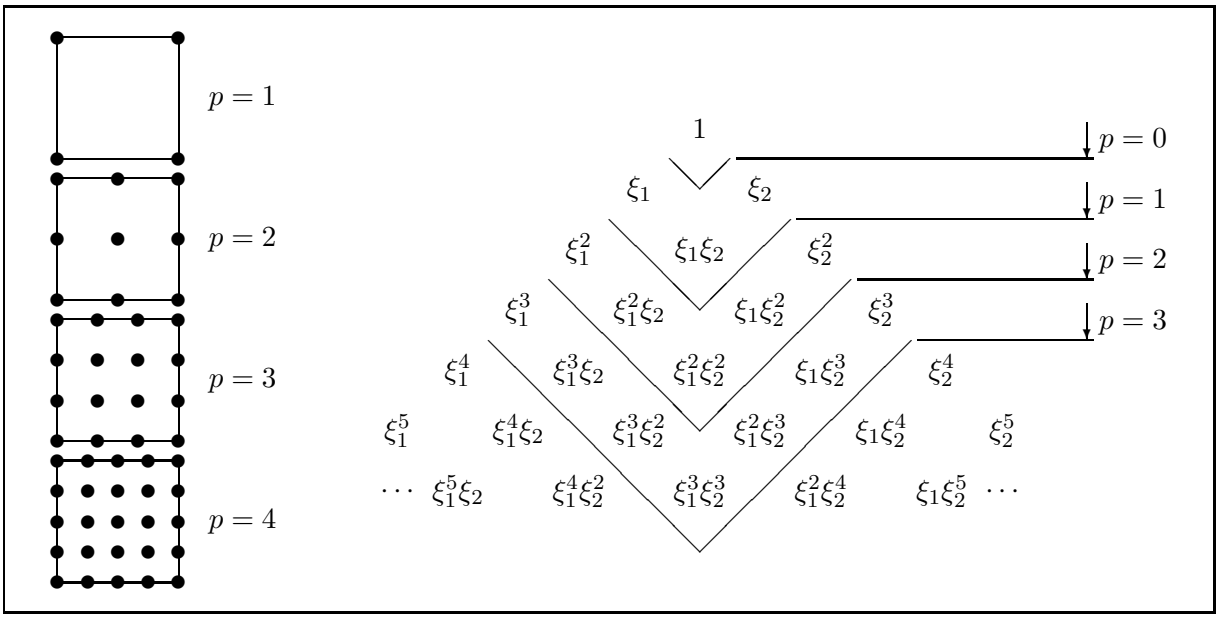


Figure 3.5: : Two-dimensional LAGRANGE ansatz polynomials of quadrangular elements in the PASCAL triangle

component. All used terms and thereby developed bilinear, biquadratic, bicubic and biquadratic LAGRANGE quadrangular elements are summed up in figure 3.5. ¿From this figure we can conclude that in case of a complete bipolynomial of the degree p, additional terms are used compared to a complete polynomial of the same degree (Figure 3.4) p which leads to an improved accuracy of quadrangular elements.

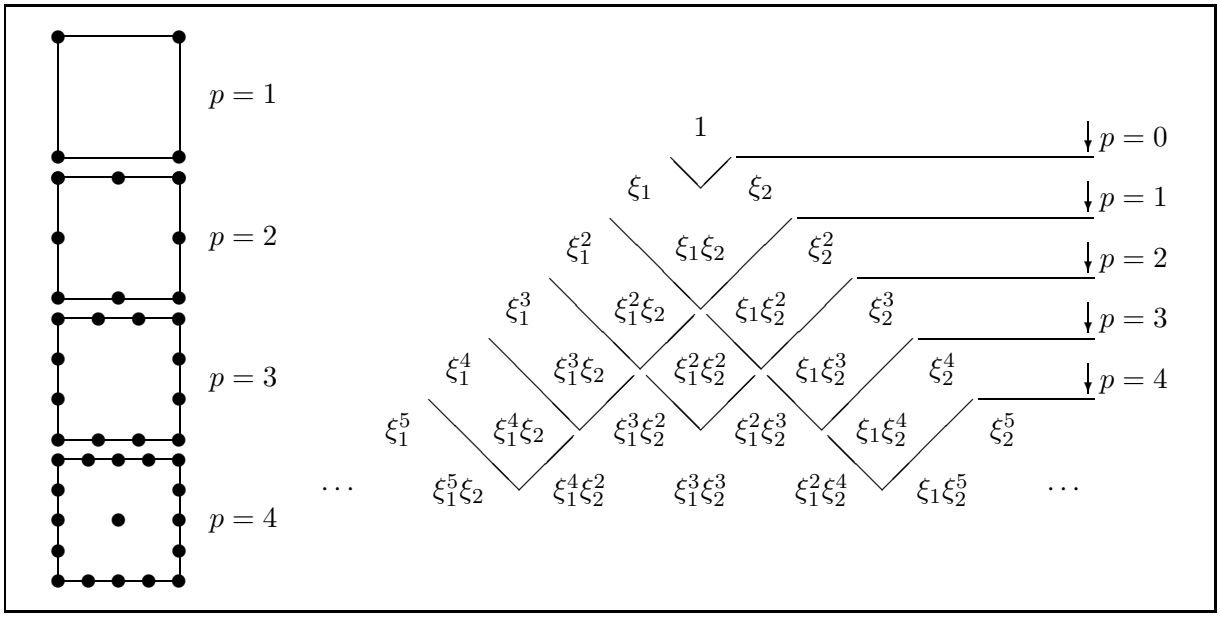


Figure 3.6: Two-dimensional Serendipity polynomials of quadrangular elements in the PASCAL triangle

3.2.3.3 Serendipity bipolynomials

Ansatz functions of the Serendipity type are generated with the help of the interpolation feature.

$$N^i(\xi_1^i, \xi_2^i) = 1 \qquad N^i(\xi_1^j, \xi_2^j) = 0 \quad \text{für } i \neq j \tag{3.32}$$

The corresponding terms in the PASCAL triangle and the resulting finite Serendipity elements are given in figure 3.6. Compared to LAGRANGE ansatz functions, a lesser number of terms is being used here, due to which a reduced accuracy can be expected compared to the corresponding LAGRANGE elements. In order to guarantee at least complete polynomials, Serendipity ansatz functions have to be furnished with an inner element node as well if the polynomial degree is greater than $p = 4$, irrespective of the fact that their number is significantly smaller than it is the case with LAGRANGE ansatz functions of the same polynomial degree.

3.3 Bilinear Lagrange element

The development of plane finite elements is elaborated with the assistance of the relatively simply generated four-noded bilinear LAGRANGE element and specified for the case of a rectangular element.

3.3.1 Ansatz functions

The ansatz function of element node one can be derived by multiplication of the one-dimensional LAGRANGE polynomials $N_1^1(\xi_1)$ and $N_2^1(\xi_2)$ corresponding to this node, which are expressed in the natural coordinate directions ξ_1 or ξ_2. These fundamentals of one-dimensional ansatz polynomials are given in equation (2.50) and in table 2.1.

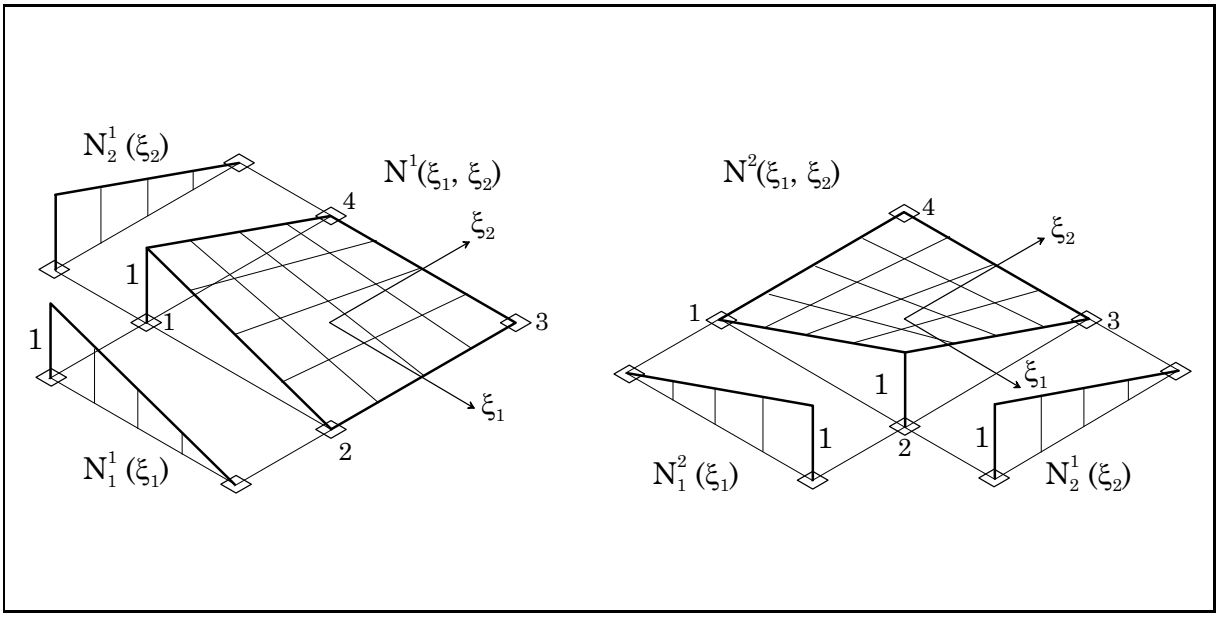


Figure 3.7: Generation of bilinear LAGRANGE ansatz functions

$$N_1^1(\xi_1) = \frac{1}{2}(1-\xi_1) \qquad\qquad N_2^1(\xi_2) = \frac{1}{2}(1-\xi_2) \tag{3.33}$$

Multiplication of $N_1^1(\xi_1)$ and $N_2^1(\xi_2)$ yields the construction presented in figure 3.7 of the ansatz function $N^1(\xi_1, \xi_2)$ for node one. The natural coordinates ξ_1 and ξ_2 are assembled in the vector $\boldsymbol{\xi} = [\xi_1 \;\; \xi_2]^T$ in order to show the dependencies.

$$\begin{aligned} N^1(\xi_1,\xi_2) &= N^1(\boldsymbol{\xi}) = N_1^1(\xi_1)\, N_2^1(\xi_2) \\ &= \frac{1}{2}(1-\xi_1)\,\frac{1}{2}(1-\xi_2) = \frac{1}{4}(1-\xi_1)(1-\xi_2) = \frac{1}{4}(1-\xi_1-\xi_2+\xi_1\xi_2) \end{aligned} \tag{3.34}$$

As we can conclude from equation (3.34), the ansatz function $N^1(\boldsymbol{\xi})$ contains constant, linear as well as bilinear segments. In figure 3.5, these segments correspond to the square in the PASCAL triangle characterised by $p = 1$. As a preliminary observation to the oncoming explanation of approximating continuous variables, it may be noted that a two-dimensional polynomial of plane elements is formed with the sum of all ansatz functions $N^i(\boldsymbol{\xi})$ that have the corresponding weights. The ansatz functions of the element nodes two to four, shown in figure 3.8, can be generated in an analogous way.

$$\begin{aligned} N^1(\boldsymbol{\xi}) &= \frac{1}{4}(1-\xi_1)(1-\xi_2) & N^3(\boldsymbol{\xi}) &= \frac{1}{4}(1+\xi_1)(1+\xi_2) \\ N^2(\boldsymbol{\xi}) &= \frac{1}{4}(1+\xi_1)(1-\xi_2) & N^4(\boldsymbol{\xi}) &= \frac{1}{4}(1-\xi_1)(1+\xi_2) \end{aligned} \tag{3.35}$$

Furthermore, in order to approximate the strains and to generate the JACOBI matrix, derivations of ansatz functions with respect to the natural coordinates ξ_1 and ξ_2 are required. These can be found as follows:

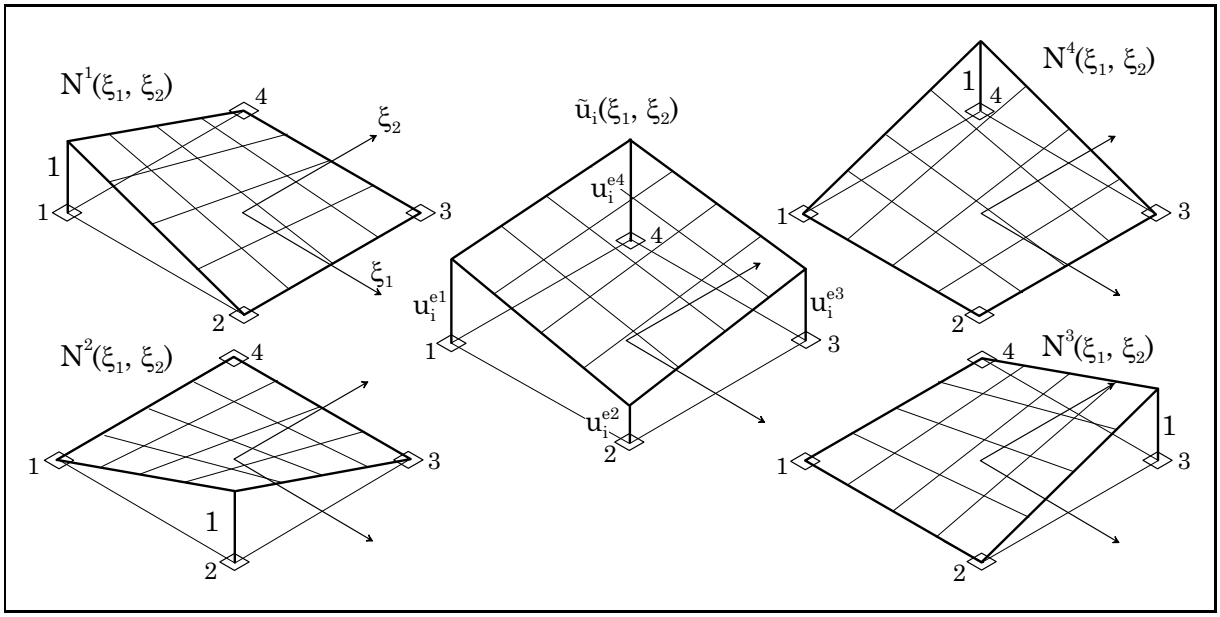


Figure 3.8: LAGRANGE ansatz functions of a plane four-noded element

$$
\begin{aligned}
\frac{\partial N^1(\boldsymbol{\xi})}{\partial \xi_1} &= N^1_{;1}(\boldsymbol{\xi}) = -\frac{1}{4}\,(1-\xi_2) & \frac{\partial N^1(\boldsymbol{\xi})}{\partial \xi_2} &= N^1_{;2}(\boldsymbol{\xi}) = -\frac{1}{4}\,(1-\xi_1) \\
\frac{\partial N^2(\boldsymbol{\xi})}{\partial \xi_1} &= N^2_{;1}(\boldsymbol{\xi}) = \frac{1}{4}\,(1-\xi_2) & \frac{\partial N^2(\boldsymbol{\xi})}{\partial \xi_2} &= N^2_{;2}(\boldsymbol{\xi}) = -\frac{1}{4}\,(1+\xi_1) \\
\frac{\partial N^3(\boldsymbol{\xi})}{\partial \xi_1} &= N^3_{;1}(\boldsymbol{\xi}) = \frac{1}{4}\,(1+\xi_2) & \frac{\partial N^3(\boldsymbol{\xi})}{\partial \xi_2} &= N^3_{;2}(\boldsymbol{\xi}) = \frac{1}{4}\,(1+\xi_1) \\
\frac{\partial N^4(\boldsymbol{\xi})}{\partial \xi_1} &= N^4_{;1}(\boldsymbol{\xi}) = -\frac{1}{4}\,(1+\xi_2) & \frac{\partial N^4(\boldsymbol{\xi})}{\partial \xi_2} &= N^4_{;2}(\boldsymbol{\xi}) = \frac{1}{4}\,(1-\xi_1)
\end{aligned}
\tag{3.36}
$$

3.3.2 Geometry

The geometry of a four-noded LAGRANGE element in physical and natural space is shown in figure 3.9. An arbitrary material point within the quadrangular element is unequivocally identifiable by its physical and natural coordinates.

$$
\boldsymbol{X} = \begin{bmatrix} X_1 & X_2 \end{bmatrix}^T \qquad\qquad \boldsymbol{\xi} = \begin{bmatrix} \xi_1 & \xi_2 \end{bmatrix}^T \tag{3.37}
$$

Positions of the corner nodes are assembled in the element position vector.

$$
\begin{aligned}
\boldsymbol{X}^e &= \begin{bmatrix} X_1^{e1} & X_2^{e1} & X_1^{e2} & X_2^{e2} & X_1^{e3} & X_2^{e3} & X_1^{e4} & X_2^{e4} \end{bmatrix}^T \\
&= \begin{bmatrix} \boldsymbol{X}^{e1\,T} & \boldsymbol{X}^{e2\,T} & \boldsymbol{X}^{e3\,T} & \boldsymbol{X}^{e4\,T} \end{bmatrix}^T
\end{aligned}
\tag{3.38}
$$

Within the scope of isoparametric approximation of geometry and element variables, the continuous position vector $\boldsymbol{X}$ is described with the help of ansatz functions $N^i(\boldsymbol{\xi})$ in natural coordinates

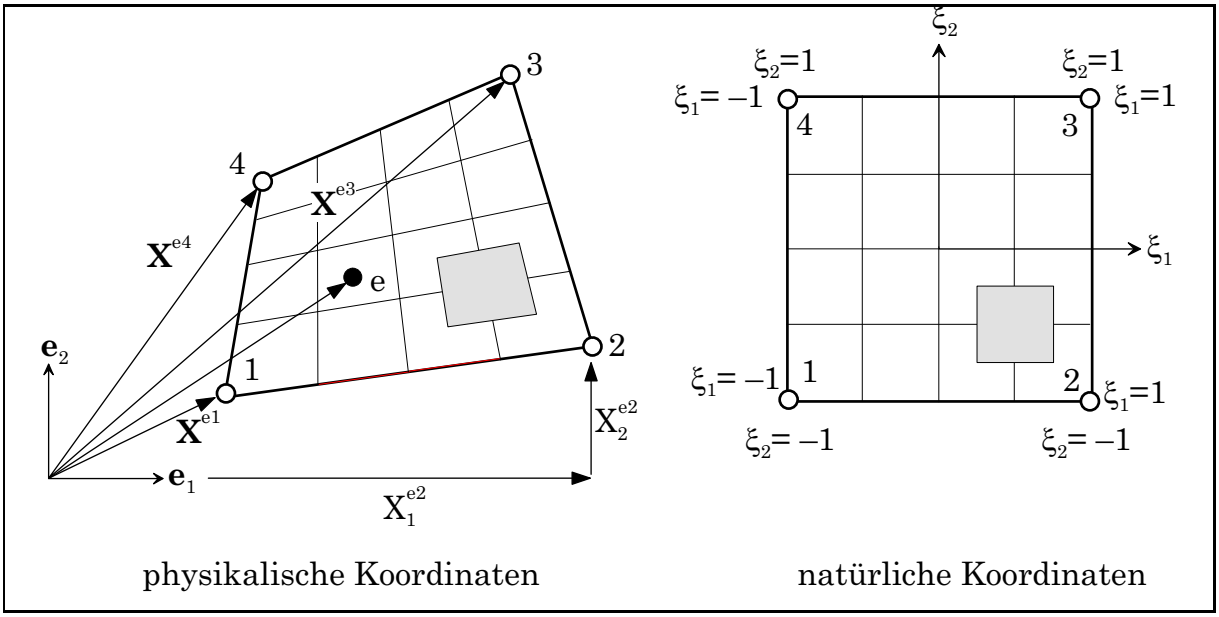


Figure 3.9: Bilinear element in physical and natural coordinates

and with the help of discrete positions of element node $\boldsymbol{X}^{ei}$,

$$\boldsymbol{X}(\xi_1, \xi_2) = \boldsymbol{X}(\boldsymbol{\xi}) \approx \tilde{\boldsymbol{X}}(\boldsymbol{\xi}) = \sum_{i=1}^{NN} \boldsymbol{X}^{ei}\, N^i(\boldsymbol{\xi}) = \sum_{i=1}^{4} \boldsymbol{X}^{ei}\, N^i(\boldsymbol{\xi}) \tag{3.39}$$

where NN generally stands for the number of element nodes. Alternatively, the continuous position vector can be approximated by means of the ansatz function matrix $\mathbf{N}(\boldsymbol{\xi})$ and with the element position vector $\boldsymbol{X}^e$.

$$\underbrace{\begin{bmatrix} \tilde{X}_1(\boldsymbol{\xi}) \\ \tilde{X}_2(\boldsymbol{\xi}) \end{bmatrix}}_{\tilde{\boldsymbol{X}}(\boldsymbol{\xi})} = \underbrace{\begin{bmatrix} N^1(\boldsymbol{\xi}) & 0 & N^2(\boldsymbol{\xi}) & 0 & N^3(\boldsymbol{\xi}) & 0 & N^4(\boldsymbol{\xi}) & 0 \\ 0 & N^1(\boldsymbol{\xi}) & 0 & N^2(\boldsymbol{\xi}) & 0 & N^3(\boldsymbol{\xi}) & 0 & N^4(\boldsymbol{\xi}) \end{bmatrix}}_{\mathbf{N}(\boldsymbol{\xi})} \underbrace{\begin{bmatrix} X_1^{e1} \\ X_2^{e1} \\ X_1^{e2} \\ X_2^{e2} \\ X_1^{e3} \\ X_2^{e3} \\ X_1^{e4} \\ X_2^{e4} \end{bmatrix}}_{\boldsymbol{X}^e} \tag{3.40}$$

Equation (3.40) describes the approximation of physical coordinates as function of natural coordinates ($NN = 4$)

$$\tilde{\boldsymbol{X}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\, \boldsymbol{X}^e \qquad\qquad \boldsymbol{X}^e = \begin{bmatrix} X_1^{e1} & X_2^{e1} & \dots & X_1^{eNN} & X_2^{eNN} \end{bmatrix}^T \tag{3.41}$$

by means of ansatz function matrix $\mathbf{N}(\boldsymbol{\xi})$, which is made up of diagonal matrices of ansatz functions $\mathbf{N}^i(\boldsymbol{\xi})$ that correspond to element nodes i.

$$\mathbf{N}^i(\boldsymbol{\xi}) = \begin{bmatrix} N^i(\boldsymbol{\xi}) & 0 \\ 0 & N^i(\boldsymbol{\xi}) \end{bmatrix} \qquad \mathbf{N} = \begin{bmatrix} \mathbf{N}^1 & \dots & \mathbf{N}^{NN} \end{bmatrix}, \; NN = 4 \tag{3.42}$$

The mapping from natural to physical coordinates according to equations (3.39) to (3.41) can be generally described by the functional relation,

$$\boldsymbol{X} = \boldsymbol{X}(\boldsymbol{\xi}) \qquad X_\beta = X_\beta(\boldsymbol{\xi}), \quad \beta = 1,2 \tag{3.43}$$

where the 'Tilda' approximation designation is omitted. The inverse mapping describes the natural coordinates as function of the physical coordinates.

$$\boldsymbol{\xi} = \boldsymbol{\xi}(\boldsymbol{X}) \qquad \xi_\alpha = \xi_\alpha(\boldsymbol{X}), \quad \alpha = 1,2 \tag{3.44}$$

3.3.3 Jacobi transformation

The calculation of the strain vector $\boldsymbol{\varepsilon}$ according to equation (3.12) requires that the displacement components be derivated with respect to physical coordinates $\boldsymbol{X}$. Since displacement components as well as approximation of the position vector within the scope of the isoparametric element concept are expressed as functions of natural coordinates, the necessary derivations with respect to physical coordinates must be obtained indirectly over the partial derivatives of equation (3.44). Applying the chain rule to equation (3.44) results in the transformation relation between derivatives with respect to physical and natural coordinates, respectively.

$$\frac{\partial}{\partial \xi_\alpha} = \frac{\partial}{\partial X_1}\frac{\partial X_1}{\partial \xi_\alpha} + \frac{\partial}{\partial X_2}\frac{\partial X_2}{\partial \xi_\alpha} = \frac{\partial}{\partial X_\beta}\frac{\partial X_\beta}{\partial \xi_\alpha} \tag{3.45}$$

This equation can alternatively be given in the matrix form with the help of the Jacobi matrix $\mathbf{J}(\boldsymbol{\xi})$.

$$\begin{bmatrix} \frac{\partial}{\partial \xi_1} \\ \frac{\partial}{\partial \xi_2} \end{bmatrix} = \begin{bmatrix} \frac{\partial X_1}{\partial \xi_1} & \frac{\partial X_2}{\partial \xi_1} \\ \frac{\partial X_1}{\partial \xi_2} & \frac{\partial X_2}{\partial \xi_2} \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial X_1} \\ \frac{\partial}{\partial X_2} \end{bmatrix} \qquad \frac{\partial}{\partial \boldsymbol{\xi}} = \mathbf{J}(\boldsymbol{\xi})\,\frac{\partial}{\partial \boldsymbol{X}} \tag{3.46}$$

The rule for derivating functions in natural coordinates with respect to physical coordinates is obtainable by inversion of equation (3.46) with $|\mathbf{J}(\boldsymbol{\xi})| > 0$.

$$\frac{\partial}{\partial \boldsymbol{X}} = \mathbf{J}^{-1}(\boldsymbol{\xi})\,\frac{\partial}{\partial \boldsymbol{\xi}} \tag{3.47}$$

The inverse Jacobi matrix $\mathbf{J}^{-1}(\boldsymbol{\xi})$ needed for this can be obtained directly by inverting the 2×2 Jacobi matrix $\mathbf{J}(\boldsymbol{\xi})$Bronstein & Semendjajew [48]),

$$\mathbf{J}^{-1}(\boldsymbol{\xi}) = \frac{1}{|\mathbf{J}(\boldsymbol{\xi})|} \begin{bmatrix} \frac{\partial X_2}{\partial \xi_2} & -\frac{\partial X_2}{\partial \xi_1} \\ -\frac{\partial X_1}{\partial \xi_2} & \frac{\partial X_1}{\partial \xi_1} \end{bmatrix} \tag{3.48}$$

where the JACOBI determinant is given by the expression

$$\boxed{|\mathbf{J}| = \frac{\partial X_1}{\partial \xi_1}\,\frac{\partial X_2}{\partial \xi_2} - \frac{\partial X_1}{\partial \xi_2}\,\frac{\partial X_2}{\partial \xi_1} \qquad (3.49)}$$

and the derivatives of physical coordinates are obtainable with respect to natural coordinates from the approximation of the position vector $\boldsymbol{X} = [X_1\ X_2]^T$ according to equations (3.39) or (3.41).

$$\frac{\partial X_\beta(\boldsymbol{\xi})}{\partial \xi_\alpha} = \sum_{i=1}^{NN} X_\beta^{ei}\, N^i_{;\alpha}(\boldsymbol{\xi}) \qquad\qquad \frac{\partial \boldsymbol{X}(\boldsymbol{\xi})}{\partial \xi_\alpha} = \boldsymbol{X}_{;\alpha}(\boldsymbol{\xi}) = \mathbf{N}_{;\alpha}(\boldsymbol{\xi})\ \boldsymbol{X}^e \qquad (3.50)$$

Formally, we can get the inverse of the JACOBI matrix by applying the chain rule to functional relation $X_\beta = X_\beta(\boldsymbol{\xi})$ given in equation (3.43)

$$\frac{\partial}{\partial X_\beta} = \frac{\partial}{\partial \xi_1}\,\frac{\partial \xi_1}{\partial X_\beta} + \frac{\partial}{\partial \xi_2}\,\frac{\partial \xi_2}{\partial X_\beta} = \frac{\partial}{\partial \xi_\alpha}\,\frac{\partial \xi_\alpha}{\partial X_\beta} \qquad (3.51)$$

Definition of the JACOBI matrix:

$$\boxed{\begin{bmatrix} \dfrac{\partial}{\partial X_1} \\ \dfrac{\partial}{\partial X_2} \end{bmatrix} = \begin{bmatrix} \dfrac{\partial \xi_1}{\partial X_1} & \dfrac{\partial \xi_2}{\partial X_1} \\ \dfrac{\partial \xi_1}{\partial X_2} & \dfrac{\partial \xi_2}{\partial X_2} \end{bmatrix} \begin{bmatrix} \dfrac{\partial}{\partial \xi_1} \\ \dfrac{\partial}{\partial \xi_2} \end{bmatrix} \qquad \frac{\partial}{\partial \boldsymbol{X}} = \mathbf{J}^{-1}(\boldsymbol{\xi})\frac{\partial}{\partial \boldsymbol{\xi}} \qquad (3.52)}$$

By means of coefficient comparison of the inverse JACOBI matrix according to equations (3.48) and (3.52), we can get the identities

$$\begin{aligned} \frac{\partial \xi_1}{\partial X_1} &= \frac{1}{|\mathbf{J}(\boldsymbol{\xi})|}\,\frac{\partial X_2}{\partial \xi_2} & \qquad \frac{\partial \xi_2}{\partial X_1} &= -\frac{1}{|\mathbf{J}(\boldsymbol{\xi})|}\,\frac{\partial X_2}{\partial \xi_1} \\ \frac{\partial \xi_1}{\partial X_2} &= -\frac{1}{|\mathbf{J}(\boldsymbol{\xi})|}\,\frac{\partial X_1}{\partial \xi_2} & \qquad \frac{\partial \xi_2}{\partial X_2} &= \frac{1}{|\mathbf{J}(\boldsymbol{\xi})|}\,\frac{\partial X_1}{\partial \xi_1} \end{aligned} \qquad (3.53)$$

A transformation relation of the surface element dA can be generated from the first of these identities in physical and natural coordinates.

$$dA = dX_1\ dX_2 = |\mathbf{J}(\boldsymbol{\xi})|\ d\xi_1\ d\xi_2 \qquad (3.54)$$

Alternatively, we can derive these transformation relations that are presented in figure 3.10 with assistance of surface elements, in physical and natural coordinates. For that purpose, vectors $d\boldsymbol{X}_1$, $d\boldsymbol{X}_2$, $d\boldsymbol{\xi}_1$ and $d\boldsymbol{\xi}_2$ spanning the surface elements are defined. The surface elements are given with the help of these definitions by the magnitude of the vector product of vectors $d\boldsymbol{X}_1$ and $d\boldsymbol{X}_2$ that is $d\boldsymbol{\xi}_1$ and $d\boldsymbol{\xi}_2$.

$$\begin{aligned} dA &= |d\boldsymbol{X}_1 \times d\boldsymbol{X}_2| &= \sin\alpha\ |d\boldsymbol{X}_1|\ |d\boldsymbol{X}_2| \\ dA_\xi &= |d\boldsymbol{\xi}_1 \times d\boldsymbol{\xi}_2| &= \sin\frac{\pi}{2}\ |d\boldsymbol{\xi}_1|\ |d\boldsymbol{\xi}_2| \quad = \quad d\xi_1\ d\xi_2 \end{aligned} \qquad (3.55)$$

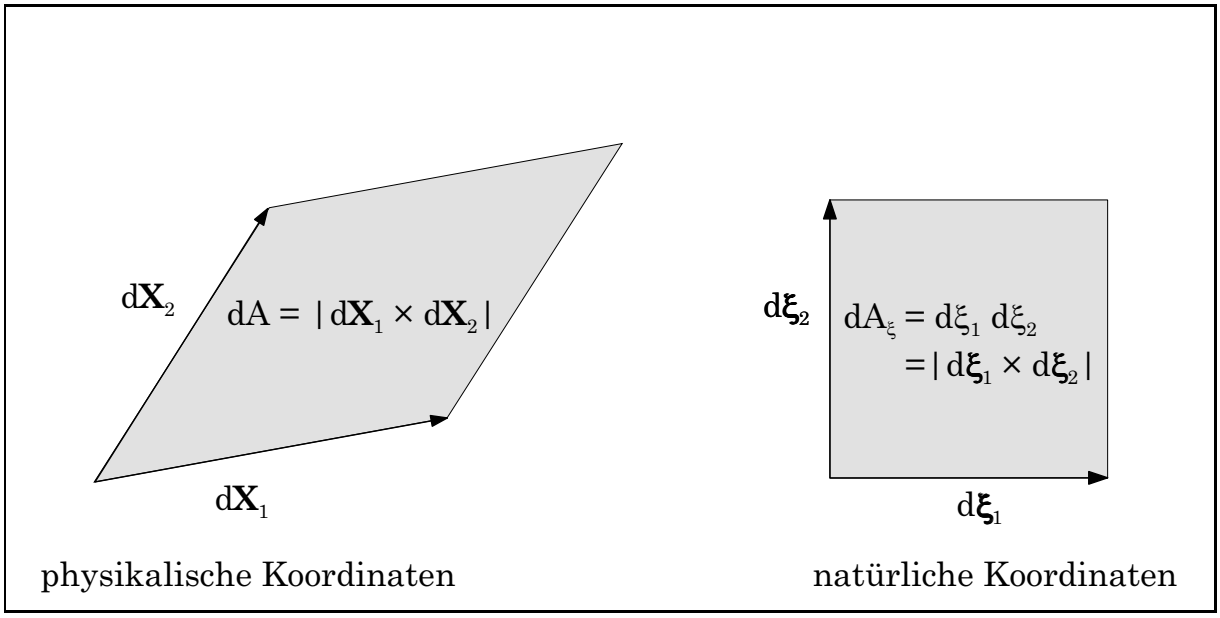


Figure 3.10: JACOBI transformation of a surface element dA in natural coordinates

Next, vectors $d\boldsymbol{X}_\beta$ are tied with vectors $d\boldsymbol{\xi}_\alpha$ with the help of equation (3.45).

$$d\boldsymbol{X}_\beta = \frac{\partial X_\beta}{\partial \xi_\alpha}\, d\boldsymbol{\xi}_\alpha = \frac{\partial X_\beta}{\partial \xi_1}\, d\boldsymbol{\xi}_1 + \frac{\partial X_\beta}{\partial \xi_1}\, d\boldsymbol{\xi}_1 \tag{3.56}$$

The square of surface element dA comes as result of a vector scalar product of $d\boldsymbol{X}_1 \times d\boldsymbol{X}_2$ (see equation (3.55)), where equation (3.56) can be introduced instead of $d\boldsymbol{X}_\beta$.

$$\begin{aligned}
dA^2 &= (d\boldsymbol{X}_1 \times d\boldsymbol{X}_2)\cdot(d\boldsymbol{X}_1 \times d\boldsymbol{X}_2) = (d\boldsymbol{X}_1 \times d\boldsymbol{X}_2)^2 \\
&= \left[\left(\frac{\partial X_1}{\partial \xi_1}\, d\boldsymbol{\xi}_1 + \frac{\partial X_1}{\partial \xi_2}\, d\boldsymbol{\xi}_2\right) \times \left(\frac{\partial X_2}{\partial \xi_1}\, d\boldsymbol{\xi}_1 + \frac{\partial X_2}{\partial \xi_2}\, d\boldsymbol{\xi}_2\right)\right]^2 \\
&= \left[\frac{\partial X_1}{\partial \xi_1}\,\frac{\partial X_2}{\partial \xi_1}\, \underbrace{d\boldsymbol{\xi}_1 \times d\boldsymbol{\xi}_1}_{\mathbf{0}} + \frac{\partial X_1}{\partial \xi_1}\,\frac{\partial X_2}{\partial \xi_2}\, d\boldsymbol{\xi}_1 \times d\boldsymbol{\xi}_2\right. \\
&\quad \left. + \frac{\partial X_1}{\partial \xi_2}\,\frac{\partial X_2}{\partial \xi_1}\, \underbrace{d\boldsymbol{\xi}_2 \times d\boldsymbol{\xi}_1}_{-d\boldsymbol{\xi}_1 \times d\boldsymbol{\xi}_2} + \frac{\partial X_1}{\partial \xi_2}\,\frac{\partial X_2}{\partial \xi_2}\, \underbrace{d\boldsymbol{\xi}_2 \times d\boldsymbol{\xi}_2}_{\mathbf{0}}\right]^2 \\
&= \underbrace{\left[\frac{\partial X_1}{\partial \xi_1}\,\frac{\partial X_2}{\partial \xi_2} - \frac{\partial X_1}{\partial \xi_2}\,\frac{\partial X_2}{\partial \xi_1}\right]^2}_{|\mathbf{J}|^2}\ \underbrace{\left[d\boldsymbol{\xi}_1 \times d\boldsymbol{\xi}_2\right]^2}_{dA_\xi^2} = |\mathbf{J}|^2\, dA_\xi^2 = |\mathbf{J}|^2\, d\xi_1^2\, d\xi_2^2
\end{aligned} \tag{3.57}$$

The JACOBI determinant here is identified by comparison to the JACOBI matrix determinant according to equation (3.49), and the surface element is obtained in natural coordinates by means of comparison to equation (3.55). The connection between the very surface elements turns out to be trivial by forming the square root of equation (3.57).

$$\boxed{dA = |\mathbf{J}(\boldsymbol{\xi})|\, d\xi_1\, d\xi_2} \tag{3.58}$$

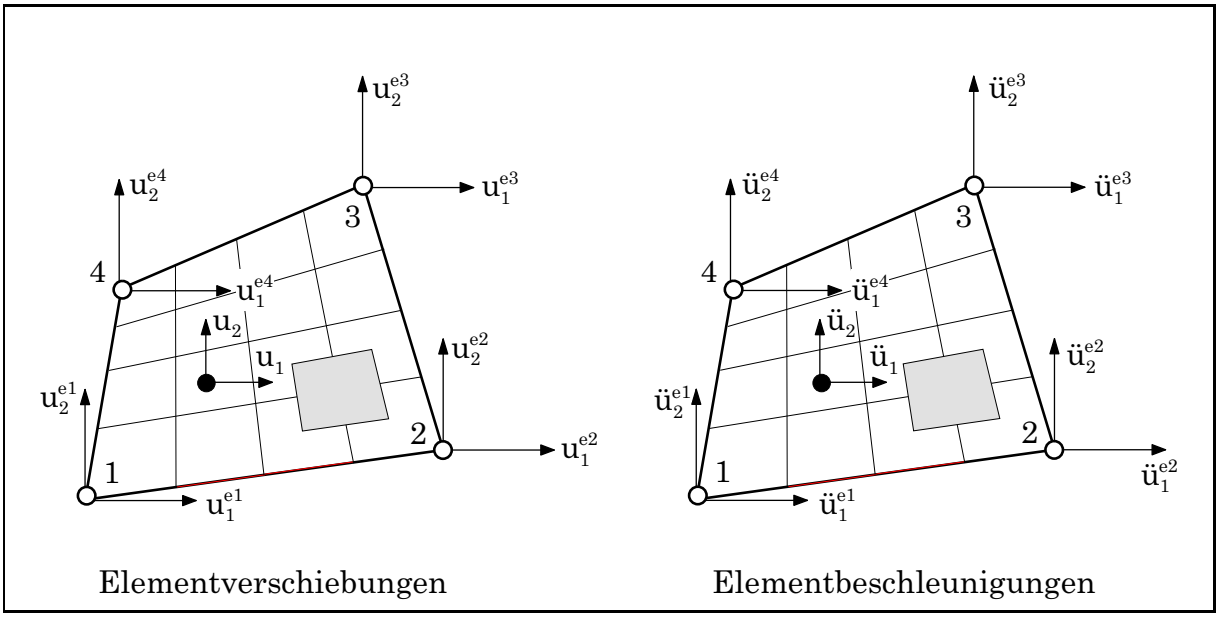


Figure 3.11: Degrees of freedom of a four-noded plane LAGRANGE element

3.3.4 Approximation of element quantities

Within the scope of the isoparametric element concept we can approximate the continuous displacements, variation and second time derivative of displacements for $NN = 4$, analogous with the approximation of the position vector in equation (3.41) (see figures 3.8 and 3.11).

$$
\begin{aligned}
\boldsymbol{u}(\boldsymbol{\xi}) &\approx \tilde{\boldsymbol{u}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \boldsymbol{u}^e & \boldsymbol{u}^e &= \left[\ u_1^{e1}\ \ u_2^{e1}\ \cdots\ u_1^{eNN}\ \ u_2^{eNN}\ \right]^T \\
\delta\boldsymbol{u}(\boldsymbol{\xi}) &\approx \delta\tilde{\boldsymbol{u}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \delta\boldsymbol{u}^e & \delta\boldsymbol{u}^e &= \left[\ \delta u_1^{e1}\ \delta u_2^{e1}\ \cdots\ \delta u_1^{eNN}\ \delta u_2^{eNN}\ \right]^T \\
\ddot{\boldsymbol{u}}(\boldsymbol{\xi}) &\approx \tilde{\ddot{\boldsymbol{u}}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \ddot{\boldsymbol{u}}^e & \ddot{\boldsymbol{u}}^e &= \left[\ \ddot{u}_1^{e1}\ \ddot{u}_2^{e1}\ \cdots\ \ddot{u}_1^{eNN}\ \ddot{u}_2^{eNN}\ \right]^T
\end{aligned}
\tag{3.59}
$$

3.3.5 Strain vector approximation

To formulate the internal virtual work, the approximation of which is given with the help of ansatz polynomials applied to displacements, and thereupon following integration of stiffness terms, the strain vector components ought to be described in natural coordinates. This is accomplished by applying the derivation rule (3.51) to the displacement strain relation in equation

(3.12).

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{\partial \xi_1}{\partial X_1}\frac{\partial}{\partial \xi_1} + \frac{\partial \xi_2}{\partial X_1}\frac{\partial}{\partial \xi_2} & 0 \\ 0 & \frac{\partial \xi_1}{\partial X_2}\frac{\partial}{\partial \xi_1} + \frac{\partial \xi_2}{\partial X_2}\frac{\partial}{\partial \xi_2} \\ \frac{\partial \xi_1}{\partial X_2}\frac{\partial}{\partial \xi_1} + \frac{\partial \xi_2}{\partial X_2}\frac{\partial}{\partial \xi_2} & \frac{\partial \xi_1}{\partial X_1}\frac{\partial}{\partial \xi_1} + \frac{\partial \xi_2}{\partial X_1}\frac{\partial}{\partial \xi_2} \end{bmatrix}}_{\mathbf{D}_{\varepsilon\xi}} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \mathbf{D}_{\varepsilon\xi}\, \boldsymbol{u} \tag{3.60}$$

With the approximation of continuous displacements according to equation (3.59), we get the approximation of the strain vector from equation (3.60)

$$\boldsymbol{\varepsilon}(\boldsymbol{\xi}) \approx \tilde{\boldsymbol{\varepsilon}}(\boldsymbol{\xi}) = \mathbf{D}_{\varepsilon\xi}(\boldsymbol{\xi})\, \tilde{\boldsymbol{u}}(\boldsymbol{\xi}) = \mathbf{D}_{\varepsilon\xi}(\boldsymbol{\xi})\, \mathbf{N}(\boldsymbol{\xi})\, \boldsymbol{u}^e = \mathbf{B}(\boldsymbol{\xi})\, \boldsymbol{u}^e \tag{3.61}$$

by means of linear mapping of the element displacement vector and the differential operator definition $\mathbf{B}(\boldsymbol{\xi})$.

$$\boxed{\tilde{\boldsymbol{\varepsilon}}(\boldsymbol{\xi}) = \mathbf{B}(\boldsymbol{\xi})\, \boldsymbol{u}^e \qquad\qquad \mathbf{B}(\boldsymbol{\xi}) = \mathbf{D}_{\varepsilon\xi}(\boldsymbol{\xi})\, \mathbf{N}(\boldsymbol{\xi}) \qquad (3.62)}$$

The differential operator $\mathbf{B}(\boldsymbol{\xi})$ is here given by applying $\mathbf{D}_{\varepsilon\xi}(\boldsymbol{\xi})$ to the matrix of ansatz functions $\mathbf{N}(\boldsymbol{\xi})$. If the operator $\mathbf{D}_{\varepsilon\xi}(\boldsymbol{\xi})$ is not applied to the entire matrix of ansatz functions but only to the submatrices $\mathbf{N}^i(\boldsymbol{\xi})$, we obtain the B-operator $\mathbf{B}_i(\boldsymbol{\xi})$ assigned to the element node i instead of B-operator $\mathbf{B}(\boldsymbol{\xi})$,

$$\mathbf{B}_i(\boldsymbol{\xi}) = \mathbf{D}_{\varepsilon\xi}(\boldsymbol{\xi})\, \mathbf{N}^i(\boldsymbol{\xi}) \qquad\qquad \mathbf{B} = \begin{bmatrix} \mathbf{B}_1 & \cdots & \mathbf{B}_{NN} \end{bmatrix}, \; NN = 4 \tag{3.63}$$

with the differential operators $\mathbf{B}_i(\boldsymbol{\xi})$ being assembled to the B-operator, in analogy with the matrix of the ansatz functions.

$$\mathbf{B}_i(\boldsymbol{\xi}) = \begin{bmatrix} \frac{\partial \xi_1}{\partial X_1}\frac{\partial}{\partial \xi_1} + \frac{\partial \xi_2}{\partial X_1}\frac{\partial}{\partial \xi_2} & 0 \\ 0 & \frac{\partial \xi_1}{\partial X_2}\frac{\partial}{\partial \xi_1} + \frac{\partial \xi_2}{\partial X_2}\frac{\partial}{\partial \xi_2} \\ \frac{\partial \xi_1}{\partial X_2}\frac{\partial}{\partial \xi_1} + \frac{\partial \xi_2}{\partial X_2}\frac{\partial}{\partial \xi_2} & \frac{\partial \xi_1}{\partial X_1}\frac{\partial}{\partial \xi_1} + \frac{\partial \xi_2}{\partial X_1}\frac{\partial}{\partial \xi_2} \end{bmatrix} \begin{bmatrix} N^i(\boldsymbol{\xi}) & 0 \\ 0 & N^i(\boldsymbol{\xi}) \end{bmatrix} \tag{3.64}$$

Finally, by applying the derivation rules we arrive at the portion of B-operator for element node i.

$$\boxed{\mathbf{B}_i(\boldsymbol{\xi}) = \begin{bmatrix} \frac{\partial \xi_1}{\partial X_1} N^i_{;1}(\boldsymbol{\xi}) + \frac{\partial \xi_2}{\partial X_1} N^i_{;2}(\boldsymbol{\xi}) & 0 \\ 0 & \frac{\partial \xi_1}{\partial X_2} N^i_{;1}(\boldsymbol{\xi}) + \frac{\partial \xi_2}{\partial X_2} N^i_{;2}(\boldsymbol{\xi}) \\ \frac{\partial \xi_1}{\partial X_2} N^i_{;1}(\boldsymbol{\xi}) + \frac{\partial \xi_2}{\partial X_2} N^i_{;2}(\boldsymbol{\xi}) & \frac{\partial \xi_1}{\partial X_1} N^i_{;1}(\boldsymbol{\xi}) + \frac{\partial \xi_2}{\partial X_1} N^i_{;2}(\boldsymbol{\xi}) \end{bmatrix} \qquad (3.65)}$$

If the components of the inverse JACOBI matrix $\xi_{\alpha,\beta}$ according to equation (3.53) are replaced

by components of the JACOBI matrix $X_{\beta,\alpha}$, it is possible to write down the B-operators $\mathbf{B}_i(\boldsymbol{\xi})$ in the following way:

$$\mathbf{B}_i(\boldsymbol{\xi}) = \frac{1}{|\mathbf{J}(\boldsymbol{\xi})|} \begin{bmatrix} \frac{\partial X_2}{\partial \xi_2} N^i_{;1}(\boldsymbol{\xi}) - \frac{\partial X_2}{\partial \xi_1} N^i_{;2}(\boldsymbol{\xi}) & 0 \\ 0 & -\frac{\partial X_1}{\partial \xi_2} N^i_{;1}(\boldsymbol{\xi}) + \frac{\partial X_1}{\partial \xi_1} N^i_{;2}(\boldsymbol{\xi}) \\ -\frac{\partial X_1}{\partial \xi_2} N^i_{;1}(\boldsymbol{\xi}) + \frac{\partial X_1}{\partial \xi_1} N^i_{;2}(\boldsymbol{\xi}) & \frac{\partial X_2}{\partial \xi_2} N^i_{;1}(\boldsymbol{\xi}) - \frac{\partial X_2}{\partial \xi_1} N^i_{;2}(\boldsymbol{\xi}) \end{bmatrix} \tag{3.66}$$

Derived in such a way, the B-operators contain matrix terms of the form $X_{\beta,\alpha}$ which again can be described as functions of ansatz function derivatives $N^j_{;\alpha}(\boldsymbol{\xi})$ and of element node coordinates X^{ej}_β according to equation (3.50).

$$\frac{\partial X_\beta}{\partial \xi_\alpha} = \frac{\partial}{\partial \xi_\alpha} \sum_{j=1}^{4} X^{ej}_\beta \; N^j(\boldsymbol{\xi}) = \sum_{j=1}^{4} X^{ej}_\beta \; N^j_{;\alpha}(\boldsymbol{\xi}) \tag{3.67}$$

3.3.6 Appproximation of internal virtual work

The internal virtual work is developed with equation (3.19) by substituting the stress vector by material law (3.16), by modifying the surface element dA and by adjusting the integration boundaries to natural coordinates $\xi_\alpha \in [-1, 1]$.

$$\delta W^e_{\text{int}} = \int_{-1}^{1}\int_{-1}^{1} \delta\boldsymbol{\varepsilon}(\boldsymbol{\xi}) \cdot \mathbf{C}\; \boldsymbol{\varepsilon}(\boldsymbol{\xi})\; |\mathbf{J}(\boldsymbol{\xi})| h \; d\xi_1 d\xi_2 \tag{3.68}$$

Approximation of internal virtual work comes from substitution of the exact strain vector by the approximated strain vector in equation (3.62)

$$\delta \tilde{W}^e_{\text{int}} = \int_{-1}^{1}\int_{-1}^{1} \delta\boldsymbol{u}^e \cdot \mathbf{B}^T(\boldsymbol{\xi})\; \mathbf{C}\; \mathbf{B}(\boldsymbol{\xi})\; \boldsymbol{u}^e\; |\mathbf{J}(\boldsymbol{\xi})| h \; d\xi_1 d\xi_2 \tag{3.69}$$

Due to the independence of the natural coordinates $\boldsymbol{\xi}$ of both element displacement vectors $\boldsymbol{u}^e$ and its variation $\delta\boldsymbol{u}^e$, both of these vectors can be exerted from the integral.

$$\delta \tilde{W}^e_{\text{int}} = \delta\boldsymbol{u}^e \cdot \int_{-1}^{1}\int_{-1}^{1} \mathbf{B}^T(\boldsymbol{\xi})\; \mathbf{C}\; \mathbf{B}(\boldsymbol{\xi})\; |\mathbf{J}(\boldsymbol{\xi})|\; h \; d\xi_1 d\xi_2\; \boldsymbol{u}^e = \delta\boldsymbol{u}^e \cdot \mathbf{k}^e\; \boldsymbol{u}^e \tag{3.70}$$

Here, we have defined the element stiffness matrix of a bilinear plane quadrangular element of an arbitrary shape.

$$\mathbf{k}^e = \int_{-1}^{1}\int_{-1}^{1} \mathbf{B}^T(\boldsymbol{\xi})\; \mathbf{C}\; \mathbf{B}(\boldsymbol{\xi})\; |\mathbf{J}(\boldsymbol{\xi})|\; h \; d\xi_1 d\xi_2 \tag{3.71}$$

The element thickness can be independent of the location in case of a plane stress state ($h = h(\boldsymbol{\xi})$) ; in case of a plane strain state, the element thickness is constant. It is customary, as was shown in chapter 3.7, to add the integrand which is evaluated and weighed at the GAUSS points and formed with the help of the B-operator, the JACOBI determinant and the element thickness, to the element stiffness matrix. The so called full integration or the 2×2 integration of a bilinear element accordingly requires four calculations of an integrand and its weighed addition.

$$\mathbf{k}^e \approx \sum_{i=1}^{2} \sum_{j=1}^{2} \alpha^i \alpha^j \; \mathbf{B}^T(\xi_1^i, \xi_2^j) \; \mathbf{C} \; \mathbf{B}(\xi_1^i, \xi_2^j) \; |\mathbf{J}(\xi_1^i, \xi_2^j)| \; h(\xi_1^i, \xi_2^j) \tag{3.72}$$

GAUSS points ξ_α^i and weighting factors α^i of numeric integration can be found in table 2.2. Apart from the full integration, a *subintegration* of the bilinear element is also introduced with only one GAUSS point in the midpoint of the element. Merely in special cases of element geometry, it is reasonable to determine the element stiffness by analytical derivation of the B-operator and analytical integration. We deal with such a special case in chapter 3.3.9, the special case being a rectangular plate element.

3.3.7 Approximation of dynamic virtual work

Besides discretization of internal virtual work, transient mechanical problems also demand discretization of dynamic virtual work. If we, in the principle of virtual displacements (equation (3.19)), replace the surface element dA and the integration boundaries in accordance with the previous chapter

$$\delta W_{\text{dyn}}^e = \int_{-1}^{1}\int_{-1}^{1} \delta \boldsymbol{u}(\boldsymbol{\xi}) \cdot \ddot{\boldsymbol{u}}(\boldsymbol{\xi}) \; |\mathbf{J}(\boldsymbol{\xi})| \; \rho \; h \; d\xi_1 d\xi_2 \tag{3.73}$$

and if we approximate the variation of displacements as well as continuous accelerations with the assistance of ansatz functions according to equation (3.59), we get the approximation of virtual work of inertial forces.

$$\delta \tilde{W}_{\text{dyn}}^e = \delta \boldsymbol{u}^e \cdot \int_{-1}^{1}\int_{-1}^{1} \mathbf{N}^T(\boldsymbol{\xi}) \; \mathbf{N}(\boldsymbol{\xi}) \; |\mathbf{J}(\boldsymbol{\xi})| \; \rho \; h \; d\xi_1 d\xi_2 \; \ddot{\boldsymbol{u}}^e = \delta \boldsymbol{u}^e \cdot \mathbf{m}^e \; \ddot{\boldsymbol{u}}^e \tag{3.74}$$

For purposes of defining the element mass matrix, we repeatedly took advantage of the fact that the element vectors $\delta \boldsymbol{u}^e$ and $\ddot{\boldsymbol{u}}^e$ are independent of natural coordinates.

$$\boxed{\mathbf{m}^e = \int_{-1}^{1}\int_{-1}^{1} \mathbf{N}^T(\boldsymbol{\xi}) \; \mathbf{N}(\boldsymbol{\xi}) \; |\mathbf{J}(\boldsymbol{\xi})| \; \rho \; h \; d\xi_1 d\xi_2 \qquad (3.75)}$$

3.3.8 Approximation of virtual work of external loads

The loads acting on a plane element can be divided into loads acting in the field and those acting at the boundaries of the field. Typical loads in the field are gravitational loads whereas actual structural loads are dominated by boundary loads such as pressure.

3.3.8.1 Volume loads

With the help of displacement variation approximation as in equation (3.59), a consistent element load vector of volume loads $\boldsymbol{b}(\boldsymbol{\xi})$ can be obtained based on external virtual work.

$$\delta\tilde{W}_{\text{ext}}^{\Omega e} = \delta\boldsymbol{u}^e \cdot \int_{-1}^{1}\int_{-1}^{1} \mathbf{N}^T(\boldsymbol{\xi})\, \boldsymbol{b}(\boldsymbol{\xi})\, |\mathbf{J}(\boldsymbol{\xi})|\, \rho\, h\, d\xi_1 d\xi_2 = \delta\boldsymbol{u}^e \cdot \boldsymbol{r}_p^e \tag{3.76}$$

Element vector of volume forces:

$$\boxed{\boldsymbol{r}_p^e = \int_{-1}^{1}\int_{-1}^{1} \mathbf{N}^T(\boldsymbol{\xi})\, \boldsymbol{b}(\boldsymbol{\xi})\, |\mathbf{J}(\boldsymbol{\xi})|\, \rho\, h\, d\xi_1 d\xi_2 \qquad (3.77)}$$

Usually, $\boldsymbol{b}$ is not given as dependent of natural coordinates $\boldsymbol{\xi}$ but as function of physical coordinates $\boldsymbol{X}$. In such a case, the position of volume load has to be transformed to natural coordinates ($\boldsymbol{b}(\boldsymbol{X}) \rightarrow \boldsymbol{b}(\boldsymbol{\xi})$).

3.3.8.2 Boundary loads

The element load vector of element boundary loads $\boldsymbol{t}^\star(\boldsymbol{\xi})h$ is derived by observation of external virtual work (see equation (3.19)). Here, the boundary Γ_σ can here be divided into four boundaries Γ_i of the element.

$$\delta W_{\text{ext}}^{\Gamma e} = \int_{\Gamma_\sigma} \delta\boldsymbol{u} \cdot \boldsymbol{t}^\star\, h\, d\Gamma_\sigma = \sum_{i=1}^{4} \int_{\Gamma_i} \delta\boldsymbol{u} \cdot \boldsymbol{t}^\star\, h\, d\Gamma_i = \sum_{i=1}^{4} \delta W_{\text{ext}}^{\Gamma_i e} \tag{3.78}$$

As an example, the proceeding for calculating the consistent equivalent loads is demonstrated for element boundary three that connects nodes three and four. As shown in figure 3.12, the element edge is characterised by the natural coordinate $\xi_2 = 1$.

$$\delta W_{\text{ext}}^{\Gamma_3 e} = \int_{\Gamma_3} \delta\boldsymbol{u}(\xi_1, 1) \cdot \boldsymbol{t}^\star(\xi_1, 1)\, h\, d\Gamma_3 \tag{3.79}$$

To integrate this integrand over element edge three, the line element $d\Gamma_3$ must be transformed into the natural coordinate $d\xi_1$. The differential line element here is analysed in physical space. Projection of line element $d\Gamma_3$ onto direction $\boldsymbol{e}_1$ yields the differential physical line element dX_1, projection onto direction $\boldsymbol{e}_2$ yields dX_2. Vice versa, the square of line element can be calculated with the PYTHAGORAS theorem.

$$d\Gamma_3^2 = dX_1^2 + dX_2^2 \tag{3.80}$$

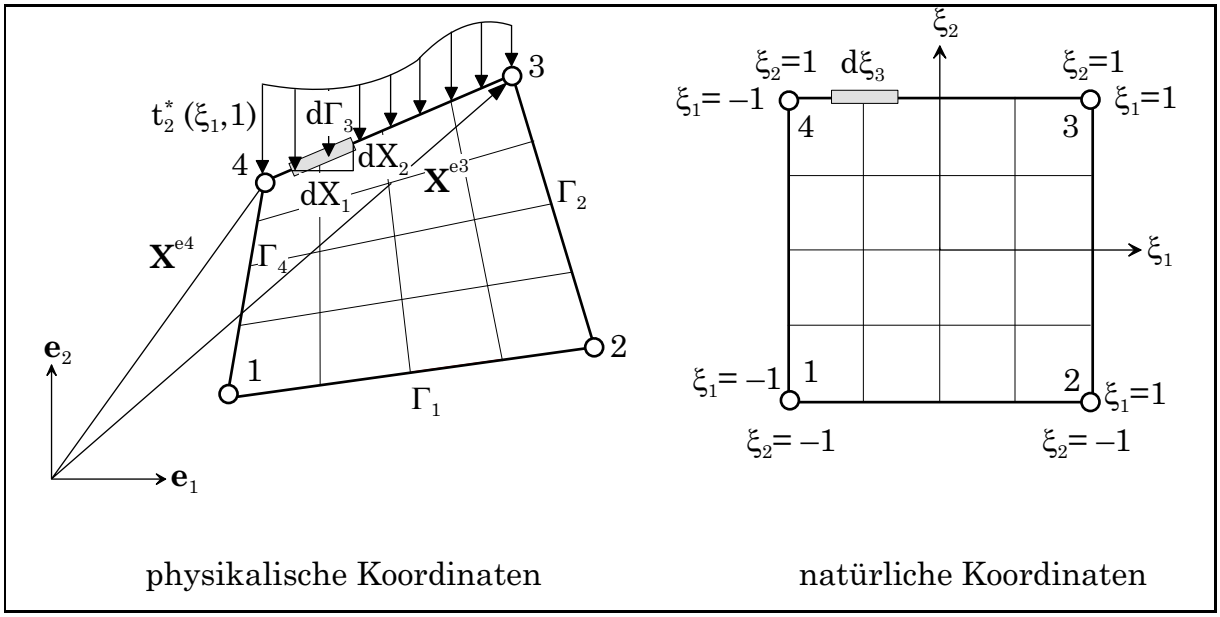


Figure 3.12: External loads of a four-noded LAGRANGE element

The total differentials dX_β for $\beta = 1, 2$ are found to be functions of total differentials in natural coordinates and in JACOBI matrix components. It should be noted here that the differential $d\xi_2$ vanishes ($d\xi_2 = 0$) at element edge three ($\xi_2 = 1$).

$$dX_\beta = \frac{\partial X_\beta}{\partial \xi_1}\, d\xi_1 + \frac{\partial X_\beta}{\partial \xi_2}\, d\xi_2 = \frac{\partial X_\beta}{\partial \xi_1}\, d\xi_1 \tag{3.81}$$

Inserting equation (3.81) into equation (3.80) yields the transformation relation of line element $d\Gamma_3$.

$$d\Gamma_3^2 = \left[\left(\frac{\partial X_1}{\partial \xi_1} \right)^2 + \left(\frac{\partial X_2}{\partial \xi_1} \right)^2 \right] d\xi_1^2 \tag{3.82}$$

If we additionally introduce the JACOBI determinant $|\mathbf{J}_3(\xi_1, 1)|$, which generally is a function of coordinate ξ_1, we can write down the transformation relation (3.82) in a familiar compact form.

$$\boxed{d\Gamma_3 = |\mathbf{J}_3(\xi_1, 1)|\, d\xi_1 \qquad |\mathbf{J}_3(\xi_1, 1)| = \left[\left(\frac{\partial X_1}{\partial \xi_1} \right)^2 + \left(\frac{\partial X_2}{\partial \xi_1} \right)^2 \right]^{\frac{1}{2}}} \tag{3.83}$$

Formally, the same transformation relation is found for line element $d\Gamma_1$, where the partial derivatives in equation (3.81) must be evaluated for natural coordinate $\xi_2 = -1$ instead of $\xi_2 = 1$. Analogous observations lead to transformation relation of line elements $d\Gamma_2$ and $d\Gamma_4$. Transformation of the line element $d\Gamma_4$ is given which is valid for both line elements.

$$d\Gamma_4 = |\mathbf{J}_4(-1, \xi_2)|\, d\xi_2 \qquad |\mathbf{J}_4(-1, \xi_2)| = \left[\left(\frac{\partial X_1}{\partial \xi_2} \right)^2 + \left(\frac{\partial X_2}{\partial \xi_2} \right)^2 \right]^{\frac{1}{2}} \tag{3.84}$$

Back to element edge three and the boundary load of bilinear LAGRANGE elements. Next, the position vector on the element edge according to equation (3.39) is described with the help of ansatz functions $N^i(\xi_1, 1),\ i = 1, 2, 3, 4$, as a basis of transformation (3.83). A reduction of ansatz functions is not necessary here since the ansatz functions corresponding to element nodes one and two vanish at the element edge three ($N^1(\xi_1, 1) = N^2(\xi_1, 1) = 0$, see equation (3.35) and figure 3.8.

$$\boldsymbol{X}(\xi_1, 1) = \sum_{i=1}^{4} \boldsymbol{X}^{ei}\ N^i(\xi_1, 1) = \sum_{i=3}^{4} \boldsymbol{X}^{ei}\ N^i(\xi_1, 1) \tag{3.85}$$

Derivative of the position vector with respect to natural coordinate ξ_1 is given by

$$\frac{\partial \boldsymbol{X}(\xi_1, 1)}{\partial \xi_1} = \sum_{i=1}^{4} \boldsymbol{X}^{ei}\ N^i_{;1}(\xi_1, 1) = \mathbf{N}_{;1}(\xi_1, 1)\ \boldsymbol{X}^e \tag{3.86}$$

and the derivative with respect to ξ_2 is zero. Horizontal and vertical components $dX_\beta,\ \beta = 1, 2$ of line element $d\Gamma_3$ are found by calculation of total differentials according to equation (3.36), where derivatives of ansatz functions according to equation (3.81) are directly used for $\xi_2 = 1$.

$$dX_\beta = \frac{\partial X_\beta(\xi_1, 1)}{\partial \xi_1}\ d\xi_1 = \left[X^{e3}_\beta\ N^3_{;1}(\xi_1, 1) + X^{e4}_\beta\ N^4_{;1}(\xi_1, 1)\right]\ d\xi_1 = \left[X^{e3}_\beta\ \frac{1}{2} - X^{e4}_\beta\ \frac{1}{2}\right]\ d\xi_1 \tag{3.87}$$

With equation (3.83), the sought JACOBI determinant $|\mathbf{J}_3(\xi_1, 1)|$ of a bilinear LAGRANGE element finally follows, related to element edge three.

$$\begin{aligned} |\mathbf{J}_3(\xi_1, 1)| &= \left[\left(X^{e3}_1 \frac{1}{2} - X^{e4}_1 \frac{1}{2}\right)^2 + \left(X^{e3}_2 \frac{1}{2} - X^{e4}_2 \frac{1}{2}\right)^2\right]^{\frac{1}{2}} \\ &= \frac{1}{2}\ \left[\left(X^{e3}_1 - X^{e4}_1\right)^2 + \left(X^{e3}_2 - X^{e4}_2\right)^2\right]^{\frac{1}{2}} \end{aligned} \tag{3.88}$$

With this we can substitute the line element $d\Gamma_3$ in equation (3.79) as well as the integration boundaries by a differential of the natural coordinate $d\xi_1$ that is by interval boundaries of the natural coordinate $\xi_1 \in [-1, 1]$.

$$\delta W^{\Gamma_3}_{\text{ext}} = \int_{-1}^{1} \delta \boldsymbol{u}(\xi_1, 1) \cdot \boldsymbol{t}^\star(\xi_1, 1)\ h\ |\mathbf{J}_3(\xi_1, 1)|\ d\xi_1 \tag{3.89}$$

The approximation of displacement variation according to equation (3.59)

$$\delta \tilde{W}^{\Gamma_3}_{\text{ext}} = \delta \boldsymbol{u}^e \cdot \int_{-1}^{1} \mathbf{N}^T(\xi_1, 1)\ \boldsymbol{t}^\star(\xi_1, 1)\ h\ |\mathbf{J}_3(\xi_1, 1)|\ d\xi_1 = \delta \boldsymbol{u}^e \cdot \boldsymbol{r}^e_{n3} \tag{3.90}$$

finally yields the consistent equivalent load of the boundary load $\boldsymbol{r}^e_{n3}$.

$$\boxed{\boldsymbol{r}^e_{n3} = \int_{-1}^{1} \mathbf{N}^T(\xi_1, 1)\ \boldsymbol{t}^\star(\xi_1, 1)\ h\ |\mathbf{J}_3(\xi_1, 1)|\ d\xi_1 \qquad (3.91)}$$

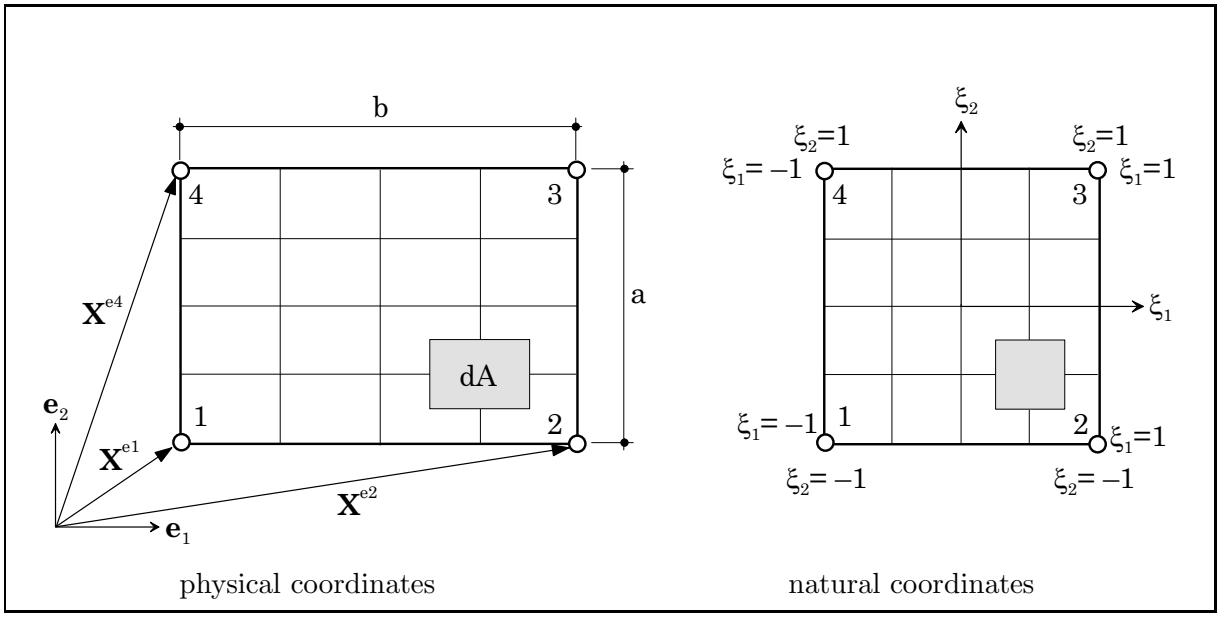


Figure 3.13: Rectangular bilinear LAGRANGE element

The summation of all correspondingly calculated equivalent loads $\boldsymbol{r}_{ni}$ for $i = 1, 2, 3, 4$ yields the consistent equivalent loads of an element.

$$\delta\tilde{W}^{\Gamma}_{\text{ext}} = \sum_{i=1}^{4} \delta\tilde{W}^{\Gamma_i}_{\text{ext}} = \sum_{i=1}^{4} \delta\boldsymbol{u}^e \cdot \boldsymbol{r}^e_{ni} = \delta\boldsymbol{u}^e \cdot \sum_{i=1}^{4} \boldsymbol{r}^e_{ni} = \delta\boldsymbol{u}^e \cdot \boldsymbol{r}^e_n \tag{3.92}$$

The integration of consistent equivalent loads according to equation (3.91) is performed mostly numerically, too. As opposed to integration of $\mathbf{k}^e$, $\mathbf{m}^e$ or $\boldsymbol{r}^e_p$, one-dimensional numerical integration is necessary here. This integration was already thoroughly clarified in chapter 2.2.6 within the scope of finite truss element development.

It may be noted here that virtual works of boundary loads of adjacent elements cancel each other out. For this reason, it is sufficient to calculate the virtual works of boundary loads at free boundaries that are not only element boundaries but also represent system boundaries.

3.3.9 Rectangular Bilinear Lagrange Element

In this chapter, the general LAGRANGE quadrangular element which has been described in the preceeding chapters will be specified into the rectangular quadrangular element described in Fig. 3.13. In the context of this specification, the geometrical description can be simplified by the respective JACOBI transformation. Equally, the calculation of the differential operator $\mathbf{B}$, the generation of the element matrices $\mathbf{k}^e$, $\mathbf{m}^e$ and the calculation of the consistent equivalent loads $\boldsymbol{r}^e_p$, $\boldsymbol{r}^e_n$ can be simplified.

3.3.9.1 Geometry

The geometry of the element is defined through equation (3.41) in the context of the isoparametrical concept of elements. In the case of the rectangular element the coordinates of the element joints are given as follows (see Fig. 3.13).

$$
\begin{aligned}
\boldsymbol{X}^{e1} &= [\ X_1^{e1} \quad X_2^{e1}\ ]^T & \boldsymbol{X}^{e3} &= [\ X_1^{e1}+b \quad X_2^{e1}+a\ ]^T \\
\boldsymbol{X}^{e2} &= [\ X_1^{e1}+b \quad X_2^{e1}\ ]^T & \boldsymbol{X}^{e4} &= [\ X_1^{e1} \quad X_2^{e1}+a\ ]^T
\end{aligned}
\tag{3.93}
$$

3.3.9.2 Jacobi Transformation

The components of the JACOBI matrix are to be calculated by equation (3.50).

$$
\frac{\partial X_\beta}{\partial \xi_\alpha} = \frac{\partial}{\partial \xi_\alpha} \sum_{i=1}^{4} N^i(\boldsymbol{\xi})\ X_\beta^{ei} = \sum_{i=1}^{4} N^i_{;\alpha}(\boldsymbol{\xi})\ X_\beta^{ei} \tag{3.94}
$$

The derivatives of the shape functions $N^i(\boldsymbol{\xi})$ to the natural coordinates ξ_α shown in equation (3.36) yield the components of the JACOBI matrix.

$$
\begin{aligned}
4\frac{\partial X_1}{\partial \xi_1} &= -X_1^{e1}\ (1-\xi_2) + (X_1^{e1}+b)\ (1-\xi_2) \\
&\quad +(X_1^{e1}+b)\ (1+\xi_2) - X_1^{e1}\ (1+\xi_2) = 2b \\
4\frac{\partial X_2}{\partial \xi_1} &= -X_2^{e1}\ (1-\xi_2) + X_2^{e1}\ (1-\xi_2) \\
&\quad +(X_2^{e1}+a)\ (1+\xi_2) - (X_2^{e1}+a)\ (1+\xi_2) = 0 \\
4\frac{\partial X_1}{\partial \xi_2} &= -X_1^{e1}\ (1-\xi_1) - (X_1^{e1}+b)\ (1+\xi_1) \\
&\quad +(X_1^{e1}+b)\ (1+\xi_1) + X_1^{e1}\ (1-\xi_1) = 0 \\
4\frac{\partial X_2}{\partial \xi_2} &= -X_2^{e1}\ (1-\xi_1) - X_2^{e1}\ (1+\xi_1) \\
&\quad +(X_2^{e1}+a)\ (1+\xi_1) + (X_2^{e1}+a)\ (1-\xi_1) = 2a
\end{aligned}
\tag{3.95}
$$

Through equations (3.95) the JACOBI matrix,

$$
\mathbf{J} = \begin{bmatrix} \frac{\partial X_1}{\partial \xi_1} & \frac{\partial X_2}{\partial \xi_1} \\ \frac{\partial X_1}{\partial \xi_2} & \frac{\partial X_2}{\partial \xi_2} \end{bmatrix} = \begin{bmatrix} \frac{b}{2} & 0 \\ 0 & \frac{a}{2} \end{bmatrix} \tag{3.96}
$$

the inverse JACOBI matrix

$$
\mathbf{J}^{-1} = \begin{bmatrix} \frac{\partial \xi_1}{\partial X_1} & \frac{\partial \xi_2}{\partial X_1} \\ \frac{\partial \xi_1}{\partial X_2} & \frac{\partial \xi_2}{\partial X_2} \end{bmatrix} = \frac{4}{a\ b} \begin{bmatrix} \frac{a}{2} & 0 \\ 0 & \frac{b}{2} \end{bmatrix} = \begin{bmatrix} \frac{2}{b} & 0 \\ 0 & \frac{2}{a} \end{bmatrix} \tag{3.97}
$$

and the JACOBI determinant can be produced.

$$
|\mathbf{J}| = \frac{1}{4}\ a\ b \tag{3.98}
$$

In contrast to the general quadrangular element, the JACOBI matrix of the rectangular element is not dependent on any specific localization.

3.3.9.3 Differential Operator $\mathbf{B}(\boldsymbol{\xi})$

The components of the differential operator $\mathbf{B}(\boldsymbol{\xi})$ can be calculated through equation (3.65), using equation (3.97). Component $B_{11}(\boldsymbol{\xi})$, e.g., can be generated this way.

$$B_{11}(\boldsymbol{\xi}) = \frac{\partial \xi_\alpha}{\partial X_1}\, N^1_{;\alpha} = \underbrace{\frac{\partial \xi_1}{\partial X_1}}_{2/b}\, N^1_{;1} + \underbrace{\frac{\partial \xi_2}{\partial X_1}}_{0}\, N^1_{;2} = -\frac{2}{b}\,\frac{1}{4}\,(1-\xi_2) = -\frac{1}{2b}\,(1-\xi_2) = B_{11}(\xi_2) \tag{3.99}$$

Further components of the B operator

$$\begin{array}{lllllllllll}
B_{11}(\xi_2) & = & -\dfrac{1}{2b}\,(1-\xi_2) & = & B_{32}(\xi_2) & \qquad B_{22}(\xi_1) & = & -\dfrac{1}{2a}\,(1-\xi_1) & = & B_{31}(\xi_1) \\
B_{13}(\xi_2) & = & \dfrac{1}{2b}\,(1-\xi_2) & = & B_{34}(\xi_2) & \qquad B_{24}(\xi_1) & = & -\dfrac{1}{2a}\,(1+\xi_1) & = & B_{33}(\xi_1) \\
B_{15}(\xi_2) & = & \dfrac{1}{2b}\,(1+\xi_2) & = & B_{36}(\xi_2) & \qquad B_{26}(\xi_1) & = & \dfrac{1}{2a}\,(1+\xi_1) & = & B_{35}(\xi_1) \\
B_{17}(\xi_2) & = & -\dfrac{1}{2b}\,(1+\xi_2) & = & B_{38}(\xi_2) & \qquad B_{28}(\xi_1) & = & \dfrac{1}{2a}\,(1-\xi_1) & = & B_{37}(\xi_1)
\end{array} \tag{3.100}$$

can be found and summarized in an analogue way.

$$\mathbf{B}(\boldsymbol{\xi}) = \frac{1}{2}\left[\begin{array}{cc|cc|cc|cc}
-\dfrac{1-\xi_2}{b} & 0 & \dfrac{1-\xi_2}{b} & 0 & \dfrac{1+\xi_2}{b} & 0 & -\dfrac{1+\xi_2}{b} & 0 \\
0 & -\dfrac{1-\xi_1}{a} & 0 & -\dfrac{1+\xi_1}{a} & 0 & \dfrac{1+\xi_1}{a} & 0 & \dfrac{1-\xi_1}{a} \\
-\dfrac{1-\xi_1}{a} & -\dfrac{1-\xi_2}{b} & -\dfrac{1+\xi_1}{a} & \dfrac{1-\xi_2}{b} & \dfrac{1+\xi_1}{a} & \dfrac{1+\xi_2}{b} & \dfrac{1-\xi_1}{a} & -\dfrac{1+\xi_2}{b}
\end{array}\right] \tag{3.101}$$

For further calculation of the element stiffness matrix it is useful to split up the B operator into partitional operators $\mathbf{B}_i$ corresponding with the element joints i, as shown in equation (3.101).

$$\mathbf{B}(\boldsymbol{\xi}) = \left[\begin{array}{cccc} \mathbf{B}_1(\boldsymbol{\xi}) & \mathbf{B}_2(\boldsymbol{\xi}) & \mathbf{B}_3(\boldsymbol{\xi}) & \mathbf{B}_4(\boldsymbol{\xi}) \end{array}\right] \tag{3.102}$$

3.3.9.4 Element Stiffness Matrix $\mathbf{k}^e$

On the basis of the generalized relation of the element stiffness matrix as shwon in equation (3.71), matrix $\mathbf{k}^e$ of the rectangular plane finite element can be calculated by supposing a constant element thickness h and by substituting the JACOBI determinant according to equation (3.98).

$$\mathbf{k}^e = \frac{1}{4}\, a\, b\, h \int_{-1}^{1}\int_{-1}^{1} \mathbf{B}^T(\boldsymbol{\xi})\, \mathbf{C}\, \mathbf{B}(\boldsymbol{\xi})\, d\xi_1 d\xi_2 \tag{3.103}$$

Analogue with the B operator, the element stiffness can be split up into sectors that correspond with the element joints.

$$\mathbf{k}^e_{ij} = \frac{1}{4}\, a\, b\, h \int_{-1}^{1}\int_{-1}^{1} \mathbf{B}_i^T(\boldsymbol{\xi})\, \mathbf{C}\, \mathbf{B}_j(\boldsymbol{\xi})\, d\xi_1 d\xi_2 \qquad \mathbf{k}^e = \begin{bmatrix} \mathbf{k}^e_{11} & \mathbf{k}^e_{12} & \mathbf{k}^e_{13} & \mathbf{k}^e_{14} \\ \mathbf{k}^e_{21} & \mathbf{k}^e_{22} & \mathbf{k}^e_{23} & \mathbf{k}^e_{24} \\ \mathbf{k}^e_{31} & \mathbf{k}^e_{32} & \mathbf{k}^e_{33} & \mathbf{k}^e_{34} \\ \mathbf{k}^e_{41} & \mathbf{k}^e_{42} & \mathbf{k}^e_{43} & \mathbf{k}^e_{44} \end{bmatrix} \tag{3.104}$$

Since the material matrix $\mathbf{C}$ is symmetrical, both for the plane strain condition ($\mathbf{C}^{ev}$) and for the plane stress condition ($\mathbf{C}^{es}$), the element stiffness matrix must be symmetrical, too.

$$\mathbf{k}^e_{ij} = \mathbf{k}^{e\,T}_{ji} \tag{3.105}$$

In order to multiply the matrices in equation (3.103), the type of plane mechanical problem has to be selected. As an example, the development of the element stiffness matrix of the panel element will be demonstrated which is characterized by the plane stress condition with the constituting equation (3.14). As a first step, the material matrix is multiplied with the B operators $\mathbf{B}_j(\boldsymbol{\xi})$.

$$\begin{aligned} \mathbf{C}^{es}\, \mathbf{B}_1(\boldsymbol{\xi}) &= \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \begin{bmatrix} -\frac{1-\xi_2}{b} & 0 \\ 0 & -\frac{1-\xi_1}{a} \\ -\frac{1-\xi_1}{a} & -\frac{1-\xi_2}{b} \end{bmatrix} \frac{1}{2} \\ &= \frac{E}{2(1-\nu^2)} \begin{bmatrix} -\frac{1-\xi_2}{b} & -\frac{1-\xi_1}{a}\nu \\ -\frac{1-\xi_2}{b}\nu & -\frac{1-\xi_1}{a} \\ -\frac{1-\xi_1}{a}\frac{1-\nu}{2} & -\frac{1-\xi_2}{b}\frac{1-\nu}{2} \end{bmatrix} \end{aligned} \tag{3.106}$$

$$\begin{aligned} \mathbf{C}^{es}\, \mathbf{B}_2(\boldsymbol{\xi}) &= \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \begin{bmatrix} \frac{1-\xi_2}{b} & 0 \\ 0 & -\frac{1+\xi_1}{a} \\ -\frac{1+\xi_1}{a} & \frac{1-\xi_2}{b} \end{bmatrix} \frac{1}{2} \\ &= \frac{E}{2(1-\nu^2)} \begin{bmatrix} \frac{1-\xi_2}{b} & -\frac{1+\xi_1}{a}\nu \\ \frac{1-\xi_2}{b}\nu & -\frac{1+\xi_1}{a} \\ -\frac{1+\xi_1}{a}\frac{1-\nu}{2} & \frac{1-\xi_2}{b}\frac{1-\nu}{2} \end{bmatrix} \end{aligned} \tag{3.106}$$

$$
\begin{aligned}
\mathbf{C}^{es}\,\mathbf{B}_3(\boldsymbol{\xi}) &= \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}\begin{bmatrix} \frac{1+\xi_2}{b} & 0 \\ 0 & \frac{1+\xi_1}{a} \\ \frac{1+\xi_1}{a} & \frac{1+\xi_2}{b} \end{bmatrix}\frac{1}{2} \\
&= \frac{E}{2(1-\nu^2)}\begin{bmatrix} \frac{1+\xi_2}{b} & \frac{1+\xi_1}{a}\nu \\ \frac{1+\xi_2}{b}\nu & \frac{1+\xi_1}{a} \\ \frac{1+\xi_1}{a}\frac{1-\nu}{2} & \frac{1+\xi_2}{b}\frac{1-\nu}{2} \end{bmatrix}
\end{aligned}
\tag{3.106}
$$

$$
\begin{aligned}
\mathbf{C}^{es}\,\mathbf{B}_4(\boldsymbol{\xi}) &= \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}\begin{bmatrix} -\frac{1+\xi_2}{b} & 0 \\ 0 & \frac{1-\xi_1}{a} \\ \frac{1-\xi_1}{a} & -\frac{1+\xi_2}{b} \end{bmatrix}\frac{1}{2} \\
&= \frac{E}{2(1-\nu^2)}\begin{bmatrix} -\frac{1+\xi_2}{b} & \frac{1-\xi_1}{a}\nu \\ -\frac{1+\xi_2}{b}\nu & \frac{1-\xi_1}{a} \\ \frac{1-\xi_1}{a}\frac{1-\nu}{2} & -\frac{1+\xi_2}{b}\frac{1-\nu}{2} \end{bmatrix}
\end{aligned}
\tag{3.106}
$$

Next, the results of equation (3.106) are pre-multiplied with the transponed B operators $\mathbf{B}_i^T(\boldsymbol{\xi})$. Multiplication with $\mathbf{B}_1^T(\boldsymbol{\xi})$ yields:

$$
\begin{aligned}
\mathbf{B}_1^T\mathbf{C}^{es}\mathbf{B}_1 &= c_1\begin{bmatrix} -\frac{1-\xi_2}{b} & 0 & -\frac{1-\xi_1}{a} \\ 0 & -\frac{1-\xi_1}{a} & -\frac{1-\xi_2}{b} \end{bmatrix}\begin{bmatrix} -\frac{1-\xi_2}{b} & -\frac{1-\xi_1}{a}\nu \\ -\frac{1-\xi_2}{b}\nu & -\frac{1-\xi_1}{a} \\ -\frac{1-\xi_1}{a}\frac{1-\nu}{2} & -\frac{1-\xi_2}{b}\frac{1-\nu}{2} \end{bmatrix} \\
&= c_1\left[\begin{array}{c|c} \frac{(1-\xi_2)^2}{b^2} + \frac{(1-\xi_1)^2}{a^2}\frac{1-\nu}{2} & \frac{(1-\xi_1)(1-\xi_2)}{ab}\frac{1+\nu}{2} \\ \hline & \frac{(1-\xi_1)^2}{a^2} + \frac{(1-\xi_2)^2}{b^2}\frac{1-\nu}{2} \end{array}\right]
\end{aligned}
\tag{3.107}
$$

$$
\begin{aligned}
\mathbf{B}_1^T\mathbf{C}^{es}\mathbf{B}_2 &= c_1 \begin{bmatrix} -\dfrac{1-\xi_2}{b} & 0 & -\dfrac{1-\xi_1}{a} \\ 0 & -\dfrac{1-\xi_1}{a} & -\dfrac{1-\xi_2}{b} \end{bmatrix} \begin{bmatrix} \dfrac{1-\xi_2}{b} & -\dfrac{1+\xi_1}{a}\nu \\ \dfrac{1-\xi_2}{b}\nu & -\dfrac{1+\xi_1}{a} \\ -\dfrac{1+\xi_1}{a}\dfrac{1-\nu}{2} & \dfrac{1-\xi_2}{b}\dfrac{1-\nu}{2} \end{bmatrix} \\
&= c_1 \left[\begin{array}{c|c} -\dfrac{(1-\xi_2)^2}{b^2} + \dfrac{1-\xi_1^2}{a^2}\dfrac{1-\nu}{2} & \dfrac{(1+\xi_1)(1-\xi_2)}{ab}\nu - \dfrac{(1-\xi_1)(1-\xi_2)}{ab}\dfrac{1-\nu}{2} \\ \hline -\dfrac{(1-\xi_1)(1-\xi_2)}{ab}\nu + \dfrac{(1+\xi_1)(1-\xi_2)}{ab}\dfrac{1-\nu}{2} & \dfrac{1-\xi_1^2}{a^2} - \dfrac{(1-\xi_2)^2}{b^2}\dfrac{1-\nu}{2} \end{array}\right]
\end{aligned}
\tag{3.107}
$$

$$
\begin{aligned}
\mathbf{B}_1^T\mathbf{C}^{es}\mathbf{B}_3 &= c_1 \begin{bmatrix} -\dfrac{1-\xi_2}{b} & 0 & -\dfrac{1-\xi_1}{a} \\ 0 & -\dfrac{1-\xi_1}{a} & -\dfrac{1-\xi_2}{b} \end{bmatrix} \begin{bmatrix} \dfrac{1+\xi_2}{b} & \dfrac{1+\xi_1}{a}\nu \\ \dfrac{1+\xi_2}{b}\nu & \dfrac{1+\xi_1}{a} \\ \dfrac{1+\xi_1}{a}\dfrac{1-\nu}{2} & \dfrac{1+\xi_2}{b}\dfrac{1-\nu}{2} \end{bmatrix} \\
&= c_1 \left[\begin{array}{c|c} -\dfrac{1-\xi_2^2}{b^2} - \dfrac{1-\xi_1^2}{a^2}\dfrac{1-\nu}{2} & -\dfrac{(1+\xi_1)(1-\xi_2)}{ab}\nu - \dfrac{(1-\xi_1)(1+\xi_2)}{ab}\dfrac{1-\nu}{2} \\ \hline -\dfrac{(1-\xi_1)(1+\xi_2)}{ab}\nu - \dfrac{(1+\xi_1)(1-\xi_2)}{ab}\dfrac{1-\nu}{2} & -\dfrac{1-\xi_1^2}{a^2} - \dfrac{1-\xi_2^2}{b^2}\dfrac{1-\nu}{2} \end{array}\right]
\end{aligned}
\tag{3.107}
$$

$$
\begin{aligned}
\mathbf{B}_1^T\mathbf{C}^{es}\mathbf{B}_4 &= c_1 \begin{bmatrix} -\dfrac{1-\xi_2}{b} & 0 & -\dfrac{1-\xi_1}{a} \\ 0 & -\dfrac{1-\xi_1}{a} & -\dfrac{1-\xi_2}{b} \end{bmatrix} \begin{bmatrix} -\dfrac{1+\xi_2}{b} & \dfrac{1-\xi_1}{a}\nu \\ -\dfrac{1+\xi_2}{b}\nu & \dfrac{1-\xi_1}{a} \\ \dfrac{1-\xi_1}{a}\dfrac{1-\nu}{2} & -\dfrac{1+\xi_2}{b}\dfrac{1-\nu}{2} \end{bmatrix} \\
&= c_1 \left[\begin{array}{c|c} \dfrac{1-\xi_2^2}{b^2} - \dfrac{(1-\xi_1)^2}{a^2}\dfrac{1-\nu}{2} & -\dfrac{(1-\xi_1)(1-\xi_2)}{ab}\nu + \dfrac{(1-\xi_1)(1+\xi_2)}{ab}\dfrac{1-\nu}{2} \\ \hline \dfrac{(1-\xi_1)(1+\xi_2)}{ab}\nu - \dfrac{(1-\xi_1)(1-\xi_2)}{ab}\dfrac{1-\nu}{2} & -\dfrac{(1-\xi_1)^2}{a^2} + \dfrac{1-\xi_2^2}{b^2}\dfrac{1-\nu}{2} \end{array}\right]
\end{aligned}
\tag{3.107}
$$

Multiplication with the transponed B operator $\mathbf{B}_2^T(\boldsymbol{\xi})$ yields:

$$
\begin{aligned}
\mathbf{B}_2^T\mathbf{C}^{es}\mathbf{B}_2 &= c_1 \begin{bmatrix} \frac{1-\xi_2}{b} & 0 & -\frac{1+\xi_1}{a} \\ 0 & -\frac{1+\xi_1}{a} & \frac{1-\xi_2}{b} \end{bmatrix} \begin{bmatrix} \frac{1-\xi_2}{b} & -\frac{1+\xi_1}{a}\nu \\ \frac{1-\xi_2}{b}\nu & -\frac{1+\xi_1}{a} \\ -\frac{1+\xi_1}{a}\frac{1-\nu}{2} & \frac{1-\xi_2}{b}\frac{1-\nu}{2} \end{bmatrix} \\
&= c_1 \left[\begin{array}{c|c} \frac{(1-\xi_2)^2}{b^2} + \frac{(1+\xi_1)^2}{a^2}\frac{1-\nu}{2} & -\frac{(1+\xi_1)(1-\xi_2)}{ab}\frac{1+\nu}{2} \\ \hline & \frac{(1+\xi_1)^2}{a^2} + \frac{(1-\xi_2)^2}{b^2}\frac{1-\nu}{2} \end{array}\right]
\end{aligned}
\tag{3.108}
$$

$$
\begin{aligned}
\mathbf{B}_2^T\mathbf{C}^{es}\mathbf{B}_3 &= c_1 \begin{bmatrix} \frac{1-\xi_2}{b} & 0 & -\frac{1+\xi_1}{a} \\ 0 & -\frac{1+\xi_1}{a} & \frac{1-\xi_2}{b} \end{bmatrix} \begin{bmatrix} \frac{1+\xi_2}{b} & \frac{1+\xi_1}{a}\nu \\ \frac{1+\xi_2}{b}\nu & \frac{1+\xi_1}{a} \\ \frac{1+\xi_1}{a}\frac{1-\nu}{2} & \frac{1+\xi_2}{b}\frac{1-\nu}{2} \end{bmatrix} \\
&= c_1 \left[\begin{array}{c|c} \frac{1-\xi_2^2}{b^2} - \frac{(1+\xi_1)^2}{a^2}\frac{1-\nu}{2} & \frac{(1+\xi_1)(1-\xi_2)}{ab}\nu - \frac{(1+\xi_1)(1+\xi_2)}{ab}\frac{1-\nu}{2} \\ \hline -\frac{(1+\xi_1)(1+\xi_2)}{ab}\nu + \frac{(1+\xi_1)(1-\xi_2)}{ab}\frac{1-\nu}{2} & -\frac{(1+\xi_1)^2}{a^2} + \frac{1-\xi_2^2}{b^2}\frac{1-\nu}{2} \end{array}\right]
\end{aligned}
\tag{3.108}
$$

$$
\begin{aligned}
\mathbf{B}_2^T\mathbf{C}^{es}\mathbf{B}_4 &= c_1 \begin{bmatrix} \frac{1-\xi_2}{b} & 0 & -\frac{1+\xi_1}{a} \\ 0 & -\frac{1+\xi_1}{a} & \frac{1-\xi_2}{b} \end{bmatrix} \begin{bmatrix} -\frac{1+\xi_2}{b} & \frac{1-\xi_1}{a}\nu \\ -\frac{1+\xi_2}{b}\nu & \frac{1-\xi_1}{a} \\ \frac{1-\xi_1}{a}\frac{1-\nu}{2} & -\frac{1+\xi_2}{b}\frac{1-\nu}{2} \end{bmatrix} \\
&= c_1 \left[\begin{array}{c|c} -\frac{1-\xi_2^2}{b^2} - \frac{1-\xi_1^2}{a^2}\frac{1-\nu}{2} & \frac{(1-\xi_1)(1-\xi_2)}{ab}\nu + \frac{(1+\xi_1)(1+\xi_2)}{ab}\frac{1-\nu}{2} \\ \hline \frac{(1+\xi_1)(1+\xi_2)}{ab}\nu + \frac{(1-\xi_1)(1-\xi_2)}{ab}\frac{1-\nu}{2} & -\frac{1-\xi_1^2}{a^2} - \frac{1-\xi_2^2}{b^2}\frac{1-\nu}{2} \end{array}\right]
\end{aligned}
\tag{3.108}
$$

Multiplication with the transponed B operator $\mathbf{B}_3^T(\boldsymbol{\xi})$ yields:

$$
\begin{aligned}
\mathbf{B}_3^T\mathbf{C}^{es}\mathbf{B}_3 &= c_1 \begin{bmatrix} \frac{1+\xi_2}{b} & 0 & \frac{1+\xi_1}{a} \\ 0 & \frac{1+\xi_1}{a} & \frac{1+\xi_2}{b} \end{bmatrix} \begin{bmatrix} \frac{1+\xi_2}{b} & \frac{1+\xi_1}{a}\nu \\ \frac{1+\xi_2}{b}\nu & \frac{1+\xi_1}{a} \\ \frac{1+\xi_1}{a}\frac{1-\nu}{2} & \frac{1+\xi_2}{b}\frac{1-\nu}{2} \end{bmatrix} \\
&= c_1 \left[\begin{array}{c|c} \frac{(1+\xi_2)^2}{b^2} + \frac{(1+\xi_1)^2}{a^2}\frac{1-\nu}{2} & \frac{(1+\xi_1)(1+\xi_2)}{ab}\frac{1+\nu}{2} \\ \hline & \frac{(1+\xi_1)^2}{a^2} + \frac{(1+\xi_2)^2}{b^2}\frac{1-\nu}{2} \end{array}\right]
\end{aligned}
\tag{3.109}
$$

$$
\begin{aligned}
\mathbf{B}_3^T\mathbf{C}^{es}\mathbf{B}_4 &= c_1 \begin{bmatrix} \frac{1+\xi_2}{b} & 0 & \frac{1+\xi_1}{a} \\ 0 & \frac{1+\xi_1}{a} & \frac{1+\xi_2}{b} \end{bmatrix} \begin{bmatrix} -\frac{1+\xi_2}{b} & \frac{1-\xi_1}{a}\nu \\ -\frac{1+\xi_2}{b}\nu & \frac{1-\xi_1}{a} \\ \frac{1-\xi_1}{a}\frac{1-\nu}{2} & -\frac{1+\xi_2}{b}\frac{1-\nu}{2} \end{bmatrix} \\
&= c_1 \left[\begin{array}{c|c} -\frac{(1+\xi_2)^2}{b^2} + \frac{1-\xi_1^2}{a^2}\frac{1-\nu}{2} & \frac{(1-\xi_1)(1+\xi_2)}{ab}\nu - \frac{(1+\xi_1)(1+\xi_2)}{ab}\frac{1-\nu}{2} \\ \hline -\frac{(1+\xi_1)(1+\xi_2)}{ab}\nu + \frac{(1-\xi_1)(1+\xi_2)}{ab}\frac{1-\nu}{2} & \frac{1-\xi_1^2}{a^2} - \frac{(1+\xi_2)^2}{b^2}\frac{1-\nu}{2} \end{array}\right]
\end{aligned}
\tag{3.109}
$$

Finally, multiplication with the transponed B operator $\mathbf{B}_4^T(\boldsymbol{\xi})$ yields:

$$
\begin{aligned}
\mathbf{B}_4^T\mathbf{C}^{es}\mathbf{B}_4 &= c_1 \begin{bmatrix} -\frac{1+\xi_2}{b} & 0 & \frac{1-\xi_1}{a} \\ 0 & \frac{1-\xi_1}{a} & -\frac{1+\xi_2}{b} \end{bmatrix} \begin{bmatrix} -\frac{1+\xi_2}{b} & \frac{1-\xi_1}{a}\nu \\ -\frac{1+\xi_2}{b}\nu & \frac{1-\xi_1}{a} \\ \frac{1-\xi_1}{a}\frac{1-\nu}{2} & -\frac{1+\xi_2}{b}\frac{1-\nu}{2} \end{bmatrix} \\
&= c_1 \left[\begin{array}{c|c} \frac{(1+\xi_2)^2}{b^2} + \frac{(1-\xi_1)^2}{a^2}\frac{1-\nu}{2} & -\frac{(1-\xi_1)(1+\xi_2)}{ab}\frac{1+\nu}{2} \\ \hline & \frac{(1-\xi_1)^2}{a^2} + \frac{(1+\xi_2)^2}{b^2}\frac{1-\nu}{2} \end{array}\right]
\end{aligned}
\tag{3.110}
$$

In equations (3.107) through (3.110) the abreviation c_1 was used. For further derivatives, the abreviation c_2 is used additionally.

$f(\xi_1,\xi_2)$	$\int_{-1}^{1}\int_{-1}^{1} f(\xi_1,\xi_2)d\xi_1 d\xi_2$	$f(\xi_1,\xi_2)$	$\int_{-1}^{1}\int_{-1}^{1} f(\xi_1,\xi_2)d\xi_1 d\xi_2$
$(1+\xi_1)^2$	$\frac{16}{3} = 4\,\frac{4}{3}$	$(1+\xi_1)(1+\xi_2)$	$4 = 3\,\frac{4}{3}$
$(1+\xi_2)^2$	$\frac{16}{3} = 4\,\frac{4}{3}$	$(1-\xi_1)(1-\xi_2)$	$4 = 3\,\frac{4}{3}$
$(1-\xi_1)^2$	$\frac{16}{3} = 4\,\frac{4}{3}$	$(1+\xi_1)(1-\xi_2)$	$4 = 3\,\frac{4}{3}$
$(1-\xi_2)^2$	$\frac{16}{3} = 4\,\frac{4}{3}$	$(1-\xi_1)(1+\xi_2)$	$4 = 3\,\frac{4}{3}$
$1-\xi_1^2$	$\frac{8}{3} = 2\,\frac{4}{3}$	$1-\xi_2^2$	$\frac{8}{3} = 2\,\frac{4}{3}$

Table 3.1: Integration von Termen der Elementsteifigkeitsmatrix

$$c_1 = \frac{E}{4\,(1-\nu^2)} \qquad\qquad c_2 = c_1\,\frac{4}{3\,a^2\,b^2}\,\frac{1}{4}abh = \frac{E\,h}{12ab\,(1-\nu^2)} \tag{3.111}$$

As a third step, integration of the terms $\mathbf{B}_i\mathbf{C}^{es}\mathbf{B}_j$ above the natural coordinates ξ_1 and ξ_2 and multiplication with factor $1/4abh$ yields the components of the element stiffness matrix $\mathbf{k}^e_{ij} = \mathbf{k}^{e\,T}_{ji}$. The corresponding integrands and their defined integral are shown in chart 3.1. It is to be noted that the integrands $(1\pm\xi_1)(1\pm\xi_2)$ correspond with the shape functions (equation (3.35)) multiplied with factor four. Looking at the shape functions in Fig. 3.8, the identity of the corresponding integrals becomes clear. Integration of equations (3.107) yields:

$$\begin{aligned}
\mathbf{k}^e_{11} &= c_2\begin{bmatrix} 4a^2+4b^2\frac{1-\nu}{2} & 3ab\frac{1+\nu}{2} \\ & 4b^2+4a^2\frac{1-\nu}{2}\end{bmatrix} = c_2\begin{bmatrix} 4a^2+2b^2(1-\nu) & 3ab\frac{1+\nu}{2} \\ & 4b^2+2a^2(1-\nu)\end{bmatrix}\\
\mathbf{k}^e_{12} &= c_2\begin{bmatrix} -4a^2+2b^2\frac{1-\nu}{2} & 3ab\nu-3ab\frac{1-\nu}{2} \\ -3ab\nu+3ab\frac{1-\nu}{2} & 2b^2-4a^2\frac{1-\nu}{2}\end{bmatrix} = c_2\begin{bmatrix} -4a^2+b^2(1-\nu) & -3ab\frac{1-3\nu}{2} \\ 3ab\frac{1-3\nu}{2} & 2b^2-2a^2(1-\nu)\end{bmatrix}\\
\mathbf{k}^e_{13} &= c_2\begin{bmatrix} -2a^2-2b^2\frac{1-\nu}{2} & -3ab\nu-3ab\frac{1-\nu}{2} \\ -3ab\nu-3ab\frac{1-\nu}{2} & -2b^2-2a^2\frac{1-\nu}{2}\end{bmatrix} = c_2\begin{bmatrix} -2a^2-b^2(1-\nu) & -3ab\frac{1+\nu}{2} \\ -3ab\frac{1+\nu}{2} & -2b^2-a^2(1-\nu)\end{bmatrix}\\
\mathbf{k}^e_{14} &= c_2\begin{bmatrix} 2a^2-4b^2\frac{1-\nu}{2} & -3ab\nu+3ab\frac{1-\nu}{2} \\ 3ab\nu-3ab\frac{1-\nu}{2} & -4b^2+2a^2\frac{1-\nu}{2}\end{bmatrix} = c_2\begin{bmatrix} 2a^2-2b^2(1-\nu) & 3ab\frac{1-3\nu}{2} \\ -3ab\frac{1-3\nu}{2} & -4b^2+a^2(1-\nu)\end{bmatrix}
\end{aligned} \tag{3.112}$$

Integration of equations (3.108) yields:

$$\begin{aligned}
\mathbf{k}^e_{22} &= c_2\begin{bmatrix} 4a^2+4b^2\frac{1-\nu}{2} & -3ab\frac{1+\nu}{2} \\ & 4b^2+4a^2\frac{1-\nu}{2}\end{bmatrix} = c_2\begin{bmatrix} 4a^2+2b^2(1-\nu) & -3ab\frac{1+\nu}{2} \\ & 4b^2+2a^2(1-\nu)\end{bmatrix}\\
\mathbf{k}^e_{23} &= c_2\begin{bmatrix} 2a^2-4b^2\frac{1-\nu}{2} & 3ab\nu-3ab\frac{1-\nu}{2} \\ -3ab\nu+3ab\frac{1-\nu}{2} & -4b^2+2a^2\frac{1-\nu}{2}\end{bmatrix} = c_2\begin{bmatrix} 2a^2-2b^2(1-\nu) & -3ab\frac{1-3\nu}{2} \\ 3ab\frac{1-3\nu}{2} & -4b^2+a^2(1-\nu)\end{bmatrix}\\
\mathbf{k}^e_{24} &= c_2\begin{bmatrix} -2a^2-2b^2\frac{1-\nu}{2} & 3ab\nu+3ab\frac{1-\nu}{2} \\ 3ab\nu+3ab\frac{1-\nu}{2} & -2b^2-2a^2\frac{1-\nu}{2}\end{bmatrix} = c_2\begin{bmatrix} -2a^2-b^2(1-\nu) & 3ab\frac{1+\nu}{2} \\ 3ab\frac{1+\nu}{2} & -2b^2-a^2(1-\nu)\end{bmatrix}
\end{aligned} \tag{3.113}$$

Integration of equations (3.109) yields:

$$
\begin{aligned}
\mathbf{k}^e_{33} &= c_2 \begin{bmatrix} 4a^2 + 4b^2\dfrac{1-\nu}{2} & 3ab\dfrac{1+\nu}{2} \\ & 4b^2 + 4a^2\dfrac{1-\nu}{2} \end{bmatrix} = c_2 \begin{bmatrix} 4a^2 + 2b^2(1-\nu) & 3ab\dfrac{1+\nu}{2} \\ & 4b^2 + 2a^2(1-\nu) \end{bmatrix} \\
\mathbf{k}^e_{34} &= c_2 \begin{bmatrix} -4a^2 + 2b^2\dfrac{1-\nu}{2} & 3ab\nu - 3ab\dfrac{1-\nu}{2} \\ -3ab\nu + 3ab\dfrac{1-\nu}{2} & 2b^2 - 4a^2\dfrac{1-\nu}{2} \end{bmatrix} = c_2 \begin{bmatrix} -4a^2 + b^2(1-\nu) & -3ab\dfrac{1-3\nu}{2} \\ 3ab\dfrac{1-3\nu}{2} & 2b^2 - 2a^2(1-\nu) \end{bmatrix}
\end{aligned}
\tag{3.114}
$$

Integration of equation (3.110) yields:

$$
\mathbf{k}^e_{44} = c_2 \begin{bmatrix} 4a^2 + 4b^2\dfrac{1-\nu}{2} & -3ab\dfrac{1+\nu}{2} \\ & 4b^2 + 4a^2\dfrac{1-\nu}{2} \end{bmatrix} = c_2 \begin{bmatrix} 4a^2 + 2b^2(1-\nu) & -3ab\dfrac{1+\nu}{2} \\ & 4b^2 + 2a^2(1-\nu) \end{bmatrix}
\tag{3.115}
$$

The above development of the stiffness components determines the element stiffness matrix. The complete element stiffness matrix $\mathbf{k}^e$ is given as follows:

$$
\mathbf{k}^e = \frac{E\,h}{12\,a\,b\,(1-\nu^2)} \cdot
\begin{bmatrix}
\begin{array}{c} 4a^2 \\ +2b^2(1-\nu) \end{array} & 3ab\dfrac{1+\nu}{2} & \begin{array}{c} -4a^2 \\ +b^2(1-\nu) \end{array} & -3ab\dfrac{1-3\nu}{2} & \begin{array}{c} -2a^2 \\ -b^2(1-\nu) \end{array} & -3ab\dfrac{1+\nu}{2} & \begin{array}{c} 2a^2 \\ -2b^2(1-\nu) \end{array} & 3ab\dfrac{1-3\nu}{2} \\
 & \begin{array}{c} 4b^2 \\ +2a^2(1-\nu) \end{array} & 3ab\dfrac{1-3\nu}{2} & \begin{array}{c} 2b^2 \\ -2a^2(1-\nu) \end{array} & -3ab\dfrac{1+\nu}{2} & \begin{array}{c} -2b^2 \\ -a^2(1-\nu) \end{array} & -3ab\dfrac{1-3\nu}{2} & \begin{array}{c} -4b^2 \\ +a^2(1-\nu) \end{array} \\
 & & \begin{array}{c} 4a^2 \\ +2b^2(1-\nu) \end{array} & -3ab\dfrac{1+\nu}{2} & \begin{array}{c} 2a^2 \\ -2b^2(1-\nu) \end{array} & -3ab\dfrac{1-3\nu}{2} & \begin{array}{c} -2a^2 \\ -b^2(1-\nu) \end{array} & 3ab\dfrac{1+\nu}{2} \\
 & & & \begin{array}{c} 4b^2 \\ +2a^2(1-\nu) \end{array} & 3ab\dfrac{1-3\nu}{2} & \begin{array}{c} -4b^2 \\ +a^2(1-\nu) \end{array} & 3ab\dfrac{1+\nu}{2} & \begin{array}{c} -2b^2 \\ -a^2(1-\nu) \end{array} \\
 & & & & \begin{array}{c} 4a^2 \\ +2b^2(1-\nu) \end{array} & 3ab\dfrac{1+\nu}{2} & \begin{array}{c} -4a^2 \\ +b^2(1-\nu) \end{array} & -3ab\dfrac{1-3\nu}{2} \\
 & & & & & \begin{array}{c} 4b^2 \\ +2a^2(1-\nu) \end{array} & 3ab\dfrac{1-3\nu}{2} & \begin{array}{c} 2b^2 \\ -2a^2(1-\nu) \end{array} \\
 & & & & & & \begin{array}{c} 4a^2 \\ +2b^2(1-\nu) \end{array} & -3ab\dfrac{1+\nu}{2} \\
\text{sym} & & & & & & & \begin{array}{c} 4b^2 \\ +2a^2(1-\nu) \end{array}
\end{bmatrix}
\tag{3.116}
$$

3.3.9.5 Element Masses Matrix $\mathbf{k}^e$

By supposing a constant element thickness h and a constant density p and by substituting the JACOBI determinant according to equation (3.98), the matrix $\mathbf{m}^e$ of the rectangular plane finite element can be calculated with the help of the general relation of the element masses matrix in equation (3.75).

$$
\mathbf{m}^e = \frac{1}{4}\,a\,b\,h\,\rho \int_{-1}^{1}\int_{-1}^{1} \mathbf{N}^T(\boldsymbol{\xi})\,\mathbf{N}(\boldsymbol{\xi})\,d\xi_1 d\xi_2
\tag{3.117}
$$

Analogue with the development of the element stiffness matrix, the element masses matrix is split up into sectors corresponding with the element joints.

$$\mathbf{m}^e_{ij} = \frac{1}{4}\, a\, b\, h\, \rho \int_{-1}^{1}\int_{-1}^{1} \mathbf{N}^{i\,T}(\boldsymbol{\xi})\, \mathbf{N}^j(\boldsymbol{\xi})\, d\xi_1 d\xi_2 \qquad \mathbf{m}^e = \begin{bmatrix} \mathbf{m}^e_{11} & \mathbf{m}^e_{12} & \mathbf{m}^e_{13} & \mathbf{m}^e_{14} \\ \mathbf{m}^e_{21} & \mathbf{m}^e_{22} & \mathbf{m}^e_{23} & \mathbf{m}^e_{24} \\ \mathbf{m}^e_{31} & \mathbf{m}^e_{32} & \mathbf{m}^e_{33} & \mathbf{m}^e_{34} \\ \mathbf{m}^e_{41} & \mathbf{m}^e_{42} & \mathbf{m}^e_{43} & \mathbf{m}^e_{44} \end{bmatrix} \tag{3.118}$$

$\mathbf{N}^i(\boldsymbol{\xi})$ represents the diagonal matrix of the shape functions $N^i(\boldsymbol{\xi})$ corresponding with the element joint i according to equation (3.42). The element masses matrix is symmetrical.

$$\mathbf{m}^e_{ij} = \mathbf{m}^{e\,T}_{ji} \tag{3.119}$$

Through equations (3.118) and (3.42), the masses matrix $\mathbf{m}^e_{ij}$ turns out as a diagonal matrix.

$$\mathbf{m}^e_{ij} = \frac{1}{4}\, a\, b\, h\, \rho \int_{-1}^{1}\int_{-1}^{1} \begin{bmatrix} N^i(\boldsymbol{\xi})\, N^j(\boldsymbol{\xi}) & 0 \\ 0 & N^i(\boldsymbol{\xi})\, N^j(\boldsymbol{\xi}) \end{bmatrix} d\xi_1 d\xi_2 \tag{3.120}$$

By filling in the shape functions $N^i(\boldsymbol{\xi})$ according to equations (3.35) or by having a close look at the shape functions in Fig. (3.35), one obtains the following identities of the surface integrals.

$$\begin{aligned}
\int_{-1}^{1}\int_{-1}^{1} N^1(\boldsymbol{\xi})\, N^2(\boldsymbol{\xi})\, d\xi_1 d\xi_2 &= \int_{-1}^{1}\int_{-1}^{1} N^2(\boldsymbol{\xi})\, N^3(\boldsymbol{\xi})\, d\xi_1 d\xi_2 = \int_{-1}^{1}\int_{-1}^{1} N^3(\boldsymbol{\xi})\, N^4(\boldsymbol{\xi})\, d\xi_1 d\xi_2 \\
&= \int_{-1}^{1}\int_{-1}^{1} N^4(\boldsymbol{\xi})\, N^1(\boldsymbol{\xi})\, d\xi_1 d\xi_2 \\
\int_{-1}^{1}\int_{-1}^{1} N^1(\boldsymbol{\xi})\, N^3(\boldsymbol{\xi})\, d\xi_1 d\xi_2 &= \int_{-1}^{1}\int_{-1}^{1} N^2(\boldsymbol{\xi})\, N^4(\boldsymbol{\xi})\, d\xi_1 d\xi_2 \\
\int_{-1}^{1}\int_{-1}^{1} N^1(\boldsymbol{\xi})\, N^1(\boldsymbol{\xi})\, d\xi_1 d\xi_2 &= \int_{-1}^{1}\int_{-1}^{1} N^2(\boldsymbol{\xi})\, N^2(\boldsymbol{\xi})\, d\xi_1 d\xi_2 = \int_{-1}^{1}\int_{-1}^{1} N^3(\boldsymbol{\xi})\, N^3(\boldsymbol{\xi})\, d\xi_1 d\xi_2 \\
&= \int_{-1}^{1}\int_{-1}^{1} N^4(\boldsymbol{\xi})\, N^4(\boldsymbol{\xi})\, d\xi_1 d\xi_2
\end{aligned} \tag{3.121}$$

The integration is immensely simplified by transforming the integrands into a product $f_1(\xi_1)f_2(\xi_2)$ and by using the already determined integrals in chart 3.1 with $\int_{-1}^{1} d\xi_\alpha = 2$.

$$
\begin{aligned}
\int_{-1}^{1}\int_{-1}^{1} N^1(\boldsymbol{\xi})\; N^1(\boldsymbol{\xi})\; d\xi_1 d\xi_2 &= \frac{1}{16}\int_{-1}^{1}\int_{-1}^{1}(1-\xi_1)^2\;(1-\xi_2)^2\; d\xi_1 d\xi_2 \\
&= \frac{1}{16}\int_{-1}^{1}(1-\xi_1)^2\; d\xi_1 \int_{-1}^{1}(1-\xi_2)^2\; d\xi_2 = \frac{1}{16}\,\frac{8}{3}\,\frac{8}{3} = \frac{4}{9} \\
\int_{-1}^{1}\int_{-1}^{1} N^1(\boldsymbol{\xi})\; N^2(\boldsymbol{\xi})\; d\xi_1 d\xi_2 &= \frac{1}{16}\int_{-1}^{1}\int_{-1}^{1}(1-\xi_1^2)\;(1-\xi_2)^2\; d\xi_1 d\xi_2 = \frac{1}{16}\,\frac{4}{3}\,\frac{8}{3} = \frac{2}{9} \\
\int_{-1}^{1}\int_{-1}^{1} N^1(\boldsymbol{\xi})\; N^3(\boldsymbol{\xi})\; d\xi_1 d\xi_2 &= \frac{1}{16}\int_{-1}^{1}\int_{-1}^{1}(1-\xi_1^2)\;(1-\xi_2^2)\; d\xi_1 d\xi_2 = \frac{1}{16}\,\frac{4}{3}\,\frac{4}{3} = \frac{1}{9}
\end{aligned}
\tag{3.122}
$$

The summary of equations (3.118) through (3.122) finally produces the element masses matrix of the rectangular plane LAGRANGE element.

$$
\mathbf{m}^e = \frac{1}{36}\; a\; b\; h\; \rho
\left[\begin{array}{c|c|c|c|c|c|c|c}
4 & 0 & 2 & 0 & 1 & 0 & 2 & 0 \\ \hline
 & 4 & 0 & 2 & 0 & 1 & 0 & 2 \\ \hline
 & & 4 & 0 & 2 & 0 & 1 & 0 \\ \hline
 & & & 4 & 0 & 2 & 0 & 1 \\ \hline
 & & & & 4 & 0 & 2 & 0 \\ \hline
 & & & & & 4 & 0 & 2 \\ \hline
 & & & & & & 4 & 0 \\ \hline
\text{sym} & & & & & & & 4
\end{array}\right]
\tag{3.123}
$$

The sum total of the components of the element masses matrix corresponds with the double mass of the elements $2\,m = 2\;a\;b\;h\;\rho$ with the two directions of movement $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$.

3.3.9.6 Consistent Element Load Vectors $\boldsymbol{r}_p^e$ and $\boldsymbol{r}_n^e$

The consistent element loads of the volume loads according to equation (3.77) together with the JACOBI determinant according to equation (3.98) and constant $h\;\rho$ turn into:

$$
\boldsymbol{r}_p^e = \frac{1}{4}\; a\; b\; h\; \rho \int_{-1}^{1}\int_{-1}^{1} \mathbf{N}^T(\boldsymbol{\xi})\; \boldsymbol{b}(\boldsymbol{\xi})\; d\xi_1 d\xi_2
\tag{3.124}
$$

For a constant internal load $\boldsymbol{b}$, e.g. as the self weight, one obtains the consistent element load vector from the integral over the shape functions (equation (3.35)) with the surface integrals over the shape functions being identical (see Fig. 3.8 and chart 3.1).

$$
\int_{-1}^{1}\int_{-1}^{1} N^1(\boldsymbol{\xi})\; d\xi_1 d\xi_2 = \frac{1}{4}\int_{-1}^{1}\int_{-1}^{1}(1-\xi_1)\,(1-\xi_2)\;\; d\xi_1 d\xi_2 = \frac{1}{4}\;4 = 1
\tag{3.125}
$$

The integration of all terms of equation (3.124) yields the consistent element load vector of the rectangular bilinear LAGRANGE element for a constant volume load $\boldsymbol{b}$ over the element surface.

$$\boldsymbol{r}_p^e = \frac{1}{4}\, a\, b\, h\, \rho \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}^T \boldsymbol{b} \tag{3.126}$$

The generation of the consistent equivalent load of surface loads is demonstrated as example for the element edge three between the element joints three and four. Moreover, the edge loads shall be linearly variable with regard to the element length. Thereby the course of the edge load can be described with the help of the surface loads at the GAUSS points $\xi_1 = -1$ and $\xi_1 = 1$ ($\boldsymbol{t}^4 \equiv \boldsymbol{t}^\star(-1,1)$ ($\boldsymbol{t}^4 \equiv \boldsymbol{t}^\star(-1,1)$ and $\boldsymbol{t}^3 \equiv \boldsymbol{t}^\star(1,1)$) and $\boldsymbol{t}^3 \equiv \boldsymbol{t}^\star(1,1)$) and $\boldsymbol{t}^3 \equiv \boldsymbol{t}^\star(1,1)$) ($\boldsymbol{t}^4 \equiv \boldsymbol{t}^\star(-1,1)$ and $\boldsymbol{t}^3 \equiv \boldsymbol{t}^\star(1,1)$) and the shape functions $N^3(\xi_1,1)$ and $N^4(\xi_1,1)$

$$\boldsymbol{t}^\star(\xi_1,1) = N^3(\xi_1,1)\, \boldsymbol{t}^\star(1,1) + N^4(\xi_1,1)\, \boldsymbol{t}^\star(-1,1) = N^3(\xi_1,1)\, \boldsymbol{t}^3 + N^4(\xi_1,1)\, \boldsymbol{t}^4 \tag{3.127}$$

The evaluation of the consistent equivalent loads according to equation (3.91) necessitates the transformation of the line element $d\Gamma_3$ with the help of the JACOBI determinant $|\mathbf{J}_3|$ according to equation (3.88). Filling in the joint coordinates according to equation (3.93) yields the element edge three corresponding with the JACOBI determinant:

$$|\mathbf{J}_3| = \frac{1}{2}\, \left[\left(X_1^{e1} + b - X_1^{e1}\right)^2 + \left(X_2^{e1} + a - X_2^{e1} - a\right)^2\right]^{\frac{1}{2}} = \frac{1}{2}\, b \tag{3.128}$$

Thus, the equivalent loads can be calculated as function of shape functions for $\xi_2 = 1$,

$$\begin{aligned} \boldsymbol{r}_{n3}^e &= \frac{1}{2}\, b\, h \int_{-1}^{1} \mathbf{N}^T(\xi_1,1)\, \left[N^3(\xi_1,1)\, \boldsymbol{t}^3 + N^4(\xi_1,1)\, \boldsymbol{t}^4\right]\, d\xi_1 \\ &= \frac{1}{2}\, b\, h \int_{-1}^{1} \mathbf{N}^T(\xi_1,1)\, \mathbf{N}(\xi_1,1)\, \boldsymbol{t}^e\, d\xi_1 \end{aligned} \tag{3.129}$$

with the vector of the surface loads

$$\boldsymbol{t}^e = \begin{bmatrix} t_1^1 & t_2^1 & t_1^2 & t_2^2 & t_1^3 & t_2^3 & t_1^4 & t_2^4 \end{bmatrix}^T \tag{3.130}$$

being defined. Both, the shape functions according to equation (3.35) and multiplication of equation (3.129) turn the equivalent loads into:

$$
\begin{aligned}
\boldsymbol{r}^e_{n3} &= \frac{1}{2}\, b\, h \int_{-1}^{1} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ N^3(\xi_1,1)\; N^3(\xi_1,1)\; t_1^3 + N^3(\xi_1,1)\; N^4(\xi_1,1)\; t_1^4 \\ N^3(\xi_1,1)\; N^3(\xi_1,1)\; t_2^3 + N^3(\xi_1,1)\; N^4(\xi_1,1)\; t_2^4 \\ N^3(\xi_1,1)\; N^4(\xi_1,1)\; t_1^3 + N^4(\xi_1,1)\; N^4(\xi_1,1)\; t_1^4 \\ N^3(\xi_1,1)\; N^4(\xi_1,1)\; t_2^3 + N^4(\xi_1,1)\; N^4(\xi_1,1)\; t_2^4 \end{bmatrix} d\xi_1 \\
&= \frac{1}{8}\, b\, h \int_{-1}^{1} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ (1+\xi_1)^2\; t_1^3 + (1-\xi_1^2)\; t_1^4 \\ (1+\xi_1)^2\; t_2^3 + (1-\xi_1^2)\; t_2^4 \\ (1-\xi_1^2)\; t_1^3 + (1-\xi_1)^2\; t_1^4 \\ (1-\xi_1^2)\; t_2^3 + (1-\xi_1)^2\; t_2^4 \end{bmatrix} d\xi_1 = \frac{1}{6}\, b\, h \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 2\; t_1^3 + t_1^4 \\ 2\; t_2^3 + t_2^4 \\ t_1^3 + 2\; t_1^4 \\ t_2^3 + 2\; t_2^4 \end{bmatrix}
\end{aligned}
\tag{3.131}
$$

In the case of a constant uniform load of the element edge three ($t_1^3 = t_1^4 = t_1$ and $t_2^3 = t_2^4 = t_2$) the surface load integrated above the length of the edge ($t_1\; b$ or $t_2\; b$) is distributed evenly among the element joints.

3.4 Rectangular biquadratic Lagrange element

The increase in the approximation order or in the polynomial degree of a four-noded plane finite element, thoroughly elaborated in chapter 3.3, results in a plane biquadratic finite LAGRANGE element with nine nodes. The decisive advantages of such an element compared to a bilinear element are

- an increased accuracy of approximation
- and the possibility of adequate discretisization of curve-edged areas.

Opposite to that, the disadvantages of such an element are

- an increased number of degrees of freedom
- and a somewhat more complex derivation of element quantities.

In the following sections we will discuss ansatz functions, approximation of geometry, approximation of displacement field, JACOBI transformation and the approximation of the strain field with the B-operator. The proceeding and the mathematical formulation for generation of element matrices and vectors is identical to that of a four-noded plane element. Therefore, a detailed presentation of these facts is abandoned here.

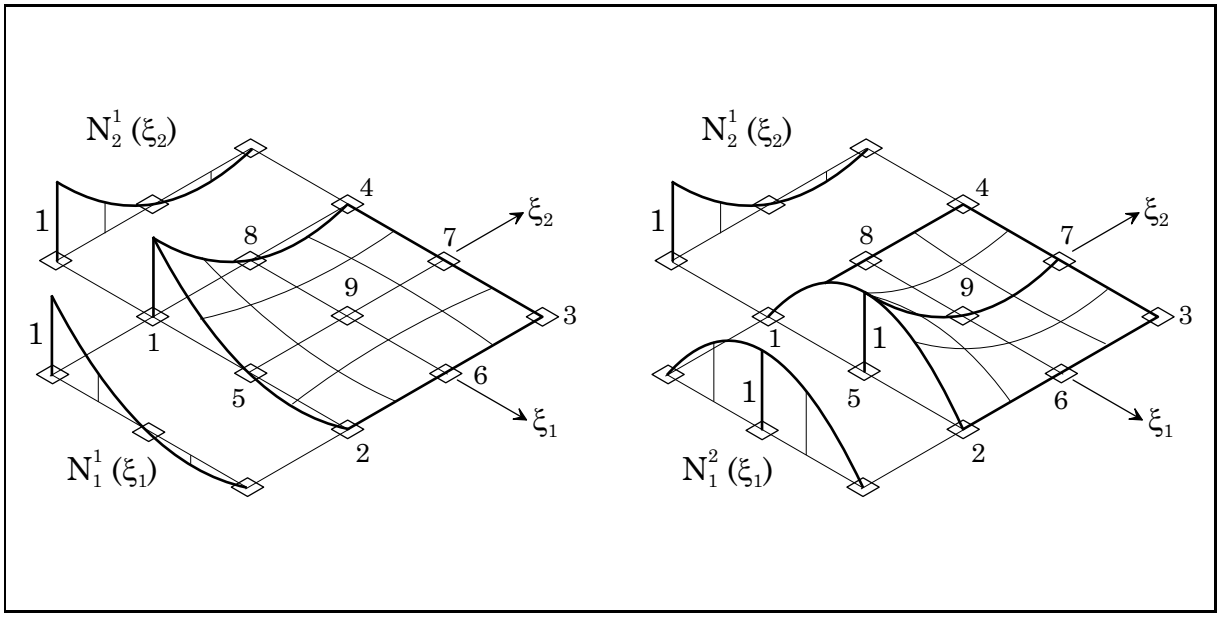


Figure 3.14: Generation of biquadratic LAGRANGE ansatz functions

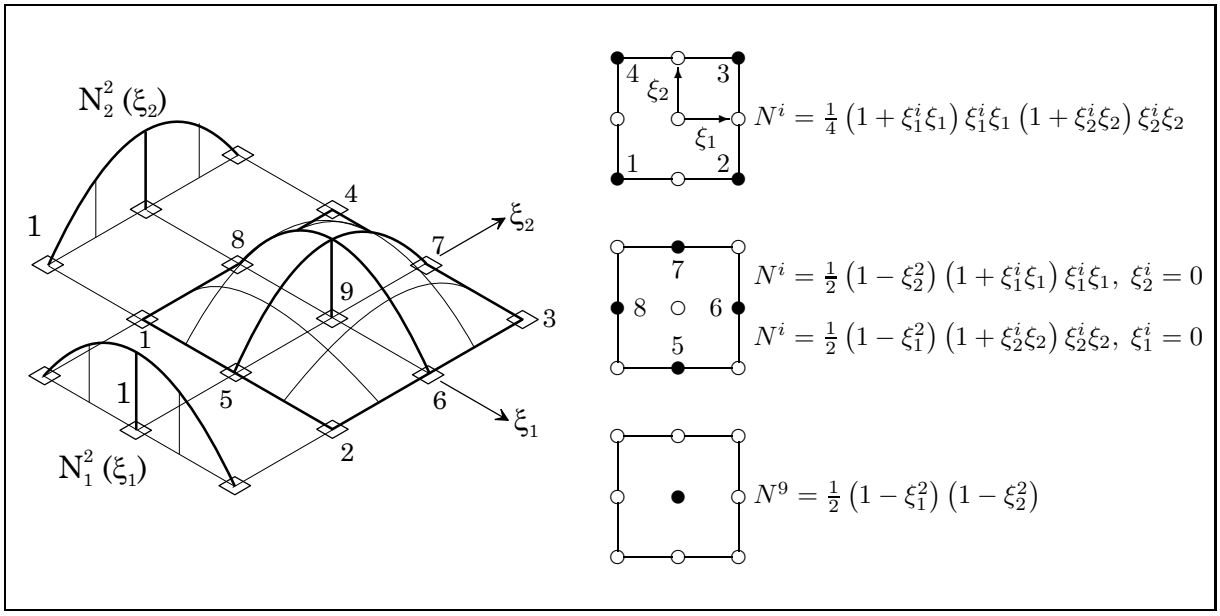


Figure 3.15: Generation of biquadratic LAGRANGE ansatz functions

3.4.1 Ansatz functions

Ansatz functions of biquadratic plane LAGRANGE elements are obtained by multiplication of possible combinations of one-dimensional quadratic ansatz functions according to table 2.1 in directions of coordinates ξ_1 and ξ_2 .The three principally different constructions of ansatz functions are shown in figures 3.14 and 3.15. For example, the ansatz function of the corner node one $N^1(\xi_1, \xi_2)$ is formed of a product of one-dimensional ansatz functions $N_1^1(\xi_1)$ and $N_2^1(\xi_2)$. The one-dimensional ansatz functions are given in table 2.1,

$$N_1^1(\xi_1) = \frac{1}{2}\,(\xi_1 - 1)\;\xi_1 \qquad\qquad N_2^1(\xi_2) = \frac{1}{2}\,(\xi_2 - 1)\;\xi_2 \tag{3.132}$$

with which the two-dimensional ansatz function $N^1(\xi_1, \xi_2)$ can be constructed.

$$\begin{aligned} N^1(\xi_1,\xi_2) &= N^1(\boldsymbol{\xi}) = N_1^1(\xi_1)\;N_2^1(\xi_2) \\ &= \frac{1}{2}\,(\xi_1 - 1)\;\xi_1\;\frac{1}{2}\,(\xi_2 - 1)\;\xi_2 \\ &= \frac{1}{4}\left[\xi_1\;\xi_2 - \xi_1^2\;\xi_2 + \xi_1^2\;\xi_2^2 - \xi_1\;\xi_2^2\right] \end{aligned} \tag{3.133}$$

Ansatz functions of node five are obtained with one-dimensional ansatzes $N_1^2(\xi_1)$ and $N_2^1(\xi_2)$ (see table 2.1).

$$N_1^2(\xi_1) = 1 - \xi_1^2 \qquad\qquad N_2^1(\xi_2) = \frac{1}{2}\,(\xi_2 - 1)\;\xi_2 \tag{3.134}$$

By switching ξ_1 and ξ_2 we get the ansatz function for node eight.

$$\begin{aligned} N^5(\xi_1,\xi_2) &= N^5(\boldsymbol{\xi}) = N_1^2(\xi_1)\;N_2^1(\xi_2) \\ &= (1-\xi_1^2)\;\frac{1}{2}(\xi_2 - 1)\;\xi_2 \\ &= \frac{1}{2}\left[-\xi_2 - \xi_1^2\;\xi_2^2 + \xi_1^2\;\xi_2 + \xi_2^2\right] \end{aligned} \qquad \begin{aligned} N^8(\xi_1,\xi_2) &= N^8(\boldsymbol{\xi}) = N_1^1(\xi_1)\;N_2^2(\xi_2) \\ &= \frac{1}{2}\;(\xi_1 - 1)\;\xi_1(1-\xi_2^2) \\ &= \frac{1}{2}\left[-\xi_1 + \xi_1^2 + \xi_1^2\;\xi_2 - \xi_1^2\;\xi_2^2\right] \end{aligned} \tag{3.135}$$

Ansatz function of the middle node nine is constructed with one-dimensional ansatz functions $N_1^2(\xi_1)$ and $N_2^2(\xi_2)$

$$N_1^2(\xi_1) = 1 - \xi_1^2 \qquad\qquad N_2^2(\xi_2) = 1 - \xi_2^2 \tag{3.136}$$

$$\begin{aligned} N^9(\xi_1,\xi_2) &= N^9(\boldsymbol{\xi}) = N_1^2(\xi_1)\;N_2^2(\xi_2) \\ &= (1-\xi_1^2)\;(1-\xi_2^2) \\ &= 1 - \xi_1^2 + \xi_1^2\;\xi_2^2 - \xi_2^2 \end{aligned} \tag{3.137}$$

As we can see from equations (3.133), (3.135) and (3.137), the ansatz functions of a biquadratic LAGRANGE element contain the terms designated with $p = 2$ in the PASCAL triangle (figure 3.5). The ansatz functions sketched in figure 3.16 are written together without prior derivation.

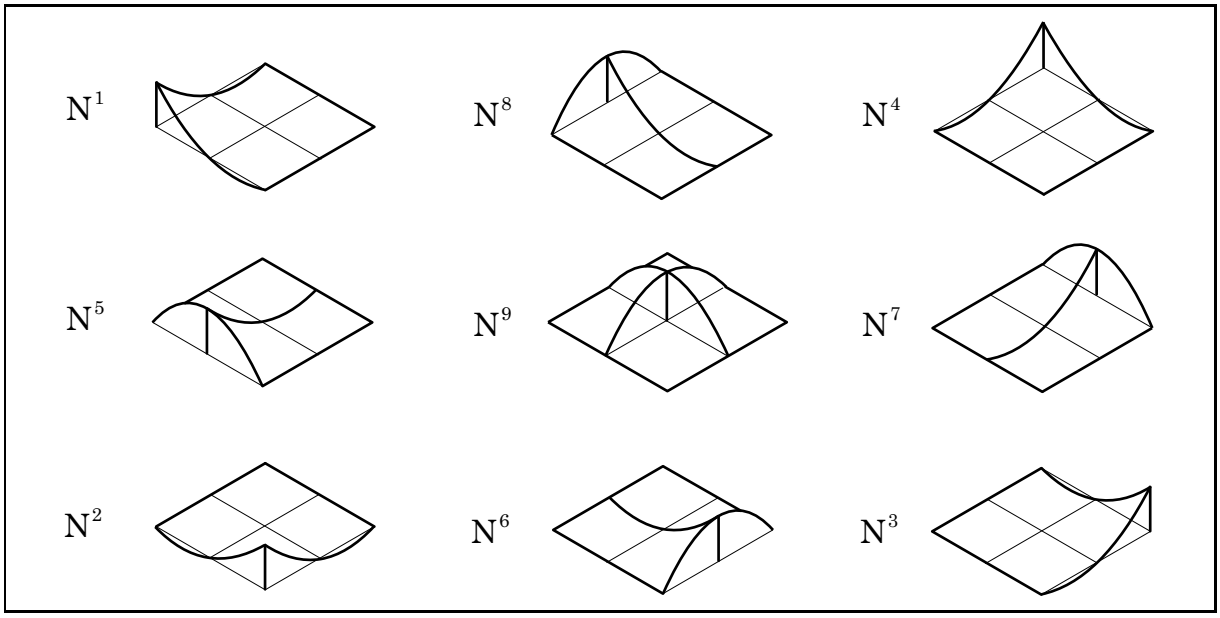


Figure 3.16: LAGRANGE ansatz functions of a nine-noded plane element

$$
\begin{aligned}
N^1(\boldsymbol{\xi}) &= \frac{1}{4}\,(1-\xi_1)\,(1-\xi_2)\,\xi_1\,\xi_2 & N^5(\boldsymbol{\xi}) &= \frac{1}{2}\,(1-\xi_1^2)\,(\xi_2-1)\,\xi_2 \\
N^2(\boldsymbol{\xi}) &= \frac{1}{4}\,(1+\xi_1)\,(\xi_2-1)\,\xi_1\,\xi_2 & N^6(\boldsymbol{\xi}) &= \frac{1}{2}\,(1+\xi_1)\,\xi_1\,(1-\xi_2^2) \\
N^3(\boldsymbol{\xi}) &= \frac{1}{4}\,(1+\xi_1)\,(1+\xi_2)\,\xi_1\,\xi_2 & N^7(\boldsymbol{\xi}) &= \frac{1}{2}\,(1-\xi_1^2)\,(1+\xi_2)\,\xi_2 \\
N^4(\boldsymbol{\xi}) &= \frac{1}{4}\,(\xi_1-1)\,(1+\xi_2)\,\xi_1\,\xi_2 & N^8(\boldsymbol{\xi}) &= \frac{1}{2}\,(\xi_1-1)\,\xi_1\,(1-\xi_2^2)
\end{aligned}
\tag{3.138}
$$

$$
N^9(\boldsymbol{\xi}) = (1-\xi_1^2)\,(1-\xi_2^2)
$$

The derivatives of ansatz functions with respect to the natural coordinates ξ_1 and ξ_2 are as follows:

$$
\begin{aligned}
N^1_{;1}(\boldsymbol{\xi}) &= \frac{1}{4}\,(1-2\,\xi_1)\,(1-\xi_2)\,\xi_2 & N^1_{;2}(\boldsymbol{\xi}) &= \frac{1}{4}\,(1-\xi_1)\,(1-2\,\xi_2)\,\xi_1 \\
N^2_{;1}(\boldsymbol{\xi}) &= \frac{1}{4}\,(1+2\,\xi_1)\,(\xi_2-1)\,\xi_2 & N^2_{;2}(\boldsymbol{\xi}) &= \frac{1}{4}\,(1+\xi_1)\,(2\,\xi_2-1)\,\xi_1 \\
N^3_{;1}(\boldsymbol{\xi}) &= \frac{1}{4}\,(1+2\,\xi_1)\,(1+\xi_2)\,\xi_2 & N^3_{;2}(\boldsymbol{\xi}) &= \frac{1}{4}\,(1+\xi_1)\,(1+2\,\xi_2)\,\xi_1
\end{aligned}
\tag{3.139}
$$

$$
\begin{aligned}
N^4_{;1}(\boldsymbol{\xi}) &= \frac{1}{4}\,(2\,\xi_1-1)\,(1+\xi_2)\,\xi_2 & N^4_{;2}(\boldsymbol{\xi}) &= \frac{1}{4}\,(\xi_1-1)\,(1+2\,\xi_2)\,\xi_1 \\
N^5_{;1}(\boldsymbol{\xi}) &= \xi_1\,(1-\xi_2)\,\xi_2 & N^5_{;2}(\boldsymbol{\xi}) &= \frac{1}{2}\,(1-\xi_1^2)\,(2\,\xi_2-1) \\
N^6_{;1}(\boldsymbol{\xi}) &= \frac{1}{2}\,(1+2\,\xi_1)\,(1-\xi_2^2) & N^6_{;2}(\boldsymbol{\xi}) &= -(1+\xi_1)\,\xi_1\,\xi_2
\end{aligned}
\tag{3.139}
$$

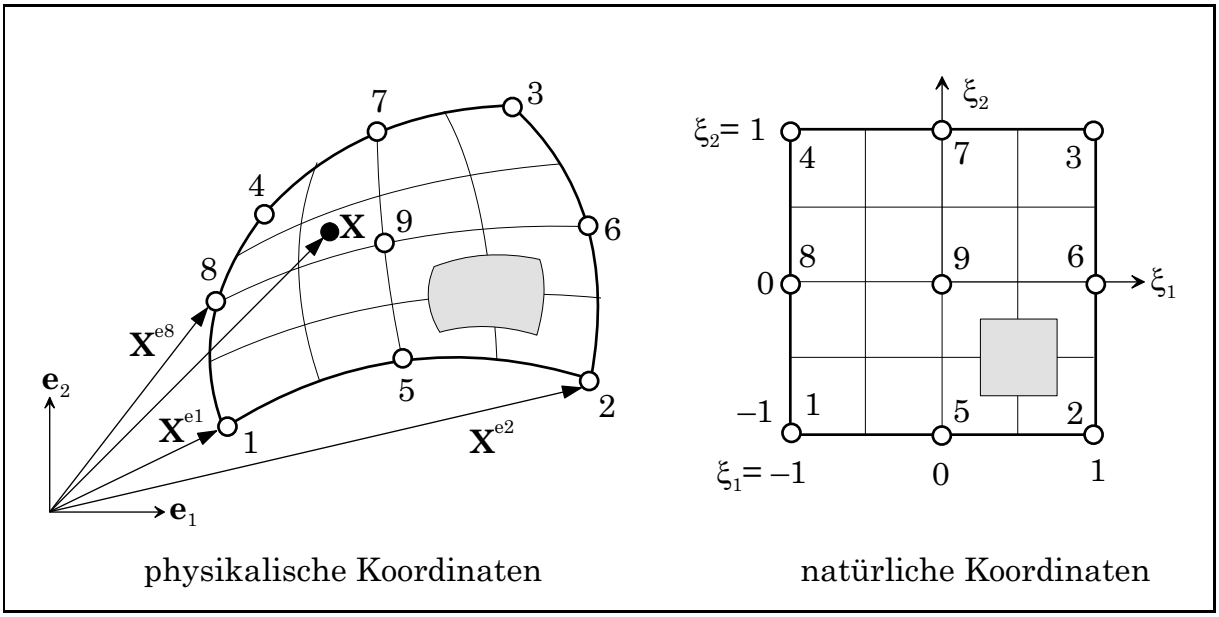


Figure 3.17: Biquadratic element in physical and natural coordinates

$$
\begin{aligned}
N^7_{;1}(\boldsymbol{\xi}) &= -\xi_1\,(1+\xi_2)\,\xi_2 & N^7_{;2}(\boldsymbol{\xi}) &= \frac{1}{2}\,(1-\xi_1^2)\,(1+2\,\xi_2) \\
N^8_{;1}(\boldsymbol{\xi}) &= \frac{1}{2}\,(2\,\xi_1-1)\,(1-\xi_2^2) & N^8_{;2}(\boldsymbol{\xi}) &= (1-\xi_1)\,\xi_1\,\xi_2 \\
N^9_{;1}(\boldsymbol{\xi}) &= 2\,\xi_1\,(\xi_2^2-1) & N^9_{;2}(\boldsymbol{\xi}) &= (\xi_1^2-1)2\,\xi_2
\end{aligned}
\tag{3.139}
$$

3.4.2 Geometry

Geometry of an element in physical and natural space is shown in figure 3.17. An arbitrary material point within a nine-noded quadrangular element is unequivocally identified by its physical $\boldsymbol{X}$ and natural $\boldsymbol{\xi}$ coordinates. The physical coordinates of this material point are determined in analogy with equation (3.41) with the assistance of the discrete node coordinate vector

$$
\boldsymbol{X}^e = \left[\ \boldsymbol{X}^{e1\,T}\ \ \boldsymbol{X}^{e2\,T}\ \ \boldsymbol{X}^{e3\,T}\ \ \boldsymbol{X}^{e4\,T}\ \ \boldsymbol{X}^{e5\,T}\ \ \boldsymbol{X}^{e6\,T}\ \ \boldsymbol{X}^{e7\,T}\ \ \boldsymbol{X}^{e8\,T}\ \ \boldsymbol{X}^{e9\,T}\right]^T \tag{3.140}
$$

and the ansatz function matrix (see equation (3.42)))

$$
\mathbf{N}^i(\boldsymbol{\xi}) = \begin{bmatrix} N^i(\boldsymbol{\xi}) & 0 \\ 0 & N^i(\boldsymbol{\xi}) \end{bmatrix} \qquad \mathbf{N}(\boldsymbol{\xi}) = \left[\ \mathbf{N}^1\ \ \mathbf{N}^2\ \ \dots\ \ \mathbf{N}^8\ \ \mathbf{N}^9\ \right] \tag{3.141}
$$

as function of the natural coordinates $\boldsymbol{\xi} = [\xi_1\ \ \xi_2]^T$ with $(NN = 9)$.

$$
\tilde{\boldsymbol{X}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \boldsymbol{X}^e \qquad \boldsymbol{X}^e = \left[\ X_1^{e1}\ \ X_2^{e1}\ \ \dots\ \ X_1^{eNN}\ \ X_2^{eNN}\ \right]^T \tag{3.142}
$$

Since the ansatz functions are quadratic, it is possible to map curved edges (see figure 3.17) and,

compared to a linear finite element achieve a significantly improved geometry approximation of two-dimensional continua with curved boundaries.

3.4.3 Jacoby transformation

The JACOBI matrix $\mathbf{J}(\boldsymbol{\xi})$, that is the inverse JACOBI matrix necessary to generate the derivatives with respect to physical coordinates by means of derivatives with respect to natural coordinates, can be computed in analogy with chapter 3.3.3 according to equations (3.46) that is (3.48) or (3.52). Here it should by all means be noted that in case of a quadratic LAGRANGE element, the physical coordinates $X_\beta(\boldsymbol{\xi})$ are defined by equations (3.141) and (3.142), which is relevant for their derivatives with respect to natural coordinates. Components of the JACOBI matrix can be calculated consequently with equation (3.50) for $NN = 9$.

$$\frac{\partial X_\beta(\boldsymbol{\xi})}{\partial \xi_\alpha} = \sum_{i=1}^{NN} X_\beta^{ei}\, N^i_{;\alpha}(\boldsymbol{\xi}) \qquad\qquad \frac{\partial \boldsymbol{X}(\boldsymbol{\xi})}{\partial \xi_\alpha} = \boldsymbol{X}_{;\alpha}(\boldsymbol{\xi}) = \mathbf{N}_{;\alpha}(\boldsymbol{\xi})\, \boldsymbol{X}^e \tag{3.143}$$

Transformation of a differential surface element dA follows according to equation (3.58) with the help of the JACOBI determinant.

3.4.4 Approximation of element quantities

The local displacements, accelerations and the variation of displacements can formally be approximated in analogy with a bilinear LAGRANGE element. Only the changed number of element free values and the correspondingly altered content of the ansatz function matrix have to be considered for evaluation of the approximation.

$$\begin{aligned}
\boldsymbol{u}(\boldsymbol{\xi}) &\approx \tilde{\boldsymbol{u}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \boldsymbol{u}^e & \boldsymbol{u}^e &= \left[\ u_1^{e1}\ \ u_2^{e1}\ \cdots\ u_1^{e9}\ \ u_2^{e9}\ \right]^T \\
\delta\boldsymbol{u}(\boldsymbol{\xi}) &\approx \delta\tilde{\boldsymbol{u}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \delta\boldsymbol{u}^e & \delta\boldsymbol{u}^e &= \left[\ \delta u_1^{e1}\ \delta u_2^{e1}\ \cdots\ \delta u_1^{e9}\ \delta u_2^{e9}\ \right]^T \\
\ddot{\boldsymbol{u}}(\boldsymbol{\xi}) &\approx \tilde{\ddot{\boldsymbol{u}}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \ddot{\boldsymbol{u}}^e & \ddot{\boldsymbol{u}}^e &= \left[\ \ddot{u}_1^{e1}\ \ddot{u}_2^{e1}\ \cdots\ \ddot{u}_1^{e9}\ \ddot{u}_2^{e9}\ \right]^T
\end{aligned} \tag{3.144}$$

The element vectors $\boldsymbol{u}^e$, $\delta\boldsymbol{u}^e$ and $\ddot{\boldsymbol{u}}^e$ are of dimension 18×1 and the ansatz functions matrix $\mathbf{N}$ is of dimension 2×18.

3.4.5 Approximation of the strain vector

Approximation of local strains is given with equation (3.62) by the linear mapping of the element displacement vector $\boldsymbol{u}^e$ with help of B-operator $\mathbf{B}(\boldsymbol{\xi})$. In case of a biquadratic plane LAGRANGE element the B-operator of dimension 3×18 is defined by the following relations (see equation

3.65), where the operator $\mathbf{B}_i(\boldsymbol{\xi})$ can alternatively be calculated according to equation (3.66).

$$\begin{aligned}
\mathbf{B}_i(\boldsymbol{\xi}) &= \begin{bmatrix} \dfrac{\partial \xi_1}{\partial X_1} N^i_{;1} + \dfrac{\partial \xi_2}{\partial X_1} N^i_{;2} & 0 \\ 0 & \dfrac{\partial \xi_1}{\partial X_2} N^i_{;1} + \dfrac{\partial \xi_2}{\partial X_2} N^i_{;2} \\ \dfrac{\partial \xi_1}{\partial X_2} N^i_{;1} + \dfrac{\partial \xi_2}{\partial X_2} N^i_{;2} & \dfrac{\partial \xi_1}{\partial X_1} N^i_{;1} + \dfrac{\partial \xi_2}{\partial X_1} N^i_{;2} \end{bmatrix} \\
\mathbf{B}(\boldsymbol{\xi}) &= \begin{bmatrix} \mathbf{B}_1 & \mathbf{B}_2 & \cdots & \mathbf{B}_8 & \mathbf{B}_9 \end{bmatrix}
\end{aligned} \tag{3.145}$$

3.4.6 Element matrices and vectors

After generating the ansatz function matrix $\mathbf{N}(\boldsymbol{\xi})$, the differential operator $\mathbf{B}(\boldsymbol{\xi})$, the JACOBI determinant $|\mathbf{J}(\boldsymbol{\xi})|$, JACOBI determinants $|\mathbf{J}_i(\boldsymbol{\xi})|$ for transformation of line elements $d\Gamma_i$ and after choosing the type of a plane mechanical problem (plane stress or plane strain state),

- - the element stiffness matrix $\mathbf{k}^e$ according to equation (3.71),
- - the element mass matrix $\mathbf{m}^e$ according to equation (3.75)
- - and the consistent element loads $\boldsymbol{r}^e_p$ and $\boldsymbol{r}^e_{ni}$ according to equations (3.77) and (3.91)

can be formulated and numerically integrated.

3.5 Biquadratic serendipity element

The considerable disadvantage of nine-noded biquadratic LAGRANGE elements compared to a bilinear element is the increased number of degrees of freedom. Especially both degrees of freedom of the middle node are increasing the linear equation system of the assembled system which has to be solved by two equations per discrete LAGRANGE element. On the other hand, the same middle node is not necessary for preservation of compatibility with the adjacent element. This means that the number of degrees of freedom can especially be reduced at system level by utilization of serendipity elements instead of LAGRANGE elements. Hence, the motivation for generation of quadratic serendipity elements with no middle node is given.

In the sections that follow we will discuss ansatz functions, approximation of geometry and of the displacement field. Since the development of the differential operator and of element matrices and vectors has to be accomplished in analogy with the four-noded or nine-noded LAGRANGE element, the presentation of this procedure is abandoned.

3.5.1 Ansatz functions

Contrary to a biquadratic LAGRANGE element, the ansatz functions of the eight-noded serendipity element cannot be generated from one-dimensional LAGRANGE interpolation functions. They are constructed directly by means of the demands on ansatz function $N^i(\boldsymbol{\xi})$ to take on the value one at node i and the value zero at node $j \neq i$ (see equation (3.32)). The generation of principally different ansatz functions of corner nodes and side middle nodes is shown in figure 3.18. Here, as an example we have developed the ansatz functions of corner node one and side middle

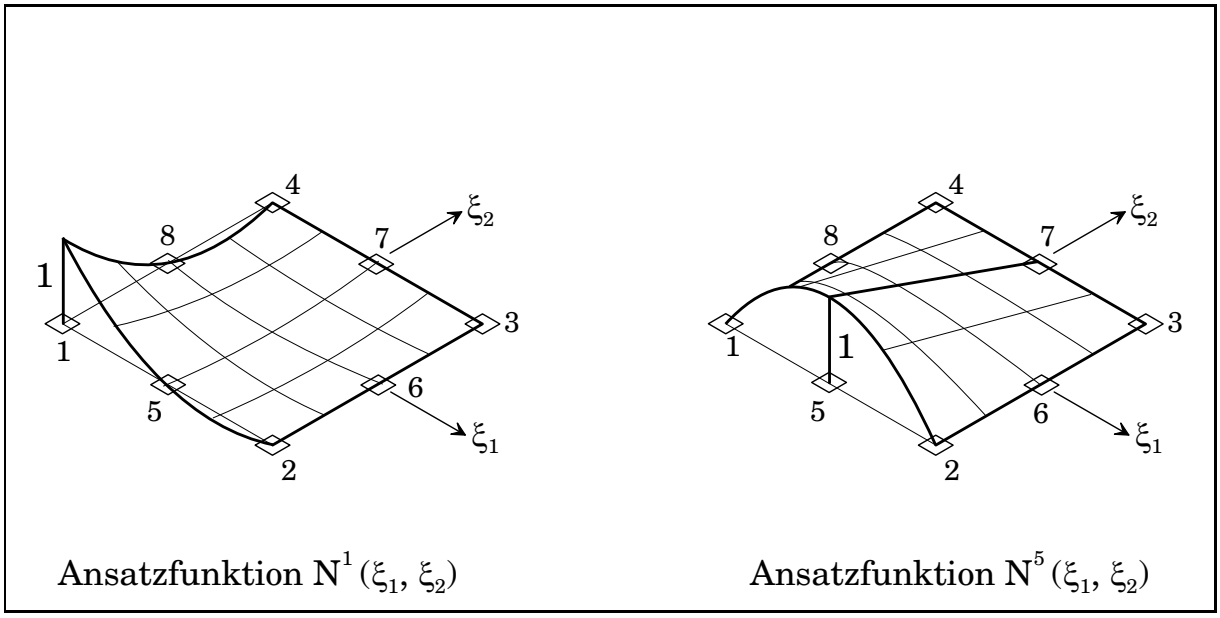


Figure 3.18: Generation of biquadratic serendipity ansatz functions

node eight. Ansatz function $N^1(\boldsymbol{\xi})$ has the value zero at boundaries for which $\xi_1 = 1$ and $\xi_2 = 1$ and at nodes five and eight, at node one the value one. If we expand the condition regarding nodes five and eight and their connecting straight line $\xi_2 = -(\xi_1 + 1)$, apart from the scaling factor c_1, the ansatz $N^1(\boldsymbol{\xi})$ is unequivocally determined.

$$N^1(\boldsymbol{\xi}) = c_1 \quad \underbrace{(1-\xi_1)}_{\text{Kante } \xi_1 = 1} \quad \underbrace{(1-\xi_2)}_{\text{Kante } \xi_2 = 1} \quad \underbrace{(1+\xi_1+\xi_2)}_{\text{Gerade } \xi_2 = -(\xi_1+1)} \tag{3.146}$$

The constant factor c_1 is the result of the condition $N^1(-1,-1) = 1$, and is $c_1 = -1/4$.

$$\begin{aligned} N^1(\boldsymbol{\xi}) &= -\frac{1}{4}\ (1-\xi_1)\ (1-\xi_2)\ (1+\xi_1+\xi_2) \\ &= -\frac{1}{4}\ \left[1 - \xi_1\xi_2 - \xi_1^2 + \xi_1^2\xi_2 + \xi_1\xi_2^2 - \xi_2^2\right] \end{aligned} \tag{3.147}$$

$N^5(\boldsymbol{\xi})$ is featured by ansatz functions that vanish at boundaries $\xi_1 = \pm 1$ and $\xi_2 = 1$. With this, the ansatz function is determined except for constant factor c_1 which can be found from the condition $N^5(0,-1) = 1$. Ansatz function $N^8(\boldsymbol{\xi})$ is found from $N^5(\boldsymbol{\xi})$ by switching places of ξ_1 and ξ_2.

$$\begin{aligned} N^5(\boldsymbol{\xi}) &= c_1\ (1-\xi_1)\ (1+\xi_1)\ (1-\xi_2) \\ &= \frac{1}{2}\ (1-\xi_1)\ (1+\xi_1)\ (1-\xi_2) = \frac{1}{2}\ (1-\xi_1^2)\ (1-\xi_2) \\ &= \frac{1}{2}\ \left[1 - \xi_2 - \xi_1^2 + \xi_1^2\xi_2\right] \\ N^8(\boldsymbol{\xi}) &= \frac{1}{2}\ (1-\xi_1)\ (1-\xi_2^2) \\ &= \frac{1}{2}\ \left[1 - \xi_1 + \xi_1\xi_2^2 - \xi_2^2\right] \end{aligned} \tag{3.148}$$

The terms in ξ_1 and ξ_2 appearing in ansatz functions N^1, N^5 and N^8 are designated by $p = 2$ in the Pascal triangle in figure 3.6. The remaining ansatz functions of a quadratic serendipity

element are written without deriving.

$$\boxed{\begin{aligned}
N^1(\boldsymbol{\xi}) &= -\frac{1}{4}(1-\xi_1)(1-\xi_2)(1+\xi_1+\xi_2) & N^5(\boldsymbol{\xi}) &= \frac{1}{2}(1-\xi_1^2)(1-\xi_2)\\
N^2(\boldsymbol{\xi}) &= -\frac{1}{4}(1+\xi_1)(1-\xi_2)(1-\xi_1+\xi_2) & N^6(\boldsymbol{\xi}) &= \frac{1}{2}(1+\xi_1)(1-\xi_2^2)\\
N^3(\boldsymbol{\xi}) &= -\frac{1}{4}(1+\xi_1)(1+\xi_2)(1-\xi_1-\xi_2) & N^7(\boldsymbol{\xi}) &= \frac{1}{2}(1-\xi_1^2)(1+\xi_2)\\
N^4(\boldsymbol{\xi}) &= -\frac{1}{4}(1-\xi_1)(1+\xi_2)(1+\xi_1-\xi_2) & N^8(\boldsymbol{\xi}) &= \frac{1}{2}(1-\xi_1)(1-\xi_2^2)
\end{aligned}} \tag{3.149}$$

It is obtrusive that ansatz functions of side middle nodes N^5, N^6, N^7 and N^8 are not biquadratic as the LAGRANGE element, but linear in one direction. It is further obtrusive that LAGRANGE and serendipity ansatz functions are identical at boundaries for which $\xi_1 \pm 1$ and $\xi_2 \pm 1$. This claim can be checked with (3.138) and (3.149). Thus, the nine-noded LAGRANGE element and the serendipity element are compatible.

The derivatives with respect to natural coordinates ξ_1 and ξ_2 are :

$$\begin{aligned}
N^1_{;1}(\boldsymbol{\xi}) &= \tfrac{1}{4}(1-\xi_2)(2\xi_1+\xi_2) & N^1_{;2}(\boldsymbol{\xi}) &= \tfrac{1}{4}(1-\xi_1)(\xi_1+2\xi_2)\\
N^2_{;1}(\boldsymbol{\xi}) &= \tfrac{1}{4}(1-\xi_2)(2\xi_1-\xi_2) & N^2_{;2}(\boldsymbol{\xi}) &= \tfrac{1}{4}(1+\xi_1)(-\xi_1+2\xi_2)\\
N^3_{;1}(\boldsymbol{\xi}) &= \tfrac{1}{4}(1+\xi_2)(2\xi_1+\xi_2) & N^3_{;2}(\boldsymbol{\xi}) &= \tfrac{1}{4}(1+\xi_1)(\xi_1+2\xi_2)\\
N^4_{;1}(\boldsymbol{\xi}) &= \tfrac{1}{4}(1+\xi_2)(2\xi_1-\xi_2) & N^4_{;2}(\boldsymbol{\xi}) &= \tfrac{1}{4}(1-\xi_1)(-\xi_1+2\xi_2)
\end{aligned} \tag{3.150}$$

$$\begin{aligned}
N^5_{;1}(\boldsymbol{\xi}) &= -\xi_1(1-\xi_2) & N^5_{;2}(\boldsymbol{\xi}) &= -\tfrac{1}{2}(1-\xi_1^2)\\
N^6_{;1}(\boldsymbol{\xi}) &= \tfrac{1}{2}(1-\xi_2^2) & N^6_{;2}(\boldsymbol{\xi}) &= -(1+\xi_1)\xi_2\\
N^7_{;1}(\boldsymbol{\xi}) &= -\xi_1(1+\xi_2) & N^7_{;2}(\boldsymbol{\xi}) &= \tfrac{1}{2}(1-\xi_1^2)\\
N^8_{;1}(\boldsymbol{\xi}) &= -\tfrac{1}{2}(1-\xi_2^2) & N^8_{;2}(\boldsymbol{\xi}) &= -(1-\xi_1)\xi_2
\end{aligned} \tag{3.150}$$

3.5.2 Geometry

The element coordinate vector $\boldsymbol{X}^e$ has a dimension reduced by two as opposed to the vector $\boldsymbol{X}^e$ of a LAGRANGE element, which corresponds to the coordinates of the middle node.

$$\boldsymbol{X}^e = \begin{bmatrix} \boldsymbol{X}^{e1\,T} & \boldsymbol{X}^{e2\,T} & \boldsymbol{X}^{e3\,T} & \boldsymbol{X}^{e4\,T} & \boldsymbol{X}^{e5\,T} & \boldsymbol{X}^{e6\,T} & \boldsymbol{X}^{e7\,T} & \boldsymbol{X}^{e8\,T} \end{bmatrix}^T \tag{3.151}$$

Geometry of a serendipity element is usually described as function of natural coordinates $\boldsymbol{\xi}$ according to equation (3.42) with the ansatz functions matrix

$$\mathbf{N}^i(\boldsymbol{\xi}) = \begin{bmatrix} N^i(\boldsymbol{\xi}) & 0 \\ 0 & N^i(\boldsymbol{\xi}) \end{bmatrix} \qquad \mathbf{N}(\boldsymbol{\xi}) = \begin{bmatrix} \mathbf{N}^1 & \mathbf{N}^2 & \cdots & \mathbf{N}^7 & \mathbf{N}^8 \end{bmatrix} \tag{3.152}$$

3.5.3 Approximation of element quantities

Element quantities, displacements, accelerations and variation of displacements are approximated in analogy with the LAGRANGE element.

$$\begin{aligned} \boldsymbol{u}(\boldsymbol{\xi}) &\approx \tilde{\boldsymbol{u}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \boldsymbol{u}^e & \boldsymbol{u}^e &= \begin{bmatrix} u_1^{e1} & u_2^{e1} & \cdots & u_1^{e8} & u_2^{e8} \end{bmatrix}^T \\ \delta\boldsymbol{u}(\boldsymbol{\xi}) &\approx \delta\tilde{\boldsymbol{u}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \delta\boldsymbol{u}^e & \delta\boldsymbol{u}^e &= \begin{bmatrix} \delta u_1^{e1} & \delta u_2^{e1} & \cdots & \delta u_1^{e8} & \delta u_2^{e8} \end{bmatrix}^T \\ \ddot{\boldsymbol{u}}(\boldsymbol{\xi}) &\approx \tilde{\ddot{\boldsymbol{u}}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \ddot{\boldsymbol{u}}^e & \ddot{\boldsymbol{u}}^e &= \begin{bmatrix} \ddot{u}_1^{e1} & \ddot{u}_2^{e1} & \cdots & \ddot{u}_1^{e8} & \ddot{u}_2^{e8} \end{bmatrix}^T \end{aligned} \tag{3.153}$$

The dimension of the element vector is by two smaller than that of an element with a middle node. Element vectors $\boldsymbol{u}^e$, $\delta\boldsymbol{u}^e$ and $\ddot{\boldsymbol{u}}^e$ are of dimension 16×1 and the ansatz function matrix $\mathbf{N}$ is of dimension 2×16.

3.6 Triangular plane finite elements

The finite triangular element is of historical importance to the finite element method. The element presented already in 1956 by TURNER, CLOUGH, MARTIN and TOPP (TURNER ET AL. [79]) has been further developed until today in order to upgrade the approximation quality and to apply the basic idea of a triangular element to other element types (plate or shell). The quality improvement in the field of plane elements elaborated here is possible by increasing the degree of ansatz polynomials. In spite of that, the quality of quadrangular elements cannot be reached with an equivalent number of degrees of freedom which is impressively demostrated by linear elements. Regardless of that, the unhindered development of triangular elements can be explained by the fact that arbitrary structures cannot be fully discretized only by quadrangles. This argument is amplified by the development of automatic mesh generators : Mesh generators for triangular meshes are currently more developed than those for quadrangular meshes. Still with an ongoing widening of the use of modern generators that deal only with quadrangles, the significance of triangular elements shall diminish.

As it will be shown, the development of triangular elements is standardized too. Merely the essential point of triangular element development, namely the generation of natural triangular coordinates and formulation of ansatz functions based on these triangular coordinates, makes it a speciality. Due to this the natural triangular coordinates are also an important aspect of this chapter which contains the development of ansatz functions and the generation of element matrices and vectors. The presentation of the development of the primal element according to TURNER ET AL. [79] within the scope of a general hierarchical triangular element rounds off this chapter.

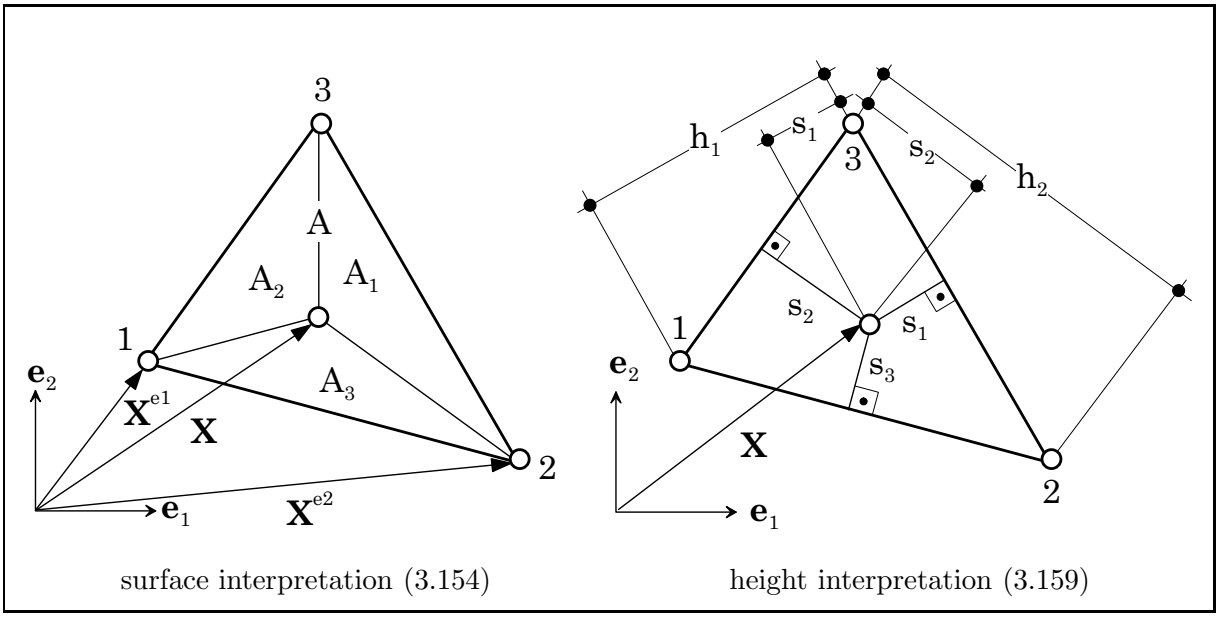


Figure 3.19: Natural coordinates of triangular finite elements

3.6.1 Natural coordinates of a triangle

In the case of the development of finite triangular elements, it is also of great advantage to formulate ansatz functions in natural coordinates on a master or unit triangle since the same ansatz functions can be used with this method for triangles of arbitrary shape and size. Natural coordinates ξ_i for $i = 1, 2, 3$ of an arbitrarily positioned material point $\boldsymbol{X}$ within the triangle are the areas of partial triangles A_i , created by connecting the point with triangle corners, divided by the entire area A (see equation 3.19).

$$\xi_i = \frac{A_i}{A} = \frac{A_i}{A_1 + A_2 + A_3} \qquad i = 1, 2, 3 \tag{3.154}$$

Here, A_i is the partial area of the triangle lying opposite to the node i. Since the sum of partial areas A_i has to be the entire area A, there exists a condition

$$\sum_{i=1}^{3} A_i = \sum_{i=1}^{3} \xi_i \, A = A \qquad \sum_{i=1}^{3} \xi_i = \xi_1 + \xi_2 + \xi_3 = 1 \tag{3.155}$$

of the natural coordinates ξ_i. This means that natural coordinates of a triangle are not independent. One coordinate can be replaced by the remaining two. Without a general restriction, the ξ_3 coordinate here is substituted by coordinates ξ_1 and ξ_2.

$$\xi_3 = \xi_3(\xi_1, \xi_2) = 1 - \xi_1 - \xi_2 \tag{3.156}$$

It remains for us to determine the interval of natural coordinates. In the case that point $\boldsymbol{X}$ corresponds to the position vector, that is coincides with the position of node i, the area A_i is equal to area A and thereby according to equation (3.154) $\xi_i = 1$. If the observed material point is placed at the element edge opposite to the node i, the partial area A_i and hence the natural coordinate is zero ($\xi_i = 0$). In view of that, the triangle can be completely described by natural coordinates varying from zero to one. To now limit $1 \geq \xi_3 \geq 0$ downwards, a conditional equation has to exist according to condition (3.156) for interval limits of natural coordinates ξ_1

and ξ_2 .

$$\xi_i \in [0,1] \qquad \xi_1 + \xi_2 \leq 1 \tag{3.157}$$

If for example the coordinate ξ_2 is chosen within the interval $[0,1]$, the interval of coordinate ξ_1 is found from condition $\xi_1 + \xi_2 \leq 1$.

$$\xi_2 \in [0,1] \qquad \xi_1 \in [0, 1 - \xi_2] \tag{3.158}$$

As an alternative to the interpretation of natural coordinates with the help of partial areas, we can interpret the coordinates ξ_i as distance s_i of a material point from the node lying opposite to the edge i divided by the corresponding height of the triangle. If we define the edge length as l_i and the element height perpendicular to edge i as h_i, as shown in figure 3.19, what follows from equation (3.154) is the alternative interpretation of natural coordinates ξ_i.

$$\xi_i = \frac{A_i}{A} = \frac{1/2\ l_i\ s_i}{1/2\ l_i\ h_i} = \frac{s_i}{h_i} \qquad i = 1,2,3 \tag{3.159}$$

3.6.2 Ansatz functions

Ansatz functions of a linear and of a quadratic plane triangular element can be generated by means of interpolation properties given in equation (3.32)

- - ansatz function $N^i(\boldsymbol{\xi}) = 1$ at node i
- - and ansatz function $N^j(\boldsymbol{\xi}) = 0$ at all other nodes $j \neq i$

or based on the PASCAL triangle and the generalised polynomial (3.30) according to the strategy discussed in chapter 2.2.2 with the help of VANDERMONDE matrix. For the generation of ansatz functions exercised here, a generalized hierarchical approach can be introduced as well based on interpolation properties. As it will be shown in the following sections, the ansatz functions are complete polynomials of the degree one or two, as marked in the PASCAL triangle in figure 3.4

3.6.2.1 Linear ansatz functions

Linear ansatz functions for a unit or a master triangle in natural coordinates are very simple to determine. For example, the ansatz function $N^1(\boldsymbol{\xi})$ shown in figure 3.20 directly comes out as the natural coordinate ξ_1. ξ_1 at node one is equal to one and at other nodes zero. The ansatz functions $N^2(\boldsymbol{\xi})$ and $N^3(\boldsymbol{\xi})$ are obtained in the same way directly in terms of the corresponding natural coordinate, where ξ_3 is replaced by the condition (3.156).

$$\begin{aligned} N^1(\boldsymbol{\xi}) &= \xi_1 \\ N^2(\boldsymbol{\xi}) &= \xi_2 \\ N^3(\boldsymbol{\xi}) &= \xi_3 = 1 - \xi_1 - \xi_2 \end{aligned} \tag{3.160}$$

Ansatz functions contain the terms in the PASCAL triangle designated with $p = 1$. Thus,

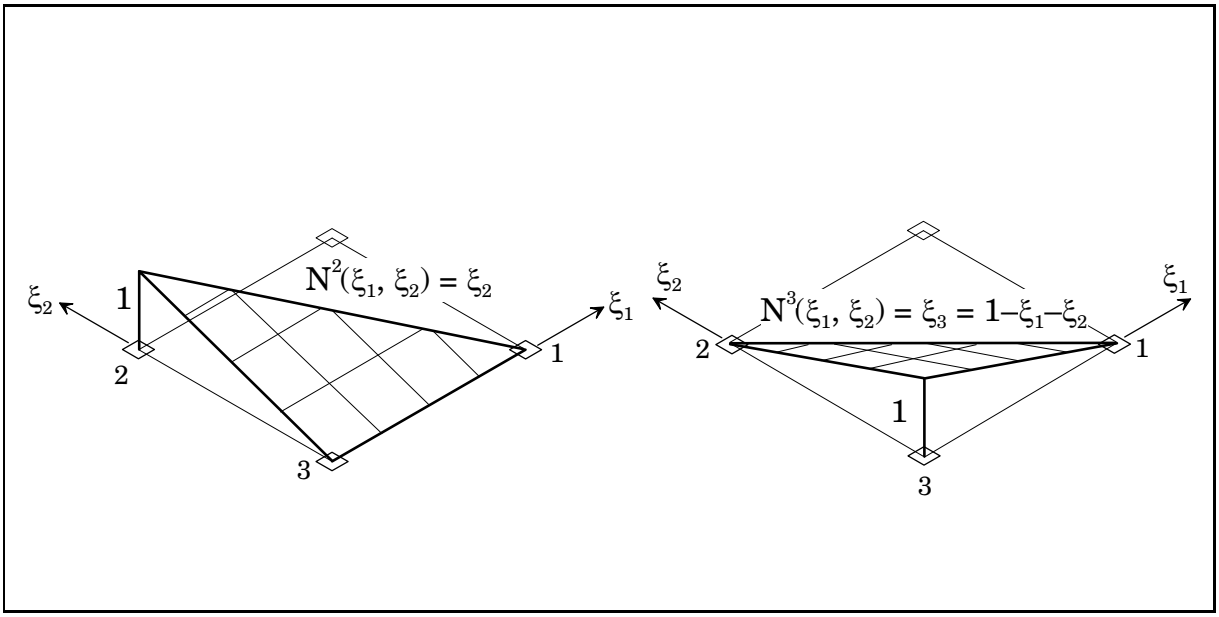


Figure 3.20: Linear ansatz function $N^1(\boldsymbol{\xi})$ of triangular finite elements

in the case of an ansatz of a three-noded finite element, we are talking about a complete two-dimensional polynomial of degree one. The derivatives of ansatz functions with respect to the independent natural coordinates are:

$$\begin{aligned} N^1_{;1}(\boldsymbol{\xi}) &= 1 & N^1_{;2}(\boldsymbol{\xi}) &= 0 \\ N^2_{;1}(\boldsymbol{\xi}) &= 0 & N^2_{;2}(\boldsymbol{\xi}) &= 1 \\ N^3_{;1}(\boldsymbol{\xi}) &= \xi_{3,1} = -1 & N^3_{;2}(\boldsymbol{\xi}) &= \xi_{3,2} = -1 \end{aligned} \tag{3.161}$$

3.6.2.2 Quadratic ansatz functions

Construction of quadratic ansatz functions should be demonstrated by means of the ansatz functions $N^1(\boldsymbol{\xi})$ and $N^4(\boldsymbol{\xi})$ and shown as examples in figure 3.21. The ansatz function $N^1(\boldsymbol{\xi})$ has to be zero at nodes two to six, meaning that conditions $N^1(0, \xi_2) = 0$ (nodes 2,3, and 5) und $N^1(1/2, \xi_2) = 0$ (node 4 and 6) have to be met. This can be fulfilled with the ansatz

$$N^1(\boldsymbol{\xi}) = c_1\ \xi_1\ \left(\xi_1 - \frac{1}{2}\right) = \xi_1\ (2\ \xi_1 - 1) \tag{3.162}$$

where the constant factor c_1 from condition $N^1(1, \xi_2) = 1$ could be determined as $c_1 = 2$. Ansatz function $N^4(\boldsymbol{\xi})$ has to be equal to one at element edges $\xi_1 = 0$ and $\xi_2 = 0$ and for $\xi_1 = \xi_2 = 1/2$.

$$N^4(\boldsymbol{\xi}) = c_1\ \xi_1 \xi_2 = 4\ \xi_1\ \xi_2 \tag{3.163}$$

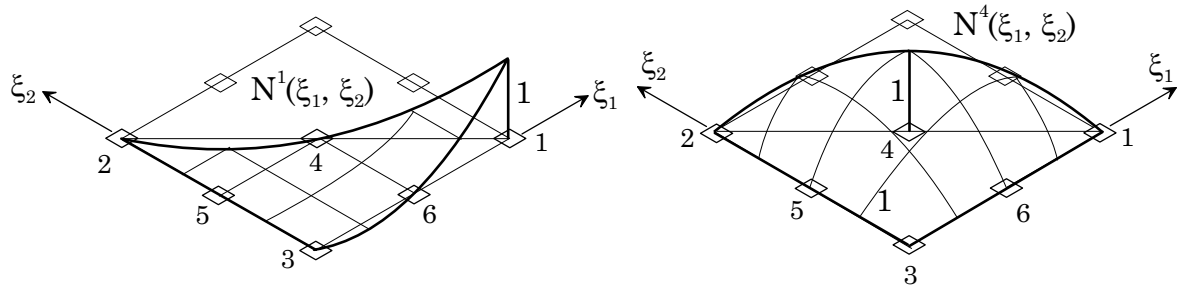


Figure 3.21: Quadratic ansatz functions $N^1(\boldsymbol{\xi})$ and $N^4(\boldsymbol{\xi})$ of triangular finite elements

Further ansatz functions can be constructed likewise having in mind that ξ_3 is replaced by condition (3.156).

$$
\begin{aligned}
N^1(\boldsymbol{\xi}) &= \xi_1(2\xi_1 - 1) & N^4(\boldsymbol{\xi}) &= 4\xi_1\xi_2 \\
N^2(\boldsymbol{\xi}) &= \xi_2(2\xi_2 - 1) & N^5(\boldsymbol{\xi}) &= 4\xi_2\xi_3 = 4\xi_2(1-\xi_1-\xi_2) \\
N^3(\boldsymbol{\xi}) &= \xi_3(2\xi_3 - 1) & N^6(\boldsymbol{\xi}) &= 4\xi_3\xi_1 = 4\xi_1(1-\xi_1-\xi_2) \\
&= (1-\xi_1-\xi_2)(1-2\xi_1-2\xi_2)
\end{aligned}
\tag{3.164}
$$

Completeness of quadratic ansatz polynomials according to Pascale triangle can be verified by multiplying of ansatz function relations in equation (3.164). The derivatives with respect to natural coordinates are the following:

$$
\begin{aligned}
N^1_{;1}(\boldsymbol{\xi}) &= 4\xi_1 - 1 & N^1_{;2}(\boldsymbol{\xi}) &= 0 \\
N^2_{;1}(\boldsymbol{\xi}) &= 0 & N^2_{;2}(\boldsymbol{\xi}) &= 4\xi_2 - 1 \\
N^3_{;1}(\boldsymbol{\xi}) &= 4(\xi_1+\xi_2) - 3 & N^3_{;2}(\boldsymbol{\xi}) &= 4(\xi_1+\xi_2) - 3 \\
N^4_{;1}(\boldsymbol{\xi}) &= 4\xi_2 & N^4_{;2}(\boldsymbol{\xi}) &= 4\xi_1 \\
N^5_{;1}(\boldsymbol{\xi}) &= -4\xi_2 & N^5_{;2}(\boldsymbol{\xi}) &= 4(1-\xi_1-2\xi_2) \\
N^6_{;1}(\boldsymbol{\xi}) &= 4(1-2\xi_1-\xi_2) & N^6_{;2}(\boldsymbol{\xi}) &= -4\xi_1
\end{aligned}
\tag{3.165}
$$

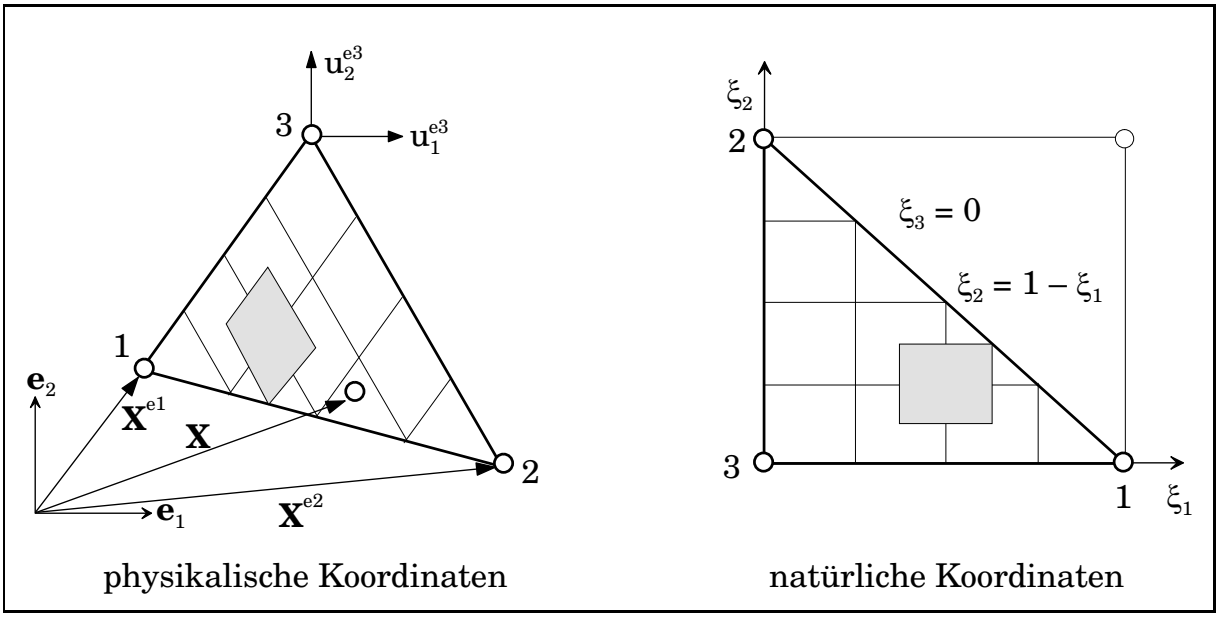


Figure 3.22: Linear triangular element (*Constant Strain Triangle*) in physical and natural coordinates

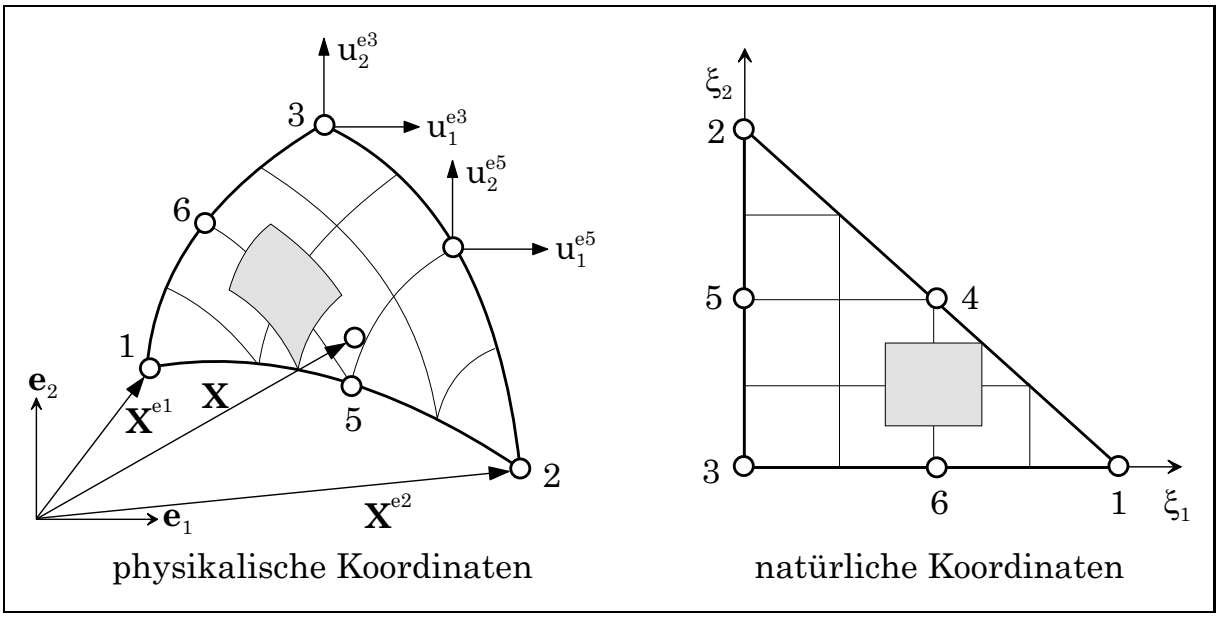


Figure 3.23: Quadratic triangular element in physical and natural coordinates

3.6.3 Isoparametric approximation of continuous quantities

The proceeding to illustrate the geometry in physical space (see figures 3.22 and 3.23) and to approximate displacements and accelerations is standardized in the isoparametric concept of an element. For an element from a family of hierarchical elements with NN element nodes we get:

$$
\begin{aligned}
\boldsymbol{X}(\boldsymbol{\xi}) &\approx \tilde{\boldsymbol{X}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \boldsymbol{X}^e & \boldsymbol{X}^e &= \left[\ X_1^{e1}\ X_2^{e1}\ \cdots\ X_1^{eNN}\ X_2^{eNN}\ \right]^T \\
\boldsymbol{u}(\boldsymbol{\xi}) &\approx \tilde{\boldsymbol{u}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \boldsymbol{u}^e & \boldsymbol{u}^e &= \left[\ u_1^{e1}\ u_2^{e1}\ \cdots\ u_1^{eNN}\ u_2^{eNN}\ \right]^T \\
\delta\boldsymbol{u}(\boldsymbol{\xi}) &\approx \delta\tilde{\boldsymbol{u}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \delta\boldsymbol{u}^e & \delta\boldsymbol{u}^e &= \left[\ \delta u_1^{e1}\ \delta u_2^{e1}\ \cdots\ \delta u_1^{eNN}\ \delta u_2^{eNN}\ \right]^T \\
\ddot{\boldsymbol{u}}(\boldsymbol{\xi}) &\approx \tilde{\ddot{\boldsymbol{u}}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \ddot{\boldsymbol{u}}^e & \ddot{\boldsymbol{u}}^e &= \left[\ \ddot{u}_1^{e1}\ \ddot{u}_2^{e1}\ \cdots\ \ddot{u}_1^{eNN}\ \ddot{u}_2^{eNN}\ \right]^T
\end{aligned}
\tag{3.166}
$$

where the ansatz function matrix is formulated also in a general manner.

$$
\mathbf{N}^i(\boldsymbol{\xi}) = \begin{bmatrix} N^i(\boldsymbol{\xi}) & 0 \\ 0 & N^i(\boldsymbol{\xi}) \end{bmatrix} \qquad \mathbf{N}(\boldsymbol{\xi}) = \left[\ \mathbf{N}^1(\boldsymbol{\xi})\ \ \cdots\ \ \mathbf{N}^{NN}(\boldsymbol{\xi})\ \right] \tag{3.167}
$$

3.6.4 Element matrices and vectors

With advancing development steps of triangular elements, a status has been reached that can be worked off in an almost standardized procedure all the way to calculation of element matrices and vectors. After the physical coordinates were described with equation (3.166) as function of natural coordinates, the JACOBI matrix $\mathbf{J}(\boldsymbol{\xi})$, its inverse $\mathbf{J}^{-1}(\boldsymbol{\xi})$ and its determinant $|\mathbf{J}(\boldsymbol{\xi})|$ can be computed according to equations (3.46), (3.48) and (3.49). With components of inverse JACOBI matrix $\mathbf{J}^{-1}(\boldsymbol{\xi})$ in the sequel it is possible to determine the differential operator $\mathbf{D}_{\varepsilon\xi}(\boldsymbol{\xi})$ in natural coordinates according to equation (3.60). The application of this operator to the matrix of ansatz functions $\mathbf{N}(\boldsymbol{\xi})$ yields according to equation (3.61), the B-operator $\mathbf{B}(\boldsymbol{\xi})$ of a generalised hierarchical triangular element the form of which is given in equation (3.145). Only the assembling of the B-operator $\mathbf{B}_i(\boldsymbol{\xi})$ has to be adjusted to the number of nodes of a triangular element.

$$
\mathbf{B}(\boldsymbol{\xi}) = \left[\ \mathbf{B}_1\ \ \cdots\ \ \mathbf{B}_{NN}\ \right] \tag{3.168}
$$

Thus, the prerequisites are created for generating the stiffness matrix, the mass matrix and the consistent load vector of an element. Compared to the quadrangular elements (equations (3.71), (3.75), (3.77) and (3.91)) in order to integrate the quantities of triangular elements the altered intervals of natural coordinates given in equations (3.158) have to be considered. In accordance with that one has to integrate from 0 to 1 in direction ξ_2 and from 0 to $1-\xi_2$ in direction ξ_1 due to bounds of a triangle surface $\xi_3 = 1-\xi_1-\xi_2 = 0$. The corresponding modification of integration boundaries in equation (3.71) eventually results in the stiffness matrix of a triangular element. [2].

$$
\mathbf{k}^e = \int_0^1 \int_0^{1-\xi_2} \mathbf{B}^T(\boldsymbol{\xi})\ \mathbf{C}\ \mathbf{B}(\boldsymbol{\xi})\ |\mathbf{J}(\boldsymbol{\xi})|\ h\ d\xi_1 d\xi_2 \tag{3.169}
$$

[2] The inverted order of integration; in ξ_1 direction from 0 to 1 and in ξ_2 direction from 0 to $1-\xi_1$ gives identical element quantities, of course

The mass matrix of a generalised triangular element is found by modification of integration boundaries in equation (3.75)

$$\mathbf{m}^e = \int_0^1 \int_0^{1-\xi_2} \mathbf{N}^T(\boldsymbol{\xi})\, \mathbf{N}(\boldsymbol{\xi})\, |\mathbf{J}(\boldsymbol{\xi})|\, \rho\, h\, d\xi_1 d\xi_2 \tag{3.170}$$

Also, the vector of consistent equivalent loads can be found with equation (3.77) as result of acting volume loads.

$$\boldsymbol{r}_p^e = \int_0^1 \int_0^{1-\xi_2} \mathbf{N}^T(\boldsymbol{\xi})\, \boldsymbol{b}(\boldsymbol{\xi})\, |\mathbf{J}(\boldsymbol{\xi})|\, \rho\, h\, d\xi_1 d\xi_2 \tag{3.171}$$

The consistent equivalent loads of a boundary load on the exemplary chosen element edge two (opposite to the node two determined by $\xi_2 = 0$) can be generated with the JACOBI determinant $|\mathbf{J}_2(\xi_1, 0)|$ calculated according to equation (3.83) with corresponding physical coordinates (equation (3.166)).

$$\boldsymbol{r}_{n2}^e = \int_0^1 \mathbf{N}^T(\xi_1, 0)\, \boldsymbol{t}^\star(\xi_1, 0)\, h\, |\mathbf{J}_2(\xi_1, 0)|\, d\xi_1 \tag{3.172}$$

As in the case of integrals over the element surface, the integration boundaries of natural triangular coordinates have to be borne in mind.

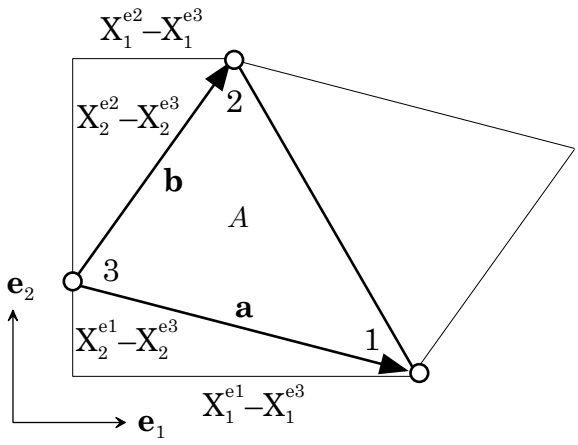
The calculation of integrals in equations (3.169) to (3.172) is performed with the help of GAUSS-LEGENDRE integration, in the way it shall be elaborated in chapter 3.7. A special part here plays the integration of consistent equivalent loads of boundary loads $\boldsymbol{r}_{n1}^e$. The integration is performed one-dimensionally which is enabled by the GAUSS-LEGENDRE integration already presented in chapter 2.2.6 where the choice of GAUSS points requires the allowing for natural triangular coordinates.

3.6.5 Constant Strain Triangle

Not solely due to its historical importance does the Constant Strain Triangle assume a special position among the finite elements as the first ever published finite continuum element. The primal element presented in 1956 by TURNER, CLOUGH, MARTIN and TOPP (TURNER ET AL. [79]) is of great value also for its didactic reasons since owing to this element the, development of a finite triangular element is impressively put on display and the approximation of physical states can be carried out with numerical methods.

For development of the *Constant Strain Triangle*, we use linear ansatz functions (3.160) in natural coordinates ξ_1 and ξ_2. An arbitrarily chosen material point within the three-noded element can be characterised by linear functions with the help of equations (3.166) und (3.167) for $NN = 3$ with the following physical coordinates X_β for $\beta = 1, 2$.

$$\begin{aligned} X_\beta &= N^1(\xi_1, \xi_2)\, X_\beta^{e1} + N^2(\xi_1, \xi_2)\, X_\beta^{e2} + N^3(\xi_1, \xi_2)\, X_\beta^{e3} \\ &= \xi_1\, X_\beta^{e1} + \xi_2\, X_\beta^{e2} + (1 - \xi_1 - \xi_2)\, X_\beta^{e3} \end{aligned} \tag{3.173}$$



Definition of vectors $\boldsymbol{a}$ und $\boldsymbol{b}$

$$\boldsymbol{a} = \begin{bmatrix} X_1^{e1} - X_1^{e3} \\ X_2^{e2} - X_2^{e3} \\ 0 \end{bmatrix} \qquad \boldsymbol{b} = \begin{bmatrix} X_1^{e2} - X_1^{e3} \\ X_2^{e2} - X_2^{e3} \\ 0 \end{bmatrix}$$

Calculation of triangle area A

$$\begin{aligned} A &= \tfrac{1}{2}\,|\boldsymbol{a} \times \boldsymbol{b}| = \tfrac{1}{2}\,(a_1 b_2 - a_2 b_1) \\ 2A &= \left(X_1^{e1} - X_1^{e3}\right)\left(X_2^{e2} - X_2^{e3}\right) \\ &\quad - \left(X_1^{e2} - X_1^{e3}\right)\left(X_2^{e1} - X_2^{e3}\right) \end{aligned}$$

Figure 3.24: Jacobi determinant of *Constant Strain Triangle*

Jacobi matrix components must be computed according to equation (3.46) with derivatives of ansatz functions according to (3.161).

$$\frac{\partial X_\beta}{\partial \xi_1} = X_\beta^{e1} - X_\beta^{e3} \qquad\qquad \frac{\partial X_\beta}{\partial \xi_2} = X_\beta^{e2} - X_\beta^{e3} \tag{3.174}$$

The Jacobi determinant is given according to equation (3.49).

$$|\mathbf{J}| = \left(X_1^{e1} - X_1^{e3}\right)\left(X_2^{e2} - X_2^{e3}\right) - \left(X_1^{e2} - X_1^{e3}\right)\left(X_2^{e1} - X_2^{e3}\right) = 2\,A \tag{3.175}$$

The Jacobi determinant equals the twofold triangle area which can be concluded from observations in figure 3.24. Components of the inverse Jacobi matrix can be obtained according to equation (3.48), similar to generating quadrangular elements.

$$\begin{aligned} \frac{\partial \xi_1}{\partial X_1} &= \frac{1}{|\mathbf{J}|}\frac{\partial X_2}{\partial \xi_2} = \frac{X_2^{e2} - X_2^{e3}}{2A} & \frac{\partial \xi_2}{\partial X_1} &= -\frac{1}{|\mathbf{J}|}\frac{\partial X_2}{\partial \xi_1} = \frac{X_2^{e3} - X_2^{e1}}{2A} \\ \frac{\partial \xi_1}{\partial X_2} &= -\frac{1}{|\mathbf{J}|}\frac{\partial X_1}{\partial \xi_2} = \frac{X_1^{e3} - X_1^{e2}}{2A} & \frac{\partial \xi_2}{\partial X_2} &= \frac{1}{|\mathbf{J}|}\frac{\partial X_1}{\partial \xi_1} = \frac{X_1^{e1} - X_1^{e3}}{2A} \end{aligned} \tag{3.176}$$

It is obtrusive that both the Jacobi determinant as well as the Jacobi matrix are independent of natural coordinates. These quantities are constant throughout the element surface. If we develope derivative transformations in differential operator $\mathbf{D}_{\varepsilon\xi}$ according to (3.60) and applied them to ansatz matrix $\mathbf{N}^i(\boldsymbol{\xi})$ of node i according to equation (3.167), the result is the general B-operator of node i.

$$\mathbf{D}_{\varepsilon\xi}\,\mathbf{N}^i(\boldsymbol{\xi}) = \mathbf{B}_i = \begin{bmatrix} \frac{\partial \xi_1}{\partial X_1} N_{;1}^i + \frac{\partial \xi_2}{\partial X_1} N_{;2}^i & 0 \\ 0 & \frac{\partial \xi_1}{\partial X_2} N_{;1}^i + \frac{\partial \xi_2}{\partial X_2} N_{;2}^i \\ \frac{\partial \xi_1}{\partial X_2} N_{;1}^i + \frac{\partial \xi_2}{\partial X_2} N_{;2}^i & \frac{\partial \xi_1}{\partial X_1} N_{;1}^i + \frac{\partial \xi_2}{\partial X_1} N_{;2}^i \end{bmatrix} \tag{3.177}$$

and with derivatives of ansatz functions according to equation (3.161), the result is the differential operator of the *Constant Strain Triangle.*

$$\mathbf{B} = \begin{bmatrix} \frac{\partial \xi_1}{\partial X_1} & 0 & \frac{\partial \xi_2}{\partial X_1} & 0 & -\frac{\partial \xi_1}{\partial X_1} - \frac{\partial \xi_2}{\partial X_1} & 0 \\ 0 & \frac{\partial \xi_1}{\partial X_2} & 0 & \frac{\partial \xi_2}{\partial X_2} & 0 & -\frac{\partial \xi_1}{\partial X_2} - \frac{\partial \xi_2}{\partial X_2} \\ \frac{\partial \xi_1}{\partial X_2} & \frac{\partial \xi_1}{\partial X_1} & \frac{\partial \xi_2}{\partial X_2} & \frac{\partial \xi_2}{\partial X_1} & -\frac{\partial \xi_1}{\partial X_2} - \frac{\partial \xi_2}{\partial X_2} & -\frac{\partial \xi_1}{\partial X_1} - \frac{\partial \xi_2}{\partial X_1} \end{bmatrix} \tag{3.178}$$

Since the components of the inverse JACOBI matrix $\partial \xi_\alpha / \partial X_\beta$ according to (3.176) are constant, the B-operator and consequently also the approximated strain vector independent of triangular coordinates are constant too. In a linear triangular element the strain vector approximation is constant throughout the element surface. If we fill equations (3.176) into equation (3.178), we get the differential operator dependent only on node coordinates (3.178) of a triangular element.

$$\mathbf{B} = \frac{1}{2A} \begin{bmatrix} X_2^{e2} - X_2^{e3} & 0 & X_2^{e3} - X_2^{e1} & 0 & X_2^{e1} - X_2^{e2} & 0 \\ 0 & X_1^{e3} - X_1^{e2} & 0 & X_1^{e1} - X_1^{e3} & 0 & X_1^{e2} - X_1^{e1} \\ X_1^{e3} - X_1^{e2} & X_2^{e2} - X_2^{e3} & X_1^{e1} - X_1^{e3} & X_2^{e3} - X_2^{e1} & X_1^{e2} - X_1^{e1} & X_2^{e1} - X_2^{e2} \end{bmatrix} \tag{3.179}$$

For the development of element stiffness matrix $\mathbf{k}^e$ according to (3.169), it follows from the constancy of the B-operator and from the JACOBI determinant that the term $\mathbf{B}^T\mathbf{C}\ \mathbf{B}\ |\mathbf{J}| = \mathbf{B}^T\mathbf{C}\ \mathbf{B}\ 2A$ is constant as well, and thereby can be extracted from the integral. If the thickness is constant as well, only the integral has to be computed

$$\int_0^1 \int_0^{1-\xi_2} d\xi_1 d\xi_2 = \frac{1}{2} \tag{3.180}$$

which yields the area of a unit or a master triangle.

$$\mathbf{k}^e = \mathbf{B}^T\ \mathbf{C}\ \mathbf{B}\ |\mathbf{J}|\ h \int_0^1 \int_0^{1-\xi_2} d\xi_1 d\xi_2 = \mathbf{B}^T\ \mathbf{C}\ \mathbf{B}\ 2\ A\ h\ \frac{1}{2} = A\ h\ \mathbf{B}^T\ \mathbf{C}\ \mathbf{B} \tag{3.181}$$

In the case of plane stress state characterised by the material matrix $\mathbf{C}^{es}$ according to equation (3.14), after partitioning of the symmetric stiffness matrix

$$\mathbf{k}^e = \begin{bmatrix} \mathbf{k}_{11}^e & \mathbf{k}_{12}^e & \mathbf{k}_{13}^e \\ \mathbf{k}_{21}^e & \mathbf{k}_{22}^e & \mathbf{k}_{23}^e \\ \mathbf{k}_{31}^e & \mathbf{k}_{32}^e & \mathbf{k}_{33}^e \end{bmatrix} \qquad \mathbf{k}_{ij}^e = \mathbf{k}_{ji}^{e\,T} \tag{3.182}$$

the stiffness matrix components can be calculated.

$$
\begin{aligned}
\mathbf{k}_{11}^e &= c_1 \begin{bmatrix} \left(X_2^{e2}-X_2^{e3}\right)^2 + \frac{1-\nu}{2}\left(X_1^{e3}-X_1^{e2}\right)^2 & \frac{1+\nu}{2}\left(X_2^{e2}-X_2^{e3}\right)\left(X_1^{e3}-X_1^{e2}\right) \\ \cdots & \left(X_1^{e3}-X_1^{e2}\right)^2 + \frac{1-\nu}{2}\left(X_2^{e2}-X_2^{e3}\right)^2 \end{bmatrix} \\
\mathbf{k}_{22}^e &= c_1 \begin{bmatrix} \left(X_2^{e3}-X_2^{e1}\right)^2 + \frac{1-\nu}{2}\left(X_1^{e1}-X_1^{e3}\right)^2 & \frac{1+\nu}{2}\left(X_2^{e3}-X_2^{e1}\right)\left(X_1^{e1}-X_1^{e3}\right) \\ \cdots & \left(X_1^{e1}-X_1^{e3}\right)^2 + \frac{1-\nu}{2}\left(X_2^{e3}-X_2^{e1}\right)^2 \end{bmatrix} \\
\mathbf{k}_{33}^e &= c_1 \begin{bmatrix} \left(X_2^{e1}-X_2^{e2}\right)^2 + \frac{1-\nu}{2}\left(X_1^{e2}-X_1^{e1}\right)^2 & \frac{1+\nu}{2}\left(X_2^{e1}-X_2^{e2}\right)\left(X_1^{e2}-X_1^{e1}\right) \\ \cdots & \left(X_1^{e2}-X_1^{e1}\right)^2 + \frac{1-\nu}{2}\left(X_2^{e1}-X_2^{e2}\right)^2 \end{bmatrix}
\end{aligned}
\tag{3.183}
$$

$$
\begin{aligned}
\mathbf{k}_{12}^e &= c_1 \begin{bmatrix} \begin{matrix}\left(X_2^{e2}-X_2^{e3}\right)\left(X_2^{e3}-X_2^{e1}\right) \\ +\frac{1-\nu}{2}\left(X_1^{e3}-X_1^{e2}\right)\left(X_1^{e1}-X_1^{e3}\right)\end{matrix} & \begin{matrix}\nu\left(X_2^{e2}-X_2^{e3}\right)\left(X_1^{e1}-X_1^{e3}\right) \\ +\frac{1-\nu}{2}\left(X_1^{e3}-X_1^{e2}\right)\left(X_2^{e3}-X_2^{e1}\right)\end{matrix} \\ \begin{matrix}\nu\left(X_1^{e3}-X_1^{e2}\right)\left(X_2^{e3}-X_2^{e1}\right) \\ +\frac{1-\nu}{2}\left(X_1^{e1}-X_1^{e3}\right)\left(X_2^{e2}-X_2^{e3}\right)\end{matrix} & \begin{matrix}\left(X_1^{e3}-X_1^{e2}\right)\left(X_1^{e1}-X_1^{e3}\right) \\ +\frac{1-\nu}{2}\left(X_2^{e2}-X_2^{e3}\right)\left(X_2^{e3}-X_2^{e1}\right)\end{matrix} \end{bmatrix} \\
\mathbf{k}_{13}^e &= c_1 \begin{bmatrix} \begin{matrix}\left(X_2^{e1}-X_2^{e2}\right)\left(X_2^{e2}-X_2^{e3}\right) \\ +\frac{1-\nu}{2}\left(X_1^{e2}-X_1^{e1}\right)\left(X_1^{e3}-X_1^{e2}\right)\end{matrix} & \begin{matrix}\nu\left(X_2^{e2}-X_2^{e3}\right)\left(X_1^{e2}-X_1^{e1}\right) \\ +\frac{1-\nu}{2}\left(X_1^{e3}-X_1^{e2}\right)\left(X_2^{e1}-X_2^{e2}\right)\end{matrix} \\ \begin{matrix}\nu\left(X_2^{e1}-X_2^{e2}\right)\left(X_1^{e3}-X_1^{e2}\right) \\ +\frac{1-\nu}{2}\left(X_1^{e2}-X_1^{e1}\right)\left(X_2^{e2}-X_2^{e3}\right)\end{matrix} & \begin{matrix}\left(X_1^{e2}-X_1^{e1}\right)\left(X_1^{e3}-X_1^{e2}\right) \\ +\frac{1-\nu}{2}\left(X_2^{e1}-X_2^{e2}\right)\left(X_2^{e2}-X_2^{e3}\right)\end{matrix} \end{bmatrix} \\
\mathbf{k}_{23}^e &= c_1 \begin{bmatrix} \begin{matrix}\left(X_2^{e3}-X_2^{e1}\right)\left(X_2^{e1}-X_2^{e2}\right) \\ +\frac{1-\nu}{2}\left(X_1^{e1}-X_1^{e3}\right)\left(X_1^{e2}-X_1^{e1}\right)\end{matrix} & \begin{matrix}\nu\left(X_1^{e2}-X_1^{e1}\right)\left(X_2^{e3}-X_2^{e1}\right) \\ +\frac{1-\nu}{2}\left(X_1^{e1}-X_1^{e3}\right)\left(X_2^{e1}-X_2^{e2}\right)\end{matrix} \\ \begin{matrix}\nu\left(X_1^{e1}-X_1^{e3}\right)\left(X_2^{e1}-X_2^{e2}\right) \\ +\frac{1-\nu}{2}\left(X_1^{e2}-X_1^{e1}\right)\left(X_2^{e3}-X_2^{e1}\right)\end{matrix} & \begin{matrix}\left(X_1^{e2}-X_1^{e1}\right)\left(X_1^{e1}-X_1^{e3}\right) \\ +\frac{1-\nu}{2}\left(X_2^{e1}-X_2^{e2}\right)\left(X_2^{e3}-X_2^{e1}\right)\end{matrix} \end{bmatrix}
\end{aligned}
\tag{3.183}
$$

The abreviation c_1 is defined as:

$$
c_1 = \frac{Eh}{4A(1-\nu^2)} \tag{3.184}
$$

Integration of the mass matrix $\mathbf{m}^e$ and of the consistent equivalent loads $\boldsymbol{r}_p^e$ and $\boldsymbol{r}_n^e$ is in its form only slightly more complex than that of the stiffness matrix since the matrix of ansatz functions $\mathbf{N}(\boldsymbol{\xi})$ varies with the position on the element surface. This means that the calculation of $\mathbf{m}^e$, $\boldsymbol{r}_p^e$ and $\boldsymbol{r}_n^e$ is standardised according to equations (3.170), (3.171) and (3.172) where the necessary integration over the unit triangle can be performed analytically and free of difficulties. The mass matrix $\mathbf{m}^e$ can be integrated with equations (3.170), (3.175) and with ansatz functions according to equation (3.160). The term $|\mathbf{J}(\boldsymbol{\xi})|\ \rho\ h = 2\ A\ \rho\ h$ can be removed from the integral and the mass matrix $\mathbf{m}^e$ can be partitioned in analogy with the stiffness matrix in equation (3.182).

$$
\mathbf{m}_{ij}^e = 2\ A\ h\ \rho \int_0^1 \int_0^{1-\xi_2} \begin{bmatrix} N^i(\boldsymbol{\xi})\ N^j(\boldsymbol{\xi}) & 0 \\ 0 & N^i(\boldsymbol{\xi})\ N^j(\boldsymbol{\xi}) \end{bmatrix} d\xi_1 d\xi_2 \tag{3.185}
$$

The integrations generally yield

$$\int_0^1 \int_0^{1-\xi_2} N^i(\boldsymbol{\xi})\, N^j(\boldsymbol{\xi})\, d\xi_1 d\xi_2 = \begin{cases} \frac{1}{12} & \forall \quad i = j \\ \frac{1}{24} & \forall \quad i \neq j \end{cases} \tag{3.186}$$

and specifically, shown for integrands $N^1(\boldsymbol{\xi})\, N^1(\boldsymbol{\xi})$ and $N^1(\boldsymbol{\xi})\, N^3(\boldsymbol{\xi})$:

$$\begin{aligned} \int_0^1 \int_0^{1-\xi_2} N^1(\boldsymbol{\xi})\, N^1(\boldsymbol{\xi})\, d\xi_1 d\xi_2 &= \int_0^1 \int_0^{1-\xi_2} {\xi_1}^2\, d\xi_1 d\xi_2 = \frac{1}{3} \int_0^1 (1-\xi_2)^3\, d\xi_2 \\ &= \frac{1}{3} \int_0^1 (1 - 3\xi_2 + 3{\xi_2}^2 - {\xi_2}^3)\, d\xi_2 = \frac{1}{3}\left(1 - \frac{3}{2} + 1 - \frac{1}{4}\right) = \frac{1}{12} \\ \int_0^1 \int_0^{1-\xi_2} N^1(\boldsymbol{\xi})\, N^3(\boldsymbol{\xi})\, d\xi_1 d\xi_2 &= \int_0^1 \int_0^{1-\xi_2} \xi_1\ (1 - \xi_1 - \xi_2)\ d\xi_1 d\xi_2 \\ &= \frac{1}{6} \int_0^1 \left(1 - 3\xi_2 + 3\xi_2^2 - \xi_2^3\right)\ d\xi_2 = \frac{1}{6}\left(1 - \frac{3}{2} + 1 - \frac{1}{4}\right) = \frac{1}{24} \end{aligned} \tag{3.187}$$

Thus, we developed the mass matrix $\mathbf{m}^e$ of a *Constant Strain Triangle.*

$$\mathbf{m}^e = \frac{1}{12}\, A\, h\, \rho \begin{bmatrix} 2 & 0 & 1 & 0 & 1 & 0 \\ & 2 & 0 & 1 & 0 & 1 \\ & & 2 & 0 & 1 & 0 \\ & & & 2 & 0 & 1 \\ & & & & 2 & 0 \\ \text{sym} & & & & & 2 \end{bmatrix} \tag{3.188}$$

By application of equations (3.171) and (3.175) and of ansatz functions according to equation (3.160), for the consistent equivalent volume load vector of volume loads we get

$$\boldsymbol{r}_p^e = \frac{1}{3}\, A\, h\, \rho \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}^T \mathbf{b} \tag{3.189}$$

3.7 Numerical integration

The developments of plane finite elements in the previous chapter mostly ended with integral expressions defining the stiffness matrix, the mass matrix or the consistent loading vectors of an element. In order to conduct the finite element method, it is necessary to solve the respective integral expressions. It is recommended to apply numerical integration, for the analytical integration is very demanding or not at all possible for the entire shape diversity of an element type. The one-dimensional numerical GAUSS LEGENDRE integration was already presented in sections where derivations regarding a spatial truss took place, although those integrals were absolutely soluble analytically. This method should be expanded here only for solutions of two-dimensional integrals. The supporting points and weighting factors of GAUSS-LEGENDRE integration of element matrices and vectors

$n \times n$	$2n-1$	$f(\xi_1, \xi_2)$	ξ_1^i	ξ_2^j	α^i, α^j
1×1	1	$\times$	$\xi_1^1 = 0$	$\xi_2^1 = 0$	$\alpha^1 = 2$
2×2	3	$\times \quad \times$ $\times \quad \times$	$\xi_1^1 = -1/\sqrt{3}$ $\xi_1^2 = 1/\sqrt{3}$	$\xi_2^1 = -1/\sqrt{3}$ $\xi_2^2 = 1/\sqrt{3}$	$\alpha^1 = 1$ $\alpha^2 = 1$
3×3	5	$\times \quad \times \quad \times$ $\times \quad \times \quad \times$ $\times \quad \times \quad \times$	$\xi_1^1 = -\sqrt{3/5}$ $\xi_1^2 = 0$ $\xi_1^3 = \sqrt{3/5}$	$\xi_2^1 = -\sqrt{3/5}$ $\xi_2^2 = 0$ $\xi_2^3 = \sqrt{3/5}$	$\alpha^1 = 5/9$ $\alpha^2 = 8/9$ $\alpha^3 = 5/9$
4×4	7	$\times \quad \times \quad \times \quad \times$ $\times \quad \times \quad \times \quad \times$ $\times \quad \times \quad \times \quad \times$ $\times \quad \times \quad \times \quad \times$	$\xi_1^1 = -0.86114$ $\xi_1^2 = -0.33998$ $\xi_1^3 = 0.33998$ $\xi_1^4 = 0.86114$	$\xi_2^1 = -0.86114$ $\xi_2^2 = -0.33998$ $\xi_2^3 = 0.33998$ $\xi_2^4 = 0.86114$	$\alpha^1 = 0.34785$ $\alpha^2 = 0.65241$ $\alpha^3 = 0.65241$ $\alpha^4 = 0.34785$

Table 3.2: GAUSS points ξ_1^i, ξ_2^j and weighting factors α^i, α^j of GAUSS-LEGENDRE quadrature

- of quadrangular elements with $\xi_1 \in [-1, 1]$, $\xi_2 \in [-1, 1]$
- and of triangular elements with $\xi_2 \in [0, 1]$, $\xi_1 \in [0, 1 - \xi_2]$

are specifically made available with different degrees of accuracy and numerical effort.

3.7.1 Quadrangular elements

To integrate a function $f(\xi_1, \xi_2)$ over a master or a unit square $\xi_1 \in [-1, 1]$, $\xi_2 \in [-1, 1]$, it is possible to apply the one-dimensional GAUSS-LEGENDRE quadrature according to equation (2.108) successively to the surface integral. If the identical number of GAUSS points n for numeric integration in ξ_1 and ξ_2 direction is used, we get the following approximation of the integral.

$$\int_{-1}^{1} \int_{-1}^{1} f(\xi_1, \xi_2) \, d\xi_1 \, d\xi_2 = \sum_{j=1}^{n} \alpha^j \int_{-1}^{1} f(\xi_1, \xi_2^j) \, d\xi_1 = \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha^i \, \alpha^j \, f(\xi_1^i, \xi_2^j) \tag{3.190}$$

The supporting points ξ_1^i, ξ_2^j and the weighting factors α^i and α^j for one-dimensional GAUSS-LEGENDRE integration should be inferred from table 2.2 accordingly. A summary of these GAUSS points and weighting factors specifically for integration over the master square, at which the natural coordinates ξ_1 and ξ_2 are placed in the interval $[-1, 1]$, is given in table 3.2. The integration of one of the two natural coordinates when applying n GAUSS points yields an exact integration of the polynomial of degree $p = 2n - 1$. If one numerically integrated over the surface of a master square, it would require n^2 GAUSS points to accurately analyse a bipolynomial of degree $p = 2n - 1$.

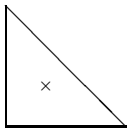
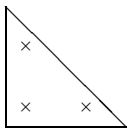
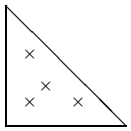
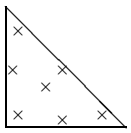
n	p	$f(\xi_1, \xi_2)$	ξ_1^i	ξ_2^j	α^i
1	1		$\xi_1^1 = 1/3$	$\xi_2^1 = 1/3$	$\alpha^1 = 1$
3	2		$\xi_1^1 = 1/6$ $\xi_1^2 = 2/3$ $\xi_1^3 = 1/6$	$\xi_2^1 = 1/6$ $\xi_2^2 = 1/6$ $\xi_2^3 = 2/3$	$\alpha^1 = 1/3$ $\alpha^2 = 1/3$ $\alpha^3 = 1/3$
4	3		$\xi_1^1 = 1/3$ $\xi_1^2 = 3/5$ $\xi_1^3 = 1/5$ $\xi_1^4 = 1/5$	$\xi_2^2 = 1/3$ $\xi_2^2 = 1/5$ $\xi_2^3 = 3/5$ $\xi_2^4 = 1/5$	$\alpha^1 = -27/48$ $\alpha^2 = 25/48$ $\alpha^3 = 25/48$ $\alpha^4 = 25/48$
7	5		$\xi_1^1 = 0.33333$ $\xi_1^2 = 0.10128$ $\xi_1^3 = 0.79742$ $\xi_1^4 = \xi_1^2$ $\xi_1^5 = 0.47014$ $\xi_1^6 = \xi_1^5$ $\xi_1^7 = 0.05971$	$\xi_2^1 = \xi_1^1$ $\xi_2^2 = \xi_1^2$ $\xi_2^3 = \xi_1^2$ $\xi_2^4 = \xi_1^3$ $\xi_2^5 = \xi_1^7$ $\xi_2^6 = \xi_1^5$ $\xi_2^7 = \xi_1^5$	$\alpha^1 = 0.22500$ $\alpha^2 = 0.12593$ $\alpha^3 = \alpha^2$ $\alpha^4 = \alpha^2$ $\alpha^5 = 0.13239$ $\alpha^6 = \alpha^5$ $\alpha^7 = \alpha^5$

Table 3.3: GAUSS points ξ_1^i, ξ_2^i and weighting factors α^i of GAUSS-LEGENDRE quadrature

3.7.2 Triangular elements

The prevailing distinction of the numerical integration of triangular elements in comparison with quadrangular elements is the fact that integration boundaries of the second integration contain the natural coordinate ξ_2. The GAUSS-LEGENDRE quadrature

$$\int_0^1 \int_0^{1-\xi_2} f(\xi_1, \xi_2)\, d\xi_1\, d\xi_2 = \frac{1}{2} \sum_{i=1}^{n} \alpha^i f(\xi_1^i, \xi_2^i) \tag{3.191}$$

is carried out according to COWPER [49] with weightings α^i and GAUSS points ξ_1^i, ξ_2^i summed up in table 3.3 (see BATHE [8] and ZIENKIEWICZ & TAYLOR [36]).

Chapter 4

Finite volume elements

The consistent continuation of the presented finite element series from one-dimensional truss elements to plane finite elements eventually comes to an end with the presentation of three-dimensional isoparametric volume elements. The volume elements present the direct discretization of three-dimensional continuum with no additional model assumptions regarding the kinetics or kinematics.

Because of the possibility to carry out the analysis of arbitrarily shaped three-dimensional elastic bodies with, theoretically seen, a relatively simple element formulation one is tempted to take the volume elements as the high point of finite elements. However, this is totally wrong. Inspite of the unsurpassed universal applicability of these elements they have rather lived in the shadows during the last decades of the finite element method development. The only problems that explicitly required a three-dimensional description of geometry or of the loads and were analysed with three-dimensional finite elements in the past were detailed problems of engineering structures. This was mostly due to the fact that for discretization of typical engineering structures such as frames or trusses, plates, or shells a notably larger number of degrees of freedom is/was necessary than in the case of discretization of structures with suitable structural elements such as bars, beams, plates, folded plate structures or shells. For frames or trusses this argument still has its validity inspite of the tremendous increase of available computer performance and memory capacity as well as of parallel computing possibilities. This becomes clear when for example a profiled girder is modelled as a beam element or when it is comparatively discretized with volume elements. The ruling out of limitations, elaborated in elementary bending theory, by three-dimensional discretisizing must be paid with multiple increase in the number of degrees of freedom and hence with increase in computation time and memory capacity.

From todays point of view, the mentioned argument regarding shells that can basically be taken for an additionally curved generalisation of discs, plates and folded plate structures has to be evaluated differently. In contemporary research, the kinematics of shells or plates are described with assistance of volume elements on equal terms. The decisive advantage of substantial simplification of element development is in contrast with the disadvantage of increased numerical effort at numerical integration of stiffness and mass matrix. Compared to discs and plates the disadvantage of increased number of element free values has to be taken into consideration and simultaneously put into perspective with performance of modern computers.

A further advantage that should not be underestimated is the possibility to carry out in a plain way strategies of *Dimension Adaptability* within the scope of volume elements. For example this means that in case of discretization of plates, as opposed to application of plate elements in the vicinity of supports or clamping points, the order of approximation in direction of plate thickness can effortlessly be increased and hence the analysis quality can be upgraded crucially.

The simple formulation of elements and putting into practice of various orders of approximation at various sections of a body or structure, together with ever-growing performance, will help volume elements reach significantly greater importance in the future than the one they possessed in past decades. Even if the development of a volume element is identical down to three-dimensional completion of plane element development, the following chapter should be dedicated to volume elements on the grounds put forward. From basically two different element types, namely

- hexahedron elements
- and tetrahedron elements,

the former one is presented here as an exponent for all three-dimensional finite elements. The 20-noded hexahedron element is developed within the scope of a hierarchical formulation of ansatz functions. Furthermore, a 16-noded special case of a hierarchical element family is separately elaborated regarding the presentable deformation states.

4.1 Fundamental equations of three-dimensional continua

The basis for the development of volume elements is given by fundamental equations of the three-dimensional continuum. These can be taken over from chapter 1 with no additional model assumptions. A repetition of essential aspects of the three-dimensional continuum such as

- formulation of strains as functions of the displacement field (kinematics)
- formulation of forces equilibrium on a differential volume element (kinetics),
- strain-stress relation (constitutive equation),
- Dirichlet and Neumann boundary conditions
- and the weak form of initial boundary value problem (principle of virtual displacements)

is already given in 1 and should therefore not be repeated here.

4.2 Finite element discretization

4.2.1 Natural coordinates

Hexahedron elements are described in the expansion of quadrangular elements, in natural coordinates

$$\xi_j = [-1, 1] \qquad j = 1, 2, 3 \tag{4.1}$$

that are assembled in the natural coordinate vector

$$\boldsymbol{\xi} = \begin{bmatrix} \xi_1 & \xi_2 & \xi_3 \end{bmatrix}^T \tag{4.2}$$

Natural coordinates again make up the master or unit cube of three-dimensional elements on which ansatz functions of arbitrarily shaped hexahedron elements can be formed.

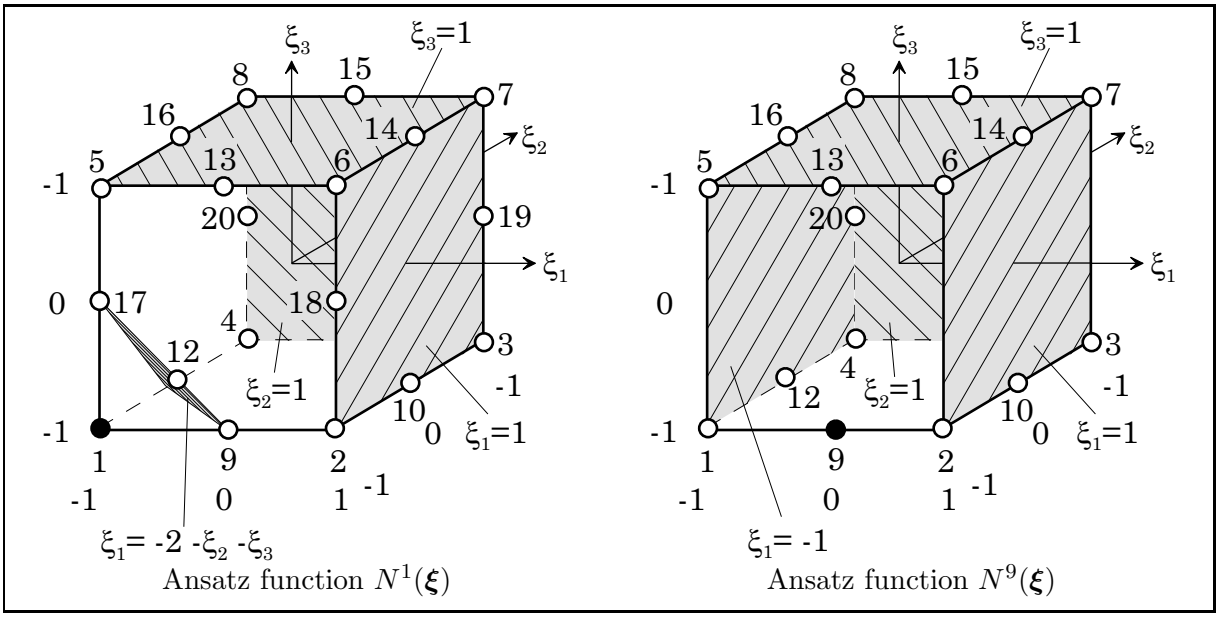


Figure 4.1: Ansatz function generation of a 20-noded volume element

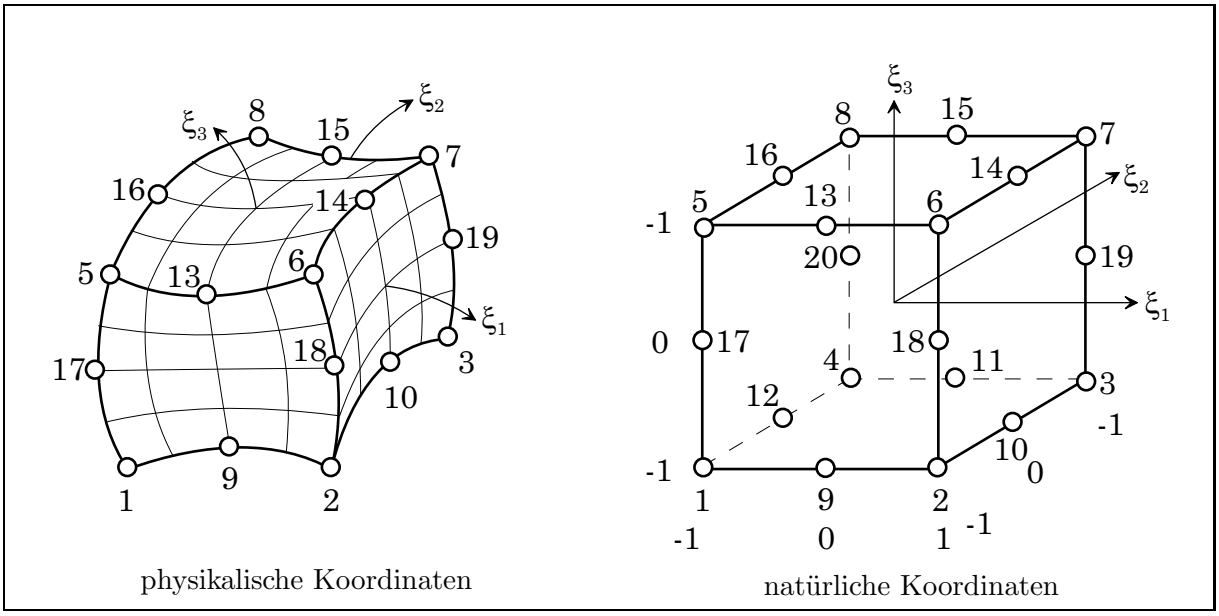


Figure 4.2: Geometry approximation and displacement field of a 20-noded volume element

4.2.2 Ansatz Functions

In analogy with generation of two-dimensional LAGRANGE elements, it is possible also in the case of three-dimensional finite elements to form ansatz functions by multiplication of three one-dimensional ansatz functions dependent on natural coordinates ξ_j for $j = 1, 2, 3$. This yields a trilinear or a triquadratic volume element depending on the ansatz order of basic one-dimensional ansatz functions. As already shown in the case of the biquadratic LAGRANGE element, an internal node is necessary to generate LAGRANGE elements of higher order. In order to be able to omit this node, a family of ansatz functions of a 20-noded finite element shown in figure 4.1 is developed that needs neither a node in the interior of volume element nor one in the centre of elements outer surfaces. These ansatz functions are generated with the help of interpolation properties

$(N^i(\boldsymbol{\xi}^i) = 1,\ i = 1, \cdots, NN$ and $N^i(\boldsymbol{\xi}^j) = 0,\ j \neq i)$. According to figure 4.1, two different basic types of ansatz functions exist : firstly, corner nodes $i = 1, \cdots, 8$ and secondly, middle nodes of edges $i = 9, \cdots, 20$. As representative of both types $N^5(\boldsymbol{\xi})$ and $N^{13}(\boldsymbol{\xi})$ are derived. Ansatz function $N^5(\boldsymbol{\xi})$ has to be equal zero at planes characterised by

$$\begin{aligned} \xi_1 &= 1 & \qquad \xi_3 &= -1 \\ \xi_2 &= 1 & \qquad \xi_1 &= -2 - \xi_2 + \xi_3 \end{aligned} \tag{4.3}$$

in natural coordinate space. With this the condition $N^5(\boldsymbol{\xi}^j) = 0$ for $j \neq 5$ is fulfilled. Thus, the ansatz function is given entirely apart from the constant factor c_1 which still must be determined.

$$N^5(\boldsymbol{\xi}) = c_1\ (1 - \xi_1)(1 - \xi_2)(1 + \xi_3)(2 + \xi_1 + \xi_2 - \xi_3) \tag{4.4}$$

With condition

$$N^5(\boldsymbol{\xi}^5) = 1 = c_1\ (1 + 1)(1 + 1)(1 + 1)(2 - 1 - 1 - 1) = -8\ c_1 \tag{4.5}$$

the unknown constant $c_1 = -1/8$ is obtained. Ansatz function $N^{13}(\boldsymbol{\xi})$ vanishes at planes marked by

$$\begin{aligned} \xi_1 &= -1 & \qquad \xi_2 &= 1 \\ \xi_1 &= 1 & \qquad \xi_3 &= -1 \end{aligned} \tag{4.6}$$

and can so be generated with consideration of interpolation property $N^{13}(\boldsymbol{\xi}^{13}) = 1$.

$$N^{13}(\boldsymbol{\xi}) = c_1\ (1 - \xi_1)(1 + \xi_1)(1 - \xi_2)(1 + \xi_3) = \frac{1}{4}\ (1 - \xi_1^2)(1 - \xi_2)(1 + \xi_3) \tag{4.7}$$

In analogy with this, all other ansatz functions of the 20-noded hexahedron element can be generated.

$$\begin{aligned} N^i(\boldsymbol{\xi}) &= -\frac{1}{8}\ (1 + \xi_1^i\xi_1)(1 + \xi_2^i\xi_2)(1 + \xi_3^i\xi_3)(2 - \xi_1^i\xi_1 - \xi_2^i\xi_2 - \xi_3^i\xi_3) & i &= 1, \cdots, 8 \\ N^i(\boldsymbol{\xi}) &= \frac{1}{4}\ (1 - \xi_1^2)(1 + \xi_2^i\xi_2)(1 + \xi_3^i\xi_3) & i &= 9, 11, 13, 15 \\ N^i(\boldsymbol{\xi}) &= \frac{1}{4}\ (1 + \xi_1^i\xi_1)(1 - \xi_2^2)(1 + \xi_3^i\xi_3) & i &= 10, 12, 14, 16 \\ N^i(\boldsymbol{\xi}) &= \frac{1}{4}\ (1 + \xi_1^i\xi_1)(1 + \xi_2^i\xi_2)(1 - \xi_3^2) & i &= 17, 18, 19, 20 \end{aligned} \tag{4.8}$$

In equation (4.8) ξ_j^i represents the natural coordinates ξ_j for $j = 1, 2, 3$ of element node i. Besides the ansatz functions, as basis of element development, their derivatives with respect to natural coordinates ξ_1, ξ_2 and ξ_3 are needed as well.

$$\begin{aligned} N^i_{;1}(\boldsymbol{\xi}) &= -\frac{1}{8}\ (1 + \xi_2^i\xi_2)(1 + \xi_3^i\xi_3)(1 - 2\xi_1^i\xi_1 - \xi_2^i\xi_2 - \xi_3^i\xi_3)\xi_1^i \quad i = 1, \cdots, 8 \\ N^i_{;2}(\boldsymbol{\xi}) &= -\frac{1}{8}\ (1 + \xi_1^i\xi_1)(1 + \xi_3^i\xi_3)(1 - \xi_1^i\xi_1 - 2\xi_2^i\xi_2 - \xi_3^i\xi_3)\xi_2^i \\ N^i_{;3}(\boldsymbol{\xi}) &= -\frac{1}{8}\ (1 + \xi_1^i\xi_1)(1 + \xi_2^i\xi_2)(1 - \xi_1^i\xi_1 - \xi_2^i\xi_2 - 2\xi_3^i\xi_3)\xi_3^i \end{aligned} \tag{4.9}$$

$$
\begin{aligned}
N^i_{;1}(\boldsymbol{\xi}) &= -\frac{1}{2}\,\xi_1(1+\xi^i_2\xi_2)(1+\xi^i_3\xi_3) \quad i = 9, 11, 13, 15 \\
N^i_{;2}(\boldsymbol{\xi}) &= \frac{1}{4}\,(1-\xi^2_1)\xi^i_2(1+\xi^i_3\xi_3) \\
N^i_{;3}(\boldsymbol{\xi}) &= \frac{1}{4}\,(1-\xi^2_1)(1+\xi^i_2\xi_2)\xi^1_3
\end{aligned}
\tag{4.9}
$$

$$
\begin{aligned}
N^i_{;1}(\boldsymbol{\xi}) &= \frac{1}{4}\,\xi^i_1(1-\xi^2_2)(1+\xi^i_3\xi_3) \quad i = 10, 12, 14, 16 \\
N^i_{;2}(\boldsymbol{\xi}) &= -\frac{1}{2}\,(1+\xi^i_1\xi_1)\xi_2(1+\xi^i_3\xi_3) \\
N^i_{;3}(\boldsymbol{\xi}) &= \frac{1}{4}\,(1+\xi^i_1\xi_1)(1-\xi^2_2)\xi^i_3
\end{aligned}
\tag{4.9}
$$

$$
\begin{aligned}
N^i_{;1}(\boldsymbol{\xi}) &= \frac{1}{4}\,\xi^i_1(1+\xi^i_2\xi_2)(1-\xi^2_3) \quad i = 17, 18, 19, 20 \\
N^i_{;2}(\boldsymbol{\xi}) &= \frac{1}{4}\,(1+\xi^i_1\xi_1)\xi^i_2(1-\xi^2_3) \\
N^i_{;3}(\boldsymbol{\xi}) &= -\frac{1}{2}\,(1+\xi^i_1\xi_1)(1+\xi^i_2\xi_2)\xi_3
\end{aligned}
\tag{4.9}
$$

4.2.3 Discretization

Discretization of position, displacements, variations of displacements and accelerations has to be formally distinguished from corresponding approximations of plane elements only by supplemented spatial third components of element vectors. For an NN-noded hexahedron element, the following approximations are obtained

$$
\begin{aligned}
\boldsymbol{X}(\boldsymbol{\xi}) &\approx \tilde{\boldsymbol{X}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \boldsymbol{X}^e & \boldsymbol{X}^e &= \left[\ X^{e1}_1\ X^{e1}_2\ X^{e1}_3\ \cdots\ X^{eNN}_3\ \right]^T \\
\boldsymbol{u}(\boldsymbol{\xi}) &\approx \tilde{\boldsymbol{u}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \boldsymbol{u}^e & \boldsymbol{u}^e &= \left[\ u^{e1}_1\ u^{e1}_2\ u^{e1}_3\ \cdots\ u^{eNN}_3\ \right]^T \\
\delta\boldsymbol{u}(\boldsymbol{\xi}) &\approx \delta\tilde{\boldsymbol{u}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \delta\boldsymbol{u}^e & \delta\boldsymbol{u}^e &= \left[\ \delta u^{e1}_1\ \delta u^{e1}_2\ \delta u^{e1}_3\ \cdots\ \delta u^{eNN}_3\ \right]^T \\
\ddot{\boldsymbol{u}}(\boldsymbol{\xi}) &\approx \tilde{\ddot{\boldsymbol{u}}}(\boldsymbol{\xi}) = \mathbf{N}(\boldsymbol{\xi})\ \ddot{\boldsymbol{u}}^e & \ddot{\boldsymbol{u}}^e &= \left[\ \ddot{u}^{e1}_1\ \ddot{u}^{e1}_2\ \ddot{u}^{e1}_3\ \cdots\ \ddot{u}^{eNN}_3\ \right]^T
\end{aligned}
\tag{4.10}
$$

with definition of $(3 \times 3NN)$ ansatz functions matrix $\mathbf{N}^i(\boldsymbol{\xi})$.

$$
\mathbf{N}^i(\boldsymbol{\xi}) = \begin{bmatrix} N^i(\boldsymbol{\xi}) & 0 & 0 \\ 0 & N^i(\boldsymbol{\xi}) & 0 \\ 0 & 0 & N^i(\boldsymbol{\xi}) \end{bmatrix} \qquad \mathbf{N}(\boldsymbol{\xi}) = \left[\ \mathbf{N}^1(\boldsymbol{\xi})\ \mathbf{N}^2(\boldsymbol{\xi})\ \cdots\ \mathbf{N}^{NN}(\boldsymbol{\xi})\ \right]
\tag{4.11}
$$

4.2.4 Jacobi transformation

Transformation of derivatives with respect to physical coordinates on the one hand and with respect to natural coordinates on the other hand can be carried out by means of JACOBI matrix.

To this transformation matrix, terms must be added corresponding to coordinates X_3 and ξ_3. The proceeding to generate the JACOBI matrix is in analogy with chapter 3.3.3,

$$\begin{bmatrix} \frac{\partial}{\partial \xi_1} \\ \frac{\partial}{\partial \xi_2} \\ \frac{\partial}{\partial \xi_3} \end{bmatrix} = \begin{bmatrix} \frac{\partial X_1}{\partial \xi_1} & \frac{\partial X_2}{\partial \xi_1} & \frac{\partial X_3}{\partial \xi_1} \\ \frac{\partial X_1}{\partial \xi_2} & \frac{\partial X_2}{\partial \xi_2} & \frac{\partial X_3}{\partial \xi_2} \\ \frac{\partial X_1}{\partial \xi_3} & \frac{\partial X_2}{\partial \xi_3} & \frac{\partial X_3}{\partial \xi_3} \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial X_1} \\ \frac{\partial}{\partial X_2} \\ \frac{\partial}{\partial X_3} \end{bmatrix} \qquad \frac{\partial}{\partial \boldsymbol{\xi}} = \mathbf{J}(\boldsymbol{\xi}) \, \frac{\partial}{\partial \boldsymbol{X}} \tag{4.12}$$

where the derivatives of physical coordinates with respect to natural coordinates can formally be generated as in equation (3.50). The rule to find derivatives of functions in natural coordinates with respect to physical coordinates can be found by observing the inverse mapping.

$$\begin{bmatrix} \frac{\partial}{\partial X_1} \\ \frac{\partial}{\partial X_2} \\ \frac{\partial}{\partial X_3} \end{bmatrix} = \begin{bmatrix} \frac{\partial \xi_1}{\partial X_1} & \frac{\partial \xi_2}{\partial X_1} & \frac{\partial \xi_3}{\partial X_1} \\ \frac{\partial \xi_1}{\partial X_2} & \frac{\partial \xi_2}{\partial X_2} & \frac{\partial \xi_3}{\partial X_2} \\ \frac{\partial \xi_1}{\partial X_3} & \frac{\partial \xi_2}{\partial X_3} & \frac{\partial \xi_3}{\partial X_3} \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial \xi_1} \\ \frac{\partial}{\partial \xi_2} \\ \frac{\partial}{\partial \xi_2} \end{bmatrix} \qquad \frac{\partial}{\partial \boldsymbol{X}} = \mathbf{J}^{-1}(\boldsymbol{\xi}) \frac{\partial}{\partial \boldsymbol{\xi}} \tag{4.13}$$

For purposes of inversion of JACOBI matrix and generation of JACOBI determinant, $|\mathbf{J}|$ one should refer to mathematical literature (e.g. BRONSTEIN & SEMENDJAJEW [48] or ZURMÜHL & FALK [82]). With the help of the JACOBI determinant, the volume element dV in physical coordinates is transformed into volume element in natural coordinates.

$$dV = |\mathbf{J}(\boldsymbol{\xi})| \; d\xi_1 \; d\xi_2 \; d\xi_3 \tag{4.14}$$

In order to generate consistent equivalent loads, of external loads further transformations of volume element surfaces and transformations of line elements, that is, element edges are compulsory. These transformation relations can be generated by expansion of transformation relations derived for surface elements (equations (3.49,3.58) (3.83)) and by an appropriate choice of parameter ξ_j. In the scope of this lecture, we should concentrate on the generation of element matrices and most of all on the presentation of possibilities of a hierarchical element and not take into consideration all possible transformations and load applications.

4.2.5 Differential Operator $\mathbf{B}(\boldsymbol{\xi})$

Differential operator $\mathbf{B}(\boldsymbol{\xi})$ is generated by applying the three-dimensional differential operator $\mathbf{D}_{\varepsilon\xi}(\boldsymbol{\xi})$ in natural coordinates to the matrix of ansatz functions. In the first development step, the derivation rules with respect to physical coordinates $\partial/\partial X_j$ for $j = 1, 2, 3$, that are in the differential operator $\mathbf{D}_\varepsilon$ of a three-dimensional continuum (equation (1.16)), have to be replaced by derivatives with respect to natural coordinates $\partial/\partial \xi_j$ with the help of the inverse JACOBI

transformation (4.13). To interpret the resulting differential operator

$$\mathbf{D}_{\varepsilon\xi}(\boldsymbol{\xi}) = \begin{bmatrix} \frac{\partial \xi_j}{\partial X_1}\frac{\partial}{\partial \xi_j} & 0 & 0 \\ 0 & \frac{\partial \xi_j}{\partial X_2}\frac{\partial}{\partial \xi_j} & 0 \\ 0 & 0 & \frac{\partial \xi_j}{\partial X_3}\frac{\partial}{\partial \xi_j} \\ \frac{\partial \xi_j}{\partial X_2}\frac{\partial}{\partial \xi_j} & \frac{\partial \xi_j}{\partial X_1}\frac{\partial}{\partial \xi_j} & 0 \\ 0 & \frac{\partial \xi_j}{\partial X_3}\frac{\partial}{\partial \xi_j} & \frac{\partial \xi_j}{\partial X_2}\frac{\partial}{\partial \xi_j} \\ \frac{\partial \xi_j}{\partial X_3}\frac{\partial}{\partial \xi_j} & 0 & \frac{\partial \xi_j}{\partial X_1}\frac{\partial}{\partial \xi_j} \end{bmatrix} \tag{4.15}$$

sum convention for components with $j = 1, 2, 3$ must be noted. Application of this operator to the ansatz function matrix $\mathbf{N}^i(\boldsymbol{\xi})$ related to node i yields the B-operator $\mathbf{B}_i(\boldsymbol{\xi})$ of a volume element

$$\mathbf{B}_i(\boldsymbol{\xi}) = \mathbf{D}_{\varepsilon\xi}(\boldsymbol{\xi})\mathbf{N}^i(\boldsymbol{\xi}) = \begin{bmatrix} \frac{\partial \xi_j}{\partial X_1}N^i_{;j}(\boldsymbol{\xi}) & 0 & 0 \\ 0 & \frac{\partial \xi_j}{\partial X_2}N^i_{;j}(\boldsymbol{\xi}) & 0 \\ 0 & 0 & \frac{\partial \xi_j}{\partial X_3}N^i_{;j}(\boldsymbol{\xi}) \\ \frac{\partial \xi_j}{\partial X_2}N^i_{;j}(\boldsymbol{\xi}) & \frac{\partial \xi_j}{\partial X_1}N^i_{;j}(\boldsymbol{\xi}) & 0 \\ 0 & \frac{\partial \xi_j}{\partial X_3}N^i_{;j}(\boldsymbol{\xi}) & \frac{\partial \xi_j}{\partial X_2}N^i_{;j}(\boldsymbol{\xi}) \\ \frac{\partial \xi_j}{\partial X_3}N^i_{;j}(\boldsymbol{\xi}) & 0 & \frac{\partial \xi_j}{\partial X_1}N^i_{;j}(\boldsymbol{\xi}) \end{bmatrix} \tag{4.16}$$

with the help of which we can form the displacement vector $\boldsymbol{u}^e$ as part of approximated strain vector $\tilde{\boldsymbol{\varepsilon}}(\boldsymbol{\xi})$.

$$\tilde{\boldsymbol{\varepsilon}}(\boldsymbol{\xi}) = \mathbf{B}(\boldsymbol{\xi})\ \boldsymbol{u}^e \qquad\qquad \mathbf{B} = \begin{bmatrix} \mathbf{B}_1 & \mathbf{B}_2 & \cdots & \mathbf{B}_{NN} \end{bmatrix} \tag{4.17}$$

4.2.6 Element Matrices

Discretization of internal virtual work according to (1.96) yields the stiffness matrix of a hierarchical hexahedron element after applying the material law (1.62), the approximation of strains according to equation (4.17) and the transformation of a differential volume element dV according to (4.14).

$$\mathbf{k}^e = \int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} \mathbf{B}^T(\boldsymbol{\xi})\ \mathbf{C}\ \mathbf{B}(\boldsymbol{\xi})\ |\mathbf{J}(\boldsymbol{\xi})|\ d\xi_1 d\xi_2 d\xi_3 \tag{4.18}$$

Mass matrix of a hexahedron element is the result of virtual work of inertial forces determined according to equation (1.96) and of displacement field discretization and acceleration variation

according to (4.10), having in mind the volume element transformation dV according to equation (4.14).

$$\mathbf{m}^e = \int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} \mathbf{N}^T(\boldsymbol{\xi})\, \mathbf{N}(\boldsymbol{\xi})\, |\mathbf{J}(\boldsymbol{\xi})|\, \rho\, d\xi_1 d\xi_2 d\xi_3 \tag{4.19}$$

Stiffness matrix as well as mass matrix are integrated by means of triple application of GAUSS-LEGENDRE quadrature with GAUSS points and weighting factors according to table 2.2.

4.2.7 Element Vectors

Consistent equivalent loads of volume loads on the one hand replace volume loads and edge loads on the other hand. Equivalent load of a volume load is obtained by discretization of volume integral of external virtual work.

$$\boldsymbol{r}_p^e = \int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} \mathbf{N}^T(\boldsymbol{\xi})\, \boldsymbol{b}(\boldsymbol{\xi})\, |\mathbf{J}(\boldsymbol{\xi})|\, \rho\, d\xi_1 d\xi_2 d\xi_3 \tag{4.20}$$

Discretization of virtual work of edge loads, which in case of volume elements can be line loads on edges or surface loads on element surfaces, gives the load vector of an element $\boldsymbol{r}_n^e$.

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